USER MANUAL

ADU Manual

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Software version: 120.2.2 or later
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ECUMASTER

Contents

1. Ecumaster ADU..... 8

1.1. Characteristics..... 9

1.2. Technical drawings..... 11

1.3. Device description..... 18

1.4. Connector..... 19

1.5. Description of the ADU AS socket 21

1.6. Description of the ADU socket..... 22

2. Installation..... 24

2.1. CAN bus..... 25

2.2. Connecting to ECU..... 27

2.3. Connecting via a CAN bus..... 27

2.4. Connecting via an RS232 serial bus..... 28

2.5. OBD 2..... 30

2.6. GPS module..... 31

2.7. Ecumaster PMU..... 35

2.8. Digital Inputs..... 36

2.9. Analog Inputs..... 37

2.10. USB Flash drive (pendrive)..... 38

2.11. Low side outputs..... 39

3. ADU Client software for Windows..... 39

3.1. First connection with the device..... 40

3.2. Firmware update..... 43

3.3. Loading a project..... 43

3.4. Appearance of the application..... 46

4. Status field..... 58

5. Switching the CAN adapter between ADU, PMU and Light Client programs..... 60

6. Pages..... 61

6.1. Default settings pg_page1..... 61

6.2. Creating a page..... 64

6.3. Page elements..... 66

6.4. Adding elements to pages..... 68

6.5. Grouping objects..... 78

6.6. Toggling between pages..... 81

6.7. Startup screen..... 83

7. Graphic objects..... 84

7.1. Gauge..... 84

7.2. Classic gauge..... 86

7.3. Bar graph..... 88

7.4. Simple indicator..... 91

7.5. Text..... 92

7.6. Time..... 95

7.7. Image..... 96

7.8. RPM Bar..... 97

7.9. Gear indicator..... 98

7.10. G-Force..... 99

7.11. Predictive time graph..... 99

7.12. Tire temperature graph..... 100

7.13. Temperature color scale..... 101

7.14. Session results..... 102

7.15. Track record table..... 102

7.16. Rectangle..... 103

7.17. Line..... 104

7.18. Circle..... 106
7.19. Grid..... 107
7.20. Textures..... 108
8. Inputs..... 111
8.1. Analog Inputs..... 111
8.2. Digital Inputs..... 114
9. Outputs..... 118
9.1. *Low side* outputs..... 118
9.2. Analog output..... 118
10. Working with CAN buses in ADU..... 120
10.1. Using pre-defined streams from *.CANX* and *.DBC* files..... 120
10.2. Built-in support for PMU..... 122
10.3. Own CAN streams – CANbus Message Object..... 124
10.4. Own CAN streams – CANbus Input..... 127
10.5. Own CAN streams – saving to a .CANX file..... 130
10.6. Sending frames using the CAN bus (CANbus Export) 130
10.7. Reserved CAN ID 134
11. CAN bus keypad support..... 134
12. Enumeration..... 140
13. Processing information in the ADU..... 146
13.1. Timers – counting time..... 146
13.2. Tables – 2D / 3D lookup tables..... 148
13.3. Switches virtual switches, counters..... 150
13.4. Numbers – mathematical channels..... 152
13.5. Functions – logical functions..... 160
14. Alarms..... 165
15. Logging channels..... 168

16. Logging to a USB storage device..... 170

17. Permanent meters.....172

17.1. Resetting / changing meter status..... 174

18. Min / max values for ECU channels.....175

19. Panels..... 176

19.1. Buttons.....176

19.2. Shift light.....177

19.3. User lights.....181

19.4. User tracks.....182

19.5. User track defined by the button..... 183

19.6. Fuel level filter..... 185

19.7. OBD 2 187

19.8. J1939..... 188

19.9. Outputs.....193

19.10. Autobrightness..... 194

19.11. Virtual fuel tank 194

19.12. Configuration..... 197

19.12.1. Brake Bias.....202

19.12.2. Delayed turning off.....202

19.13. Protection..... 203

19.14. Log..... 204

19.15. CANbus / Serial Setup.....204

19.16. Autosaved channels..... 206

19.17. Channel Simulator.....206

19.18. Channel report.....206

20. Timing.....207

20.1. Configuration of time measurement with a beacon.....208

20.2. Configuration of time measurement with a GPS....208
21. Data analysis....209
22. Appendix A - List of built-in tracks....218
23. Appendix B - How-to Configure the Hill Climb....220
23.1. Description....220
23.2. Document history....224
24. Appendix C - How to Configure PMU CAN Stream....225
24.1. Description....225
24.2. Document history....228
25. Appendix D - How-to Merge Projects....229
25.1. Description....229
25.2. Resolving potential errors....230
25.3. Document history....232
26. Appendix E - How-to Set Up Keyboards with Encoders....233
26.1. Description....233
26.2. Example....236
26.3. Document history....238
27. Appendix F - How-to Use Virtual Fuel Tank....239
27.1. Description....239
27.2. Example....243
27.3. Document history....249
28. Appendix G - How-to Configure Autosaved Channels....250
28.1. Description....250
28.2. Document history....252
29. Appendix H - How-to Log Data to a USB drive....253
29.1. Introduction....253
29.2. Wiring....253

29.3. Logging Configuration & Management..... 255
29.4. Preparing a USB drive.....257
29.5. Reviewing and Exporting Logs.....259
29.6. Troubleshooting..... 259
29.7. Document history.....261
30. Document history.....262

1. Ecumaster ADU

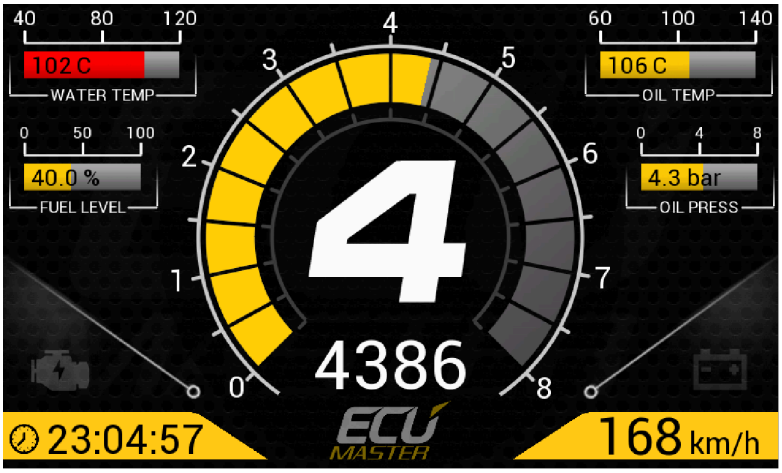
ECUMASTER ADU (Advanced Display Unit) is an universal motorsport display. Unlike other similar devices on the market, this ADU delivers very high adaptability of user displayed information and a flexible input processing system. Featuring two CAN buses, the display is able to easily communicate with other devices (e.g. ECU, GPS, ABS, etc.), and it allows external sensors to be connected to eight analog inputs (e.g. pressure sensors, temperature sensors) and eight digital inputs (e.g. frequency sensors, beacons etc.).

Furthermore, the device can transmit information about the condition of the vehicle via a set of 15 RGB LEDs, fully controlled by the driver (e.g. information about low oil level, progressive shift light, etc.).

To improve user comfort in various lighting conditions, a high-quality LCD display has been used with a brightens of 600 cd/m^2 (for the 5” display) and 1000 cd/m^2 (for the 7” display). To reduce reflections and glare, the display is coated with an anti-reflective layer. To adapt to changing light conditions, a light intensity sensor is mounted on the front of the housing to automatically adjust the brightness of the display to the surroundings.

The ADU can also be used as a central data logger. All information collected by the display can be saved on a USB storage device (Flashdrive, Pendrive) with speeds of up to 500 Hz/channel. The dates and times of the files are provided by a built-in real-time clock. Data saved to the USB storage device can then be analysed using ADU Client software or Ecumaster DATA MASTER software.

The device also allows track lap times to be recorded (using an external GPS), it features predictive timing and facilitates subsequent analysis of times across different sectors using data analysis software.



1.1. Characteristics

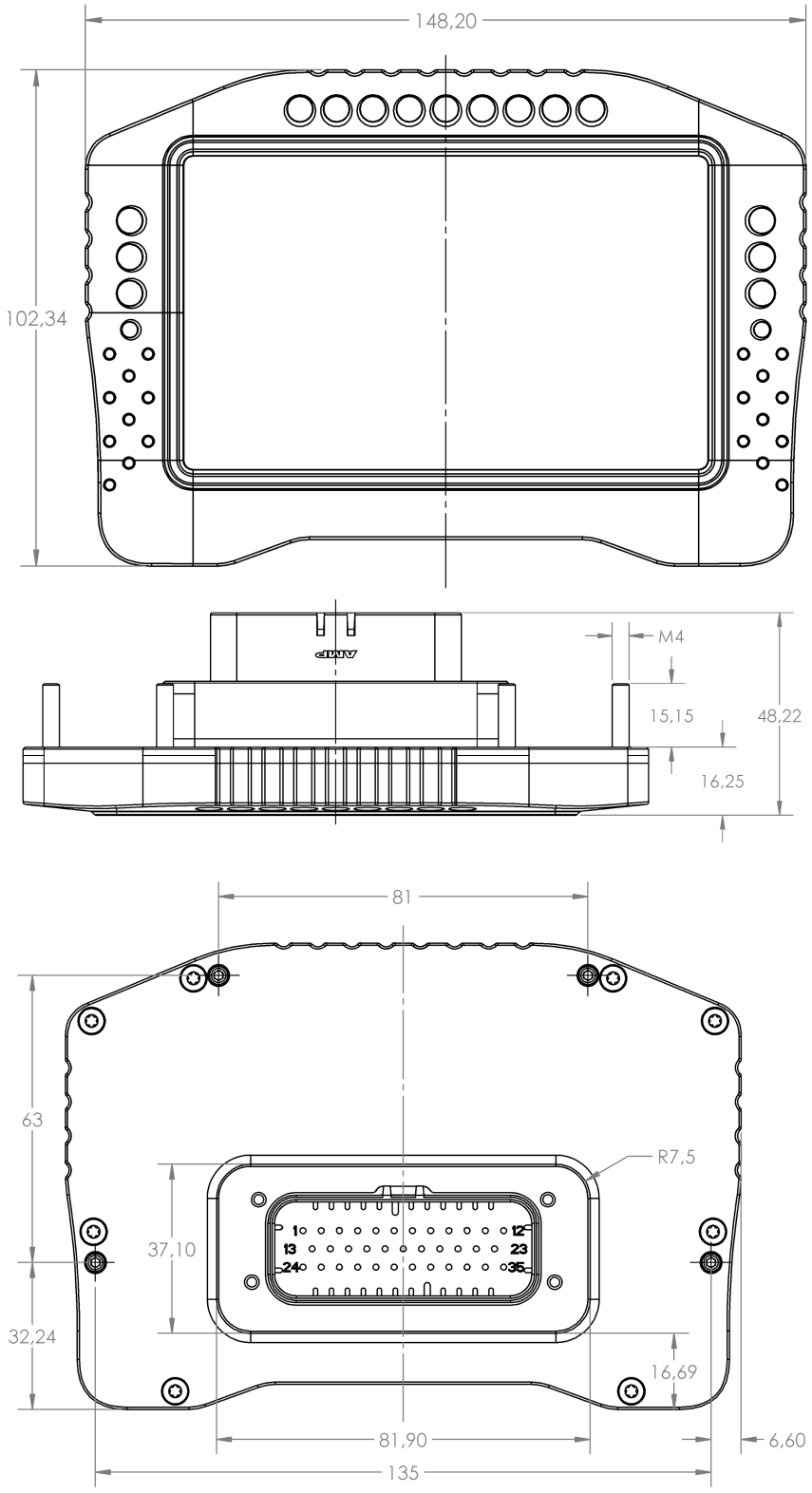
General	
Working temperature range	ACEQ100 (-40 – 85°C)
CPU	32 bit, *automotive*, 90 MIPS
Reverse polarity protection	Yes, internal
Working voltage range	6– 22 V, robust supply protection compliant with ISO 7637
Housing	Anodized aluminium, CNC-manufactured
Tightness class	IP60 for ADU 5/7 Rev.1 (manufactured 2018-2020) IP65 for ADU 5/7 Rev.2 (manufactured from 2021-) IP65 for ADU 5" AS (manufactured from 2025-) IP60 for ADU 7" AS Rev.1 (manufactured from 2018-2025) IP65 for ADU 7" AS Rev.2 (manufactured from 2025-)
Connector	1 x 35 AMP *automotive*
Communication with PC	Via CAN bus. Required USB - CAN interface: • ECUMASTER USBtoCAN • Peak PCAN-USB or Peak PCAN-LAN, • Kvaser
Display type	TFT 800x600
Display brightness	5” - 600 cd/m^2, 7” - 1000 cd/m^2
Inputs / Outputs	
Analog inputs	8 inputs, 10 bit resolution, 0-5 V, (programmed) switched pull-up / pull-down 4K7 resistor
Digital inputs	8 digital inputs, programmed input sensitivity (VR, Hall), switched internal pull-up 4K7 resistors used for engine speed sensors, Flex Fuel, wheel speeds, turbocharger rotor speed
Outputs	2 low side type outputs (switch to ground), up to 2 A current limit PWM support (10 Hz – 1000 Hz)
+5V output	500 mA, monitored
Communication	
CAN Interface	2 x CAN2.0 A/B, 125, 250, 500 (default for CAN2), 1000 kbps (default for CAN1)

CAN termination	CAN1 - none, CAN2 - software-selectable
CAN streams	User-defined
Serial connector	RS232 Rx/Tx Protocols: AiM, Ecumaster, Hondata Kpro, AEM, GEMS, Athena GET, AUTRONIC, MoTec M4 DATA SET 5
USB	For logging to external USB storage device
Other	
Light Emitting Diodes	15 very bright RGB LEDs
Accelerometer / Gyroscope	3D accelerometer + 3D gyroscope for vehicle dynamics analysis
Real-time clock	Yes, supported by a built-in battery
Light sensor	Yes, for automatic correction of brightness
Temperature sensor	Yes, for monitoring device temperature
Keyboard support	Yes, Ecumaster, Grayhill, MoTeC / RaceGrade, Emtron

1.2. Technical drawings

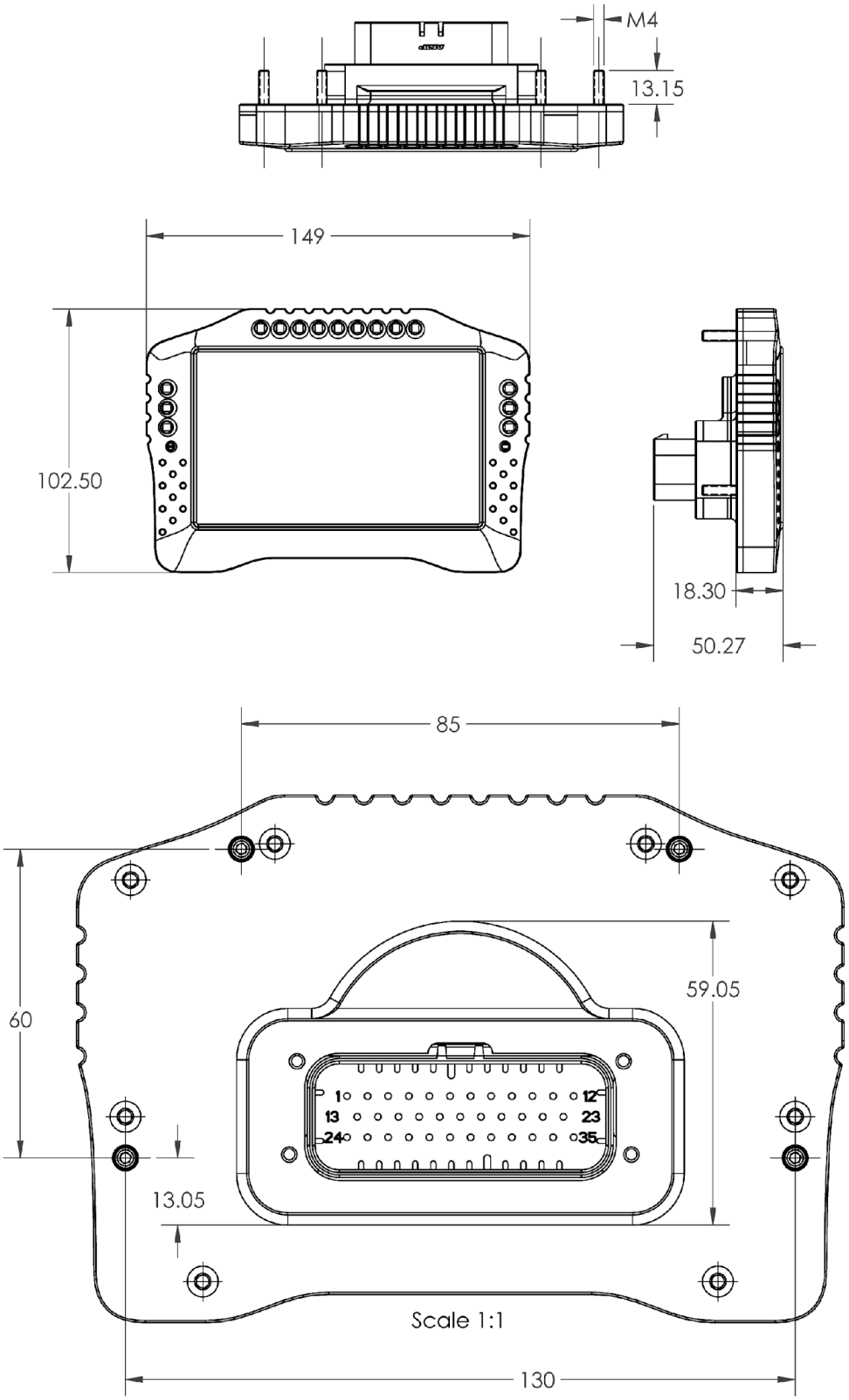
ADU 5 Rev.2

(all dimensions in mm)



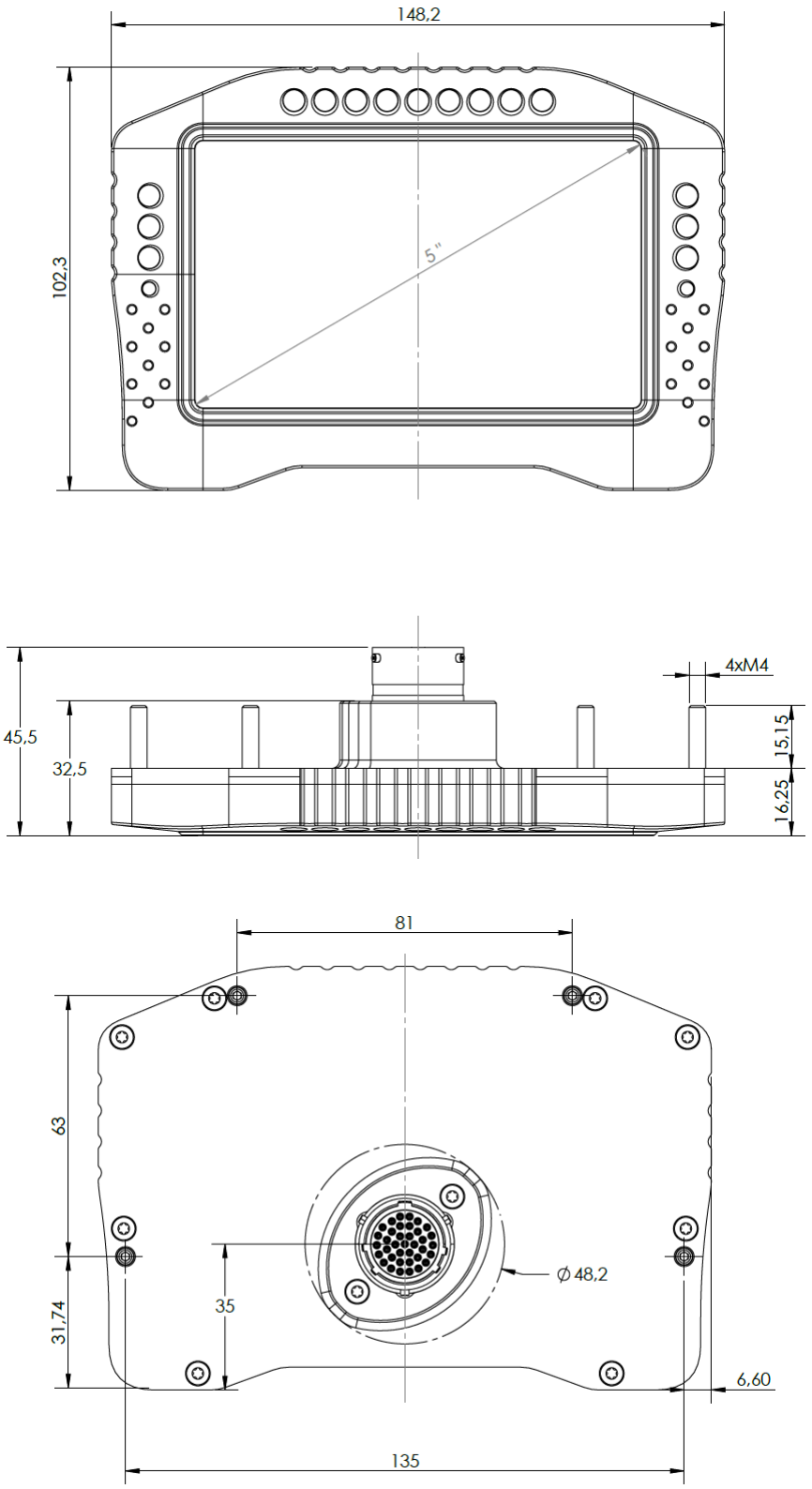
ADU 5 Rev.1 (manufactured 2018-2020, currently out of production)

(all dimensions in mm)



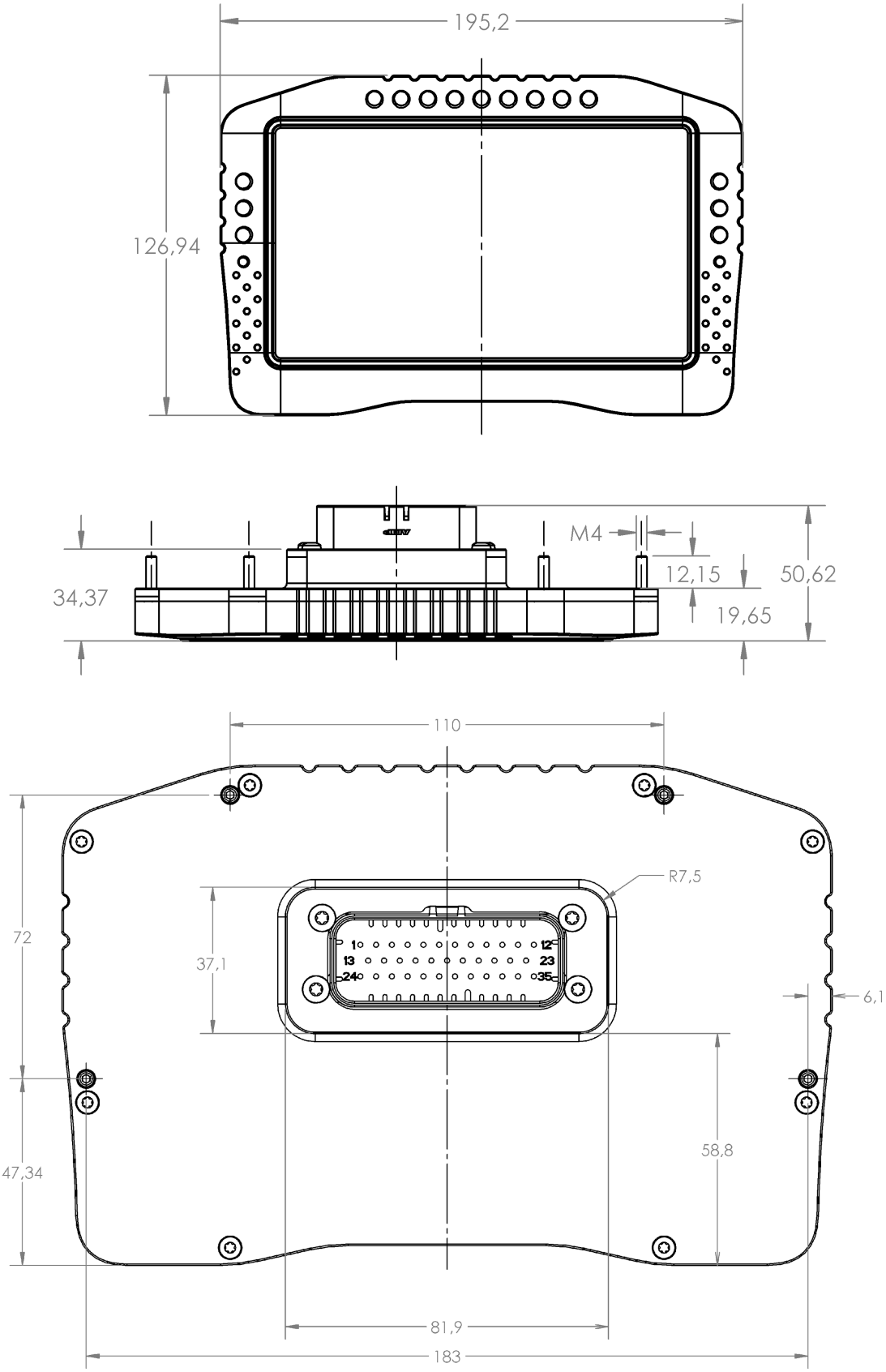
ADU 5 AS

(all dimensions in mm)



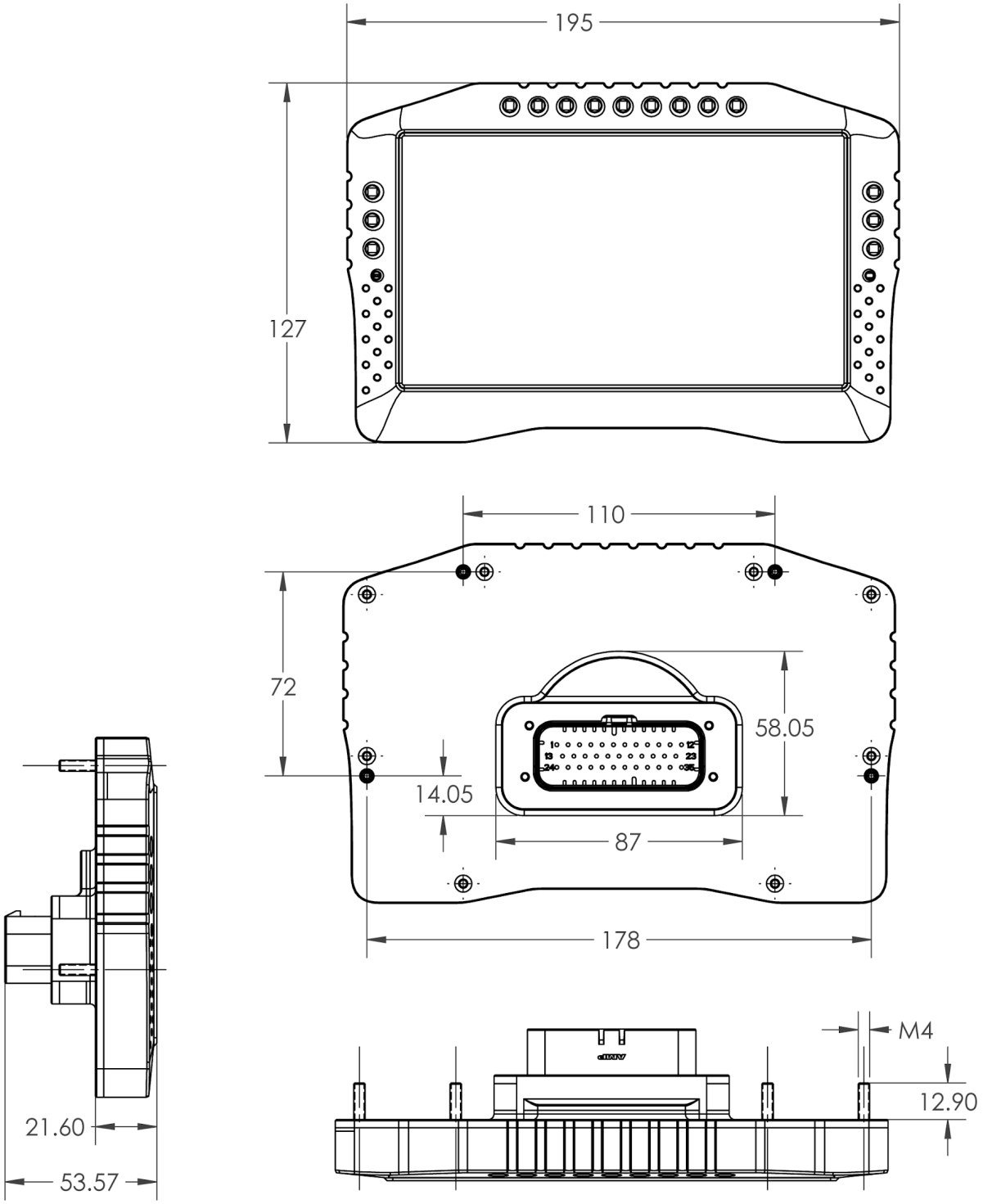
ADU 7 Rev.2

(all dimensions in mm)



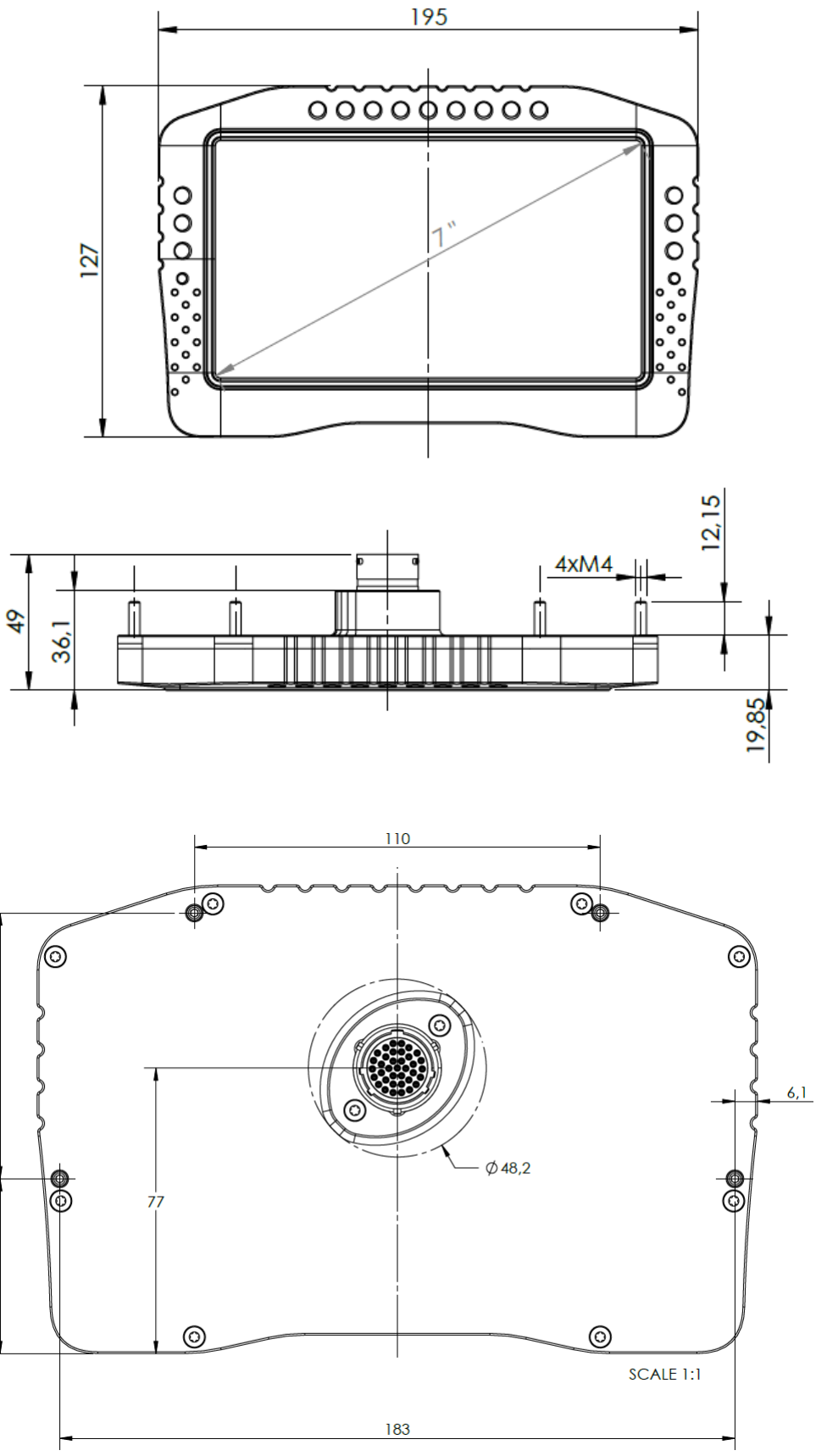
ADU 7 Rev.1 (manufactured 2018-2020, currently out of production)

(all dimensions in mm)



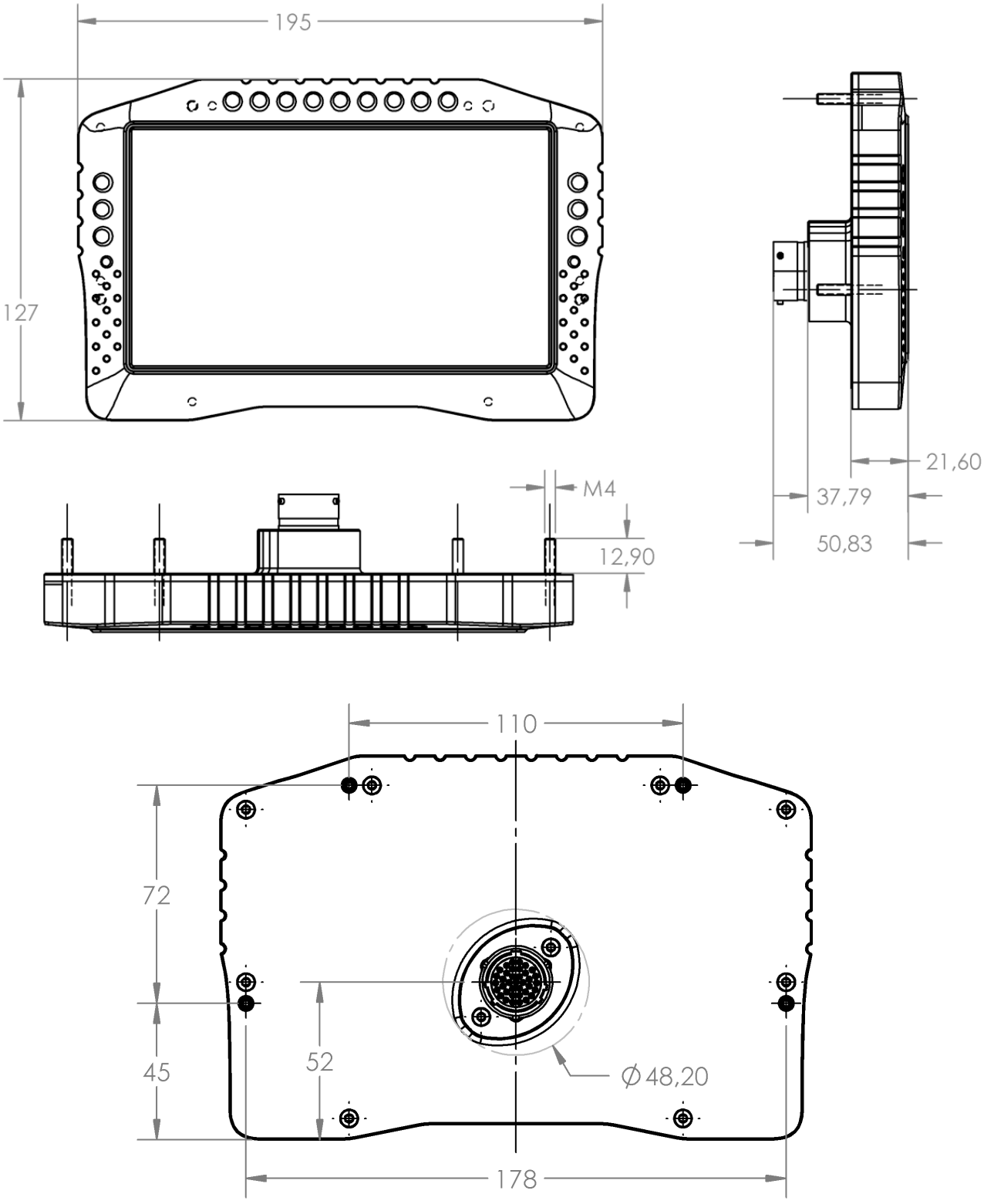
ADU 7 AS Rev.2

(all dimensions in mm)



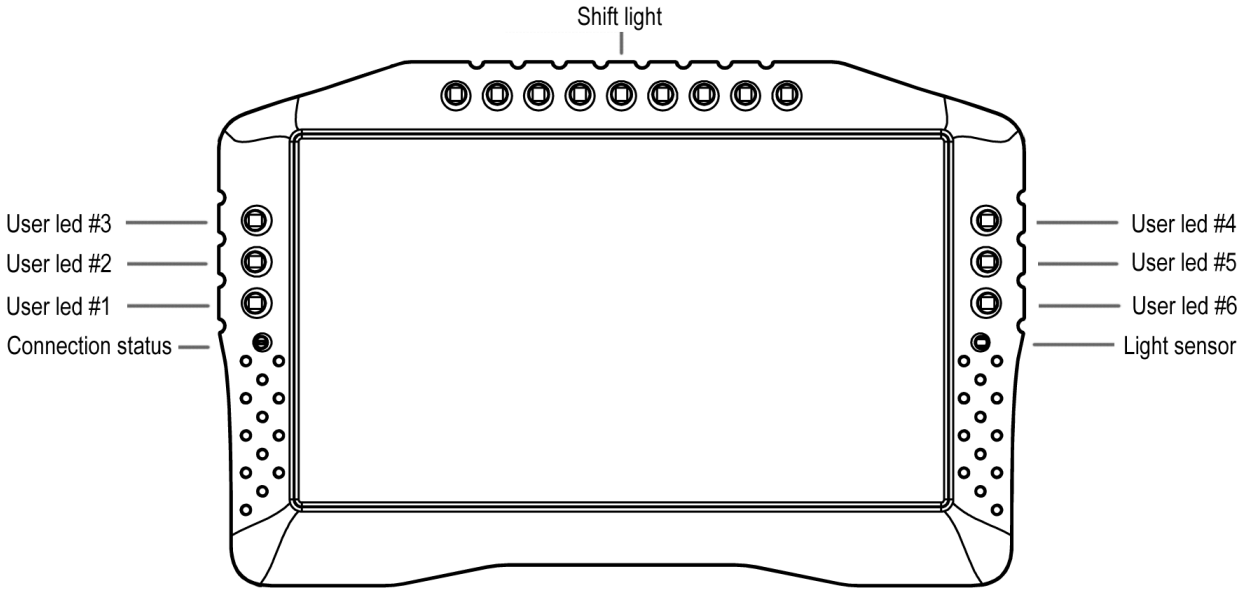
ADU 7 AS Rev.1 (manufactured 2018-2025, currently out of production)

(all dimensions in mm)



1.3. Device description

Front view

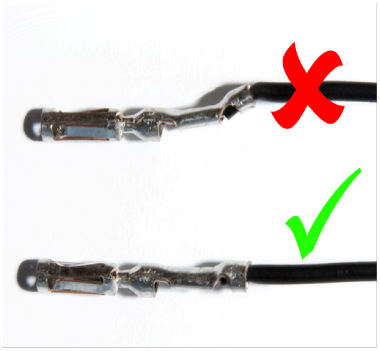


Position	Description
Connection status	LED showing communication with a PC Flashes green when communicating.
Light sensor	The light sensor is used for automatic control of display and light-emitting diodes brightness
User LED #1– #6	RGB LEDs that can be controlled by user functions (e.g. alarms, indicators, etc.).
Shift light	Gear change indicator (user configurable)

1.4. Connector

The display features a 35-pin AMPSEAL connector located at the rear of the housing. It is used to connect the power supply, CAN buses and additional sensors or buttons.
A complementary plug and terminals are included. Use the appropriate Crimper Tool to crimp the terminal's clamp around the cable. We do not recommend soldering the wire to the terminal!
Compatible terminals: AMP 770520-1.

Step 1 – crimp the terminals



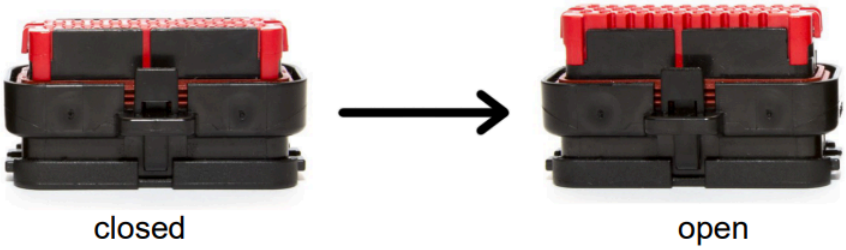
Correct crimping of the terminal's clamp is crucial to avoid problems when placing the terminal in a plug and its subsequent locking. When crimping, do not use excessive force and avoid deforming the terminal. The clamp part of the terminal which holds the cable's insulation should be straight, smooth and rounded and have a diameter equal to or smaller than the head of the terminal.

Step 2 – prepare the plug

To place the terminals into the plug, you must unlock the red part (the lock) and set it to open position. To unlock the red part of the plug, pry the black teeth on both sides of the plug using a sharp tool (Picture 1), and gently pull the lock. It should eject for about 0.5 cm (Picture 2). The teeth prevent total removal of the red lock from the plug (do not disassemble completely the plug!). In this position the plug is open and terminals can be plugged in.

Picture 1

Picture 2



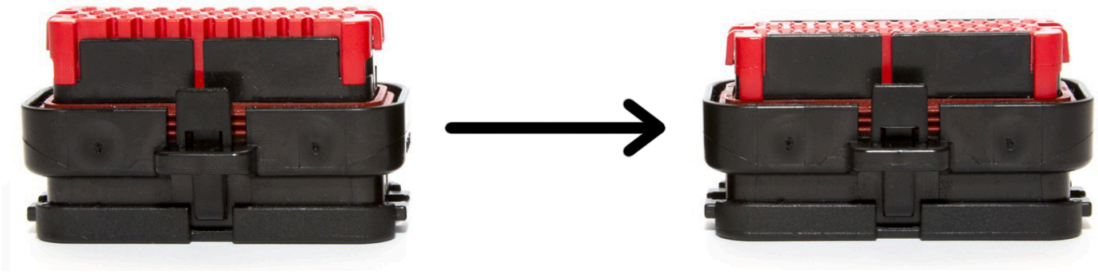
Step 3 – plug in the terminals

To place a terminal into the plug, insert a terminal until you hear and feel a click (the terminal should easily pinch through the silicon gasket).

> **Warning:**
> Do not force the terminal forward after you hear and feel the click – if it goes too deep, it will not be possible to close the red lock

Step 4 – close the lock

After all the terminals are correctly inserted, set the red lock to close position.



open closed

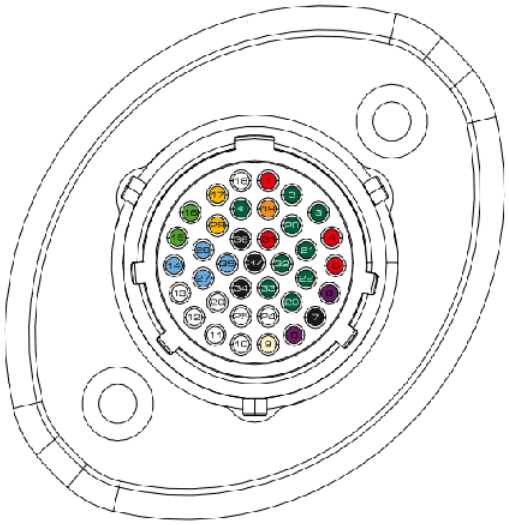
After the plug is locked inspect is any of the terminals don't stick out of the red lock, the tips of the terminals should be aligned with the surface of the lock.

Removal of the terminal

To remove a terminal from the plug set the lock to open position (check Step 2). Then grasp the cable close to the plug, turn it left about half a turn, then turn it right while pulling gently outside.

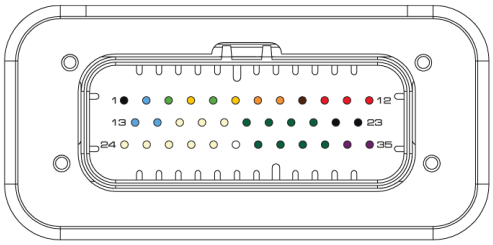
Video reference presenting the correct installation of the terminals in the AMPSEAL connector: www.youtube.com/watch?v=uXTkm_XV2OY

1.5. Description of the ADU AS socket



1	Battery 12V	14	USB.VBUS	27	USB.DM
2	Frequency/ digital in D1	15	CAN1.H	28	USB.DP
3	Frequency/ digital in D2	16	CAN2.H	29	CAN1.L
4	+5V output	17	CAN2.L	30	RS232.TXD
5	+5V output	18	Analog output	31	Switched 12V
6	AUX2	19	RS232.RXD	32	Frequency/ digital in D6
7	Power ground	20	Frequency/ digital in D3	33	Frequency/ digital in D8
8	AUX1	21	Frequency/ digital in D4	34	Sensor ground
9	Analog in A2	22	Frequency/ digital in D5	35	USB.GND
10	Analog in A1	23	Frequency/ digital in D7	36	Sensor ground
11	Analog in A3	24	Analog in A4	37	Ground
12	Analog in A5	25	Analog in A6		
13	Analog in A7	26	Analog in A8		

1.6. Description of the ADU socket



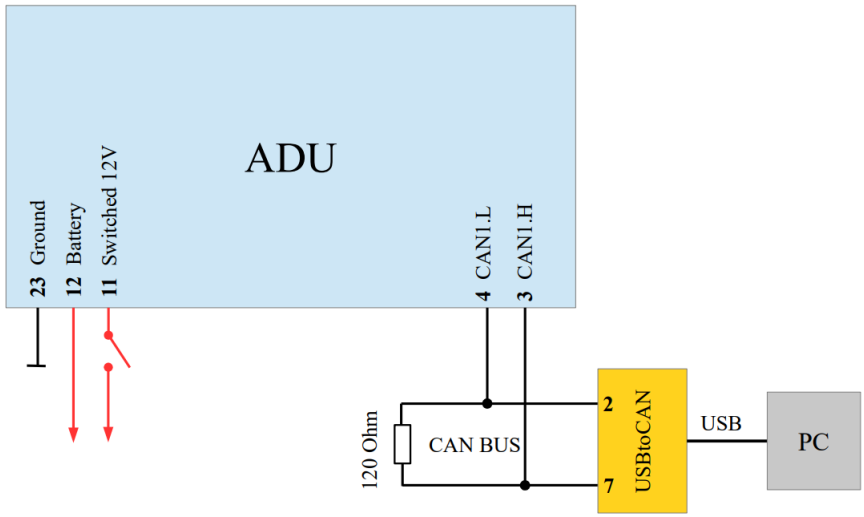
1	USB.GND	13	USB.DM	24	Analog in 8
2	USB.VBUS	14	USB.DP	25	Analog in 7
3	CAN1.H	15	Analog in 5	26	Analog in 6
4	CAN1.L	16	Analog in 3	27	Analog in 4
5	CAN2.H	17	Analog in 1	28	Analog in 2
6	CAN2.L	18	Frequency/ digital in 8	29	Analog output
7	RS232.RXD	19	Frequency/ digital in 6	30	Frequency/ digital in 7
8	RS232.TXD	20	Frequency/ digital in 4	31	Frequency/ digital in 5
9	Sensor ground	21	Frequency/ digital in 2	32	Frequency/ digital in 3
10	+5V output	22	Power ground	33	Frequency/ digital in 1
11	Switched 12V	23	Ground	34	Aux 1
12	Battery 12V			35	Aux 2

Terminal	Description
1. USB.GND	Ground cable for the USB port
2. USB.VBUS	VBUS signal for the USB port
3. CAN1.H	CAN H signal for CAN bus 1
4. CAN1.L	CAN L signal for CAN bus 1
5. CAN2.H	CAN H signal for CAN bus 2
6. CAN2.L	CAN L signal for CAN bus 2
7. RS232.RXD	RXD (reception) signal for serial RS232 bus Used for serial connection to EMU Classic, Classic Hondata, Autronic SM4, AEM, GEMS, Athena GE and other devices supporting the AIM protocol.
8. RS232.TXD	TDX signal (transmission) for serial RS232 bus
9. Ground sensor	Ground for external sensors (e.g. oil pressure sensor)
10. +5V output	+5 V power supply for external sensors. Maximum current consumption 400 mA
11. Switched 12V	+12 V signal to turn on the device. The device is supplied by terminal 12 (Battery 12 V).

Terminal	Description
12. Battery 12V	Power supply for the device and real-time clock power when the device is turned off (no signal at the 12 V input 11). If the 12 V Battery fails to supply power, the real-time clock will be powered by an internal battery of the device.
13. USB.DM	D- signal to the USB port
14. USB.DP	D+ signal to the USB port
15. Analog in 5	Analog input 5. Measuring range 0–5 V
16. Analog in 3	Analog input 3. Measuring range 0–5 V
17. Analog in 1	Analog input 1. Measuring range 0–5 V
18. Digital in 8	Digital input 8
19. Digital in 6	Digital input 6
20. Digital in 4	Digital input 4 The input can also support a digital oil level / temperature sensor
21. Digital in 2	Digital input 2 The input can also support a Flex fuel sensor.
22. Power ground	Device ground used by AUX outputs
23. Ground	Device ground
24. Analog in 8	Analog input 8. Measuring range 0–5 V
25. Analog in 7	Analog input 7. Measuring range 0–5 V
26. Analog in 6	Analog input 6. Measuring range 0–5 V
27. Analog in 4	Analog input 4. Measuring range 0–5 V
28. Analog in 2	Analog input 2. Measuring range 0–5 V
29. Analog out	Analog output 0– 5 V
30. Digital in 7	Digital input 7
31. Digital in 5	Digital input 5
32. Digital in 3	Digital input 3 The input supports signal from an AIM beacon
33. Digital in 1	Digital input 1 The input supports signals from a crankshaft / camshaft position sensor (RPM)
34. Aux 1	*Low side* output. Maximum current 3 A
35. Aux 2	*Low side* output. Maximum current 3 A

2. Installation

To start the device and to establish communication with a PC, connect the device's power supply and the USB2CAN interface to the CAN1 bus.
This bus has a fixed speed of 1 Mbps and one of its functions is communication with a PC.



The above figure shows the “minimal” configuration enabling communication with the device.

The CAN bus is connected via a special USBtoCAN interface. The software is compatible with the following interfaces:

- Ecumaster USBtoCAN (www.ecumaster.com/products/usb-to-can/)
- Peak Systems PCAN-USB and PCAN-LAN (www.peak-system.com)
- Kvaser USBcan (www.kvaser.com)

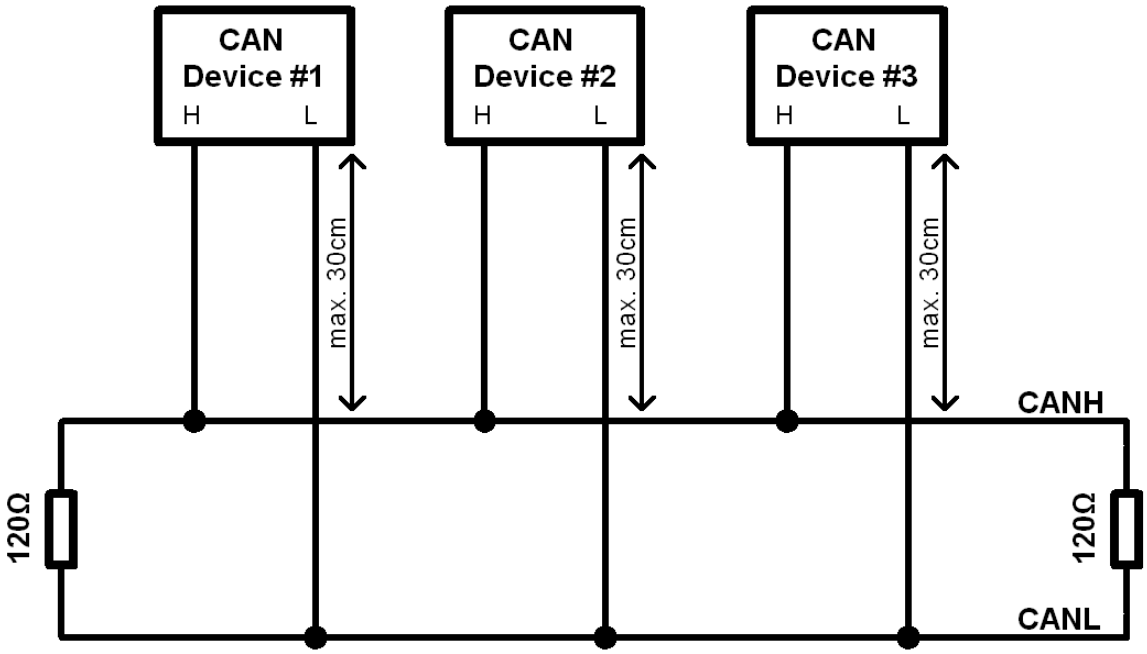
All these interfaces feature DB9 connectors, where CANL and CANH signals are on terminals No. 2 and No. 7.
The diagram also includes a 120 ohm terminator which is necessary for the bus to operate correctly (for more information about the terminators go to the CAN bus section).

Important:
Do not connect the +5 V output from the CAN interface to +5 V ADU!

2.1. CAN bus

The CAN (*Control Area Network*) bus was developed for communication in automotive applications. Its construction is very simple (only two wires) and its immunity to interference is very high. In modern vehicles, dozens of different electronic modules may communicate via the CAN bus.
The ADU device is equipped with two CAN buses: CAN1 bus is used for communication with a PC (requires an additional interface), as well as with other CAN devices.

Data frames are sent using a network, the topology of which is shown on the following diagram:



In *automotive* applications, typical data rates on a CAN bus are 1 Mbps, 500 kbps and 250 kbps. The following conditions must be met for each speed:

For 1 Mbps:

- the maximum length of the connection cable between the bus and the node must not exceed 30 cm
- the maximum bus length is 40 m
- maximum number of nodes is 30

For 500kbps:

- the maximum length of the connection cable between the bus and the node must not exceed 30 cm
- the maximum bus length is 100 m
- maximum number of nodes is 30

Regardless of the speed, the CAN bus requires termination in the form of 120 ohm resistors at both ends. In the case of CAN1 bus, there is no internal termination resistor, so external termination is required. CAN2 bus is equipped with a software-controlled termination resistor. Additionally, all the connections within the bus must be made using twisted pair wires.
It is important that the data transfer speed on a single bus has to be identical for all devices.

Important:
Failure to follow these rules will lead to the malfunctioning of the CAN bus and communication problems.

The CAN frame comprises an identifier (ID), number of transmitted bytes (DLC) and the data itself. Depending on the bus type, the identifier may be 11 bits (0x0-0x7ff) or 29 bit (0x0-0x1fffffff). The number of data bytes can range from 0 to 8

ID	DLC	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7

Below is a sample CAN frame of the CAN switch board.

ID	DLC	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
0x666	8	Analog#1(mV)		Analog#2 (mV)		CALPOT 1	CAL POT 2	Switch mask	Heartbeat

Parameter	Description
Analog#1	Voltage for the analog input #1 0–5000 mV, big endian
Analog#2	Voltage for the analog input #2 0–5000 mV, big endian
Switch mask	Bitmask of pressed buttons (1 means pressed)
CAL POT #1	Discrete value of the position of the rotary switch connected to analog input #1
CAL POT #2	Discrete value of the position of the rotary switch connected to analog input #2
Heartbeat	The counter increments its value by 1 after sending each frame

2.2. Connecting to ECU

To be able to display and log data from the control unit, connect the ADU to the ECU's CAN bus or serial port.

When using a CAN bus, you can connect the ECU to CAN1 (where the ECU CAN bus speed is 1 Mbps) or to CAN2 bus where you can define the speed of the CAN bus.

OEM engine management units usually use 500 kbps (ECUs with a 250 kbps bus are rarer), which forces connection to CAN2 bus.

Some engine managements used in motor sports do not have a CAN bus, but are equipped with a serial output (RS232). This is the case for EMU, CLASSIC EMU or Hondata. ADU supports the following serial formats: AIM, Ecumaster serial protocol, Ecumaster EMU Classic EDL1 protocol, Hondata serial protocol, AEM, GEMS, Athena GET, Autronic SM4, MoTec M4 DATA SET5. For additional information see the ***CANbus / Serial Setup*** chapter.

For OEM controllers equipped with an OBD 2 over a CAN bus (all cars after 2008) it is possible to use the OBD connector to read the basic engine operation parameters. For more information, see the ***OBD 2*** chapter.

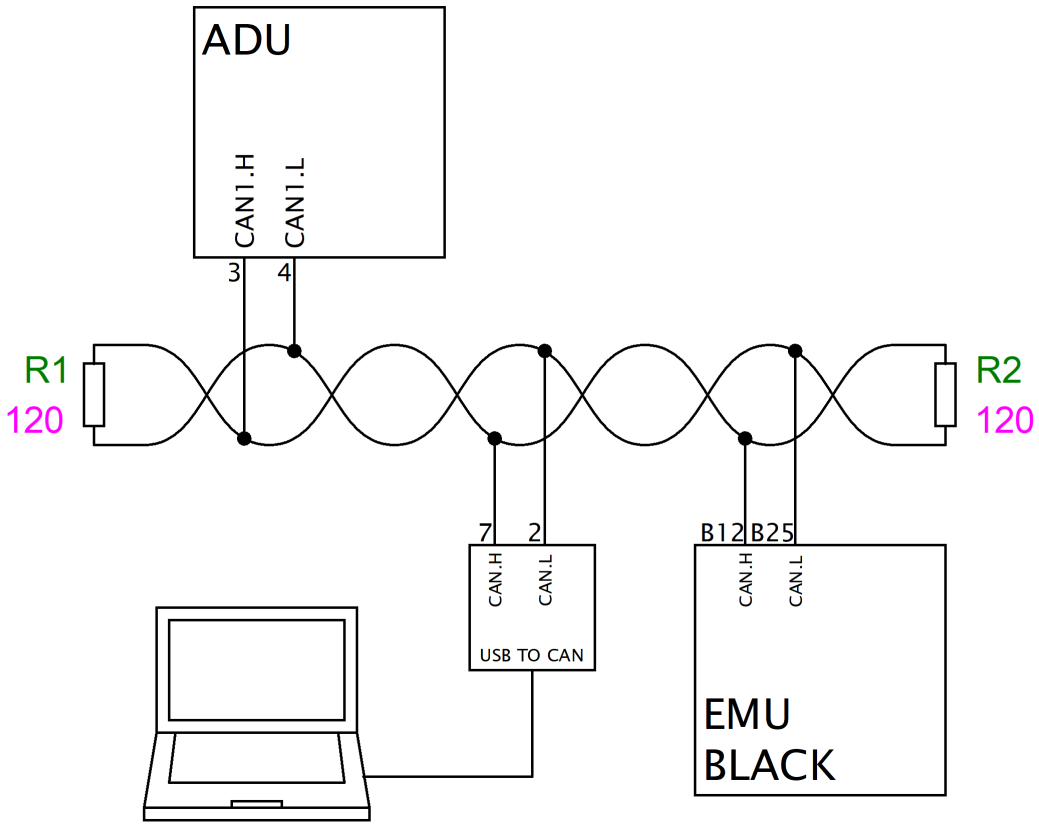
External sensors (e.g. TPS, temperature and oil / fuel pressure sensors, crankshaft position sensor, etc.) may be connected to digital and analog inputs of the ADU. As a result, the user can monitor and log parameters that are not supported by the original engine controller.

2.3. Connecting via a CAN bus

The following diagram shows an example connection of the ECU to the ADU using CAN1 bus.

Important:
The CAN bus speed of the engine controller must be 1 Mbps, as this is the only speed supported by the CAN1 ADU bus.

If the CAN bus speed is different than 1 Mbps or if you want to isolate the ECU from the ADU CAN1 bus, please use the CAN2 bus. The following diagram shows an example connection of an EMU BLACK.



Important:
Care must be taken to ensure correct topology and proper bus termination. For more information see the *CAN bus* section.

Detailed information on connecting specific ECU brands to ADU can be found in the application notes at www.ecumaster.com/products/adu/.

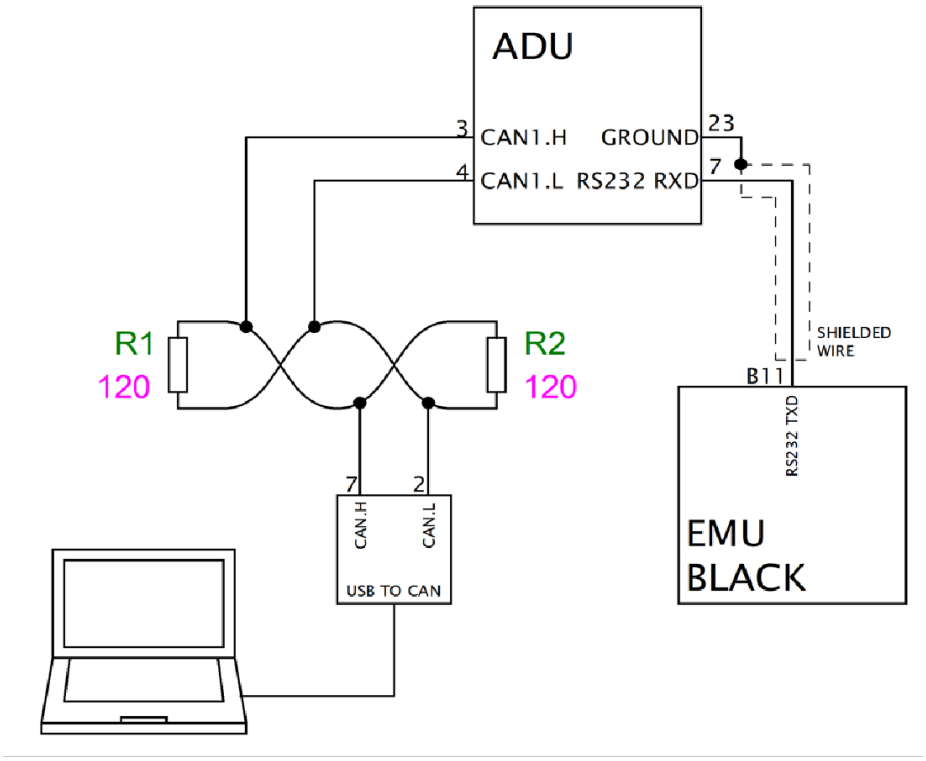
Information about channel configuration with CAN stream data can be found in the ***CANbus Inputs*** section.

2.4. Connecting via an RS232 serial bus

The ADU display features a built-in serial bus (RS232). In order to connect the ECU via a serial bus connect the ECU (Tx) output to the ADU (Rx) input.

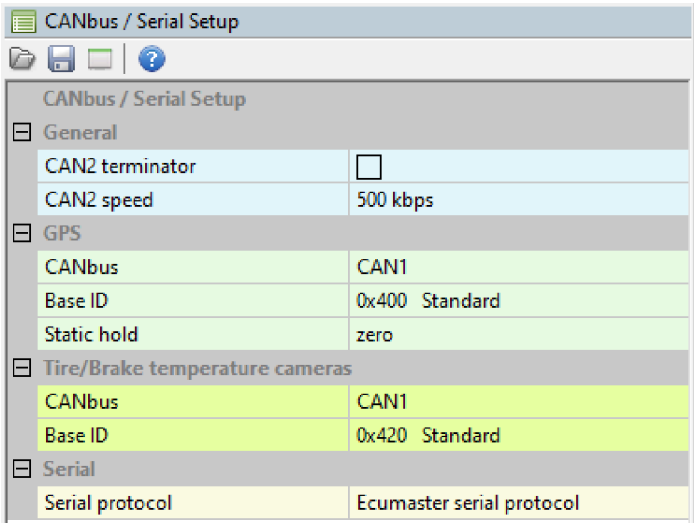
Important:
To transfer data via serial communication please use a shielded cable.

The diagram below shows an example of such a connection.



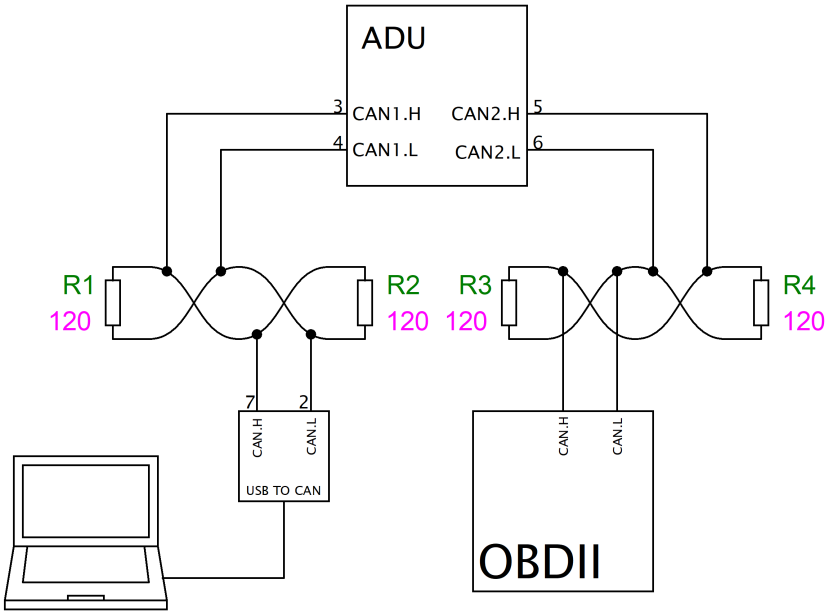
Detailed information on connecting specific ECU brands to ADU can be found in the application notes at www.ecumaster.com/products/adu/.

To configure the serial protocol (in accordance with the connected ECU), select the appropriate protocol in the *CANbus / Serial* window (*Ecumaster serial protocol, Ecumaster EMU Classic EDL1 protocol, AIM, Hondata, Autronic SM4, AEM, GEMS, Athena GET, MoTec M4 DATA SET5*).

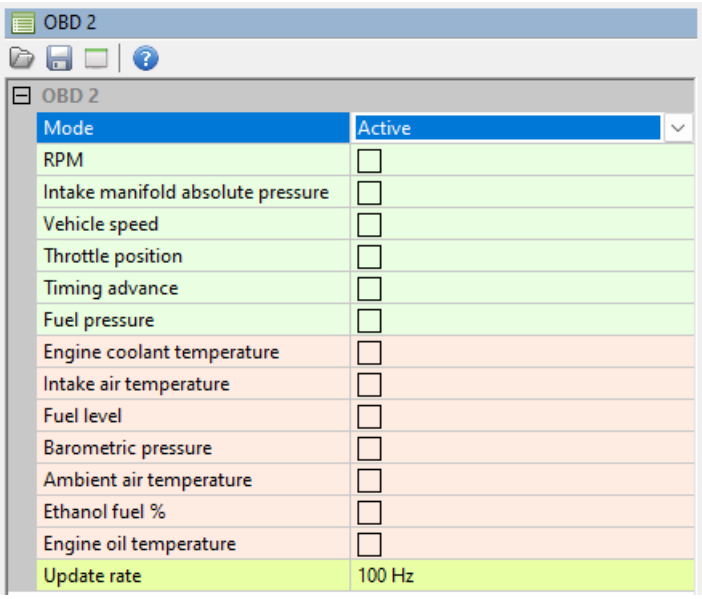


2.5. OBD 2

The following diagram shows a connection of the CAN2 bus to the OBD 2 bus of the vehicle.



In the *CANbus / Serial Setup* window, set the correct CAN bus speed (usually 500 kbps).



ADU offers compatibility with the ISO15765/SAE J2480 standard for OBD communication, providing 11-bit, 250 Kbaud, and 11-bit, 500 Kbaud variants.

Select active or passive mode. In the active mode, the device actively requests real-time data on selected parameters from the vehicle's systems. In passive mode, the device only reads information queried by another device operating in active mode. Passive mode is essential when multiple devices are connected to OBD, as allowing more than one device to send queries

simultaneously would disrupt communication. Therefore, only one device can actively query the vehicle's systems at a time, ensuring efficient and reliable communication.

Then, select the respective channels in the OBD 2 window. The channels are divided into two groups: fast (green) and slow (orange). Fast channels use 66% of the bandwidth (*update rate)* while slow channels use 34%. This means, for example, that with an Update rate of 100 Hz (data is downloaded by OBD 2 at 100 Hz), the RPM channel will be refreshed 66 times per second (66 Hz). If an additional channel is selected (e.g. *Throttle position*), the frame rate of both channels will be 33 Hz. It should be noted that not all ECUs are capable of refreshing data at 100 Hz and it may be necessary to reduce the *Update rate*.

Important:
Not all logging channels found in the OBD 2 window are available for all ECUs.

There is also the possibility of mixed parameter reading for serial controllers, e.g. *throttle position* and *RPM* can be read directly from the CAN bus and the other parameters via OBD 2. For more information see the ***CANbus Inputs*** section.

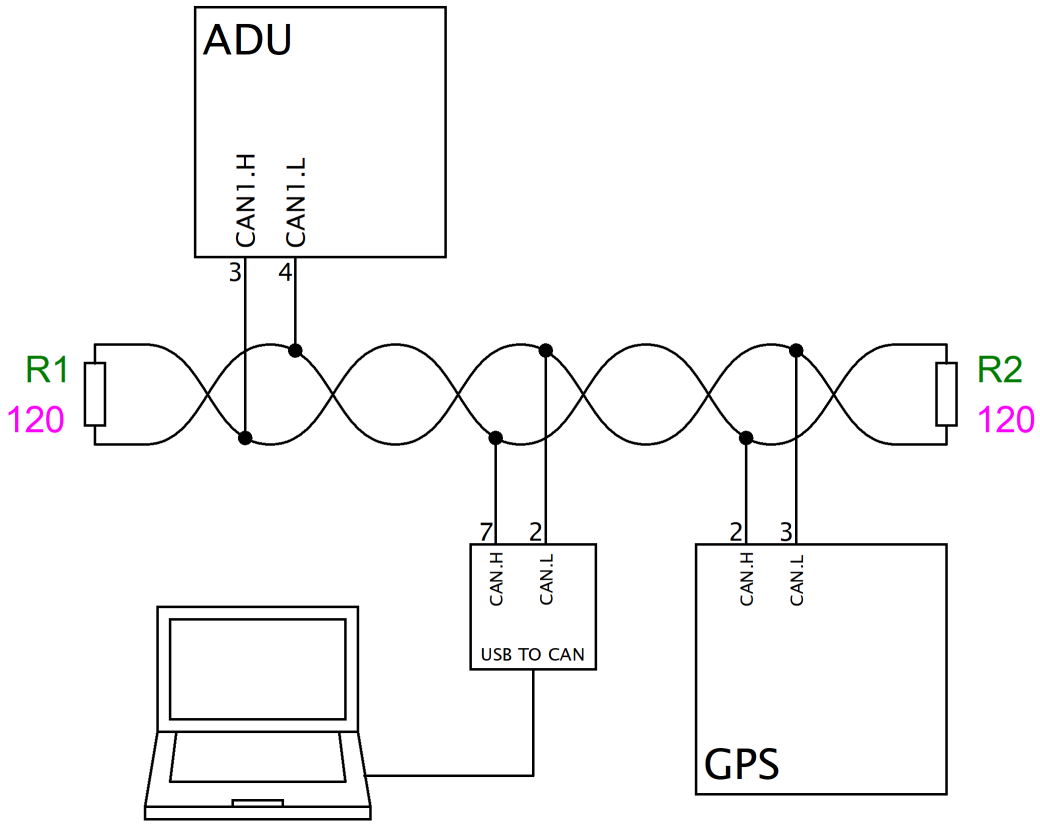
2.6. GPS module

The ADU display makes it possible to use the *Ecumaster GPS to CAN* module to measure lap times on a track and allows data analysis based on the vehicle's position on the track.
To fix the current position of the vehicle, the module uses information from GPS / Glonass satellites, a built-in accelerometer and a gyroscope. The position is refreshed with a frequency of 20 Hz or 25 Hz dependent on GPS module version.

Note:
The ADU only supports the **Ecumaster GPS to CAN** module and does not support other GPS modules or interfaces.

The CAN bus speed of the GPS module is factory set to 1 Mbps. This is why it is recommended to connect the GPS module to CAN1 bus.

A sample connection diagram is shown below:



In the *CANbus / Serial Setup* options, select the CAN bus to which the GPS module is connected (*GPS / CANbus*).

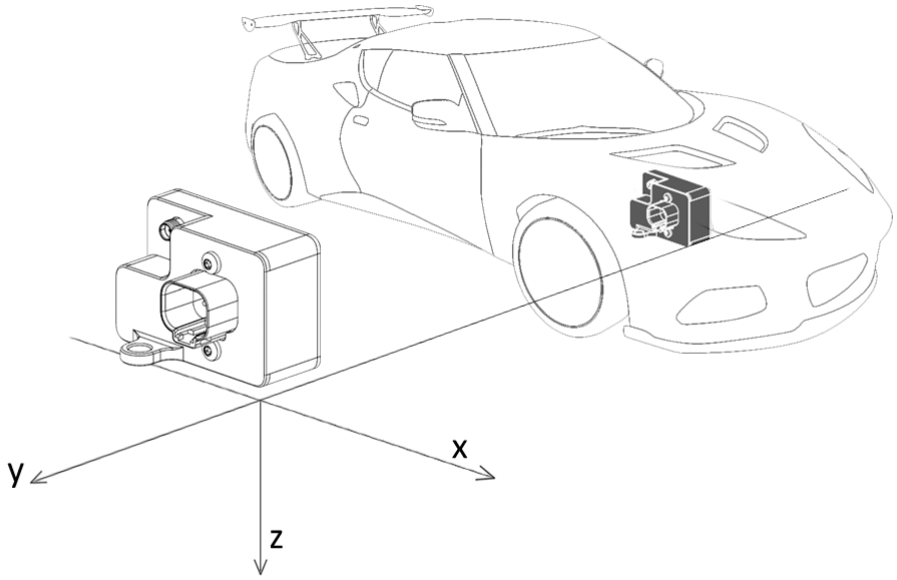
The method of installing the GPS module to the vehicle's body and its subsequent calibration are very important. Due to the use of both gyroscope and accelerometer in order to obtain more accuracy the GPS module should be mounted using the supplied anti-vibration pads.

GPS V1 (20 Hz)

The orientation of the module does not matter, however, after installation, it must be calibrated. Calibration is automatic when the vehicle travels the first few hundred meters. If the position of a calibrated module is changed, a short calibration drive is required.
For more information about measuring lap times with the GPS module, see the ***Lap Times*** chapter.

GPS V2 (25 Hz)

The orientation of GPS V2 should be as presented:



No additional calibration is required.

Available GPS channels

Channel	Description
gps.latitude	Latitude
gps.longitude	Longitude
gps.height	Height above sea level in metres
gps.status	

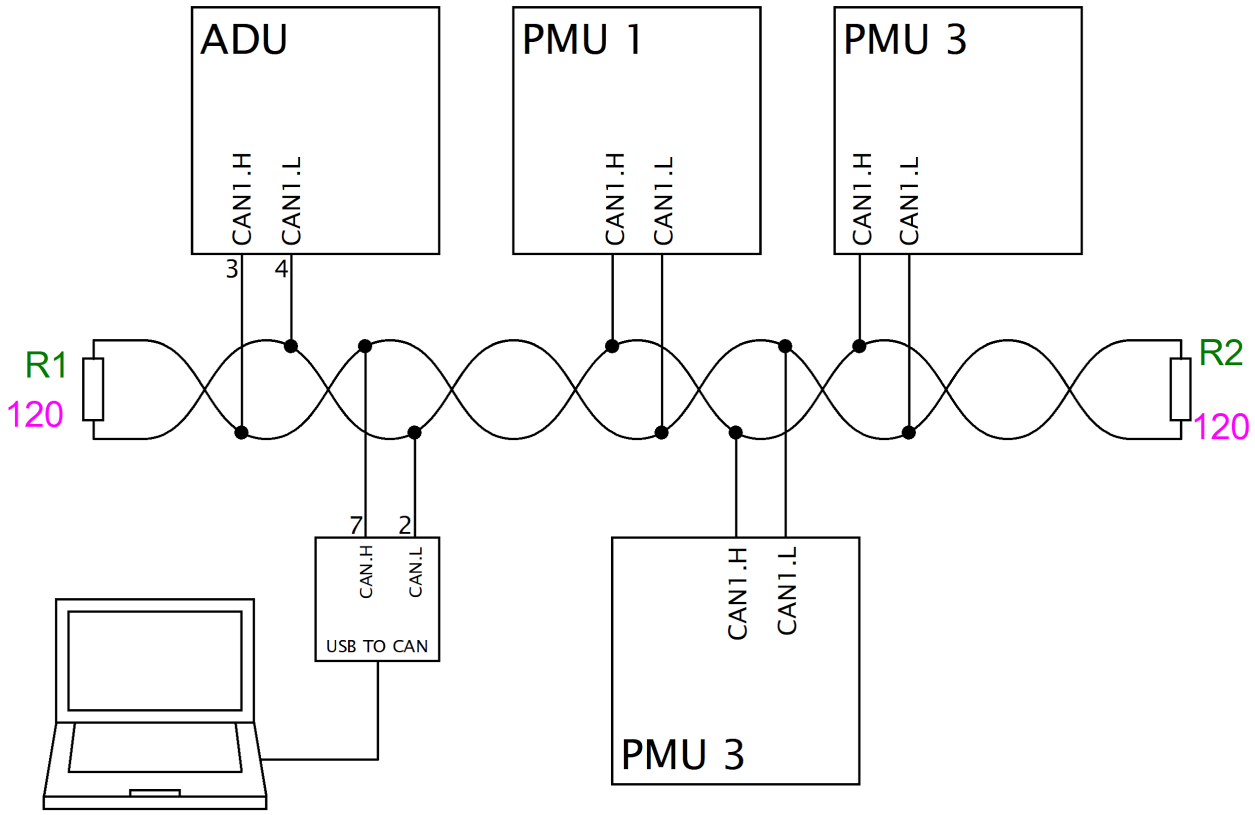
0	**Disconnected**	no GPS module data
1	**No fix**	unable to fix position
2	**IMU**	position is fixed using the built-in accelerometer and gyroscope – (only GPS V1)
3	**GPS 2D**	position is fixed in 2D space using GPS / Glonass satellite data
4	**GPS 3D**	position is fixed in 2D space using data from GPS or Glonass satellites
5	**GPS + IMU**	position is fixed in 3D space and corrected using the built-in accelerometer (highest accuracy) – (only GPS V1)

<table>
<tr><th>Channel</th><th>Description</th></tr>
<tr><td>gps.fusionStatus</td><td>
<table>
<tr><td>0</td><td>Initialisation</td><td>Initialisation and calibration of inertial sensors – (only GPS V1)</td></tr>
<tr><td>1</td><td>Fusion</td><td>the device uses inertial sensors to correct the vehicle’s position – (only GPS V1)</td></tr>
<tr><td>2</td><td>Suspend</td><td>temporary error of inertial sensors – (only GPS V1)</td></tr>
<tr><td>3</td><td>Disabled</td><td>error of inertial sensors; they are not taken into account when determining the vehicle’s position</td></tr>
</table>
</td></tr>
<tr><td>gps.speed</td><td>Vehicle speed in km/h</td></tr>
<tr><td>gps.headingMotion</td><td>Vehicle motion direction. When the vehicle is skidding, the headingMotion value will be different from the headingVehicle value.</td></tr>
<tr><td>gps.headingVehicle</td><td>The direction in which the front of the vehicle is facing. When the vehicle is skidding, the headingMotion value will be different from the headingVehicle value.</td></tr>
<tr><td>gps.accX</td><td>Longitudinal acceleration (longitudinal g)</td></tr>
<tr><td>gps.accY</td><td>Lateral acceleration (lateral g)</td></tr>
<tr><td>gps.accZ</td><td>Vertical acceleration (vertical g)</td></tr>
<tr><td>gps.gyroX</td><td>Angular velocity of the vehicle’s longitudinal axis</td></tr>
<tr><td>gps.gyroY</td><td>Angular velocity of the vehicle’s lateral axis</td></tr>
<tr><td>gps.gyroZ</td><td>Angular velocity of the vehicle’s vertical axis</td></tr>
<tr><td>gps.noise</td><td>Average satellite signal noise level. The lower the value the better.</td></tr>
<tr><td>gps.numSatelites</td><td>Umber of satellites used to fix position</td></tr>
</table>

2.7. Ecumaster PMU

If connecting the ADU to a system with a PMU, it is recommended to connect CAN1 of the PMU to CAN1 bus of the display. This will allow easy communication both with a PC and between the devices.

A sample connection diagram is shown below.

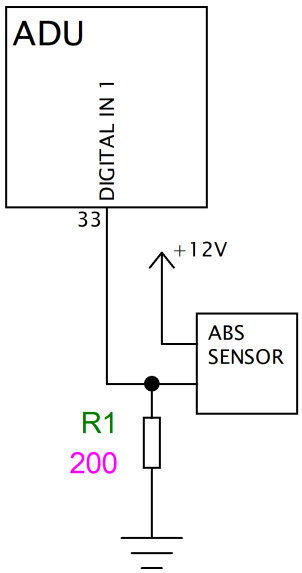


2.8. Digital Inputs

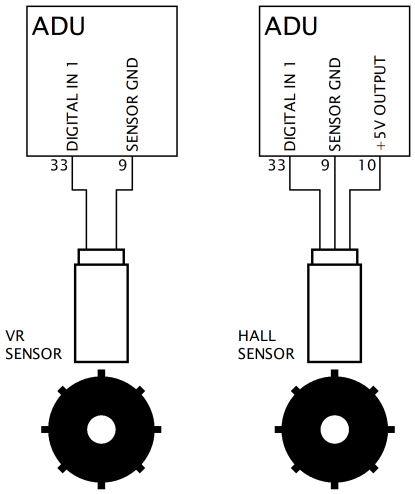
Digital inputs are used to process frequency signals (e.g. from the engine speed sensor, the turbocharger sensor, the Flex Fuel sensor), beacon signals and to connect switches.

It is possible to read signals from inductive sensors (VR sensors), Hall's sensors, optical sensors. These inputs are equipped with built-in switchable 2k2 (up to +5 V) pull-up resistors, which enables connecting Hall sensors or switches shorted to ground.

ABS sensors (magnetoresistive)

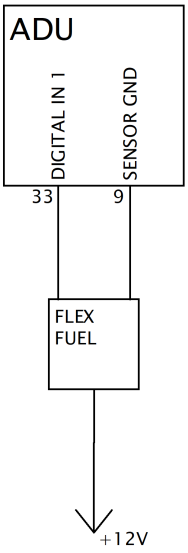


Crankshaft position sensor (Hall's or VR sensor)

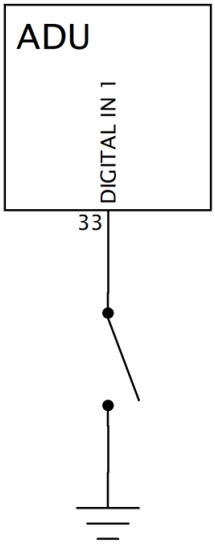


A shielded cable is recommended for a VR sensor.

Flex Fuel sensor



Switch



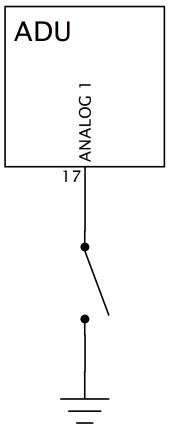
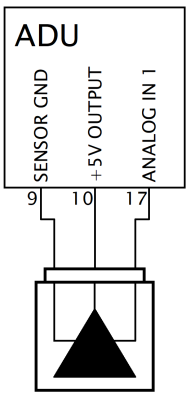
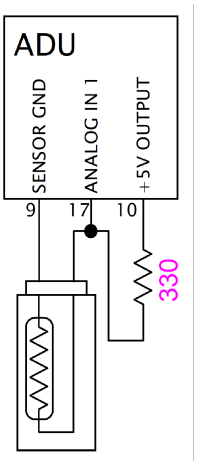
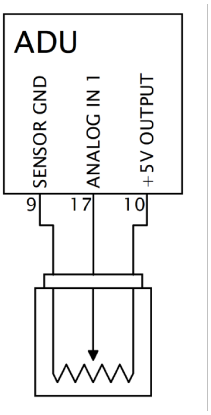
2.9. Analog Inputs

Analog ADU inputs are able to process voltage in the range of 0-5 V (voltages above 5 V are read as 5 V) with a frequency of 500 Hz.

The primary use of these inputs is to display and log signals from analog sensors such as pressure sensors (oil, fuel, water, etc.) or RTDs (e.g. oil temperature sensor, fuel level sensor).

These analog inputs can also be used to connect *switches*.

The voltage of all analog inputs can be transferred to other devices (e.g. EMU, BLACK) via the CAN bus.

<table>
<tr>
<td>Switch
 
</td>
<td>Oil pressure sensor
 
</td>
</tr>
<tr>
<td>Oil temperature sensor
 
</td>
<td>Potentiometer
 
</td>
</tr>
</table>

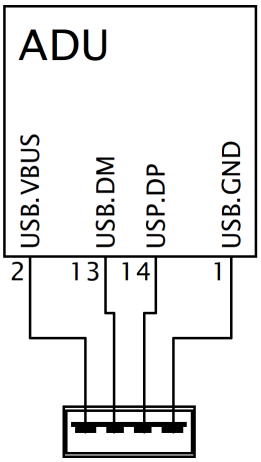
Important:

If using a **12V switch**, connect it to a **digital input** instead of an analog input. While it is technically possible to use an analog input, it is not recommended in such cases.

2.10. USB Flash drive (pendrive)

To log parameters to an external storage device, connect a USB storage device to the ADU. This makes it possible to use popular pen drives. Built-in real time clock (battery powered) is used to set the file dates correctly. The supported file system is FAT32. File size depends on the number of channels defined and the save frequency. For more information on data logging go to the Appendix H - How-to Log Data to a USB drive *(on page 253)*. It is recommended to use quality brand flash memory sticks compatible with the USB 3.0 standard (e.g. Sandisk Ultra, Sony USM8W3, USM16W3, USM32W3), with a maximum capacity of 32 GB. Using USB storage devices with poor parameters can lead to interrupt the recording of logged data.

A sample USB socket connection diagram is shown below.

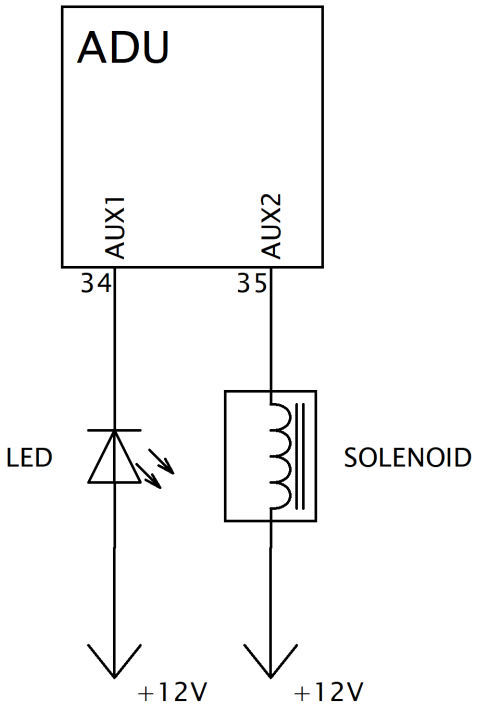


Important:

D+ and D- cables should be twisted pair wires and the entire USB bundle must be shielded.

2.11. Low side outputs

Built-in low side outputs (shorted to ground) can be used to connect external loads (e.g. LEDs, solenoid valves) depending on the functions defined by the user (e.g. radiator fan activation, signalling). They are also PWM-capable, allowing precise control of connected devices.



3. ADU Client software for Windows

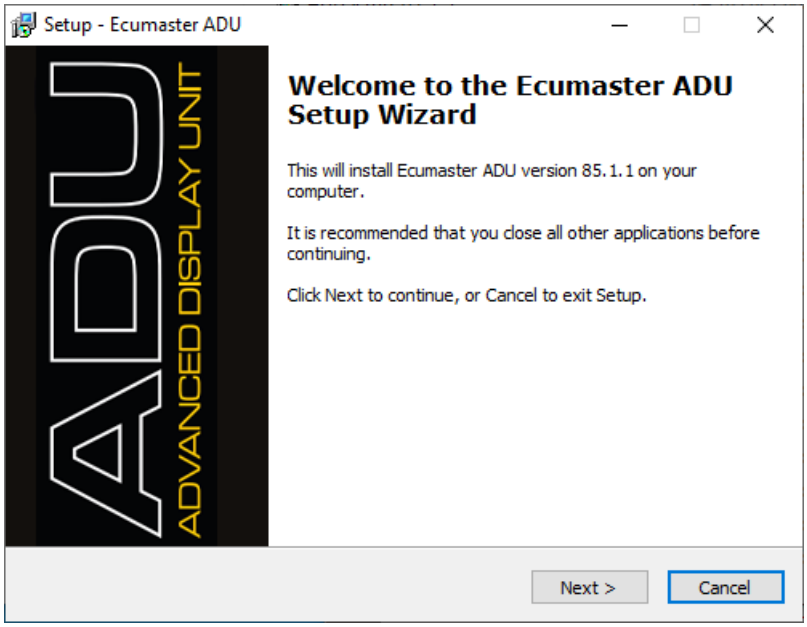
To set up the ADU dashboard, it is necessary to install software for Windows. Official software can be found at www.ecumaster.com/products/adu/.

Unofficial, test versions of the software (with additional features) are also available. They can be downloaded from www.ecumaster.com/testVersions.html. Test versions of the software may contain errors! If you have problems with the new software, please contact us at bugs@ecumaster.com. Ecumaster recommends the use of official software.

Hardware requirements to run the software:

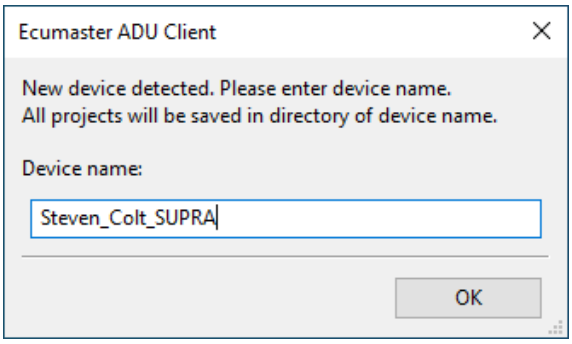
- Windows XP, VISTA, 7, 8, 10, 11 (32 and 64 bits)
- minimum screen resolution of 1366x768
- Open GL compatible graphics card
- 2GB RAM
- USB port

The following window appears during software installation. Follow the instructions to complete the installation.



3.1. First connection with the device

Once the software has been installed on a PC, upon the first connection a window will appear asking for the device name.



A subdirectory in users' Documents / ADU / folder will be created under this name, where all settings relating to the device will be stored.

Client and Firmware:

Each ADU is equipped with pre-installed embedded (internal) software that ensures its functioning. It is referred to as ***Firmware***.

Client software is used to operate the ADU using a PC.

The Client software installation file contains the latest Firmware versions for the ADU.

Numbering of Client and Firmware versions:

Client version number format:

FirmwareConfiguration.FirmwareMinor.ClientFixNumber (e.g. 51.0.2)

or

FirmwareConfiguration.FirmwareMinor - if ClientFixNumber is zero (e.g. 51.0).

- FirmwareConfiguration, or master version - changes when new settings are made for the Firmware visible in the ADU Client.
- FirmwareMinor, i.e. the minor version - changes when a Firmware fix is published, but WITHOUT making new settings in the ADU Client.
- ClientFixNumber, i.e. Client software fix number - changes each time only the Client is updated.

Firmware version number format:

FirmwareConfiguration.FirmwareMinor (e.g. 51.0)

It should be noted that the Client software version e.g. 83.0:

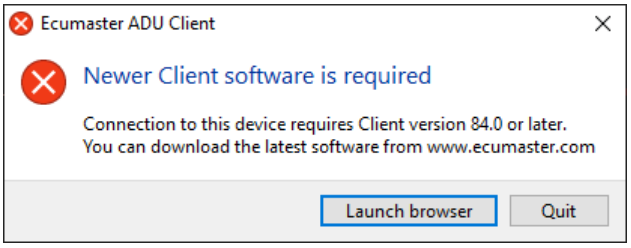
- supports any Firmware from the 83.x family: 83.0, 83.1, 83.2 etc.;
- supports all older Firmware versions, e.g. 50.0, 51.1, 82.1, 82.0, 81.0 etc.;
- It does NOT support higher FirmwareConfiguration version numbers e.g. 84.0, 84.1.

In other words, the latest ADU Client version supports all ADUs that were previously produced. It is possible to update the factory-installed firmware to the latest version or install an older version at any time.

Checking the software version on first connection:

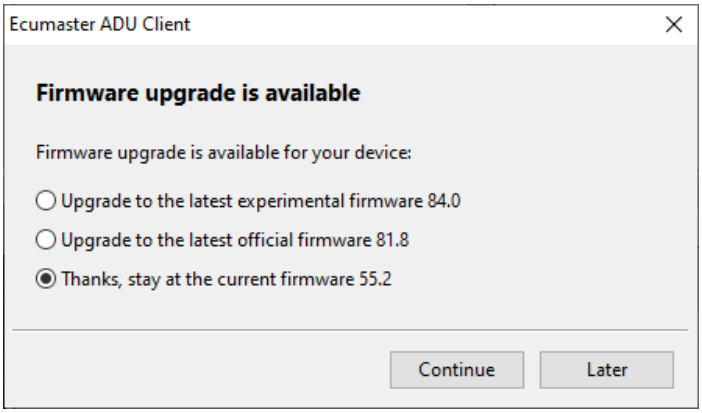
Depending on the Firmware version on the ADU and the Client software version installed on the PC, messages recommending a firmware update may appear the first time you connect to the device:

1. When the Firmware version of the device is higher than the version of the Client software installed on the PC, a message will appear indicating that ***Newer Client software is required***.



   Client software version should be the same or higher than that of the *Firmware* running on the device.

2. When the Firmware version of the device is lower than that of the Client, a message will appear indicating that newer Firmware is available for the ADU (***Firmware upgrade is available***). If an experimental version of the Client software is installed on the PC, three suggestions will appear in a message:



   - upgrade to the latest experimental version,
   - upgrade to the latest official version (but older than the listed experimental version),
   - keep the existing version.

Otherwise, only two options will be available:

- upgrade to the latest official version,
- keep the existing version.

To check the version of the *Firmware* currently installed on the device and the version of the *Client* software, select ***Help / About*** from the main menu. The following message window will appear.

3.2. Firmware update

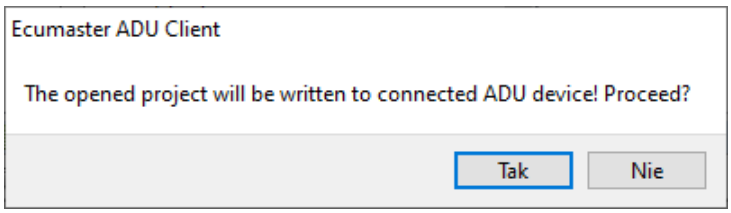
If necessary, the Firmware version can be changed at any time in the ADU. To do this, select ***File / Upgrade firmware...*** from the main menu and then select the relevant firmware version from the list.

3.3. Loading a project

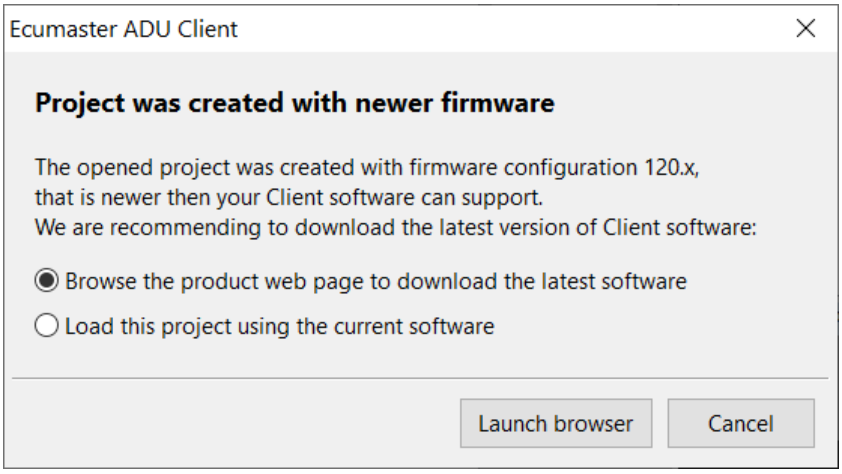
When opening a finished project, different messages may appear depending on:

- the configuration in which the file was created (*firmware configuration*);
- the *Client* software version installed on the PC;
- the *Firmware* version on the ADU;
- the connection status of the ADU or the "*Offline*" operating mode.

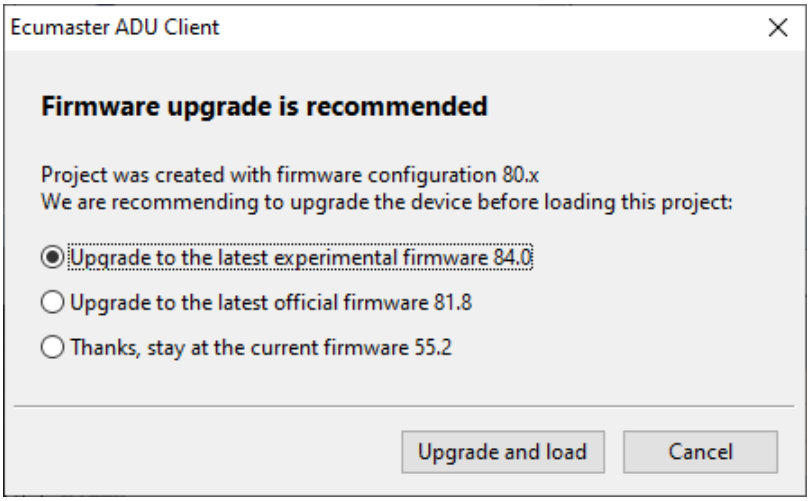
If the device is connected while opening a project, a message may appear indicating that the project being opened will be saved on the connected ADU.



If a project being opened was created using (*firmware configuration*) not supported by the Client software version, a window may appear recommending that a newer version of the Client software be downloaded. This window can appear both when the ADU is connected and when working offline.

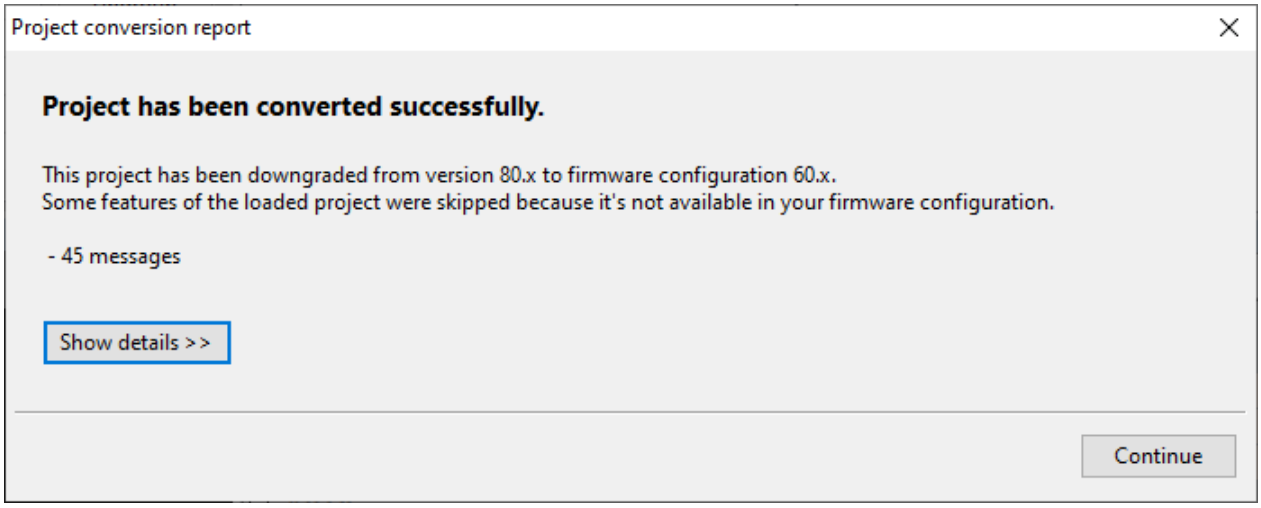


If a project was created using a firmware configuration not supported by the version of Firmware on the ADU, a window may appear recommending a Firmware update before loading the project.

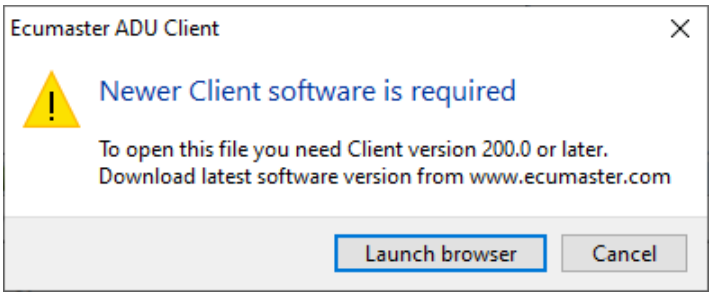


If the software was installed in an experimental (test) version, the message window will offer three choices: update to the latest experimental version, update to the latest official version or keep the existing version.
If an official version of the software was installed, you will be presented with only two choices: upgrade to the latest official version or keep the existing version.

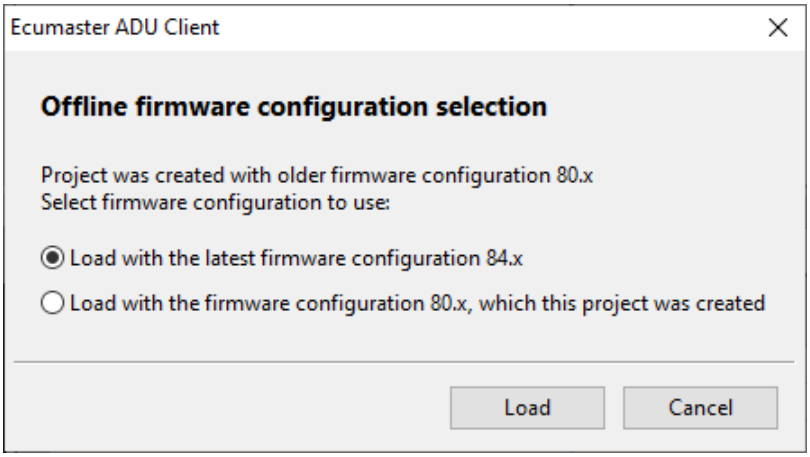
If a project created using a newer configuration is loaded into a device with older configuration software, some functions of the project will not be available because they are not supported by the older configuration.



Opening a project created using a newer configuration and saved in a format other than the format supported by the installed Client software, will cause a message to appear stating that newer software needs to be installed, along with information on the minimum version required to open the file. This window can appear both when the ADU is connected and when working offline.

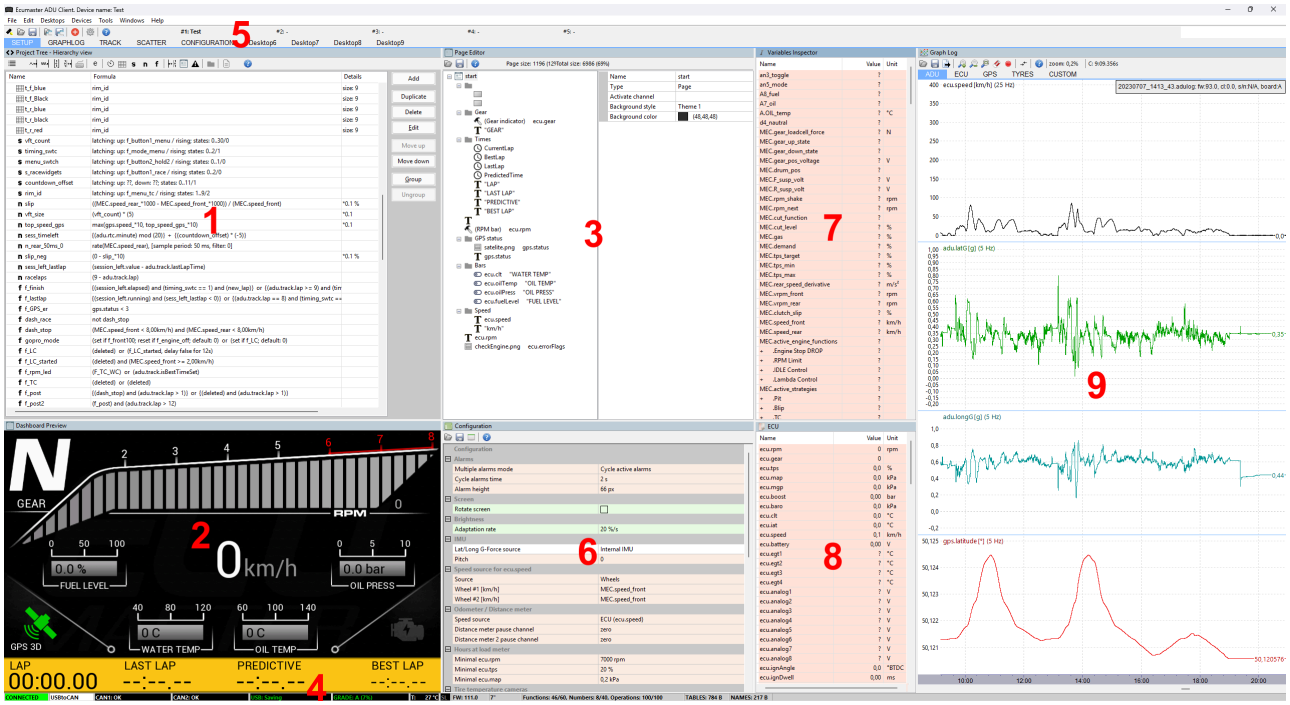


If there is no connection to the ADU and an attempt is made to open a project created using an older configuration than the one supported by the version of the Client software installed on the PC, a window will appear, prompting the user to select a software configuration.



3.4. Appearance of the application

Once the application has been installed and started, the screen should look as in the picture below:



The ***Project Tree*** window is the main window where a project is defined **(1)**. Use this window to define all project elements. Click ***Add*** to add a new element.

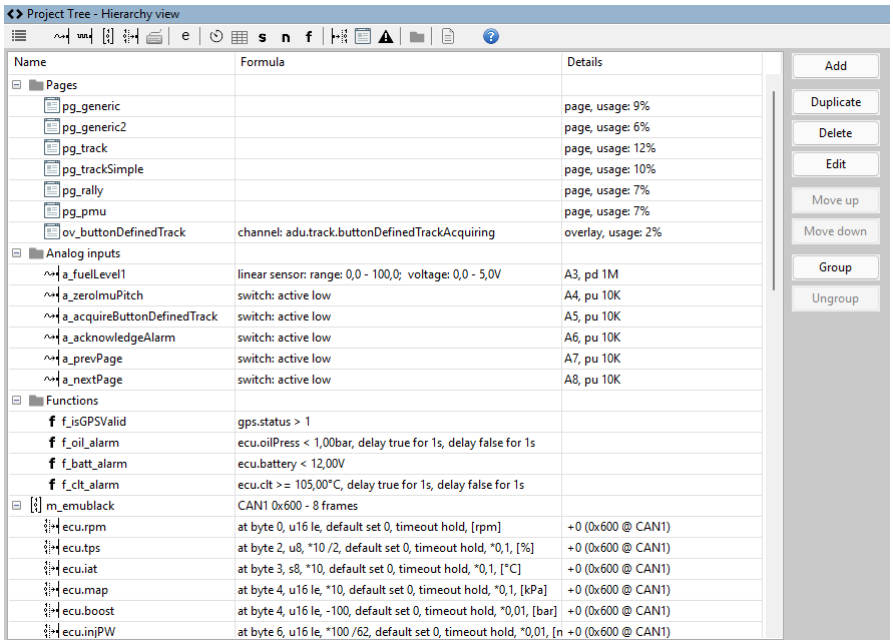
The following selection options will appear:

- ***Analog Input*** – an analog input, where all of its Parameters can be defined, e.g. the input number, the name of the variable, the type, pull-up etc.
- ***Digital Input*** – a digital input, were its properties can be defined, e.g. the type, sensitivity, name of the variable, pull-up, etc.
- ***CANbus Message Object*** – a CAN message element, where an incoming CAN frame can be defined.
- ***CANbus Input*** – an element defining variables incoming in CAN frames
- ***CANbus Keyboard*** – an element defining the keyboard
- ***Enumeration*** – an element to assign a numeric value to the desired text or color, in order to display it on the screen
- ***Timer*** – an element used for counting time
- ***Table*** – an element defining the table that can be used to transform data (e.g. transform an analog input voltage into temperature).
- ***Switch*** – an element defining a switch
- ***Number*** – an element defining a mathematical function in order to convert variable values

- ***Function*** – an element for creating complex logical functions
- ***CANbus Export*** – an element for sending CAN frames with variable and constant values
- ***Page*** – a single page displaying data. If more pages are defined, they can be toggled between using both functions and the buttons.
- ***Alarm*** – an element displaying the defined alarms, irrespective of the current page
- ***Group*** – a function for grouping elements; it allows a hierarchy to be introduced into a project in an easy way.
- ***Import .CANX / .DBC*** file – this function is intended for downloading predefined CAN streams for different devices (e.g. EMU BLACK, MoTeC M1 etc.)

When adding different elements to the project, it is recommended to use the ***Group*** element, which allows elements to be grouped into logical sets. You should also make sure to assign correct names to elements. This will facilitate project management in the future.
You can also duplicate project elements using the ***Duplicate*** button.

A sample project is shown below.



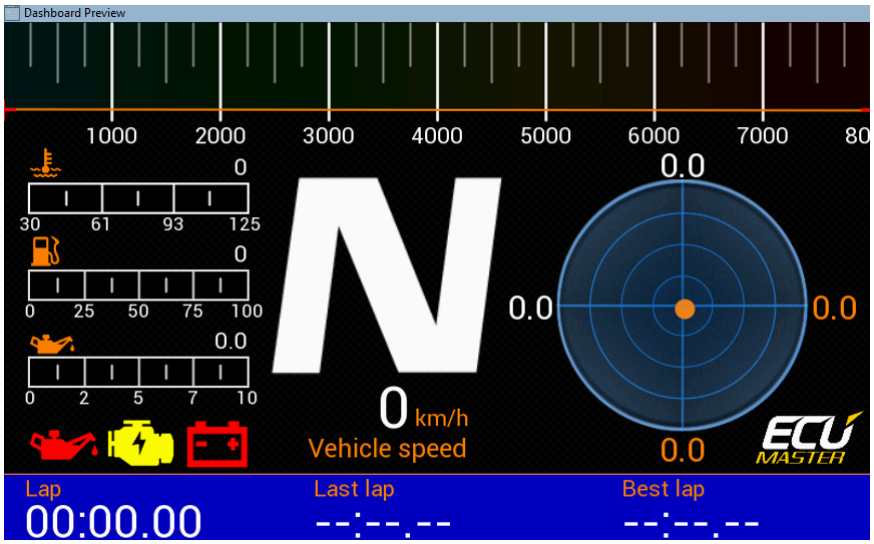
A feature available in the upper-left corner of the *Project Tree* panel is the *Change view* icon (key: F4), which allows users to quickly change how project elements are displayed. Available view modes include:

- **Hierarchy view** – default view using user-defined groups
- **Types view** – sorts elements by their type (Tables, Numbers, Functions, etc.)
- **CANbus view** – displays only CAN-related elements, grouped by CAN bus (CAN1 / CAN2)
- **Display view** – shows only created pages and alarms.

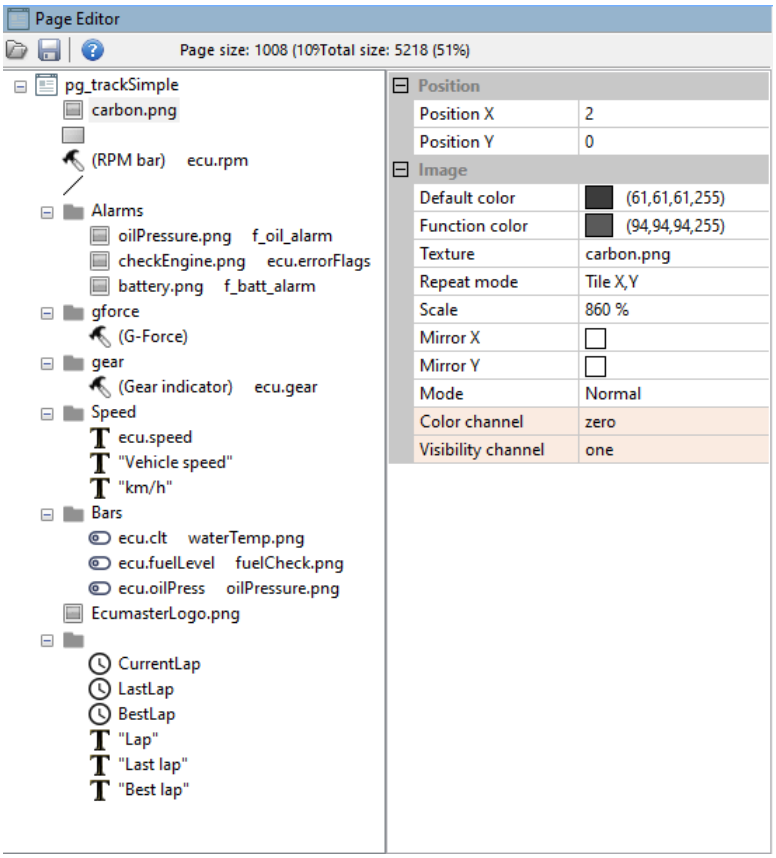
A selected page may be displayed and edited using the page preview screen **(2)**.

When the application is connected via the CAN to USB interface to the ADU, all changes made to a page are transmitted to the device in real time.

The ***CTRL+K*** shortcut pressed in an active Dashboard Preview screen saves the current page image (800x480) to a png file.

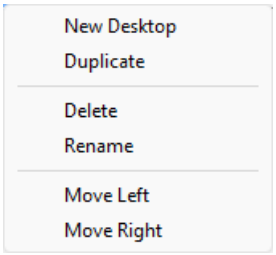


A page editor **(3)** is available in the page preview screen. It may be used to add and parametrize graphic page elements. For more information on pages see the Pages *(on page 61)* chapter.



The status bar is another important element of the interface **(4)**. It contains useful information on the status of the connected device. For a detailed description see the *Status field* chapter.

Tabs are a very important element of the application **(5)**. They may be used to create your own sets of windows to facilitate and accelerate handling of the software. Right-clicking a tab brings up the following menu:

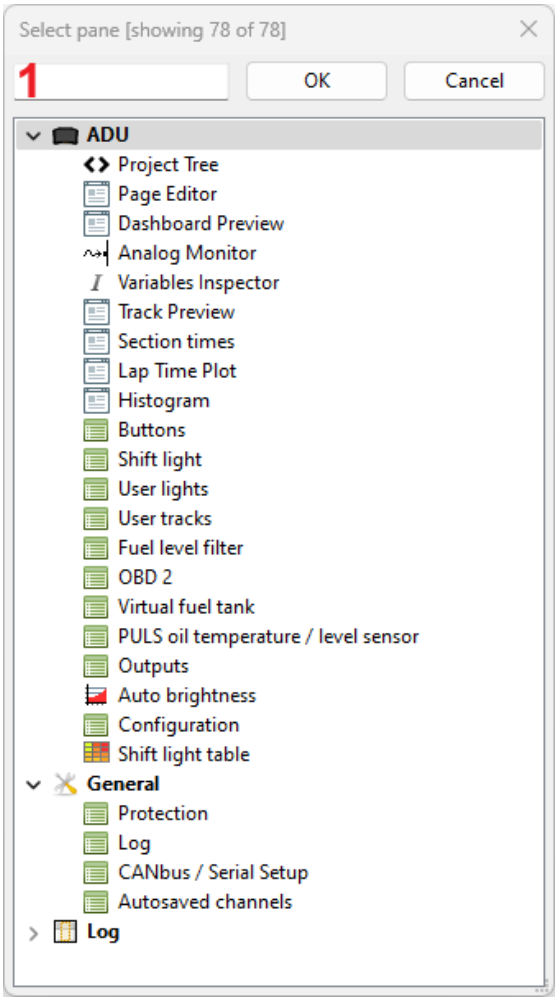


New Desktop	Create a new tab. A tab will always be created at the end.
Duplicate	Duplicate a tab. This option creates a new tab and copies the content of a highlighted tab into it.
Delete	Delete a tab.
Rename	Rename a tab.
Move Left	Move a tab to the left
Move Right	Move a tab to the right

Tabs are saved on the disc after pressing the **F2** button (*Make permanent*) or upon leaving the application. They can also be saved to a file and transferred to another PC. To this end, select the *Desktops > Save desktops template option*.

To return to the default settings select *Desktops > Open desktop template* and load the *programDefault.adulayout* file.

Panels, together with all types of parameters, are another element of the interface. The device may be configured using these. To add a new parameter panel, press **F9** (or click on the + icon on the toolbar). A window with all available parameters will pop up.



The search can be accelerated by entering the desired option in the filter field (1).

Clicking on the selected panel will open it in the desktop. New panels are always displayed on the right side of the desktop. They can be moved by left-clicking on the title bar and dragging it to a new position. To remove a panel from the desktop, right-click its bar. A menu will appear from which the delete option (*Close panel*) must be selected.

There are different panel types. The first type are the configuration panels discussed above. Another type are panels for previewing variables, such as the ***Variables Inspector*** **(7)**, log channel preview **(8)** or the graphic log **(9)** showing the course of logging channels over time.

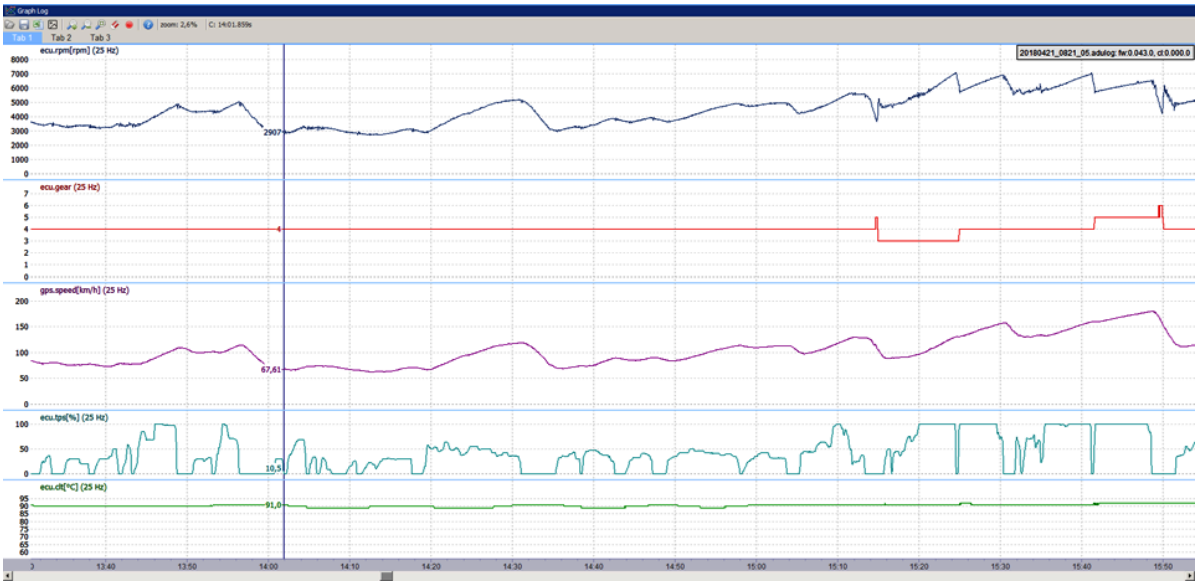
The ***Variable Inspector*** panel is used to view values of variables defined in the device. These variables include e.g. Functions, Numbers, CANbus inputs etc.

Variables Inspector

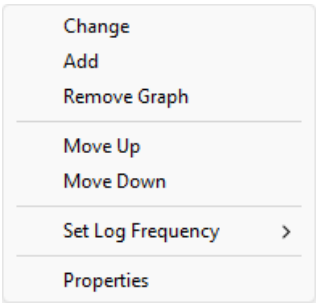
Name	Value	Unit
a_fuelLevel1	0,0	
a_zerolmuPitch	0	
a_acquireButtonDefinedTrack	0	
a_acknowledgeAlarm	0	
a_prevPage	0	
a_nextPage	0	
c_ecu_dbwPos	0,0	%
c_ecu_dbwTrgt	0,0	%
c_ecu_tcDrpmRaw	0	
c_ecu_tcDrpm	0,0	bar
c_ecu_tcTrqRdc	0	%
c_ecu_pitLTrqRdc	0	%
c_ecu_boostTarget	0	kPa
c_ecu_pwm1DC	0	%
c_ecu_atMode	0	
f_isGPSValid	0	
f_oil_alarm	1	
f_batt_alarm	1	
f_clt_alarm	0	

If a value is not a number but the ***?*** symbol, it means that the logging function for this channel is deactivated. To activate logging or change the log frequency for a given channel, click the right mouse button on a given variable and select ***Set Log Frequency*** and then the desired frequency from the menu that pops up. It is possible to display values of variables of a given class by opening the following windows: ***Analog Inputs*** (displays the analog input values), ***Digital Inputs*** (displays digital input values), ***CANbus Message Objects***, ***CANbus Inputs*** (variables from CAN bus), ***Tables*** (values from the tables), ***Numbers*** (mathematical function values) or ***Functions*** (logical function values).

The logging channel ***Graph Log*** is another panel type. It can be used to display the desired channels and preview them using the time function.

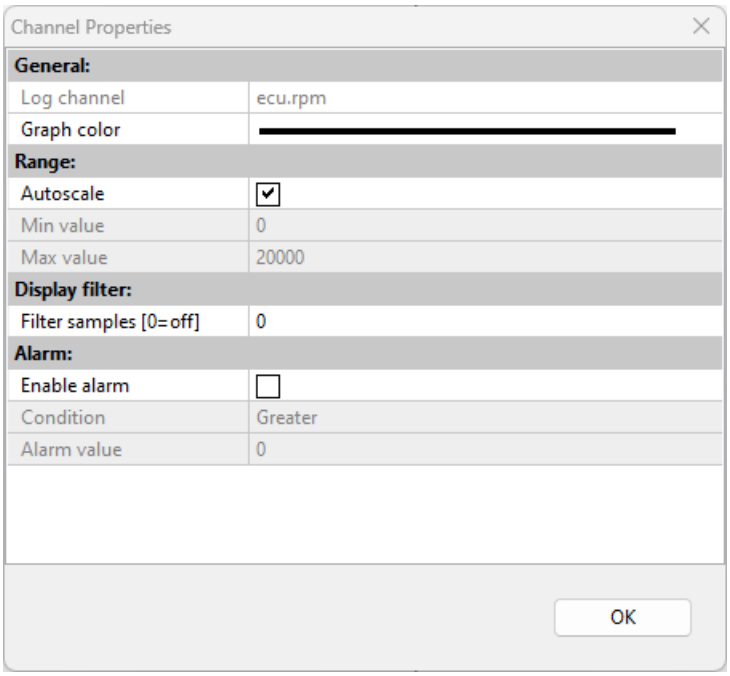


If you right-click the log area, the following menu will appear.



In this menu, new logging channels can be added (***Add***), existing logging channels can be removed (***Remove Graph***), the channel can be changed (***Change***), the position of the graph can be changed (***Move Up, Move Down***), as well as the logging frequency (***Set Log Frequency***) and the display settings for the channel (***Properties***).

Just like the main desktop of the application, the log panel has tabs used to open different logging channel groups (e.g. engine, track, etc.). These tabs work exactly the same way as the main desktop tabs.



By selecting ***Properties*** from the menu, you can access the display settings for a logging channel, such as the color of the displayed line (***Graph color***) or the range of values for a given channel (***Min and max value***). The ***Autoscale*** option results in an automatic calculation of the values range based on the log data. It is also possible to use the ***Filter samples*** option, It determines the number of samples from which the value at a given point will be determined. The 0 value means no filtering.

The panel toolbar contains icons for:

- reading the log file from the disk (***Open log***) - it is possible to append several logs for analysis (***Append data***) or replace the currently open logs with others (***Replace data***);
- saving the log file to disk (***Save log***);
- saving the displayed channels in files (***Export to text format***) with the following formats:

Export to CSV	Export to a **CSV** file with the possibility of changing the settings: ***Export frequency:*** ***Same as channel***– export frequency including actual logging frequency for individual channels ***1Hz***, ***5Hz***, ***25Hz***, ***50Hz***, ***125Hz***, ***250Hz***, ***500Hz*** - selection of export frequency, the same for all exported channels. If the export frequency is higher than the logging frequency of the individual channels, the missing data will be completed by the previous value or by interpolation (***Interpolate***). ***Decimal / Column* Separator:** ***system / ";"*** – the decimal separator is either a comma or a full stop (depending on system settings), the column separator is a semicolon. ***"." / ","*** – the decimal separator is a full stop, the column separator is a comma. The resulting file may be displayed in a ***Preview*** window ***Table*** - presents data in the form of a table. ***Raw output***- shows the contents of the resulting .csv file.
Export to PNG	Export to a **PNG** file
Export to VBO	Export to a **VBO** file (channels displayed in the log with the required channel package for the Vbox)

- It is possible to save only the selected part of the log. Ensure that the log being saved (at the time of export) is not in ***Compare laps*** mode.
- (***Zoom In***, ***Zoom Out***, ***Zoom Extents***) change of scale;
- ***Clear log***;
- ***Pause / Resume log***;

• comparing data from two selected laps of the same track by overlapping them (***Compare laps***).

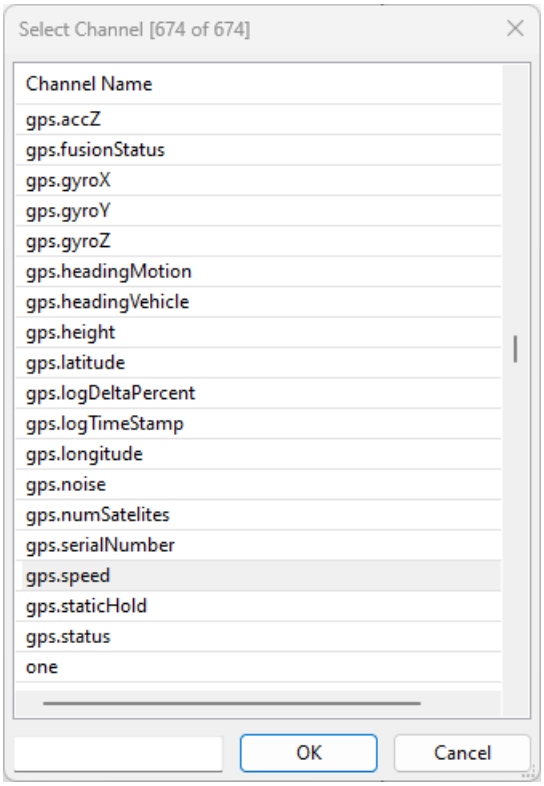
Enable	A checked tick box indicates that comparison mode is enabled.
Lap red	Lap selected for comparison - its graph is displayed in red on the *Graph Log.*
Lap blue	Lap selected for comparison - its graph is displayed in blue on the *Graph Log.*
X axis	The X axis is defined by the lap time ***Time*** or the distance from the start (track distance travelled) ***Distance.***
Speed source	Calculations that take into account the speed of the vehicle are required to compare the two laps. Speed data can be taken from the ***ecu.speed*** channel (preferred option) or ***gps.speed***.

The ***cl.timeSlip*** channel is used to show the time differences (in seconds) of the laps being compared at each point on the track. The lap displayed in red is the reference point (marked as zero), while the lap displayed in blue takes on negative values when the section of the track is faster and positive values when it is slower.

The graphic log window can also be operated using the keyboard. The following is a list of shortcut keys.

Right / left arrow	Moves a log right or left. To move a log faster, hold the shift key while pressing the arrow keys.
Up arrow or q	Decreases the time base (increasing a log)
Down arrow or a	Increases the time base (decreasing a log)
Z	Adjusts the increase to a selected log area
Mouse scroll	Time base change
Left mouse button	Area selection
Middle mouse button	Moves the log area

A channel selection window appears when changing or adding a channel to a graph. You can speed up the channel search by entering the channel name in the bottom field of the window. Available channels will then be displayed. For example, if you enter the word **gps**, only channels containing the word **gps** will be displayed.



Some of the functions are available in the application menu. The following is a description of all available Menu functions.

File	
Open project...	Open a previously saved project (***CTRL + O***)
Save project	Save to last opened / saved file (***CTRL + S***)
Save project as...	Save to a new file (***CTRL + SHIFT + S***)
Merge project	Select specific elements from one project and integrate them into current project. For more details, see: Appendix D - How-to Merge Projects *(on page 229)*
Import log...	Import a log from a memory stick (***SHIFT + F4***)
Show full creen	Full screen mode. This increases the screen space available to the application (***CTRL + F***)
Upgrade firmware...	Change the internal software of a device
Restore to defaults	Restores a device to the default settings Deletes all settings
Make permanent	Saves changes to the Flash memory of a device Additionally, a file containing the current settings is saved to the *MyDocuments / ADU / DeviceName / QuickSave* directory (***F2***).
Exit	Exit the application. The desktop arrangement is saved upon exiting (***ALT + X***).
Edit	
Undo	Undo the last operation performed (***CTRL+Z***)
Redo	Redoing a previously undone operation (***CTRL+Y***)
Show undo list	Displays a window with all operations performed.
Desktops	
Restore desktops	Reads desktop configurations from the following file: *MyDocuments / ADU / Default / desktops.adulayout*
Store desktops	Saves desktop configurations to the following file: *MyDocuments / ADU / Default / desktops.adulayout*
Open desktop templates...	Reads desktop configuration from a selected file. This allows to transfer configurations between PCs.
Save desktop templates...	Saves desktop configurations to a selected file. This allows to transfer configurations between PCs.

Add new panel	Adds a new panel to the desktop (***F9***)
Replace panel	Replaces an existing panel with another (***SHIFT + F9***)
Switch to desktop	Switch to any selected desktop
Previous desktop	Switches to the previous desktop (***CTRL + PGUP***)
Next desktop	Toggle to the next desktop (***CTRL + PGDWN***)
Devices	
Device selector	If one or more ADU devices are connected a panel enabling toggling between the devices will pop up. After switching to a device, the data between the PC and the device will be automatically synchronized. The names of particular devices can be found on the right hand side of the application's toolbar. The currently connected device is shown in bold.
Set device #n	Automatic toggling to the connected device no. #n. After switching to a device, the data between the PC and the device will be automatically synchronized. The names of particular devices can be found on the right hand side of the application's toolbar. The currently connected device is shown in bold type (***CTRL+SHIFT+1 to 5***).
Set device name	Assigns a name to a connected ADU device
Reboot device	Resets a connected device (***CTRL + SHIFT +R***)
Reconnect	Re-establish communication with the device (***CTRL + SHIFT + B***)
Receive log file	Reads logs from a USB storage device connected to the PC (***SHIFT+F4***)
Set real time clock	Sets the real-time clock of ADU according to the current PC time. This time is used to date the files of a log saved into an external USB storage device. It can also be displayed on the screen of a device.
Generate pinout	It generates an .html file with device port documentation (it shows the terminals in use and the functions assigned to them). It also generates MOBs for CAN1 and CAN2
Send data to ADU	Sends data to ADU and restarts all functions.
Tools	
Texture manager dialogue	Displays a dialogue for managing the dashboard textures (graphics). You will find more information regarding texture management further in the manual.
Customize keys	Change the shortcut keys

Reset track data	Resets all times for a given track. It is also possible to delete track data using an external button connected to an ADU device.
Set meters	Sets the vehicle mileage
Analog sensor zeroing	Resets analog sensors
Memory report	Displays a window with information on the current usage and the amount of free memory.
Logged Channels	Displays a dialogue window with a list of all log channels and their frequency. Current size of the log data is visible at the bottom of the window (number of channels and bites) (***F8***).
Project Tree	Displays the project window (***SHIFT+F7***).
Analog Monitor	Displays the analog channel monitoring window (***SHIFT + F10***).
Variables Inspector	Displays the variable monitoring window (***SHIFT + F11***).
Options	Displays a dialogue window with the application options ***2D tables color*** – color map 2D ***3D tables color scheme*** – color scheme for 3D maps ***Auto save logs*** – automatic saving of logs onto the disc ***Use mouse wheel to zoom on Graph Log*** – log scaling function using the mouse wheel

4. Status field

The status field contains important information on the status of a connected device.

Connection status	Specifies, whether a device is connected.
CAN adapter	Shows the CAN to USB interface type. The following interface types are supported: • **USBtoCAN** - ECUMASTER interface • **PCAN-USB** - Peak System interface • **Kvaser** - Kvaser interface
CAN1 status	The status of the CAN1 bus from the USB to CAN interface
CAN2 status	The status of the CAN2 bus read from CAN controller of the ADU display
USB logger state	Pendrive save status

USB buffer usage	Information about the quality of the pendrive (from A to F) and the buffer status
Board temperature	Device temperature
Saving log in progress	Log auto-save status
FW Version	Firmware version
Device type	Device type - 5” or 7”
Used resources	The number of functions used
Tables	Memory available for 2D and 3D user maps
Names	Memory available for element names (*CANbus Inputs*, *Functions* etc.)

If the CAN bus (1 or 2) status differs from OK, it means errors along the bus.

Explanation of CAN statuses of the ECUMASTER USBtoCAN adapter

Status	Typical cause of the problem
OK	CAN bus fully functional, no errors
stuff	Not all devices on the CAN bus send frames at the same speed (wrong speed of device along the CAN bus).
form	Not all devices on the CAN bus send frames at the same speed.
bitrec	No terminator on the CAN bus.
bitdom	CANL and CANH are short-circuited.
bit	Two devices send frames with the same ID but with different DLC / DATA fields.
ack	Interface is the only device on the CAN bus, no other devices. Or: CANL or CANH is disconnected from other equipment. Or: CAN and CANH are interchanged.
Offline	The programme operates in Offline mode - there is no access to the CAN bus.

5. Switching the CAN adapter between ADU, PMU and Light Client programs

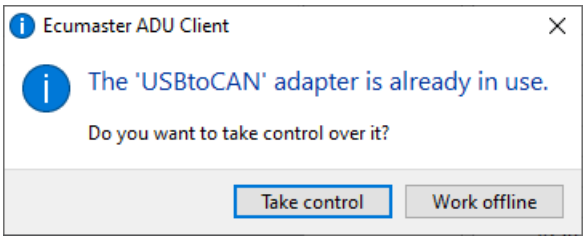
Ecumaster devices connected to the CAN bus communicate with the PC via a CAN adapter: USBtoCAN, Peak or Kvaser.
Ecumaster software used to configure the devices: ADU Client, PMU Client and Light Client.

From software version:

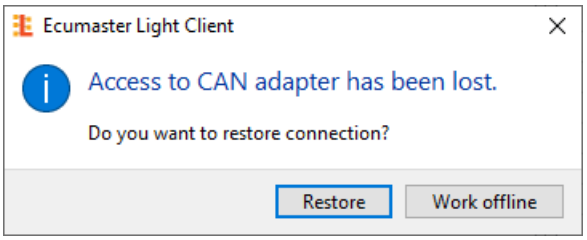
- ADU Client - 52.4
- PMU Client - 44.0
- Light Client - 1.7

it is possible to switch easily between programmes without having to close one to open another.

When opening a programme (e.g. ADU Client) whilst another programme is already active (e.g. Ecumaster Light Client), a questions automatically appears on whether the new programme (ADU Client) should take control of the USBtoCAN adapter.



The programme that was in use (Ecumaster Light Client) will then lose connection and become inactive.
When the programme is called back to the foreground, a message will automatically appear indicating that the previously lost connection can be restored.



6. Pages

Pages are the basic element defining the image displayed. The ADU allows multiple pages to be defined (their number depends on their complexity and the amount of a device memory engaged). You can switch between pages using an external button, rotary switch or function (e.g. a different page is displayed when the car is stationary than when it is moving).

There are 3 types of pages:

- ***Page*** – a standard page.
- ***Overlay*** – a page which can be overlaid on another page.
- ***Overlay with background*** – a page which can be overlaid on any other page covering it up altogether.

Additionally, an alert or warning can be displayed irrespective of the page (more information in the *Alarms and Warnings* chapter).

6.1. Default settings pg_page1

In the ADU, you can use the prepared default settings stored as **pg_page1**.

This project defines pages, analog inputs, functions, CAN streams for emu black and PMU1 and alarms.

Analog inputs (*Analog inputs*):

- **A3** - fuel level sensor (***a_fuelLevel1***)
- **A4** - display tilt reset button (***a_zeroImuPitch***)
- **A5** - button marking the start/ finish line on a closed track (***a_acquireButtonDefinedTrack***)
- **A6** - alarm acknowledgement button (***a_acknowledgeAlarm***)
- **A7** - button to change the page to the previous page (***a_prevPage***)
- **A8** - button to change the page to the next page (***a_nextPage***)

Standard CAN Stream:

- CANbus Message Object for emublack ***(m_emublack***)
- CANbus Message Object for PMU1 (***m_pmu1***)

Alarms:

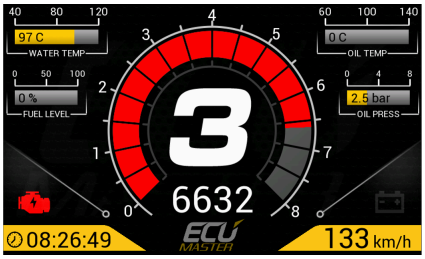
- ***al_battery*** - alarm indicating the state of charge of the battery with the engine running (if below 13 V a “Battery voltage too low” message will be displayed)

Functions:

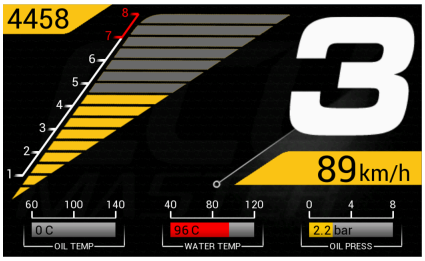
- ***f_isGPSValid***- function used to change the color of the satellite icon located on the *pg_track* page depending on the quality of the GPS signal
- ***f _ oil_alarm***- function used to light up the oil pressure light located on *pg_trackSimple* when oil pressure drops below 1 bar
- ***f_batt_alarm*** - function used to light up the battery light located on *pg_generic* and *pg_trackSimple* when the charge drops below 12 V, and to light up the led
- ***f_clt_alarm*** - function that can be used to signal the coolant temperature alarm

Pages:

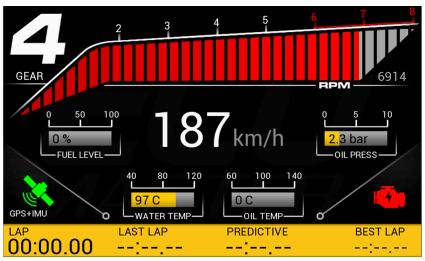
- ***pg_generic*** - general page



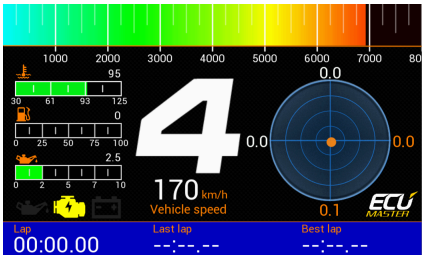
- ***pg_generic2*** - optional general page



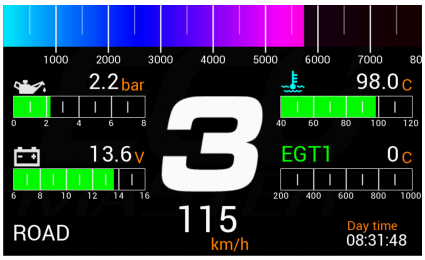
- ***pg_track*** - page prepared for track races, shows running times



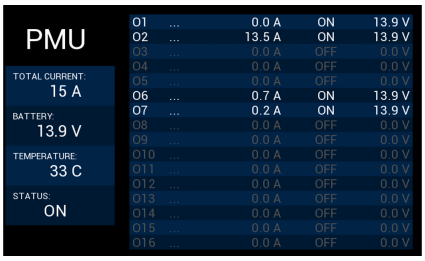
- ***pg_trackSimple*** - optional page for track races



- ***pg_rally*** - page prepared for rallies



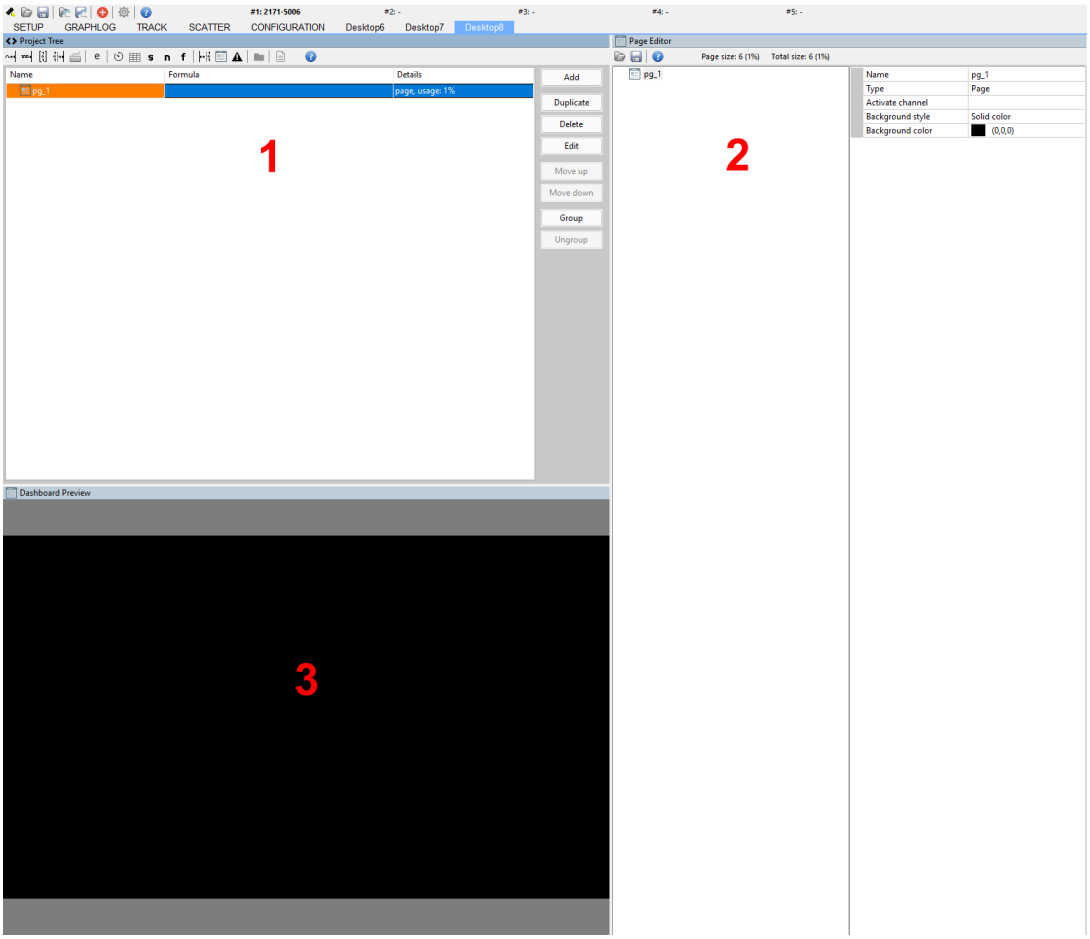
- ***pg_pmu*** - page displaying PMU information



- ***ov_buttonDefinedTrack*** - a page displayed for a short time after defining a new track, with information about the length of the track and setting the start / finish line

6.2. Creating a page

To add a page, click *Add* in *Project Tree* and select *Page*. A ***pg_1*** (1) page should appear in the project.

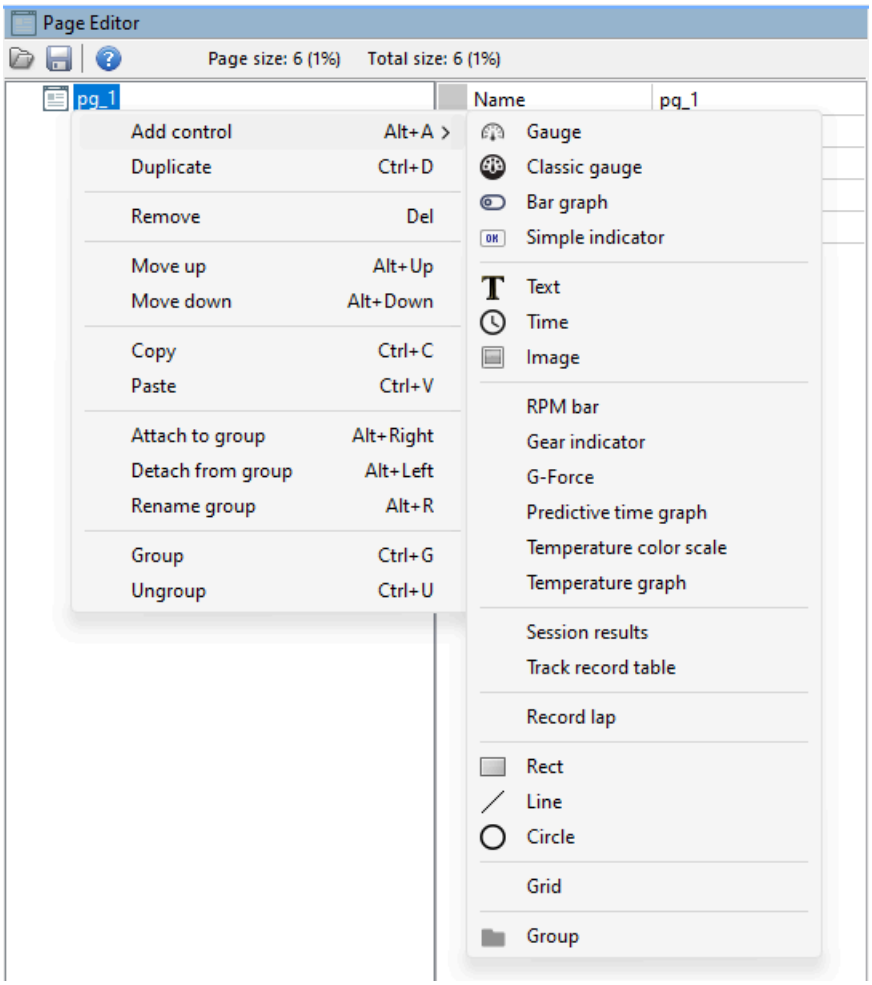


The currently selected page with its elements and configuration should appear in ***Page Editor*** (2). At this point we don't have any elements defined for an example page yet, but it is possible to set the following page attributes:

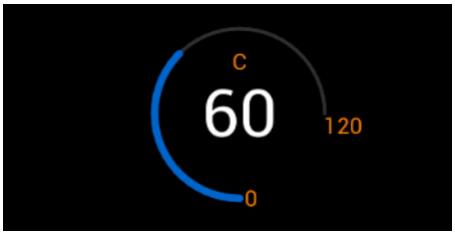
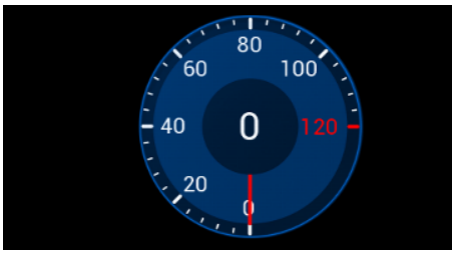
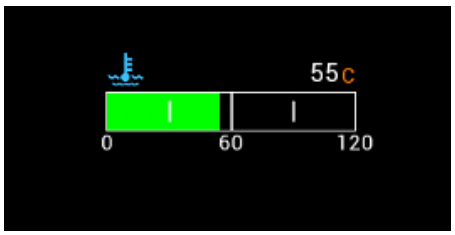
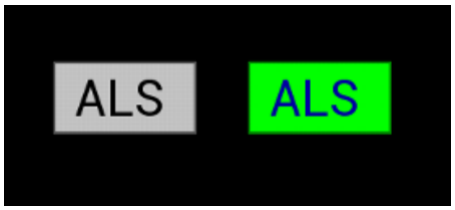
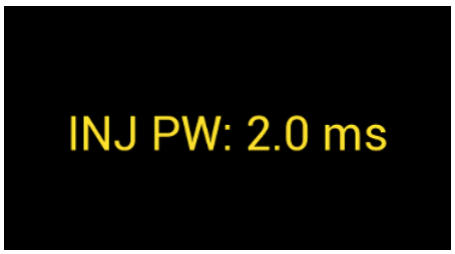
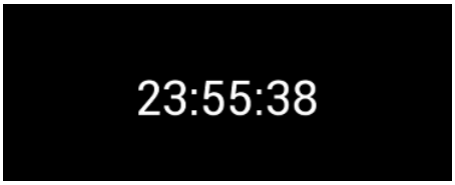
Attribute	Description
Name	Page name in ***Project view***
Type	Page type: • ***Page*** – a standard page • ***Overlay*** – a page than can be overlaid on another page • ***Overlay with background*** – a page similar to the ***Overlay*** page, but with a non-transparent background
Activation channel	Name of the channel or function that can automatically activate the page

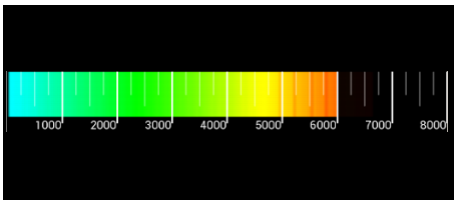
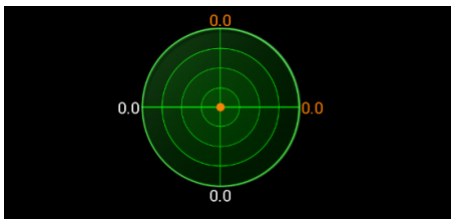
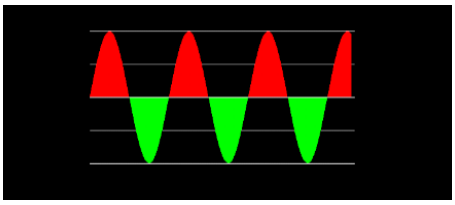
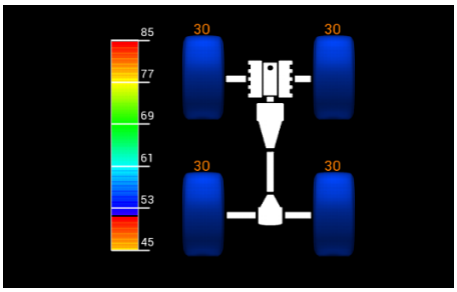
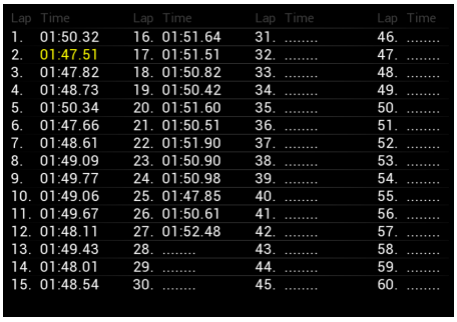
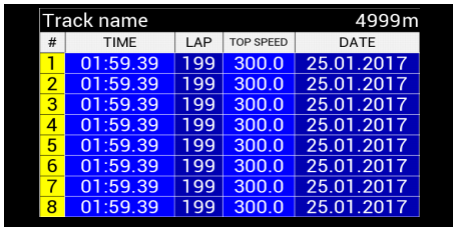
Attribute	Description
Background style	Background style: • ***Solid color*** - Background filled with a defined ***Background color*** • ***Theme*** - graphic background, predefined in the device memory. There are several types of backgrounds to choose from: Theme 1, Theme 2, Theme 3, Race track, Race track mirrored
Background color	The color of a page background

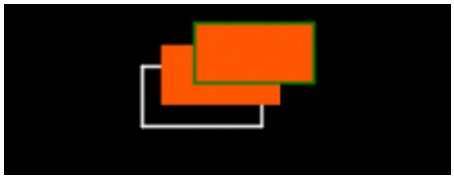
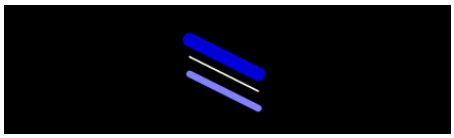
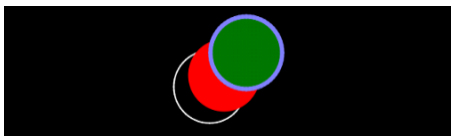
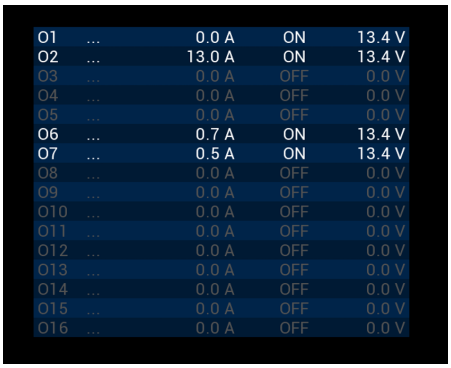
To add an element to a page right-click the page name in the page editor and select ***Add control*** and the desired element from the menu. You can also open the menu by pressing ***Alt + A***.



6.3. Page elements

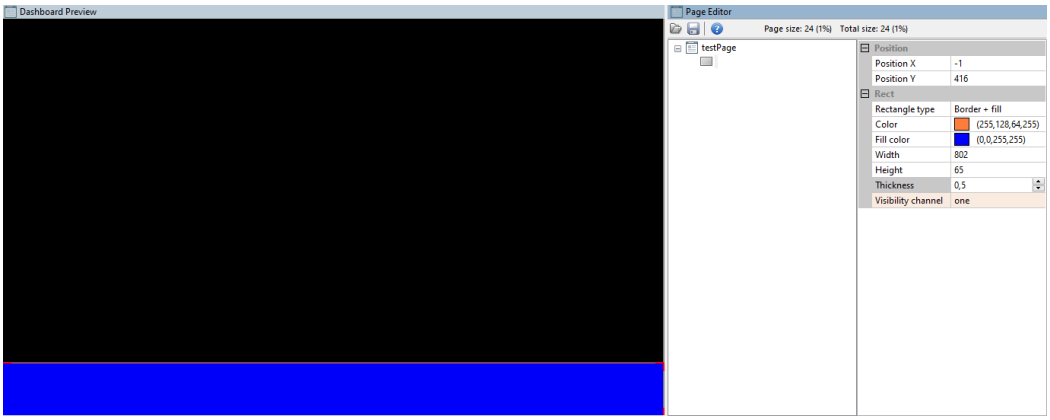
Element	Description	Preview
Gauge	A round gauge to display a numerical value and its visualization using a circle segment.	
Classic gauge	A classic gauge resembling a car gauge. It allows to display a numerical value, as well as to indicate it using a hand on its dial.	
Bar graph	An indicator displaying values in the form of a moving bar (either horizontal or vertical). It is also possible to display an icon symbolising the measured value.	
Simple indicator	It allows to display two texts in two different color sets that you can switch between using the log channel or functions.	
Text	An indicator allowing to display text in different font sizes and colors. You can additionally attach a log channel (e.g. injection time, cooling liquid temperature, etc.).	
Time	Allows to display formatted time. You can choose from the following times: real time, lap time, last lap time, best time and session time.	
Image	Allows to add graphical elements such as background, logo, icon on a page. You can use the graphics available in the device or load one of your own from a file.	

Element	Description	Preview
RPM bar	This indicator is dedicated for displaying the engine speed. There are four ways in which it can be displayed - a horizontal bar, a curved bar, a vertical triangular bar, and a round gauge.	
Gear indicator	Indicates the currently selected gear and uses a special, enlarged font	
G-Force	The g-force indicator uses an internal accelerometer of the device	
Predictive time graph	A graph displaying the current difference between a lap time and the best time. Green indicates the better time, while red the worse one.	
Temperature graph and Temperature color scale	Displays tire or brake disc temperature from thermal imaging cameras and indicates their maximum temperature. The indicator can also display the temperature in the form of horizontal bars with a temperature gradient (16 values for each tire).	
Session results	It allows to display last 60 lap times, the best of which is highlighted in a different color. These times aren't saved in the device memory and reset after powering down.	
Track record table	An indicator showing 8 best times scored on a given track. These times are saved in the device memory.	

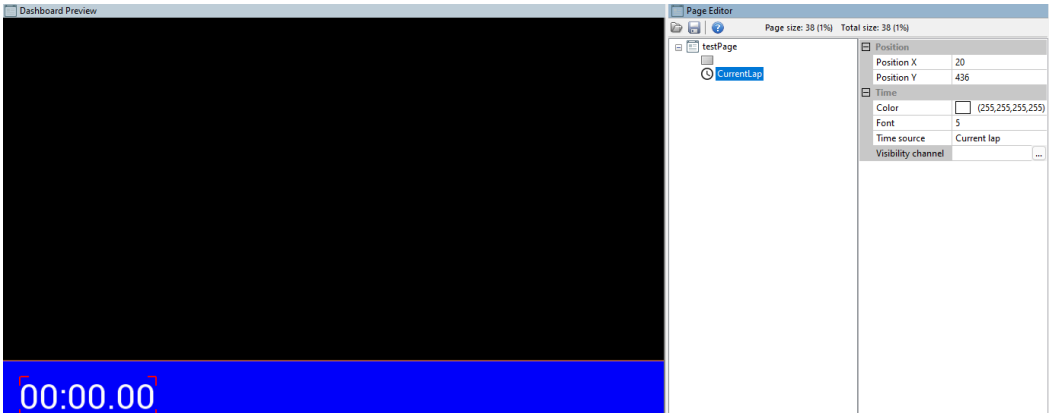
Element	Description	Preview
Rect	Displays a rectangle in the form of a frame or a filled-in shape.	
Line	Displays a line with the selected color and thickness.	
Circle	Draws a circle of any color, with or without a filling.	
Grid	A table displaying values defined by separate channels in each cell	

6.4. Adding elements to pages

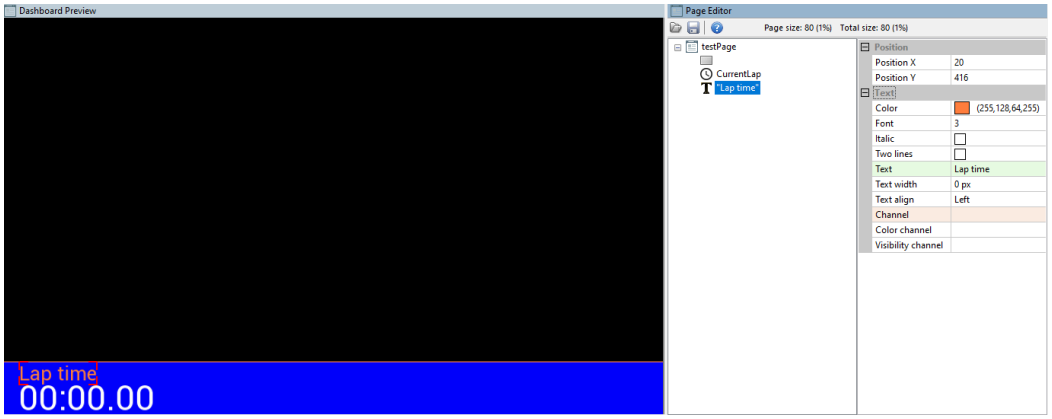
The following text will discuss the process of creating a sample page. Let's start with adding a ***Rect*** type object to a page. By clicking the object with the left mouse button, we can move it. If you are connected to an ADU device, the object will also move around on the display. You can change the ***Rect*** object parameters in the ***Page Editor*** window. After entering the parameters as shown in the image below, the page should look as follows:



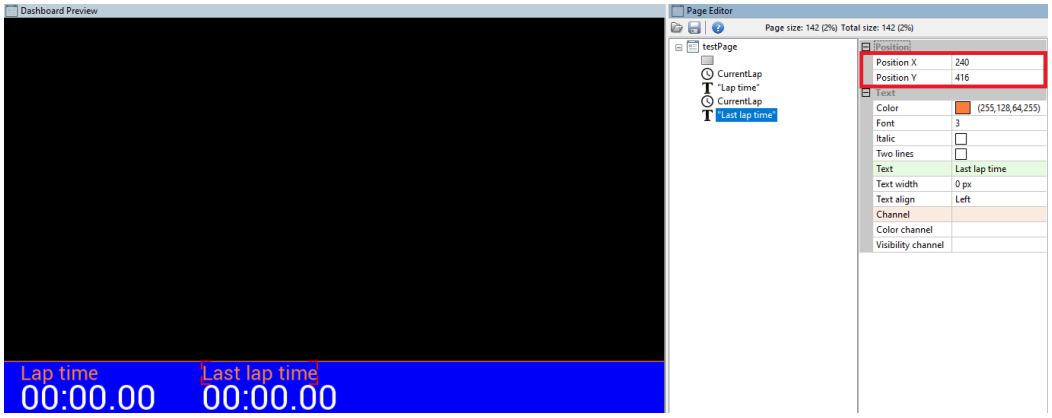
We will add information on the current lap time in the next step. To do this, a ***Time*** object should be added.



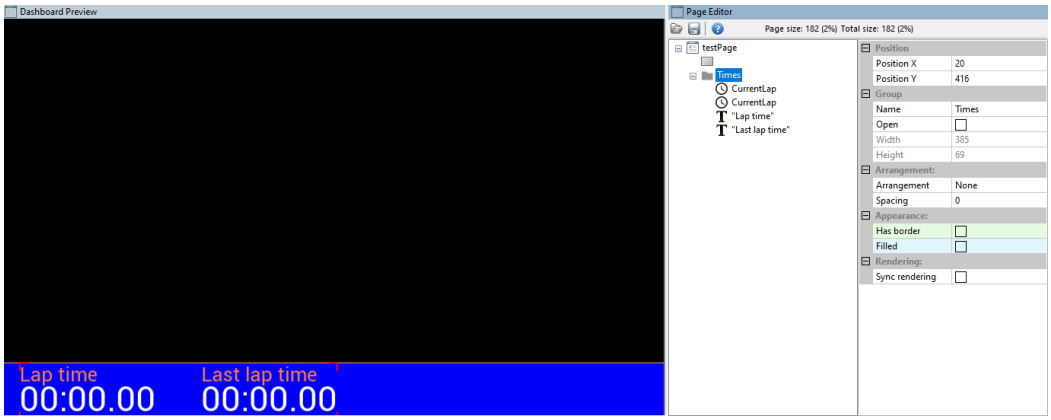
We will use a ***Text*** object to add a description explaining what the inserted time indicator relates to. You can enter the text to be displayed on the screen in the ***Text*** field. You can also choose its color. This object allows to display both the static text and the channel and variable values, which will be discussed in more detail further in the chapter.



It is possible to duplicate objects on a page. To do this, select an object in the page editor and press CTRL+D. In this example *we* will duplicate the *Current lap* and the *Lap time* objects to show the last lap time next to it.



To accurately set position on a page, you can enter values into the *Position X* and *Position Y* fields. Grouping objects is a very useful function. Thanks to it, it is possible to work on an entire group (e.g. by moving it around the page). It is also possible to automatically arrange the position of objects relative to each other (it will be discussed in more detail further on). To group objects you should select them in the editor and press *CTRL+G.* You can name a group to facilitate its further editing. In our example we will create a group out of text and times and let's name it *Times.*

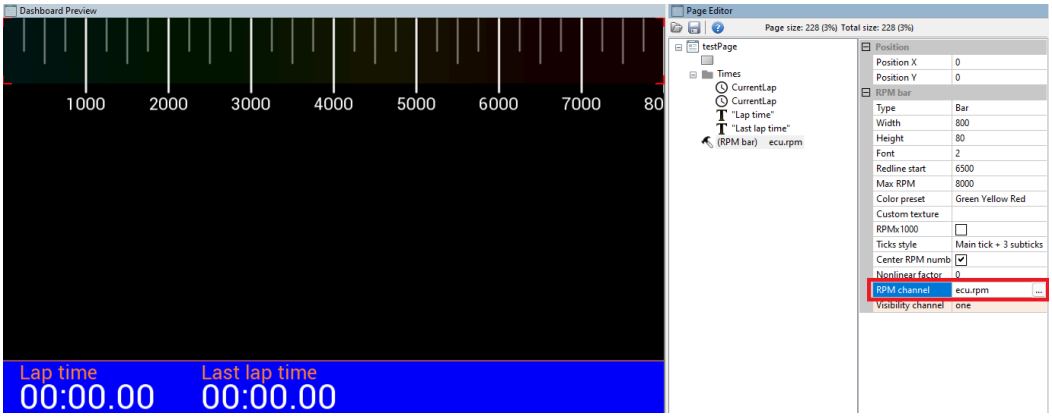


Another element that we will add is the engine speed indicator. To do this, the ***RPM Bar*** object should be added to a page.

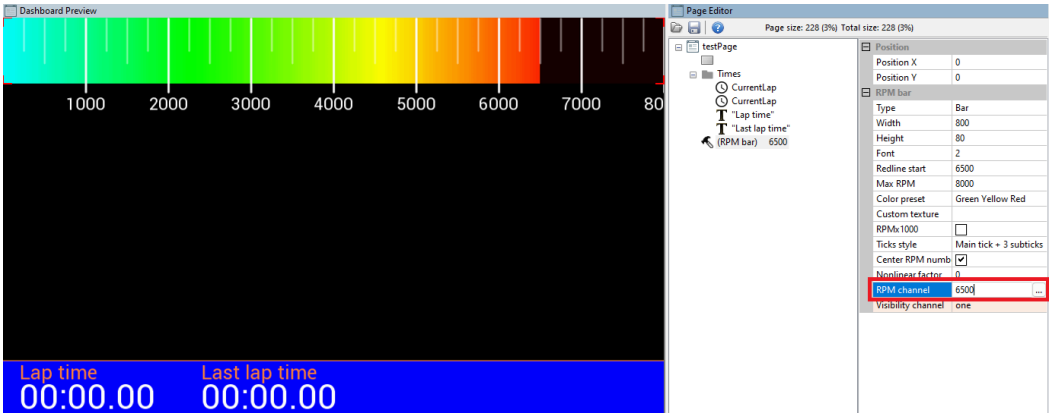
When adding an object that will display a value that changes over time, an important configuration element is the choice of channel that defines that value. This could be information from the CAN bus, a function, an analog or digital input value or a constant value. In the case of RPM, the information will come from the engine management sent via the CAN bus or serial communication. Most channels for ECUs available on the market are similar (such as RPM, lambda, coolant temperature, etc.), and they are represented in the ADU by *ecu.** channels.

Upon loading a CANX file suitable for the connected ECU or configuration of a serial connection, some ***ecu.**** channels will display the values sent by the ECU. For more information about connecting engine control units available on the market see application notes. There is also information about the available log channels. If the ECU sends specific channels not included in the ***ecu.**** channels, they are assigned unique names. They can be previewed in ***Project Tree*** or found in an application note. One of the parameters of the ***RPM Bar*** object is the ***RPM channel*** field. The name of the channel must be entered there. You can also click the '**...**' icon or press the **F4** button to open a window showing all available channels. Entered the '***ecu***' phrase in the search field will cause all available channels containing the word ecu to be display.

A set of ***ecu.**** channels has the advantage that previously prepared pages will operate properly independently of the connected ECU (RPMs will always be ***ecu.rpm***, the cooling liquid temperature will be ***ecu.clt***, etc.).

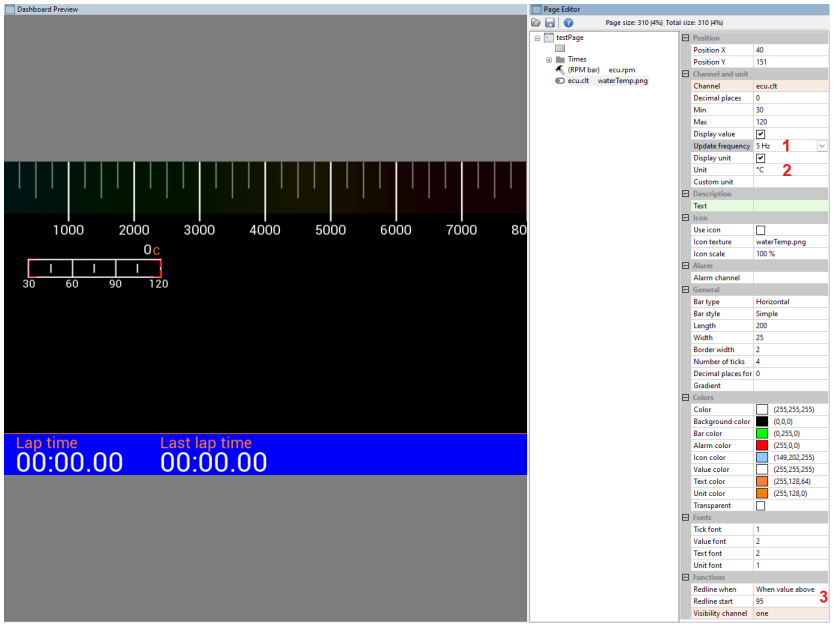


To test an indicator you can enter a numerical value (only integer values) in the ***Channel*** field. In our case, when we type the value 6500 in the ***RPM channel*** field, the engine speed indicator should look as in the picture below.

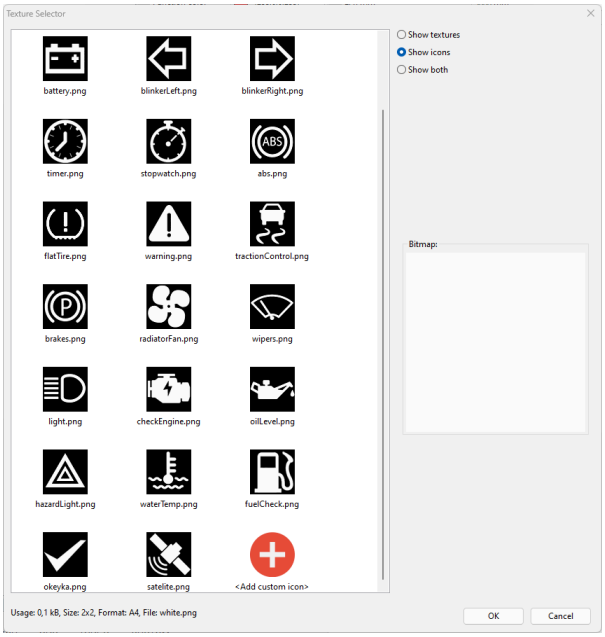


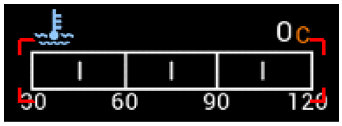
In the next step we will add a horizontal cooling liquid temperature (CLT) indicator to our page. For this purpose we will use the ***Bar graph*** control, which has much more options than those used so far. ***Update frequency*** (1) is a very important parameter. It limits the refresh rate of the value on the screen to a predetermined frequency. An ADU device refreshes the screen at a frequency of 50 Hz. The value of the channels which display quickly changing values (e.g. pulsating oil pressure) are very difficult to read as they change very frequently. Limiting the refresh rate allows you to impose a kind of filter on them. Measurement units are another important function. In our case, we want to display temperature in degrees Celsius. If we change the units to Fahrenheit in the ***Units*** field, the displayed values will immediately be converted accordingly. Another important function is “redline” *(3)*, which allows to change the indicator appearance when a predetermined value is exceeded. In the ***redlineWhen*** field you can select the conditions, while in the ***redlineStart*** field -

the value at which the appearance of the indicator will change. The “redline” appearance is defined in the ***colorRedline*** field.



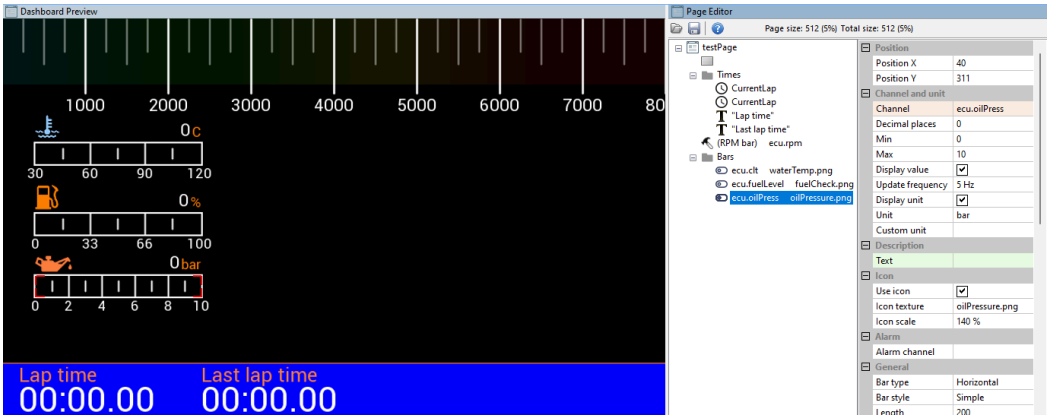
It is possible to display an icon in the ***Bar graph indicator field***. To that end, select the ***Use*** icon option and select the desired icon from the ***Icon texture*** field (by clicking the ‘...’ icon). A graphics selection window will pop up from which you will be able to select or load your own icon (you can find more information on icons further in the manual). After selecting the *waterTemp.png* icon the indicator will look as shown below.



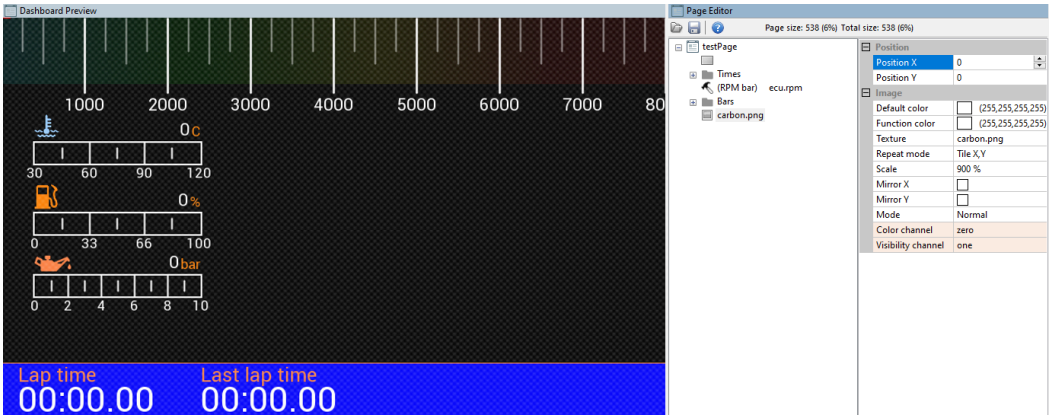


In the next step we will copy the created indicator twice in order to create a fuel level and an oil pressure indicator. After copying, set the new position of the indicator, select the appropriate

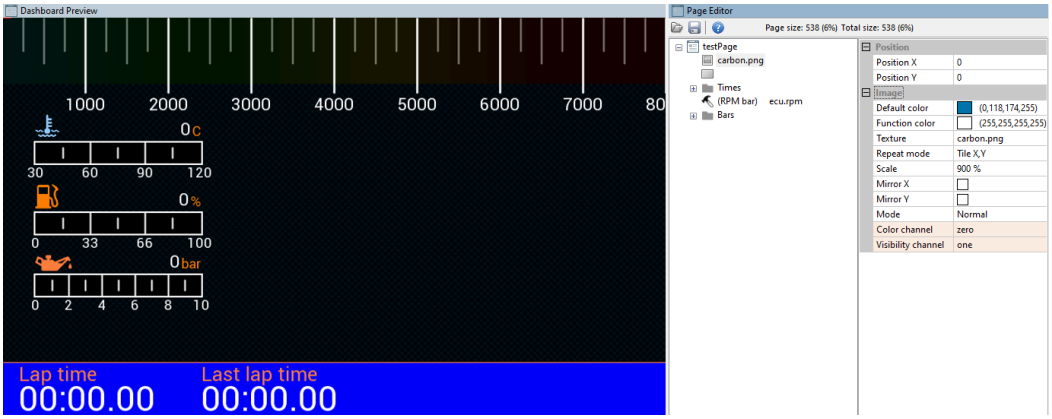
icons and assign the appropriate channels in the Channel field. These will be ***ecu.fuelLevel*** for the fuel level and ***ecu.oilPress*** for the oil pressure, respectively. It is also worthwhile to group the indicators to facilitate their editing in the future. After copying the page should like as below.



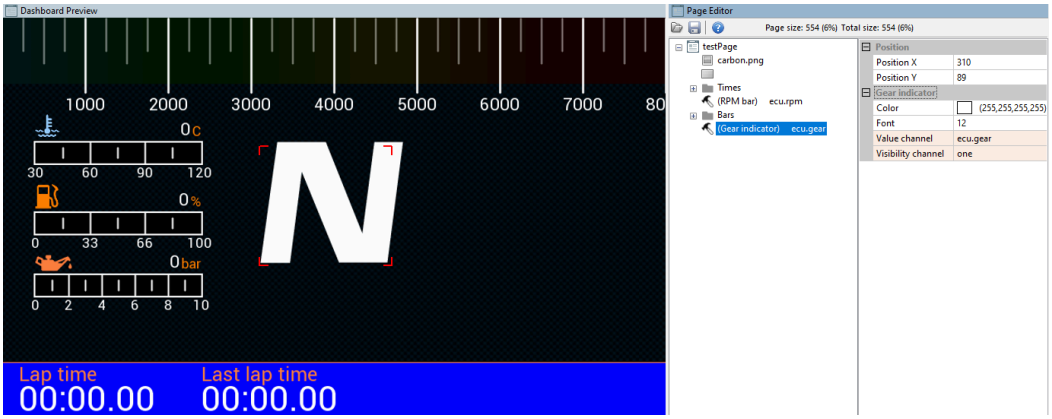
It is also possible to add background texture. For the purposes of our example we will create a carbon fibre-style background. To do this, use the ***Image*** object, and then select the *carbon.png* texture for it. To cover the entire surface with it, select the ***Repeat mode Tile X, Y*** option and set ***Scale*** to 900% to spread the texture over the entire screen. Another option to create a background is to select a predefined background in the page configuration (***Background style***).



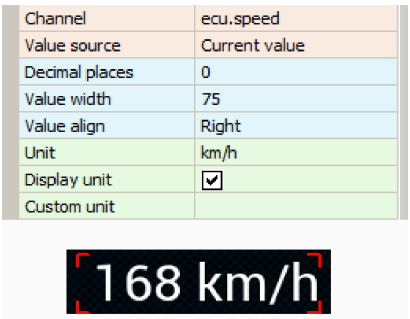
Note that our texture is displayed “over” the existing elements. To move it “under” it should be placed as the first element of the page. The order of drawing the elements is determined by their order on the list. To change the object position on the list you should select the element and then move it using the combination of ALT + up arrow or ALT + down arrow. Let's also change the background color to dark blue.



To insert the indicator for the currently displayed gear, use ***Gear indicator***. You should select a channel (***ecu.gear*** in our case) and a font size. The indicator for the currently selected gear comes with a special font for displaying large digits.

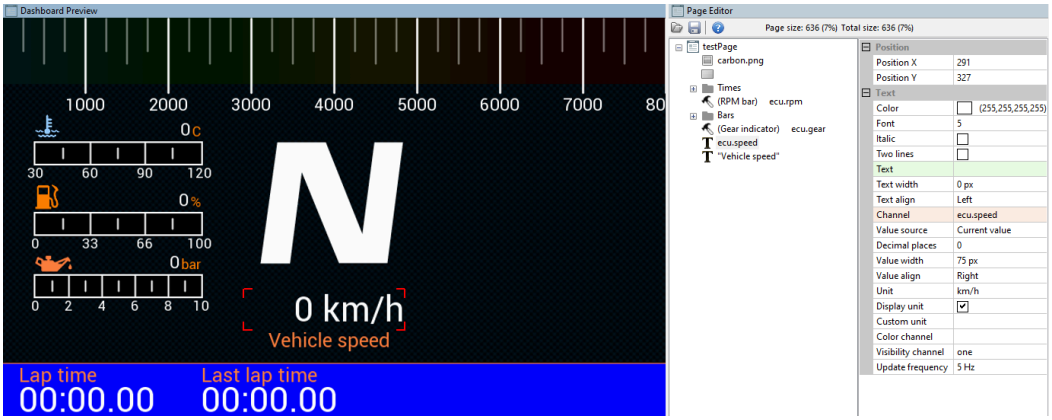


Under the gear indicator we will now display a numerical value representing the speed of the vehicle.

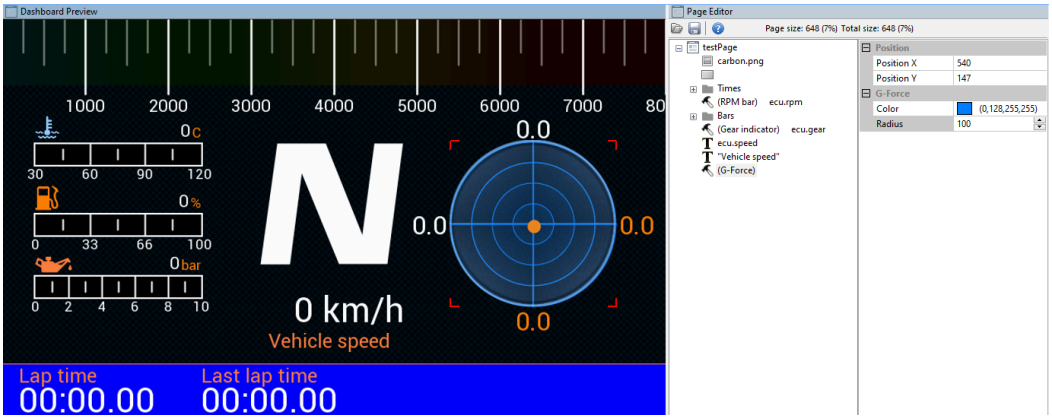


We will add the ***Text*** indicator for this purpose. In addition to displaying static texts, it allows to display the values of the channels and the variables. Enter the name of the channel or a variable in the ***Channel*** field. When a channel is selected, its unit will be shown, which can also be displayed together with a numerical value. Defining the maximum width (in pixels) of the displayed value is very important when it comes to formatting text. In the example below, the speed will be displayed in km/h. To prevent the text from shifting after changing the numerical value (e.g. From 7 km/h

to 11 km/h) we define the text width of the displayed value at 75 pixels. To change the unit from km/h to mph, select it from the ***Unit*** field. The conversion of the value to adjust it to the selected units will be done automatically. In the example below, let's also add a description under the speed explaining what the indicator shows.



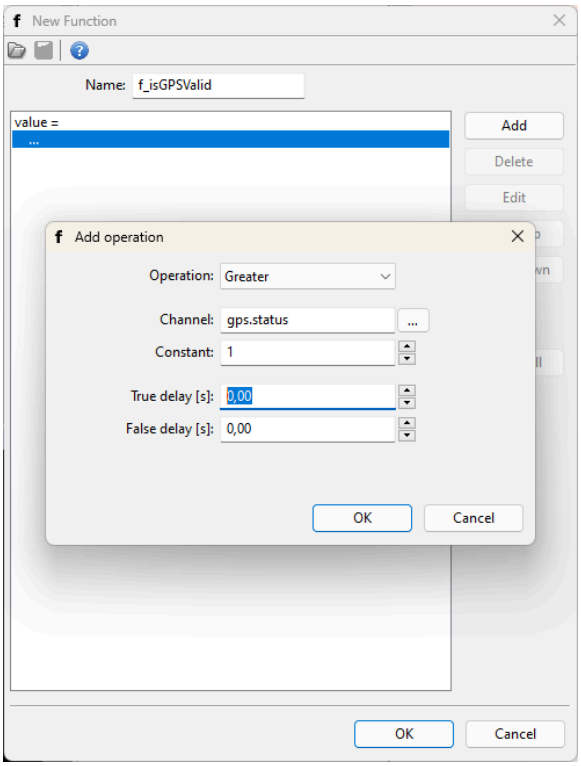
Another element that we will place on the page will be a G-force indicator (***G-Force***). It shows the current G-force acting on the vehicle. The ADU is equipped with a built-in accelerometer which, when installed in a vehicle, may require calibration (setting the 0 g point). You will find more information about the accelerometer further in the manual.

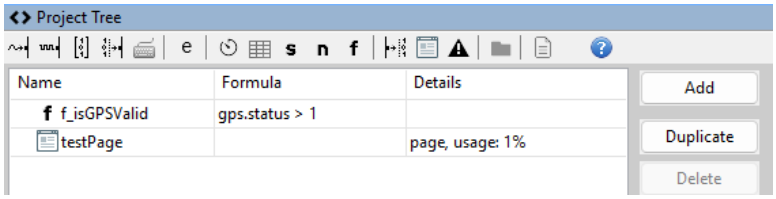


When using a GPS module, it is worth ensuring that the user can see its status. To this end, we will build a GPS status module. We will use two channels: *gps.status*, showing the mode of operation of the GPS, and *gps.numSatelites*, showing how many satellites the module is currently using.

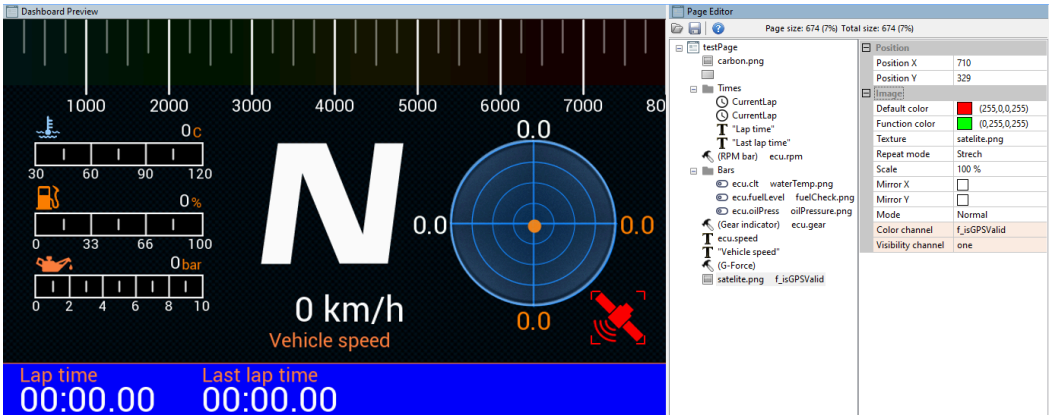
To create the indicator, we will first use the satellite icon (*satellite.png*) and place it on the page using the ***Image*** indicator. The indicator can also be used to change colors depending on the channel / function assigned. Let's set red as the default color when the GPS is not working and green when the GPS is working correctly. We also have to create a function that will check if the value of the ***gps.status*** channel is greater than 1. 0 value means that the GPS is disconnected, and value 1 means that it is not synchronized with the satellites. Value 2 means that the GPS

has fixed the position. We add the function in the ***Project Tree*** using the ***Add*** (function) button. Let's call it *f_isGPSValid* and add a condition using the ***Add*** button. In the ***Operation*** field we choose ***Greater***, and in the ***Channel*** value - *gps.status.* Value to be compared (***Constant***) is set as 1. In this way, we have created the condition f_isGPSValid = gps.status > 1. You can find more information regarding the functions further in the manual.



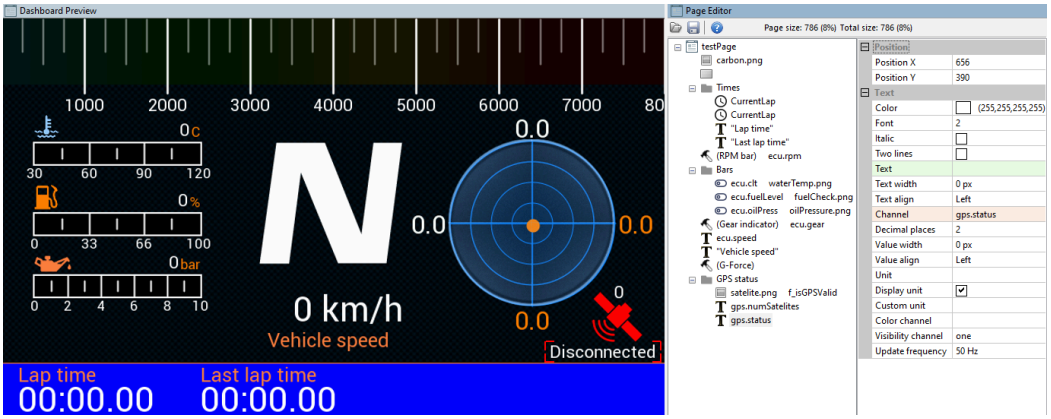


Enter ***f_isGPSValid*** in the ***ColorChannel*** field of our satellite icon. We can also test the color change option by entering there the value 0 (basic color) or 1 (alternative color).



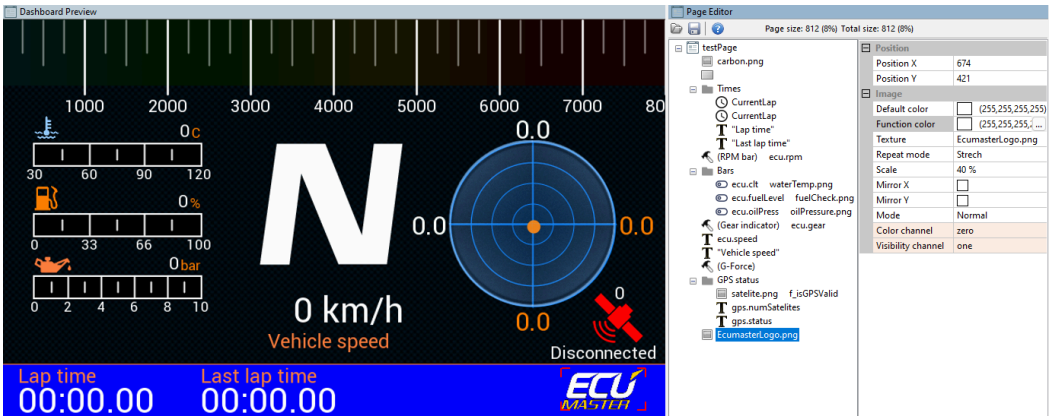
We will place two text fields over and under the icon. The upper field will indicate the number of satellites (*gps.numSatelites* channel), while the lower field will display the current GPS module status. In the case of the *gps.status* channel, the text field will automatically transform its value

into text (e.g. disconnected, gps 3D, etc.). We will additionally group all elements of indicators into the GPS status group.



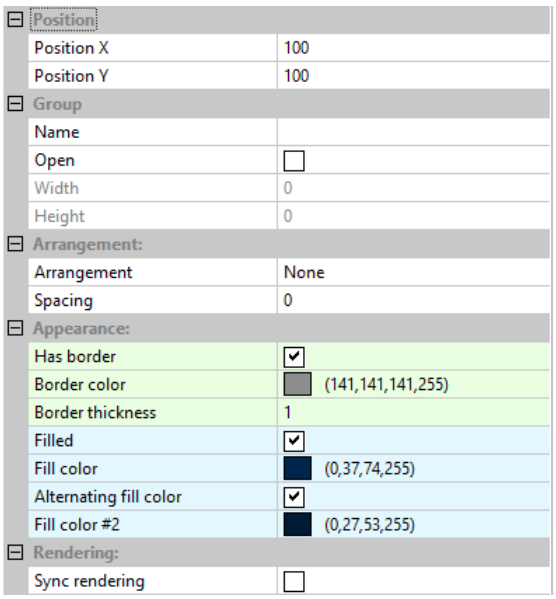
The last element to be added to the page will be the company logo. In the example we will use the logo of Ecumaster embedded in the device, however, you can add your own graphics and

display the logo of your own company. You can find more information about this further in the manual. In order to add graphics, select the ***Image*** indicator and then the desired logo (in the ***Texture*** field).



6.5. Grouping objects

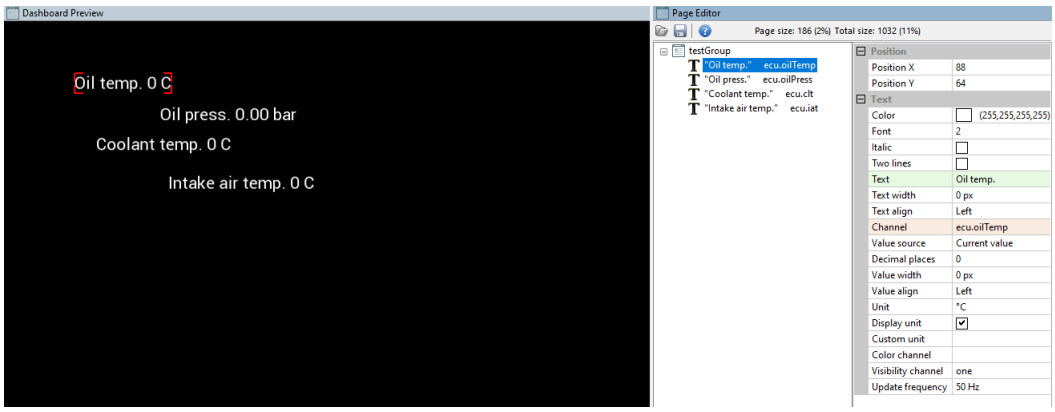
The ***Group*** object has two functions. It makes it possible to group objects for ease of management, and also makes it possible to automatically arrange objects within a group. All group object options are shown on the adjacent figure. Some settings are hidden by default and appear after activating a given option (e.g. when ***Has border*** is selected)

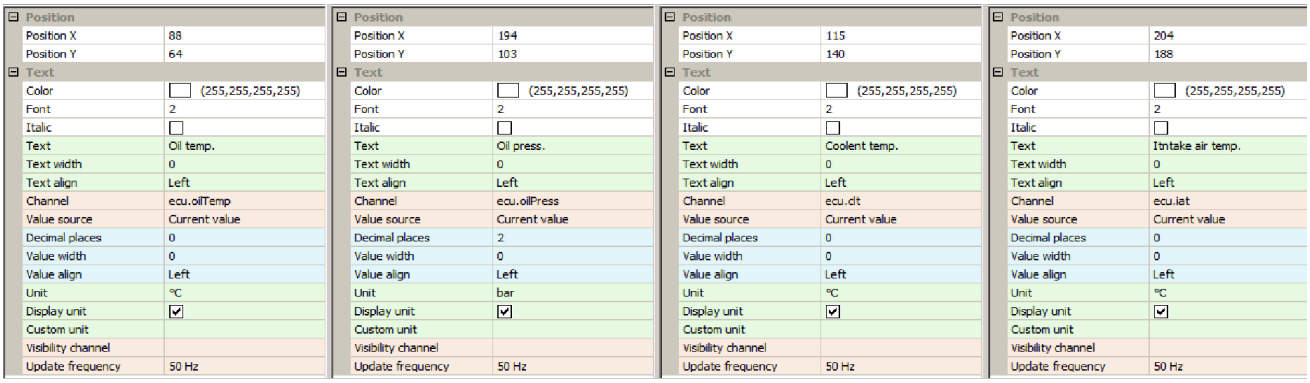


Parameter	Description
Position X, Y	Group position on the page Changing the position value affects the position of all objects in the group.
Name	Group name
Open	A group that is open (*Open* box checked) allows manipulating the position of the sub-objects on the preview screen. When a group is closed, it is only possible to move the entire group.
Width	Shows the width of the entire group (in pixels).
Height	Shows the height of the entire group (in pixels).
Arrangement	Parameter enabling automatic arrangement of group elements: • ***None*** – no arrangement of elements • ***Horizontal*** – horizontal arrangement of elements • ***Vertical*** – vertical arrangement of elements
Spacing	Additional distance between group objects being arranged (in pixels)
Has border	Enables a frame to be drawn around the group.
Border color	Frame color around the group

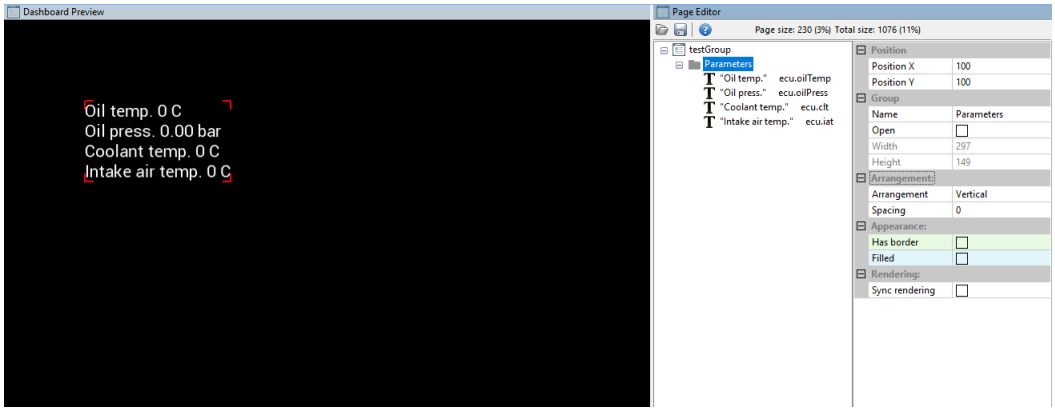
Parameter	Description
Border thickness	Frame thickness in pixels
Filled	Enables the filling of fields created when arranging group elements.
Fill color	The fill color
Alternating fill color	Enables an alternative fill color. The fill will alternately use the color defined in ***Fill color*** and in ***Alternating fill color***.
Fill color #2	Alternative fill color
Sync rendering	

To demonstrate the possibilities of arranging elements using a group, we will create an example table displaying engine performance. In our example, we will display values for oil pressure *(ecu.oilPress*), oil temperature *(ecu.oilTemp)*, coolant temperature (*ecu.clt*) and intake air temperature *(ecu.iat*).

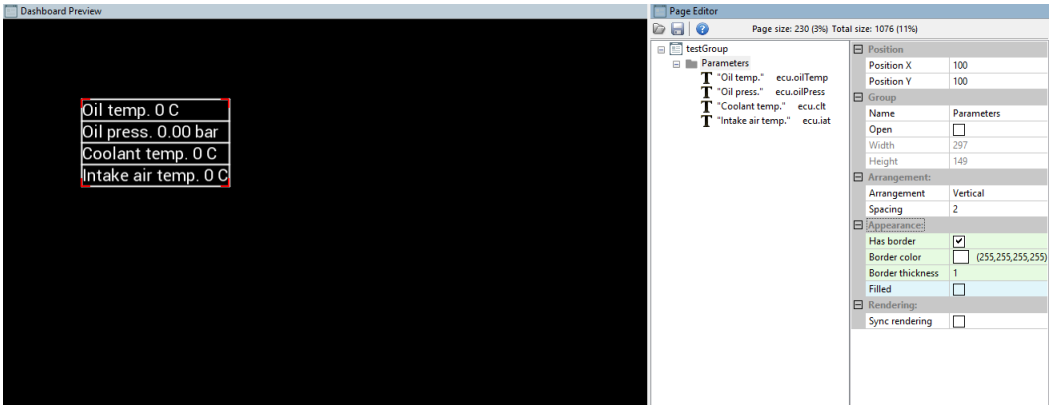




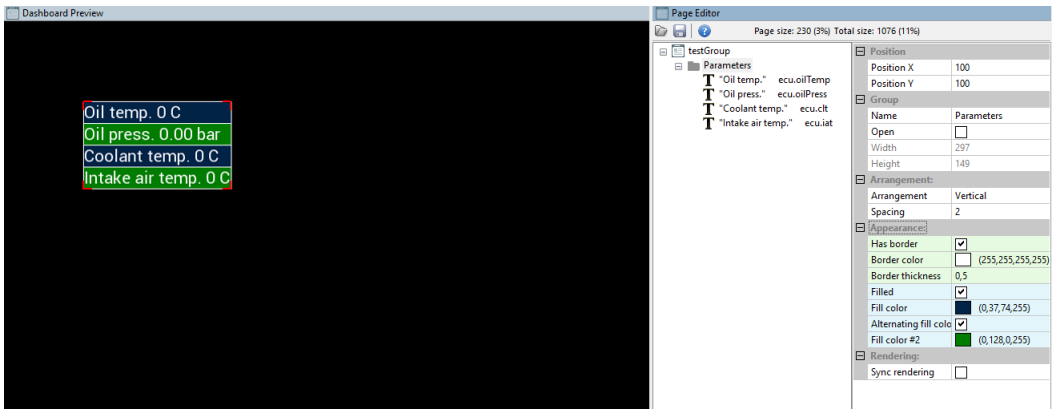
Let's select all 4 text objects and choose ***Group*** (CTRL+G). In the group options, let's set the name to '*Parameters*' and in the ***Arrangement*** option select *Vertical.* This will automatically arrange the objects along the vertical axis. Using the ***Spacing*** parameter, we can increase the distance between elements. The page should look as follows.



As you can see, the text has been tidied up, but it is still not readable because the displayed numerical values are in different places. To solve this problem, set a fixed text width in all ***Text*** objects, creating two columns: one for the description and one for the value. To do this, select all four ***Text*** objects and enter 160 in the ***Text width*** field. This will change the width in all selected objects. In addition, let's set the ***Has border*** parameter in our group and select white as border color.



Using the ***Filled*** option, we can change the background color of our 'table'.

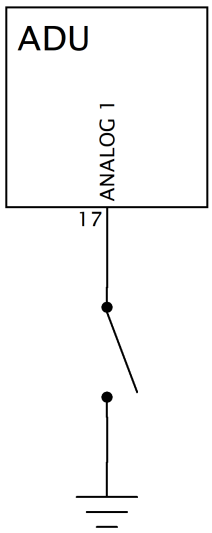


To remove a group from the page (leaving the objects), select it and choose ***Ungroup*** (CTRL+U) from the context menu (right mouse button). To exclude objects from a group, select them and choose ***Detach from group*** from the context menu (ALT+Left arrow). To attach an object to an

existing group, move the object in the hierarchy under the group of interest and then select ***Attach to group*** (ALT + Right arrow).

6.6. Toggling between pages

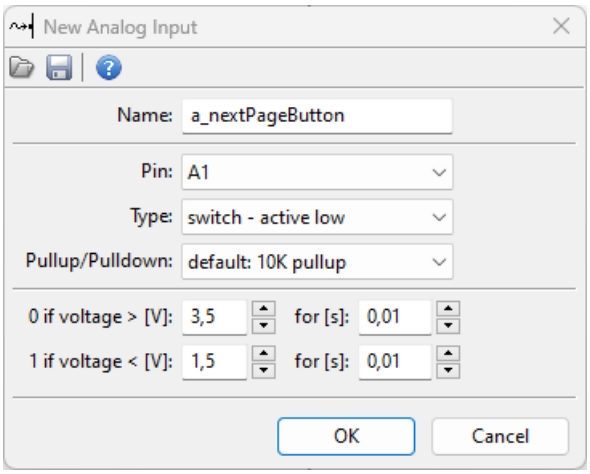
There are two methods for toggling pages. The first one employs a button than can be connected directly to the ADU (using analog or digital inputs) or to another device (e.g. a CAN switch board) and then sent to ADU using the CAN bus.



The buttons that will be used to toggle pages (*Next page* and *Previous page*) are defined in the *Buttons* panel.
A second method of toggling pages is to use a rotary switch (*Rotary switch*). You can define a function and assign it to the Activation channel attribute for each of its positions.
Page toggler (*Rotary switch*) can be defined in the *Configuration > Changing pages > Page selector channel panel.*
Page selector takes control of the keys *Next page* and *Prevous page* defined in the *Buttons* panel.

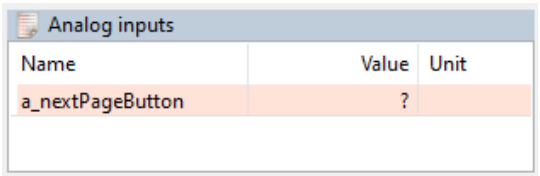
The following example shows how to connect and configure a page toggle button. The button is connected to the analog input Analog #1 and shortened to ground.
In the *Project Tree* window the button must be defined as an analog input. To do this, press ***Add*** and then select *Analog Input*.

The following window should pop up:

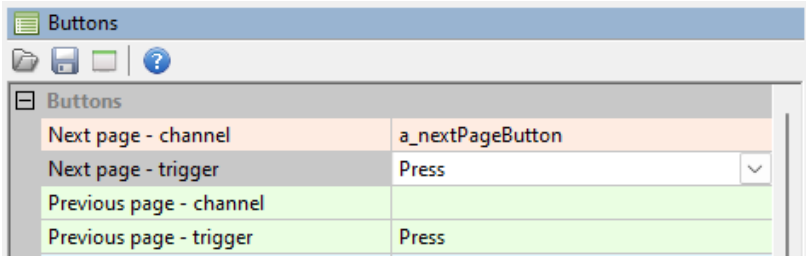


Name - determines the name of the input that will be visible in the project. In the Pin field define the analog input to which the button is connected (Analog #1 in our case). Choose *Switch - active low* (which means that the button is activated at a low state). You also need to connect a 4K7 pull-up resistor.

To preview the button state, open the *Analog Inputs* panel, in which it is possible to track all *Analog Inputs*). The value of the button should be 0 for an unpressed button and 1 for a pressed button.



A button defined in this way should be assigned in the *Buttons* panel to the option of toggling a page to the next one *(Next page - channel*). It should be noted that when the button on the last page is pressed, the view will go to page one. You can use the buttons only to switch between the Page type pages. *Overlay* pages are ignored.



Procedure for connecting a second button and assigning it to the *Previous page* switch functions the same.

Another method for toggling between the pages is to use the *Activation channel* page attribute. It is available after selecting a page in the *Page editor* together with the attributes such as page name, background color or type.

Overlays are the main application of this type of toggling. They make it possible to overlay (another page) over the displayed page whenever another event occurs. To make use of this possibility, a function must be created that activates the overlay. It will be visible as long as the function result differs from 0.

6.7. Startup screen

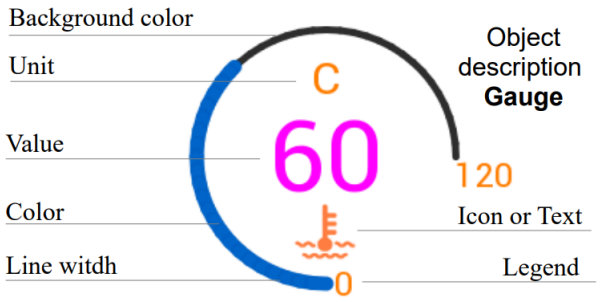
A user can create a startup screen that will appear immediately after switching the device on and will be displayed for a time defined by the user. To do this, it is required to configure the *Startup Screen* parameters in the *Configuration* panel.

Parameter	Description
Enable	It activates the startup screen.
Texture	The texture displayed on the startup screen (centred)
Scale	The scale of the displayed texture
Duration	Time during which the start-up screen will be displayed
Color	The color of the displayed texture
Background color	The color of the background

7. Graphic objects

7.1. Gauge

This object allows to display a numerical value using a circular segment. You can also add a numerical value, a description, an icon and the units in which values will be expressed to the gauge.

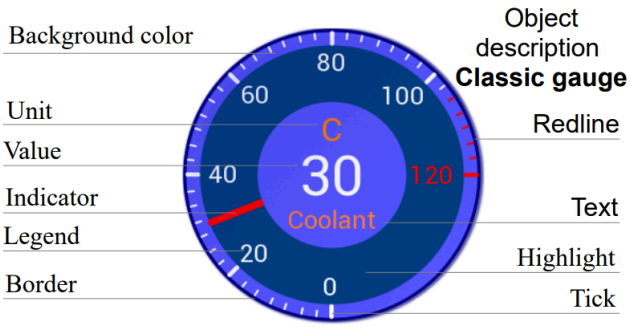


Parameter	Description
Position X,Y	Object position on the page The reference point is the upper-left corner of a rectangle inscribed into the object.
Channel	Name of a channel or a variable that will be displayed in the ***Value*** field and displayed on the gauge. In this field you can also enter a numerical value without a decimal separator to test the operation of the gauge.
Decimal places	The number of decimal places displayed for ***Value*** and ***Legend***
Min, Max	The minimum and maximum value displayed by the indicator
Display value	This parameter allows to hide the ***Value*** displayed
Update frequency	The update frequency on the ***Value*** parameter screen. The screen is updated 50 times per second (50 Hz). With fast-changing variables, reading is very difficult at this frequency. The parameter allows to decrease the update frequency (e.g. to 5 Hz), which makes the ***Value*** parameter easier to read.
Display unit	Option for displaying units (***Unit***). The units will be displayed above the ***Value*** parameter.
Unit	Selection of in which the value will be presented (e.g. kPa, Bar, Psi). You can define your own units in the ***Custom*** unit field. To this end, select ***User*** in the ***Unit*** field.

Parameter	Description
Custom unit	This field is used for entering a user-defined measurement units. To use it, select ***User*** in the Unit field.
Text	Defines the text displayed below the ***Value*** parameter.
Use icon	Parameter for displaying an icon instead of text
Icon texture	Allows to select an icon from the texture menu. You can add a custom icon (more information about managing textures can be found further in the manual).
Icon scale	The icon size is automatically adapted to the indicator size. It is possible to change the size using the ***Icon Scale*** parameter.
Alarm channel	A channel or allowing to change the indicator color when its value differs from 0. The alarm color is defined in the ***Alarm Color field***.
Radius	The indicator diameter in pixels.
Line width	The indicator line width in pixels. Decimal values are possible.
Angle offset	The initial angle of the first indicator (0 in the above example).
Angle span	The range in which the indicator is to be divided - in degrees (in the example above it is 270 degrees).
Line Color	Indicator line color
Value color	Displayed value color
Text color	Text or icon color
Unit color	Measurement units color
Alarm color	Color to which the indicator and the displayed value will change if the value of the variable used in ***Alarm channel*** differs from 0.
Background color	Indicator background line color
Value font	Font size used for displaying values
Text font	Font size used for displaying text
Unit font	Font size used for displaying measurement units
Legend font	Font size of the legend

7.2. Classic gauge

This object allows to display a numerical value in a similar way as in classic car gauges. The object displays a numerical value and also indicates it using a hand and its position on a clock face.



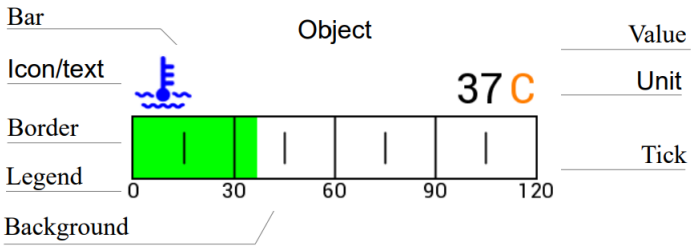
Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Channel	Name of a channel or a variable that will be displayed in the ***Value*** field and shown on the gauge. In this field you can also enter a numerical value without a decimal separator to test the operation of the gauge.
Decimal places	The number of decimal places displayed for ***Value*** and ***Legend***
Min, Max	The minimum and maximum value displayed by the indicator
Display value	This parameter allows to hide the ***Value*** displayed
Update frequency	The refresh frequency on the ***Value*** parameter screen. The screen is refreshed 50 times per second (50 Hz). Reading fast-changing variables is very difficult at this frequency. The parameter allows to decrease the refresh frequency (e.g. to 5 Hz), which makes the ***Value*** parameter easier to read.
Display unit	Option for displaying units (***Unit***). The units will be displayed above the ***Value*** parameter.
Unit	Selection of the unit in which the value will be expressed (e.g. kPa, Bar, Psi). You can define your own unit in the ***Custom unit*** field. To this end, select ***User*** in the Unit field.
Custom unit	This field is used for entering a user-defined measurement units. To use it, select ***User*** in the ***Unit field***
Text	Defines the text displayed below the ***Value*** parameter.

Parameter	Description
Use icon	Parameter for displaying an icon instead of text
Icon texture	Allows to select an icon from the texture menu. You can add your own icon (for more information regarding texture management see further sections of this manual).
Icon scale	The icon size is automatically adapted to the indicator size. It is possible to change the size using the ***Icon Scale*** parameter
Alarm channel	A channel or allowing to change the indicator color when its value differs from 0. The alarm color is defined in the ***Alarm Color field***.
Radius	The indicator diameter in pixels.
Border Size	The width of the line surrounding the indicator in pixels. Decimal values are possible.
Highlight percent	Determines how much of the indicator will be filled with the ***Highlight*** color. 0 means that the entire indicator will be filled with the ***Background color***.
Num ticks	Value defining the number of main parts into which the indicator area will be divided. For each part a respective value and indicator in the form of a section will be displayed.
Num subticks	Value defining the number of divisions between the main indicators
Indicator width	Width of pointer showing value at a given time
Angle span	The range in which the indicator is to be divided - in degrees (in the example above it is 270 degrees).
Angle offset	The initial angle of the first indicator (0 in the above example).
Color	The color of the value, legend and indicators displayed
Border color	The indicator frame color
Highlight color	The highlight area color
Indicator color	The hand color
Background color	The color of the background
Text color	Text or icon color
Unit color	Measurement units color

Parameter	Description
Alarm color	Color to which the indicator and the displayed value will change if the value of the variable used in ***Alarm channel*** differs from 0.
Tick font	The size of the font used for displaying the legend
Value font	Font size used for displaying values
Text font	Font size used for displaying text
Unit font	Font size used for displaying measurement units
Redline when	The condition for displaying the indicator in a “redline” mode. ***Never*** - never display in the “redline” mode. ***When value above*** – display the bar in ***Redline color*** when the ***Value*** parameter is greater than the ***Redline start*** value. ***When value above*** – display the bar in ***Redline color*** when the ***Value*** parameter is lower than the ***Redline start*** value.
Redline start	This parameter defines the value for a condition defined by the ***Redline when*** parameter

7.3. Bar graph

This indicator allows to display values in the form of a moving bard (either horizontal or vertical). It is also possible to display an icon symbolising the measured value.



Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Channel	Name of channel or variable that will be displayed in the ***Value*** field and shown on the gauge. In this field you can also enter a numerical value without a decimal separator to test the operation of the gauge.
Decimal places	The number of decimal places displayed for ***Value*** and ***Legend***

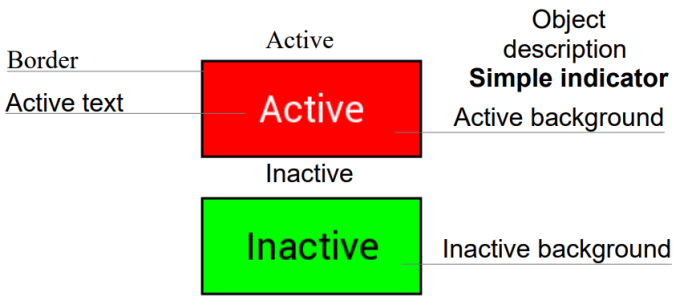
Parameter	Description
Min, Max	The minimum and maximum value displayed by the indicator
Display value	This parameter allows to hide the ***Value*** displayed
Update frequency	The refresh frequency on the ***Value*** parameter screen. The screen is refreshed 50 times per second (50 Hz). Reading fast-changing variables is very difficult at this frequency. The parameter allows to decrease the update frequency (e.g. to 5 Hz), which makes the ***Value parameter easier to read***
Display unit	Option for displaying units (***Unit***). The units will be displayed above the ***Value*** parameter.
Unit	Selection of the unit in which the value will be expressed (e.g. kPa, Bar, Psi). You can define your own unit in the ***Custom unit*** field. To this end, select ***User*** in the Unit field.
Custom unit	This field is used for entering a user-defined measurement units. To use it, select ***User*** in the Unit field
Text	Defines the text (description) displayed along the indicator.
Use icon	Parameter for displaying an icon instead of text
Icon texture	Allows to select an icon from the texture menu. You can add a custom icon (more information about managing textures can be found further in the manual).
Icon scale	The icon size is automatically adapted to the indicator size. It is possible to change the size using the ***Icon Scale*** parameter
Alarm channel	A channel or allowing to change the indicator color when its value differs from 0. The alarm color is defined in the ***Alarm Color*** field.
Bar type	Indicator type ***Horizontal***, ***Vertical***
Bar style	Indicator style: ***Simple*** - in the form of a movable bar with graduation, ***Simple without decoration*** - in the form of a movable bar without graduation, ***Style 1*** - in the form of a movable bar under the number axis
Length	Indicator length
Width	Indicator width
Border width	The width of the line surrounding the indicator (in pixels)

Parameter	Description
Number of ticks	Value defining the number of main parts into which the indicator area will be divided. For each part a respective value and indicator will be displayed. An additional section is drawn automatically between the indicators.
Decimal places for tick	This value defines the number of decimal places drawn for the indicator description
Color	Indicator frame color
Background color	Indicator background color
Bar color	Indicator bar color
Alarm color	Color of the indicator if the value of the variable used in ***Alarm channel*** differs from 0.
Icon color	Icon color
Value color	Displayed value color
Text color	Text or icon color
Unit color	Measurement units color
Redline color	The indicator bar color when the displayed value meets the ***Redline condition***.
Transparent	Decides if the indicator background is to be displayed.
Tick font	The size of the font used for displaying the legend
Value font	Font size used for displaying values
Text font	Font size used for displaying text
Unit font	Font size used for displaying measurement units
Redline when	The condition for displaying the indicator in “redline” mode. ***Never*** - never display in the “redline” mode. ***When value above*** – display the bar in ***Redline color*** when the ***Value*** parameter is greater than the ***Redline start*** value. ***When value below*** – display the bar in ***Redline color*** when the ***Value*** parameter is lower than the ***Redline start*** value.
Redline start	This parameter defines the value for a condition defined by the ***Redline when*** parameter

Parameter	Description
Visibility channel	Name of channel or variable that will control object visibility. 0 means that the object will be hidden, a value different from zero or no assigned channel means that the object will be visible.

7.4. Simple indicator

This indicator allows to display the indicator of the function attached (e.g. ALS). It makes it possible to assign two different texts in two different background colors depending on the value of the function attached in the ***Channel active*** *field.*

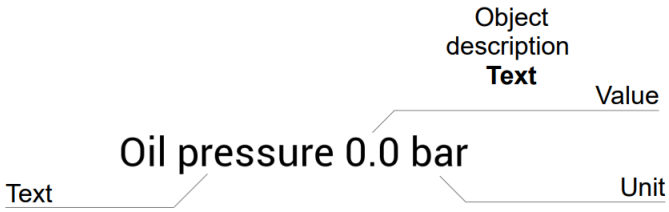


Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Channel	Name of channel or variable that will be displayed in the ***Value*** field and shown on the gauge. In this field you can also enter a numerical value without a decimal separator to test the operation of the gauge.
Width	Indicator width
Height	Indicator height
Border width	The width of the line surrounding the indicator in pixels
Style	Background type: solid or textured of your choice
Texture	Texture / icon selection
Background color default	Inactive indicator background color
Background color active	Active indicator background color
Text color default	Color of the text displayed by the inactive indicator

Parameter	Description
Text color active	Color of the text displayed by the active indicator
Border color	The indicator frame color
Text	Text displayed by the inactive indicator
Active text	Text displayed by the active indicator
Text font	Font size used for displaying text
Active channel	Channel or function or variable defining whether the indicator is to be displayed in inactive mode (0) or active mode (a value different from zero).
Visibility channel	Name of channel or variable that will control object visibility. A value of 0 and no channel assigned means that the object will be hidden, a value different means that the object will be visible.

7.5. Text

This indicator allows to display text with content read from a log channel or function together with a measurement unit. The user can also hide the object using a function or a log channel.



Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Color	Displayed text color
Font	Displayed font size
Italic	Enables italics.
Two lines	Text can be displayed on two lines
Text	A field containing the displayed text
Text width	The area width used by the ***Text align*** functions. For example, by entering 800 (screen width), and setting the position X as 0 and

Parameter	Description
	selecting appropriate alignment, you can display the text to the left side of the screen, in the centre of the screen and to the right side of the screen.

Parameter	Description
Text align	Defines text alignment in the ***Text width*** area. ***Left*** – align the text to the left ***Center*** – text centering ***Right*** – align the text to the right
Channel	Name of channel or variable that will be displayed in the ***Value*** field and shown on the gauge. In this field you can also enter a numerical value without a decimal separator to test the operation of the gauge.
Value source	Specifies the value displayed for the selected channel: ***Current*** – the current channel value ***Min value*** – the minimum registered value ***Max value*** – the maximum registered value
Decimal places	The number of decimal places displayed for the Value parameter
Value width	The width of the area of the displayed value used by the ***Value align*** function
Value align	Defines text alignment in the ***Value width area***. ***Left*** – align the text to the left ***Center*** – text centering ***Right*** – align the text to the right
Display unit	Option for displaying units (***Unit***). The units will be displayed above the ***Value*** parameter.
Unit	Selection of the unit in which the value will be presented (e.g. kPa, Bar, Psi). You can define your own unit in the ***Custom unit*** field. To this end, select User in the ***Unit*** field.
Custom unit	This field is used for entering a user-defined measurement units. To use it, select ***User*** in the Unit field.
Color channel	Name of the channel or variable that will control the color of the text displayed. If the channel / function value is 0, the color will be defined in the ***Color field***. Otherwise red will be used. If the selected channel has colors assigned by enumeration, these will be the master colors.
Visibility channel	Name of the channel or variable that will control text visibility. 0 means that the text will be hidden, a value different from zero or no assigned channel means that the text will be visible.

Parameter	Description
Update frequency	The update frequency on the ***Value*** parameter screen. The screen is refreshed 50 times per second (50 Hz). Reading fast-changing variables is very difficult at this frequency. The parameter allows to decrease the update frequency (e.g. to 5 Hz), which makes the ***Value*** parameter easier to read.

7.6. Time

This indicator allows to display the built-in “clocks” of the device, such as the real time, lap time, best lap time etc.

Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Color	Displayed text color
Font	Displayed font size
Time source	***RTC*** – the device internal clock time. To set the RTC clock time, select the ***Devices / Set real time clock*** from the application menu. ***Current lap*** – current lap time ***Last lap*** – last lap time ***Best lap*** – best lap time ***Session time*** – an internal clock measuring the current session time. You can reset it using the external button. ***Predictive time*** – predicted lap time based on the best lap time on a given track. Requires a GPS module.
Visibility channel	Name of the channel or variable that will control text visibility. 0 means that text will be hidden, a value different from zero or no assigned channel means that text will be visible.

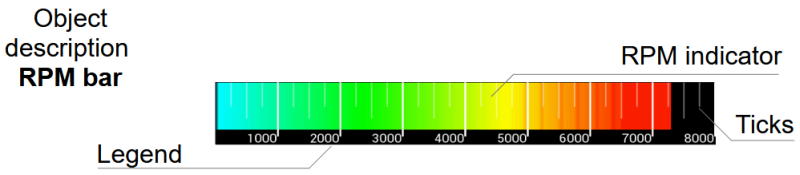
7.7. Image

This object allows a texture (image) or icon to be displayed on the screen and edited in terms of the following parameters: scaling, color selection, mirroring and repetition. In addition to the textures contained in the device memory, you can upload own textures / icons.

Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Default color	Displayed text color
Function color	Displayed font size
Texture	Allows to select a texture and an icon that will be displayed
Repeat mode	Texture repetition mode. ***Stretch*** – allows to stretch the texture out (***Scale*** parameter). ***Tile X*** – repeats texture along the X axis. ***Tile Y*** – repeats texture along the Y axis. ***Tile X,Y*** – repeats texture along the X and Y axes. To replicate a texture for ***Tile*** mode twice, set the Scale parameter to 200%. To replicate it four times, enter 400% and so on.
Scale	This parameter determines the scale of a texture or the number of its copies for ***Tile*** modes
Mirror X/Y	It allows to produce a mirror image of a texture along the X and / or Y axis.
Mode	Texture drawing mode. In the ***Screen*** mode a texture is displayed by adding it to the current image. In the ***Normal*** mode a texture overwrites the current image, having regard for the alpha channel. You can find more information regarding the textures further in the manual.
Color channel	Name of the channel or variable that will control the color of the texture displayed. If the channel / function value is 0, the color will be defined in the ***Default color*** field. Otherwise the color of the ***Function color*** will be used.
Visibility channel	Name of the channel or variable that will control text visibility. 0 means that text will be hidden, a value different from zero or no assigned channel means that text will be visible.

7.8. RPM Bar

This object allows to display the engine speed in the form of a horizontal bar or a round indicator.

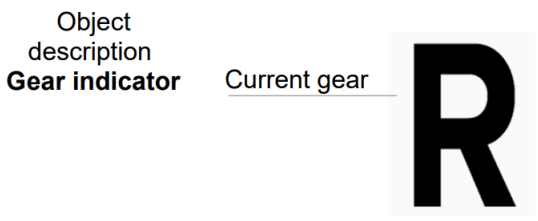


Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Type	The type of the engine speed indicator displayed: ***Classic*** – displayed in the form of a “curved” bar ***Bar*** – displayed in the form of a horizontal bar (as shown in the above visualisation) ***Round*** – displayed in the form of a round gauge ***Vertical*** - displayed as a vertical "triangular" bar
Color	Determines the color of the bar indicating the engine speed for ***Classic***, ***Round*** and ***Vertical*** type indicators.
Redline color	Determines the color of the bar indicating the engine speed for ***Classic***, ***Round*** and ***Vertical*** type indicators, when revs exceed the ***Redline start*** value.
Background color	For a ***Classic*** and ***Vertical*** type indicators determines the background color of the indicator.
Show redline on bar	For a ***Classic*** and ***Vertical*** indicator types highlights the redline zone.
Width	For a ***Bar*** type indicator determines the length of the indicator.
Height	For a ***Bar*** type indicator determines the height of the indicator.
Font	For a ***Bar*** type indicator determines the font size of the graduation description.
Redline start	A value specifying the beginning of the engine speed threshold area
Max RPM	The maximum engine speed value displayed on the indicator
Color preset	For a ***Bar*** type indicator, it determines the color gradient that will be used for the pointing bar.

Parameter	Description
Custom texture	For a ***Bar*** type indicator, if ***Custom texture*** is selected in ***Color preset***, it determines the texture selection for the pointing bar.
RPM x 1000	For a ***Bar*** type indicator, it determines whether the engine speed is displayed in the legend in full or divided by one thousand.
Ticks style	For a ***Bar*** type indicator determines whether additional graduation lines are to be displayed (***Ticks***). ***No ticks*** - No division ***Main tick only***- main division lines only ***Main tick + 1 subtick*** - one additional line between each compartment ***Main tick + 3 subticks*** - three additional lines between each compartment
Center RPM number	For a ***Bar*** type indicator displays each graduation value centrally under the dividing lines
Nonlinear factor	Non-linear graduation for a ***Classic iBar*** indicator
RPM Channel	Channel or function containing the current speed
Visibility channel	Name of channel or variable that will control object visibility. A value of 0 and no channel assigned means that the object will be hidden, a value different means that the object will be visible.

7.9. Gear indicator

This object displays the currently selected gear. The object has been provided with a specially prepared font of increased size containing numbers and the letters R (Reverse) and N (Neutral). The respective displayed gear values are as follows: -2 for park, -1 for reverse, 0 for idle, 1 for first gear, 2 for second gear and similarly up to gear 8.

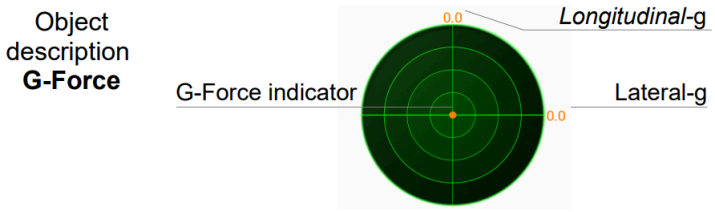


Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.

Parameter	Description
Color	Displayed gear color
Font	Displays font size (maximum size is: 15)
Value channel	Channel or function containing the currently selected gear
Visibility channel	Name of channel or variable that will control object visibility. 0 means that the object will be hidden, a value different from zero or no assigned channel means that the object will be visible.

7.10. G-Force

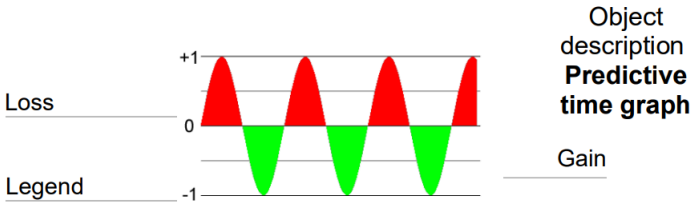
The G-Force object indicates the current G-force acting on the vehicle using acceleration data from a selected source (default: internal accelerometer). The *adu.latG* channel records lateral overloads acting on the vehicle (left / right), and the *adu.longG* channel records longitudinal overloads (braking / acceleration). The internal accelerometer requires calibrating after installing the device. This can be done using a button defined in the *Buttons / IMU* pitch zeroing panel or manually in the *Configuration / IMU Pitch panel.*



Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Color	Displayed gear color
Size	Indicator diameter

7.11. Predictive time graph

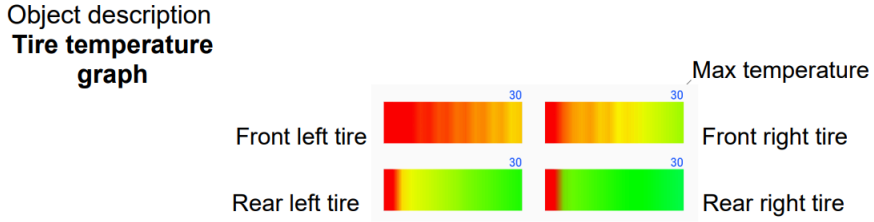
The ***Predictive time graph*** object indicates the time difference between the best reference lap and the current position on the track. This object requires a GPS module and a correct configuration of the racing track.



Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Width	Object width
Height	Object height
Gain color	The color of the graph when the current time is better than the reference time
Loss color	The color of the graph when the current time is worse than the reference time
Lines color	The graph line color
Font color	The color of the font for describing the axes
Font size	The font size
Time range	Time range: +/-0.5; +/-1 or +/-2

7.12. Tire temperature graph

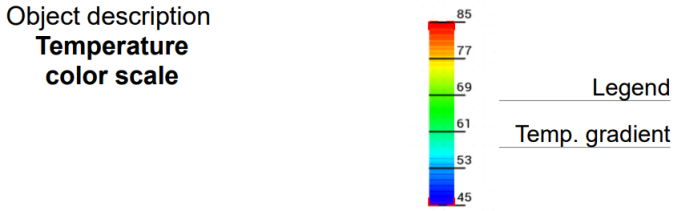
The ***Temperature graph*** object displays the temperature gradient of tyres or brake discs recorded by thermal imaging cameras. The tire temperature can be presented as gradients or tires. Configuration of the measuring range of the cameras can be found in the ***ADU / Configuration / Tire temperature cameras*** panel and in the ***ADU / Configuration / Brake temperature cameras panel***.



Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Style	Object display style: ***Tires*** – temperature displayed as tires ***Bar*** – temperature displayed as gradients
Source	***Tire temperature cameras*** ***Brake temperature cameras***
Length	Object length (parameter available only for the ***Bar*** style)
Breadth	Object width (parameter available only for the ***Bar*** style)
Spacing X	The distance on the horizontal axis between gradients representing tires (parameter available only for the Bar style)
Spacing Y	The distance on the vertical axis between gradients representing tires (parameter available only for the Bar style)
Scale	Object size (parameter available only for the ***Tires*** style)
Rotate	For the ***Bar*** style, displays temperature gradients as vertical bars

7.13. Temperature color scale

The ***Temperature color scale*** object displays the temperature gradient and the temperature values assigned to a given color. Configuration of the measuring range of the cameras can be found in the ***ADU > Configuration > Tire temperature cameras*** and ***ADU > Configuration > Brake temperature cameras panels***.



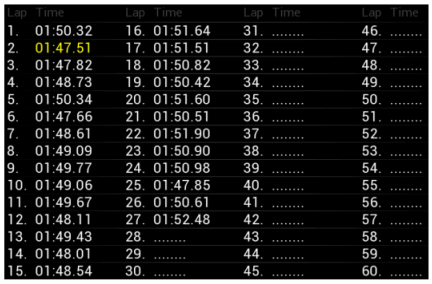
Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Source	***Tire temperature cameras*** ***Brake temperature cameras***

Parameter	Description
Scale	Defines the gradient size.
Legend color	Defines the description color.

7.14. Session results

The ***Session results*** object allows the times of sixty individual laps to be displayed, highlighting the best laps in a different color.

Object description
Session results



Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Color	Defines the color of the text in the table.
Best lap color	Defines the color of the best lap time displayed.

7.15. Track record table

The ***Track record table*** object allows to display eight best times for a given racing track. A track is recognized using a GPS position. You can find more information regarding the GPS further in the manual.

Object description
Track record table

First column

First row

Track name				4999m
#	TIME	LAP	TOP SPEED	DATE
1	01:59.39	199	300.0	25.01.2017
2	01:59.39	199	300.0	25.01.2017
3	01:59.39	199	300.0	25.01.2017
4	01:59.39	199	300.0	25.01.2017
5	01:59.39	199	300.0	25.01.2017
6	01:59.39	199	300.0	25.01.2017
7	01:59.39	199	300.0	25.01.2017
8	01:59.39	199	300.0	25.01.2017

Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
1st column bkgrd color	First column background color
1st column text color	First column text color
Table bkgrd color 1	Background color for odd columns
Table bkgrd color 2	Background color for even columns
Table text color	Table text color
1st row bkgrd color	First row background color
1st row text color	First row text color

7.16. Rectangle

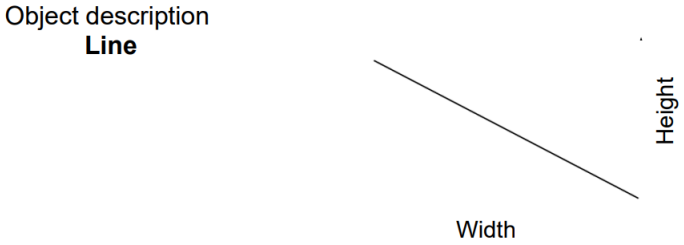
The ***Rectangle*** object allows to draw a rectangle on a page. You can define the width of the frame line and the fill color.

Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Rectangle type	The type of the displayed rectangle. ***Border*** – displays only the rectangle border ***Border + fill***– displays the rectangle border and fill ***Only fill***- displays only the rectangle fill
Color	Border color
Fill color	Fill color
Width	Rectangle width
Height	Rectangle height

Parameter	Description
Thickness	Width of the border in pixels
Visibility channel	Name of channel or variable that will control object visibility. 0 means that the object will be hidden, a value different from zero or no assigned channel means that the object will be visible.

7.17. Line

The ***Line*** object is used to draw lines on pages.

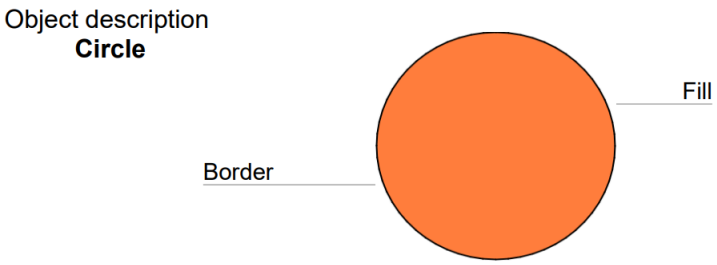


Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Color	Border color
Width	Distance between the beginning and the end of the line along the X axis
Height	Distance between the beginning and the end of the line along the X axis
Thickness	Width of the border in pixels
Style	Type of line: ***Simple*** - a line for which the beginning and end can be arbitrarily specified on the plane ***Horizontal gradient one side*** - horizontal line, colour graduation from the left side ***Horizontal gradient two sides*** - horizontal line, colour graduation from both ends ***Vertical gradient one side*** - vertical line, colour graduation from above ***Vertical gradient two sides***- vertical line, colour graduation from both ends

Parameter	Description
Visibility channel	Name of channel or variable that will control object visibility. 0 means that the object will be hidden, a value different from zero or no assigned channel means that the object will be visible.

7.18. Circle

The ***Circle*** object is used to draw circles on pages. You can define in it the width of the lines and the fill color.

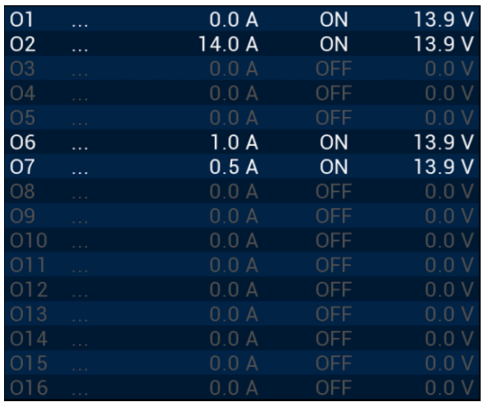


Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Circle type	The type of the displayed rectangle. ***Border*** – displays only the rectangle border ***Border + fill*** - displays border and fill ***Only fill*** - displays only the fill
Color	Border color
Fill color	Fill color
Thickness	Width of the border in pixels
Radius	Circle radius
Visibility channel	Name of channel or variable that will control object visibility. 0 means that the object will be hidden, a value different from zero or no assigned channel means that the object will be visible.

7.19. Grid

The ***Grid*** object displays a table which shows values defined by separate channels in each cell.

Object description
Grid

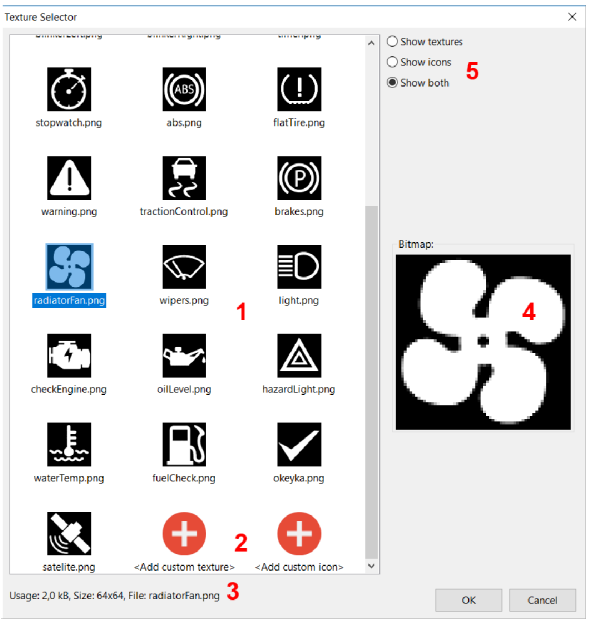


Parameter	Description
Position X,Y	Object position on a page The reference point is the upper-left corner of a rectangle inscribed into the object.
Row count	Number of rows
Font	The font size
Active	Specifies whether the column will be displayed
Column width	Width of column concerned
Prefix	Prefix for line numbering
Suffix	Suffix for line numbering
Start index	Initial value of line numbering
Name #1-#16	Description of lines from 1 to 16
Decimal places	Decimal places for the displayed values in a given column (A to C)
Text aling	Column text alignment
Channel #1-#16	The channel from which the values displayed in the specified row (from 1-16) and column (from A-C) will originate
Default text color	Primary text color
Fill color	Row background color
Alternating fill color	Alternating row background colors
Fill color #2	Alternative row background color

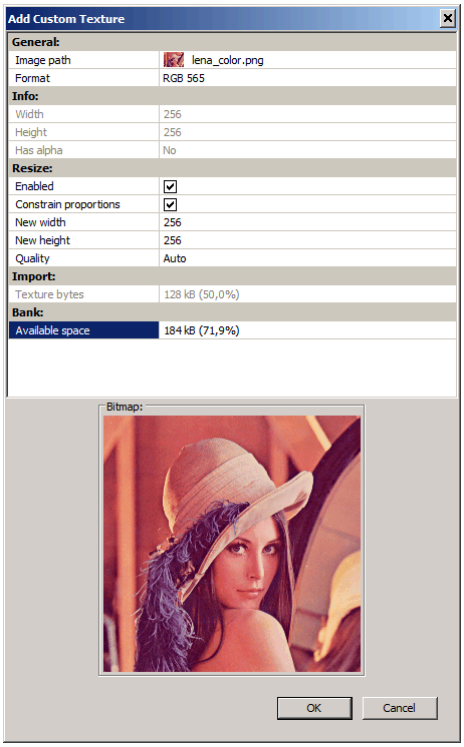
Parameter	Description
Channel #1-#16	The name of the channel or variable that will control the color of the displayed text for the corresponding line (from 1-16). If the channel / function value is 0, the color defined in the ***Default text Color*** field will be applied. Otherwise red will be used. If the selected channel has colors assigned by enumeration, these will be the master colors.

7.20. Textures

Textures are bit maps (images) defining the graphics that we can display using the Image object. The ADU device has built-in textures dedicated to displaying as icons or backgrounds. The user is also able to add their own textures (e.g. backgrounds, icons, etc.).



The textures are managed using the ***Texture Manager*** (***Menu > Tools > Texture manager dialogue***). This window is displayed also when you select a texture for the objects that display it (e.g. Image, Bar graph, etc.) In the ***Texture manager*** window, it is possible to preview the available textures (1). After selecting a texture in the lower part of the window information about the selected object (3) will be displayed along with its preview (4). In addition, a filter option is available, which allows only textures or only icons to be displayed (5). An icon differs from a texture in that it has to have a fixed size: 64x64 pixels. To add your own icon or texture, select ***Add custom texture*** or ***Add custom icon***. After selecting a file with a graphics (.png .jpg and .bmp formats are supported) a dialogue window will be displayed that will enable you configure the texture.



Parameter	Description
Image path	File name on the disc
Format	The texture target format. The format determines the quality of the texture and the amount of the device memory it takes up. Detailed information about the formats can be found on the next page.
Width, height	Information about the size of the source bit map
Has alpha	Information whether the bit map has a separate alpha channel.
Resize enabled	Allows scaling of the source texture.
Constrain proportions	When activated, this option automatically keeps the proportion of the texture during scaling.
New width, height	The new width and the new height of the texture
Quality	The filtering that will be used during scaling. Auto filtering is recommended.
Texture bytes	The size of the texture in the memory of the device.
Available space	The available free memory for textures

Format	Description	Preview
A1 **8kB**	1-bit format. A pixel of the texture can assume only 2 values. It takes up less memory.	
A2 **16kB**	2-bit format. A pixel of the texture can assume 4 values.	
A4 **32kB**	4-bit format. A pixel of the texture can assume 16 values. This format is recommended for icons. It guarantees a good quality while using little memory.	
A8 **64kB**	8-bit format. A pixel of the texture can assume 256 values.	
Indexed RGBA8 **64kB**	An 8-bit color format with an alpha channel. The texture is quantized to 256 unique colors. This format is suitable for colour textures (e.g. a company logo). The advantage is low memory consumption, while the disadvantage is low rendering efficiency and visible deterioration in smooth tonal transitions.	
RGB 565 **128 kB**	A 16-bit color format without an alpha channel. It has a very good quality and rendering efficiency. A flow of this format is that it uses a lot of memory.	
ARGB 1555 **128 kB**	A 16-bit color format with an alpha channel. It has a very good quality and rendering efficiency. A flow of this format is that it uses a lot of memory.	

8. Inputs

The ADU is equipped with 8 analog inputs and 8 digital inputs.

8.1. Analog Inputs

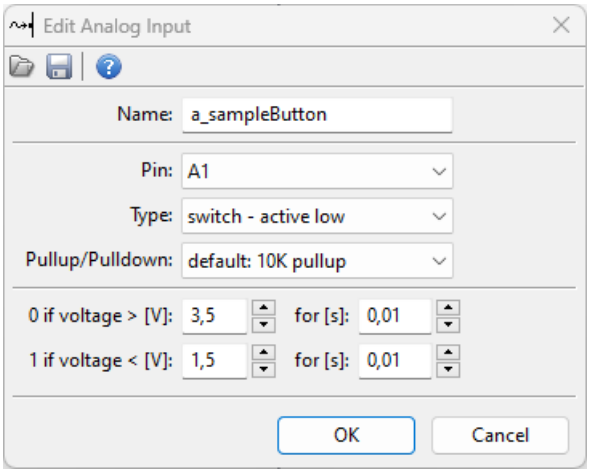
The analog inputs are used for measuring the voltage on the sensors (e.g. oil pressure sensor) or as inputs for the buttons. To add an analog input, add the ***Analog Input*** element in the ***Project Tree***. The configuration window includes the following options:

Parameter	Description
Name	Name of the analog input that will be used as the channel name in the project
Pin	Number of the analog input to which the configuration relates
Type	The function that the analog input is to perform: ***Switch – active low*** – the analog input will function as a switch (button) activated by a low state. ***Switch – active high*** – the analog input will function as a switch (button) activated by a high state. ***Rotary switch*** – the input assumes a value consistent with the position of the Rotary switch. The number of positions of the rotary switch is defined as ***Min value / Max value***. ***Linear analog sensor*** – the input is used for measuring voltage (as a *[**Unit**]* select *[Voltage]*) or any type of linear sensors (e.g. MAP sensor). ***Calibrated analog sensor*** – the input is used for measuring the values from sensors with a non-liner scale (e.g. temperature sensors NTC / PTC). A 2D map is used to define the values.
Pull-up / Pull-down	A function for activating the internal 4K7 resistor connected to ground (Pull-down) or +5 V (Pull-up). These resistors are activated mainly when buttons are connected to analog inputs. For a button activated by a low state, you should activate Pull-up 4K7 and the button should connect the analog input to ground. For analog sensors or for measuring voltage, choose the 1M Pulldown option.
Quantity / Unit	For ***Linear*** and ***Calibrated analog sensor*** inputs, the parameter defines the physical value to be measured and its unit.

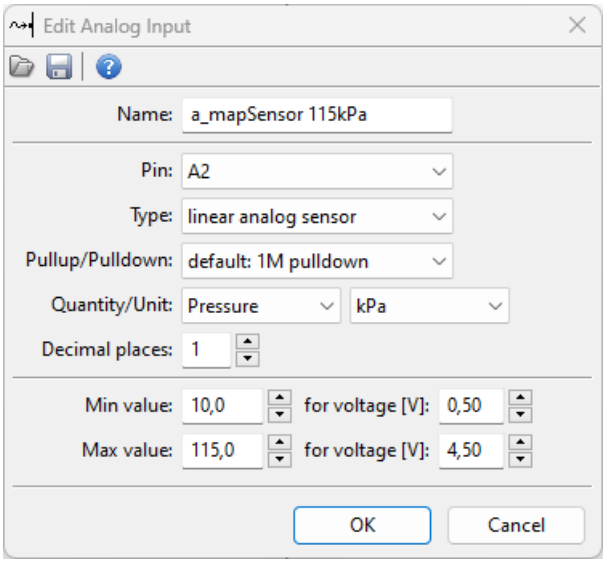
Parameter	Description
Decimal places	Defines the number of the decimal places for the measured value for the ***Linear*** and ***Calibrated analog sensor*** input type.
1 if voltage > [V]	Defines the voltage representing the value 1 for the ***Switch*** input type. For the condition to be satisfied, the voltage must be greater than the indicated value. The voltage value must be maintained for the time defined as the parameter "for [s]"
0 if voltage < [V]	Defines the voltage representing the value 0 for a ***Switch*** input type. In order for this condition to be fulfilled, the voltage needs to be smaller than that defined by time in the [s] field.
Min value for voltage	For ***Linear*** analog sensor inputs this value defines the minimum sensor value for the defined *Voltage*.
Max value for voltage	For ***Linear*** analog sensor inputs this value defines the maximum sensor value for the defined *Voltage.*

Example configurations:

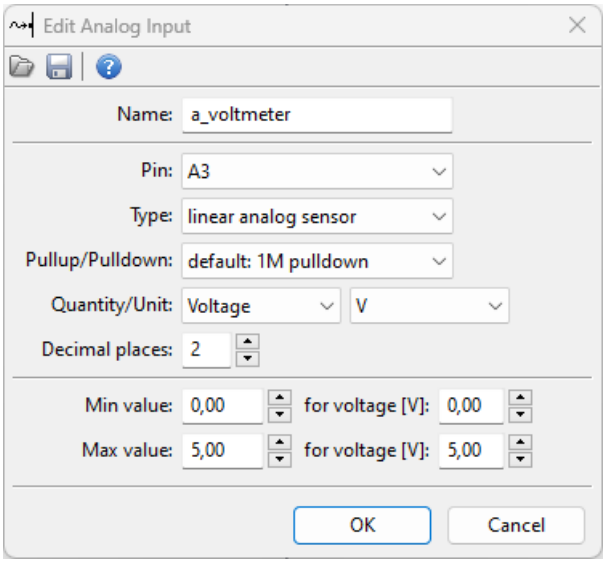
A configuration for a button connected to ground and to the analog input 1. The *a_sampleButton* value will be 0 when the button is not pressed and 1 when it is.



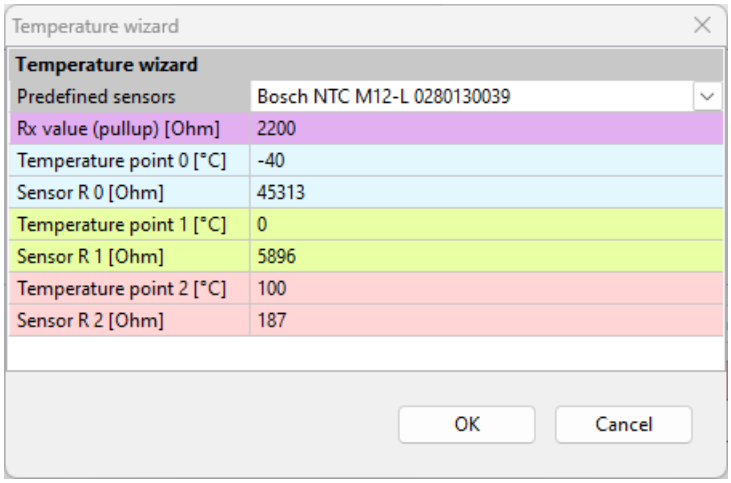
A configuration for the pressure sensor in the intake manifold (MAP) connected to the analog input 2. The *a_mapSensor115kPa* value will assume values ranging from 10.0 kPa to 115.0 kPa



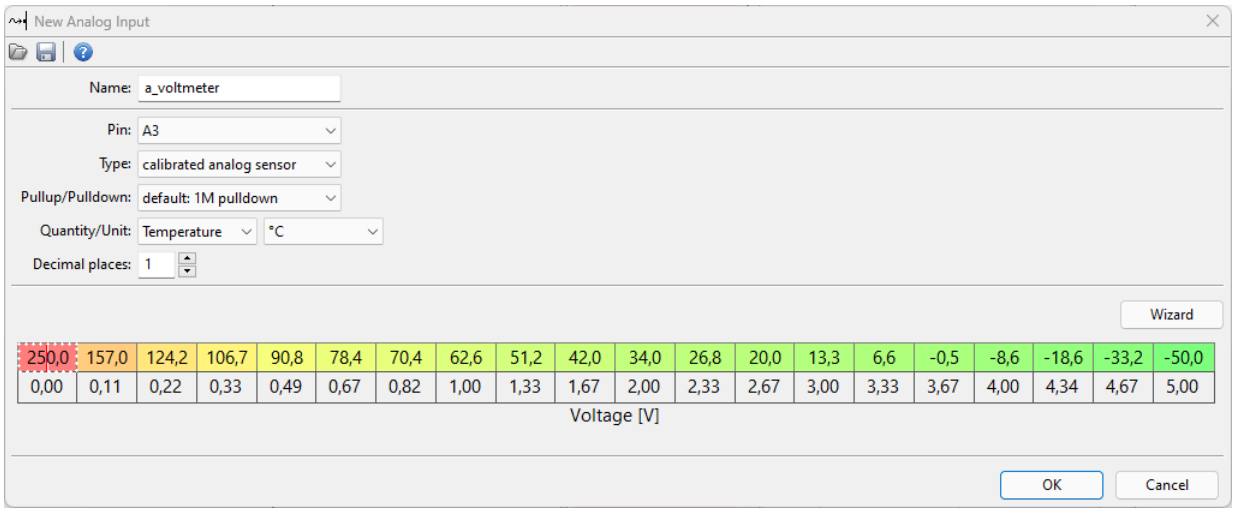
A configuration for measuring a voltage of 0-5 V for a signal connected to the analog input 3. The *a_voltmeter* value will assume values ranging from 0.00 V to 5.00 V.



The easiest way to calibrate non-linear temperature sensors is to use the *Wizard.* After pressing the *Wizzard* button a window will pop up, allowing to define a sensor using three temperature values and three corresponding sensor resistance values. You can select a defined sensor in the ***Predefined sensor*** field.



The ***Rx value*** field indicates the value of the pull-up resistor used when connecting the sensor.
If the sensor characterization data is correct, a 2D map describing the sensor characteristics will be generated automatically:



If a calibration map is created manually, values can be entered into individual cells. To change the table size, right click on it and select one of the *Modify bins* options.
The values of analog inputs can be viewed in the *Analog monitor* panel, where you can find the channel value, its voltage and information about a pull-up resistor connected.

8.2. Digital Inputs

Digital Inputs are used for transforming digital signals such as signals from the crankshaft position sensor, wheel speed sensor or ethanol sensor (*FlexFuel*). These inputs can also be used as inputs for buttons connected to ground.

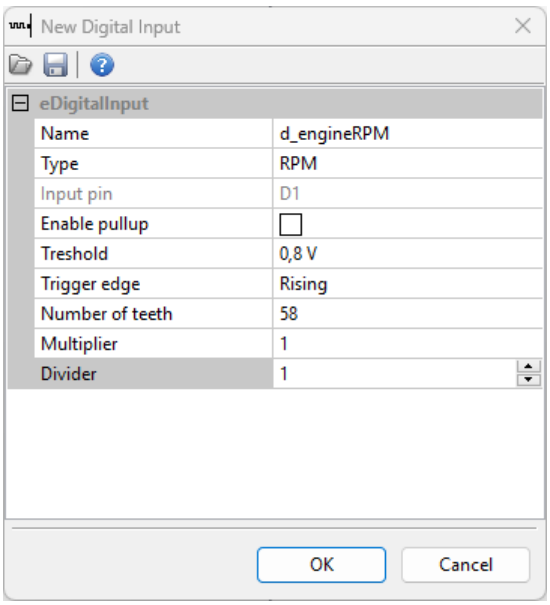
To add an digital input, in ***Project Tree*** add the ***Digital Input*** element. The configuration window includes the following options:

Parameter	Description
Name	The name of the digital input that will be used as the channel name in the project
Input pin	Number of the digital input to which the configuration relates
Type	The function that the digital input is to perform: ***Switch – active low*** – the digital input will function as a switch (button) activated by a low state. ***Switch – active high*** – the digital input will function as a switch (button) activated by a high state. ***Frequency*** – the digital input will measure the signal frequency. ***RPM*** – the digital input will decode the signal from the crankshaft / camshaft position sensor in order to measure the vehicle engine speed. Only the D1 input can be used to measure engine speed. ***Flex Fuel*** – a digital input used for reading the content of ethanol in fuel and the fuel temperature from the FlexFuel sensor. Only the D2 input is compatible with the FlexFuel sensor. Values read from the sensor are stored in the following channels: adu.ff.ethanolContent, adu.ff.fuelTemperature, adu.ff.sensorStatus ***Beacon*** - digital input used to decode the signal from an AIM's beacon. The D2 input is compatible with the AIM beacon. ***PULS oil sensor*** - digital input used to read out the digital signal from the PULS type oil level and temperature sensor. This sensor can only be connected to the D4 input. Values read from the sensor are stored in the following channels: adu.puls.level, adu.puls.temperature, adu.puls.status. An example of a PULS sensor working with an ADU: 6PR 010 497-05 Range: 18-118.8 mm Connection: 1: +12 V, 2: ground, 3: signal The sensor requires the "Enable pull-up" and the "Threshold" to 2.5 V options to be enabled. The sensor's measuring range is configured in the "PULS oil temperature" > "level sensor" window.
Enable pull-up	Allows to activate an internal pull-up resistor 4k7 for a given input.

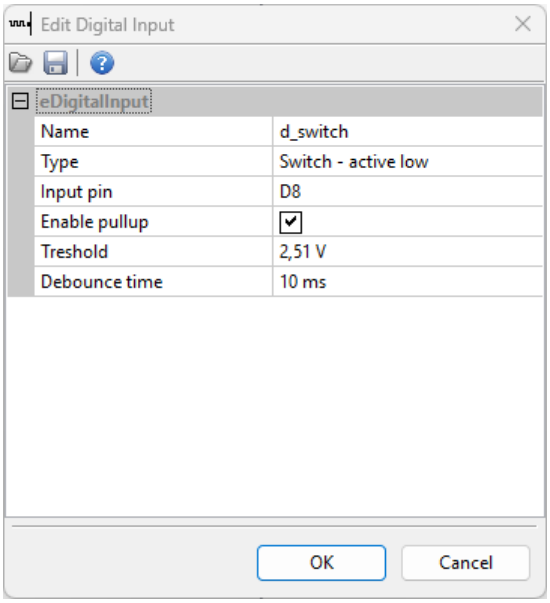
Parameter	Description
Threshold	A reference voltage which, when exceeded, results in a change of the input state from 0 to 1 (and the other way around). For inductive sensors, this will be less than 1 V. For Hall / optical sensors it will be 2.5 V.
Debounce time	For a ***Switch*** type input, this parameter determines the time needed to stabilize the switch contacts.
Trigger edge	The signal edge to be used when transforming a signal.
Number of teeth	For an RMP signal - a physical number of teeth on a toothed wheel used by the crankshaft / camshaft position sensor. For a 60-2 toothed wheel the number will be 58, for a 12+1 wheel - 13, etc.
Multiplier	The value by which input frequency (Frequency) or engine speed (RPM) will be multiplied. It allows calibration of values such as turbocharger speed or vehicle speed.
Divider	The value by which input frequency (Frequency) or engine speed (RPM) will be divided. It allows calibration of values such as turbocharger speed or vehicle speed.

Example configurations:

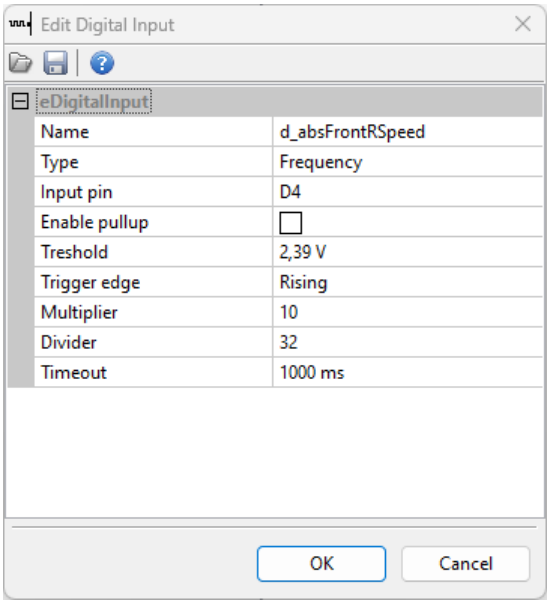
Engine speed (RPM) read-out configuration from the inductive camshaft and 60-2 toothed wheel position sensor. In the case of the engine speed read-out, the sensor must be connected to *Digital Input 1*.



Configuration of the status read-out of the button connected to *Digital Input 8*. The button must connect the signal to ground. The *d_switch* variable assumes the value of 0 when the button is not pressed and 1 when it is.



Configuration of the wheel speed read-out from the ABS sensor connected to *Digital Input 4*. The *d_absFrontRSpeed* variable value equals the input frequency multiplied by 10 and divided by 32 (multiplier, divider).



9. Outputs

The ADU device is equipped with two Low-side type outputs (connected to ground) witch a load capacity of up 2 amps. Additionally, an analog output with a signal ranging from 0 to 5 V is available, which can be used to send the voltage signal to another device.

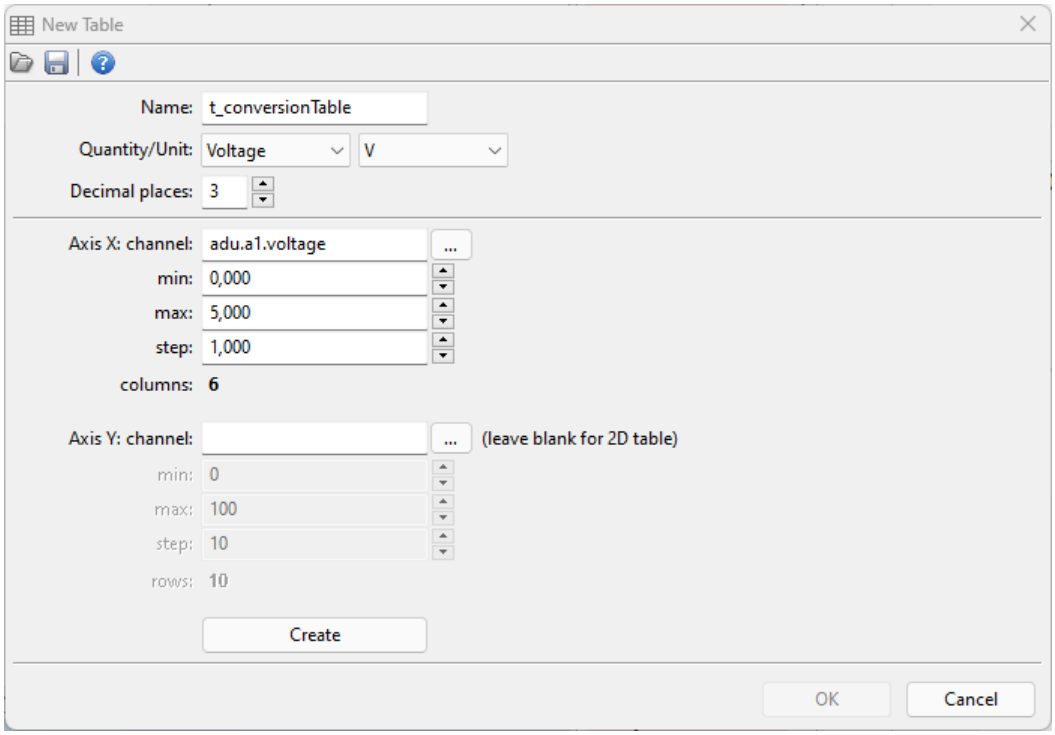
9.1. *Low side* outputs

The configuration of low-side outputs is available in the ***Outputs*** panel. The user can choose between two output channels, ***Aux1.channel*** and ***Aux2.channel***, where it is possible to define the variables / functions that will control the output. When their value is 0, the output is inactive (open). When the value is different from 0, the output is active (shorted to ground).

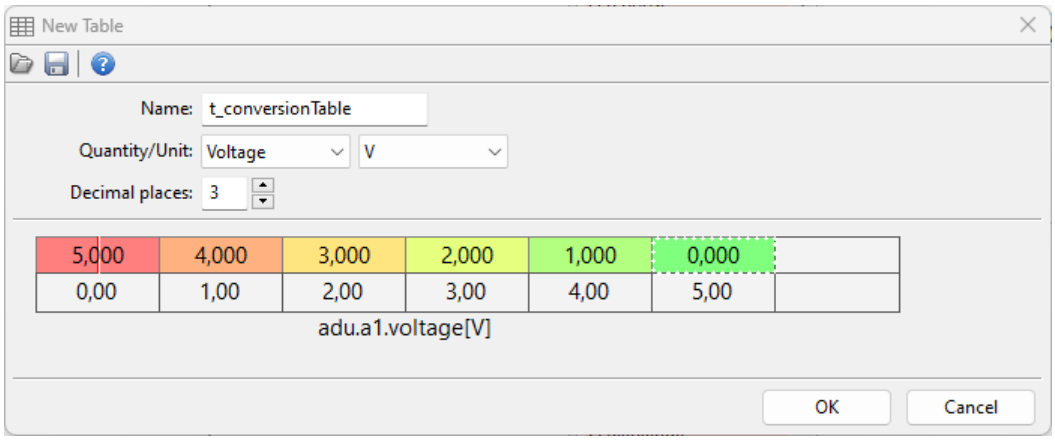
Additionally, the **Aux1** and **Aux2** outputs support **PWM signal generation** with a frequency range of **10 Hz to 1000 Hz** and a duty cycle that can be adjusted based on the value of any selected channel.

9.2. Analog output

The analog output control channel is available in the ***Outputs*** panel as ***AOout.channel***. It should be assigned a variable representing a value in V with a 0.001 V accuracy, resulting in the actual value ranging from 0 to 5000 (0-5 V).
The following example will show how to change a signal from the 0-5 V analog input to a 5-0 V signal and how to set the current voltage at the ***Analog output***.
We will use a 2D table for this purpose. To create it, select the ***Add*** button and then ***Table***. The configuration dialog should look like the following:



A 2D Table will be created after pressing the ***Create*** button. The table should be modified so that for 0 V the value read from the table is 5 V and for 5 V it is 0 V.

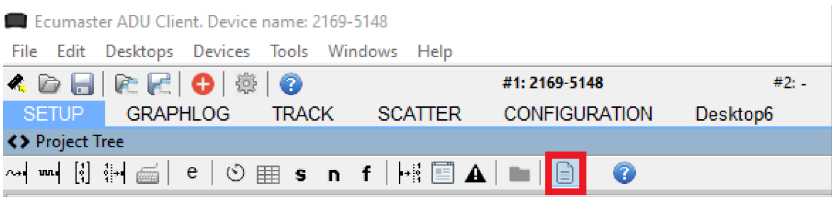


The *t_conversionTable* table value should be assigned to the ***AOut.channel*** field in the Outputs panel.

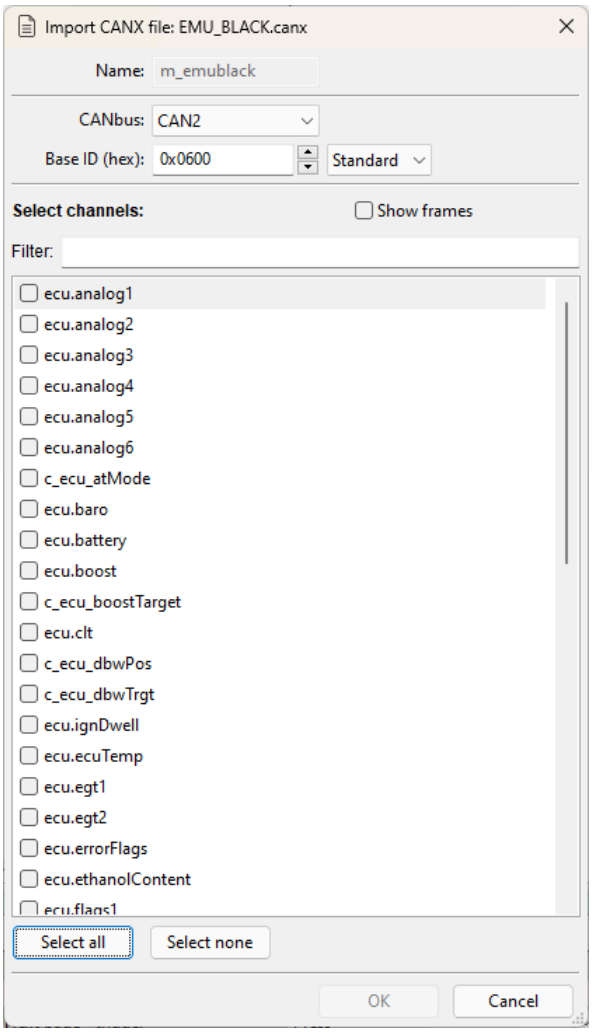
10. Working with CAN buses in ADU

10.1. Using pre-defined streams from *.CANX* and *.DBC* files

The simplest way to work with the CAN bus is to use pre-defined streams from ***.CANX*** and ***.DBC*** files. Streams in files with the *CANX* extension were supplied together with the ADU Client software.



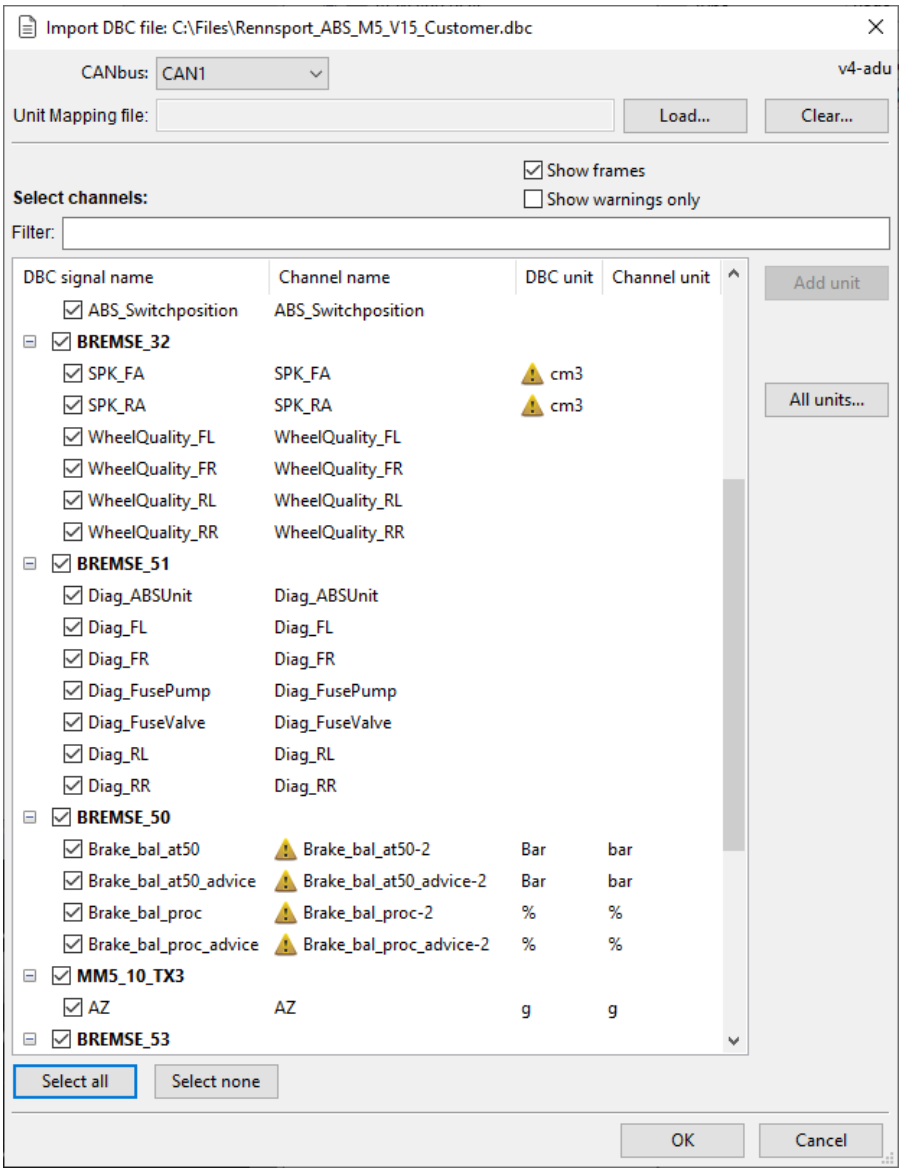
Having selected an icon from the taskbar or choosing the ***Project Tree*** > ***Add*** > ***Import .CANX / .DBC file*** option and pointing to the *.CANX* or *.DBC* file, a window with the import settings will open.



First, choose the CAN bus from which data will be received. The ADU device has two CAN buses: CAN1 and CAN2. Next select the channels to be imported. You can use the filter to select individual channels or select all using the '***Select all***' button. You should remember that the ADU device supports up to 150 CAN channels on both buses.

The following warnings may appear when importing a *.DBC* file:

- about units not defined in the ADU;
- about channels already existing under a particular name.

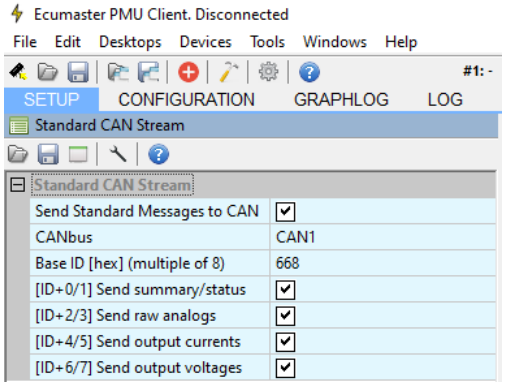


After confirming with the OK button, the selected channels will be added to the ***Project Tree***. In addition, one or more ***CANbus Message Object*** responsible for receiving frame groups will be created.

10.2. Built-in support for PMU

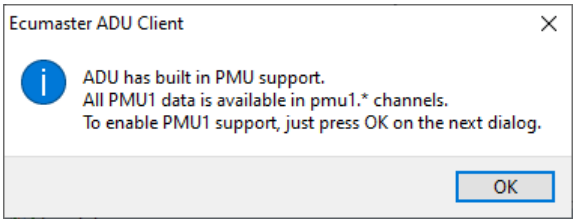
The PMU is the vehicle's intelligent energy management unit. All information about the status of the outputs, voltage and current and analog inputs can be recorded and transmitted via the CAN bus to the display. Several PMUs can be connected on the CAN bus. The ADU is equipped with default inputs for the three PMUs, so that the user does not need to use any of the 150 *CANbus inputs* to monitor channels from the PMUs. Each PMU comprises of 87 channels.

In order for the ADU to receive data correctly, the dedicated CAN Stream in the ***Ecumaster PMU Client*** software must be configured accordingly for each connected PMU. Use the **F9** key to open the ***Select panel*** window and then select ***Standard CAN Stream***. In the window that opens, tick the ***Send Standard Massage to CAN*** box, select the appropriate CAN bus and define the corresponding CAN ID. The dedicated CAN IDs are as follows: for PMU16 #1 - 0x668, for PMU16 #2 - 0x670, for PMU16 #3 - 0x678, PMU24 #1 - 0x668, for PMU24 #2 - 0x678, for PMU24 #3 - 0x688. For more details on the PMU CAN stream, check: Appendix C - How to Configure PMU CAN Stream *(on page 225)*.

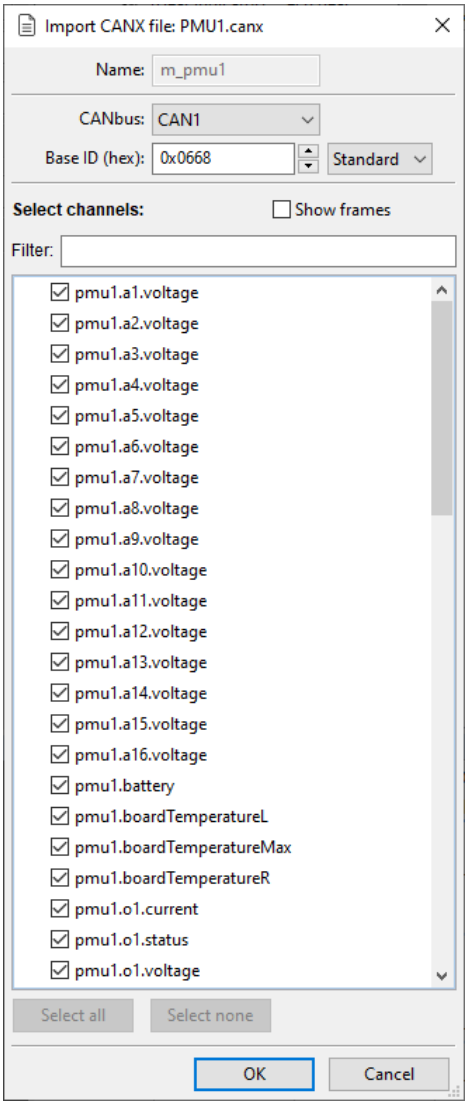


Receiving information

In ***Ecumaster ADU Client***, select the appropriate PMU#.canx file from the list in the Open CANX / DBC file window. Upon confirmation of the selection, the following message will be displayed: "The ADU has built-in PMU support. All PMU# data is available in pmu#.* channels. To enable PMU# support, press OK in the next dialog box."

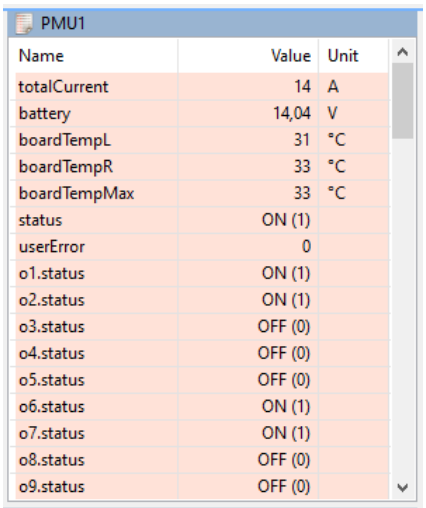


This window will show the list of channels and the pre-defined default CAN bus and CAN ID settings in the PMU, which can be edited.



Once the settings have been accepted, a new Message Object for the selected PMU will appear in the Project Tree.

To keep track of pmu channel status information, you can open a separate Log panel for each PMU.



10.3. Own CAN streams – CANbus Message Object

Access to a CAN bus in the ADU device is completely open. You can create your own streams or modify the existing ones supplied together with the programme.

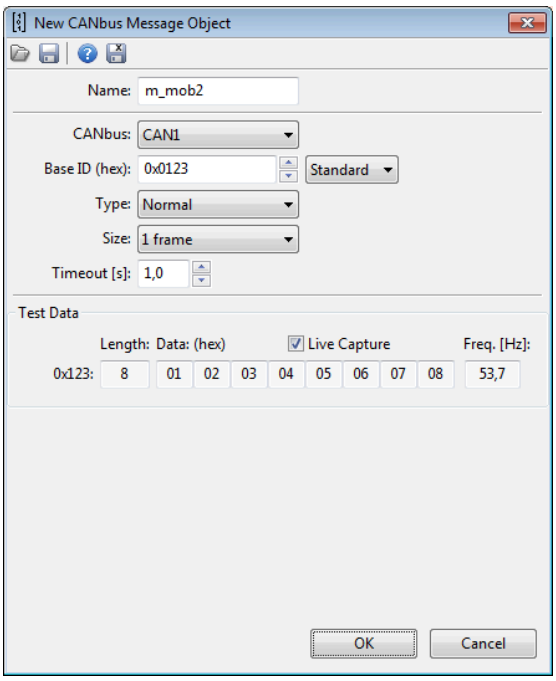
Configuration begins with the creation of a ***CANbus Message Object (MOB***) element in ***Project Tree***. Each MOB receives 1, 2, 4 or 8 CAN frames. After choosing a CAN bus you should select a base ID (***Base ID***), as well as the type (***Type***) and the number of received frames – the ***Size*** parameter. If the device in connected, a preview of the stream in real time (***Live Capture***) will be displayed. It facilitates diagnostics and speeds up work.

Important:
For ***Live Capture*** preview to work properly, active logging is required - logging cannot remain in pause mode!

Important:
Frame CAN IDs in the ADU Client are always presented in hexadecimal notation (they usually begin with the *0x* prefix, which is a symbol of the hexadecimal notation).

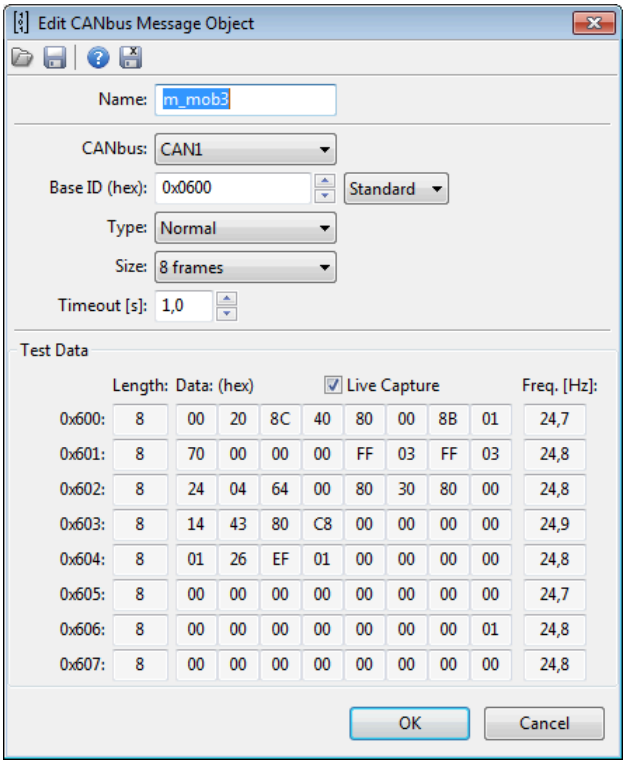
The following are examples of MOB configurations.

Receiving 1 frame ID 0x123 Standard



- **Base ID:** 0x123 Standard
- **Type:** Normal
- **Size:** 1 frame

Receiving 8 frames from the range of ID 0x600–0x607 Standard



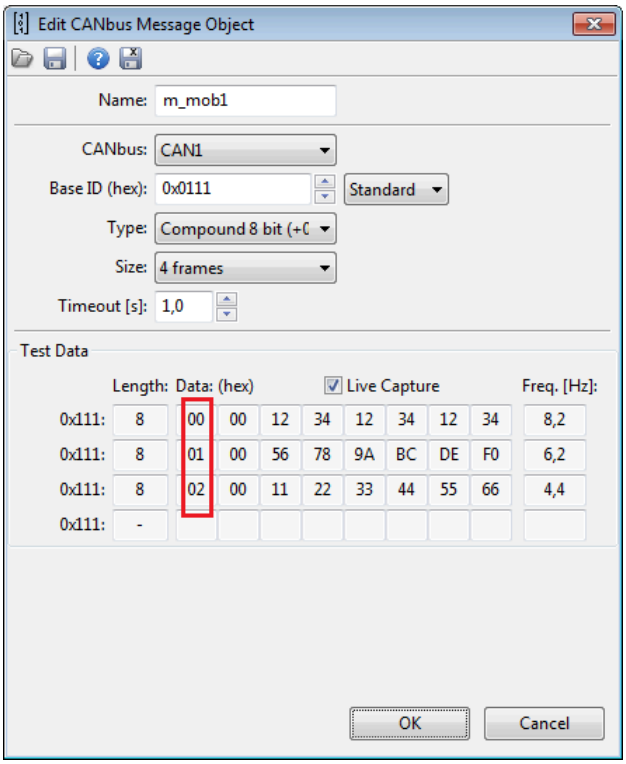
- **Base ID:** 0x600 Standard
- **Type:** Normal
- **Size:** 8 frames

Remember that for regular frames (Type: ***Normal***) Base ID must be divisible by (***Size***) without a remainder. For example, using one MOB, it is possible to receive 8 frames in the range 0x600-607.However, it is not permitted to receive frames in the range 0x601-608 using one MOB. If this is the case, this range should be divided into two MOBs of 0x600 and 0x608.

To check if the Base ID is suitable it is enough to verify the last digit in the hexadecimal notation:

- If the number is 0 or 8, then Base ID is divisible by 8 (Size: 8)
- If the number is 0, 4, 8, C, then Base ID is divisible by 4 (Size: 4)
- If the number is 0, 2, 4, 6, 8, A, C, E, then Base ID is divisible by 2 (Size: 2)
- Each number is divisible by 1 (Size: 1)

Receiving three Compound 8 bit frames from ID 0x111 Standard

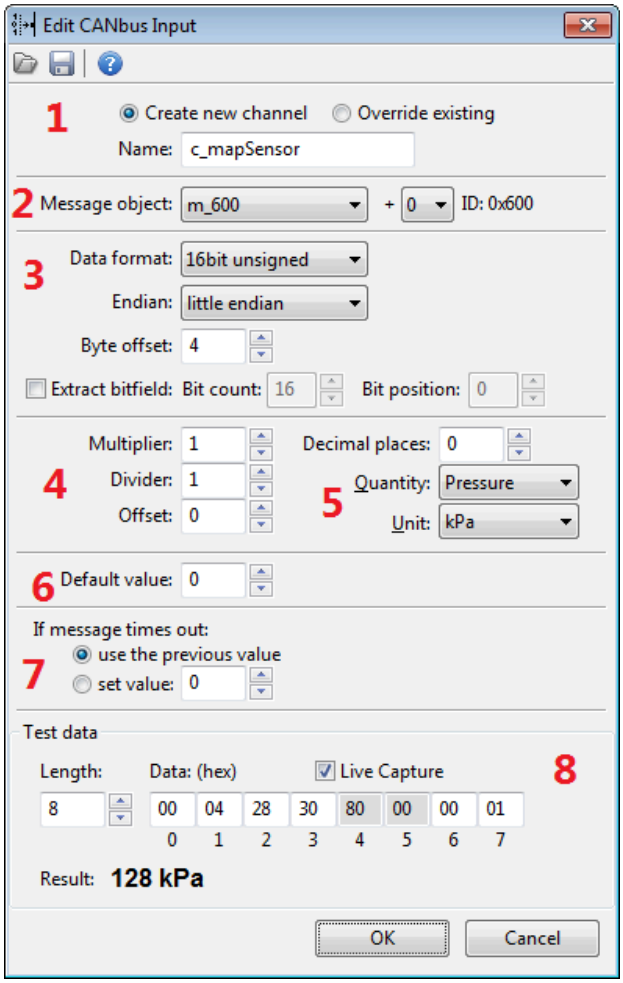


- **Base ID:** 0x111 Standard
- **Type:** Compound 8 bit (+0)
- **Size: 4** frames

For Compound frames the Base ID address doesn't have to be divisible by Size. This results from the fact that the communication takes place using only one CAN ID.
Compound frames (for Multiplexer types '4 bit', '8 bit' or '16 bit' contain an index in the first 4, 8 or 16 bits (according to Compound type respectively). For Multiplexer type 'Custom' index length (in bits), position and endianess is freely defined. In the dialogue window nearby you can see the first byte in a sequence 00, 01, 02.

10.4. Own CAN streams – CANbus Input

After a ***CAN Message Object*** has been created, you can start defining ***CANbus Input*** channels.



First, decide whether you will be creating a new channel (option: ***Create new channel***) or whether an existing channel shall be overridden (option: ***Override existing***) (1).
To create a new channel, choose a unique name so that it can be identified.
To use a channel name already in use, select an existing channel from the “ecu.” group. (e.g. ***ecu.rpm***). In addition, the decimal places and unit must be matched with the selected channel.
Next, select the previously prepared MOB, and select offset in frames from the drop-down list marked „+”. The scope of this parameter depends on the selected ***Size*** in the parameters of the used ***Message Object*** (2).

Next step is setting the ***Byte offset*** parameter (3). It marks the location of the values in question in the CAN frame (0– 7).

You should select the interpretation of the number:

- ***signed / unsigned Signed***– a number with a sign (it can receive positive and negative values, as well as zero). An example of such value is the value from the engine coolant temperature sensor. ***Unsigned*** – positive numbers or zero. For example engine speed (RMP).
- 8 bit / 16 bit– number width in bits; 1 byte or 2 bytes, respectively
  - **signed** 8 bit – range of numbers -128–127
  - **unsigned** 8 bit – range of numbers 0–255
  - **signed** 16 bit – range of numbers -32768–32767
  - **unsigned** 16 bit – range of numbers 0– 65535
- ***big endian / little endian*** – the "sequence" of bytes for 16–bit numbers. It shows how a number stored in two consecutive bytes shall be interpreted. E.g. numbers 0x12, 0x34 can be interpreted as 0x1234 for the ***big endian*** or 0x3412 for the ***little endian***.
- You can also define "***Extract bitfield***", i.e. take only a part of an 8- or 16-bit number. For example, to check the setting of a bit of a 0x80 mask the following settings should be used: Bit count: 1, Bit position: 7.

The ***Custom*** data format allows the exact width and position of the information stored in the CAN frame to be determined. The information can occupy a maximum of 16 bits, but these can be taken from 3 bytes. The bit numbering is compatible with Kvaser Database Editor 2.

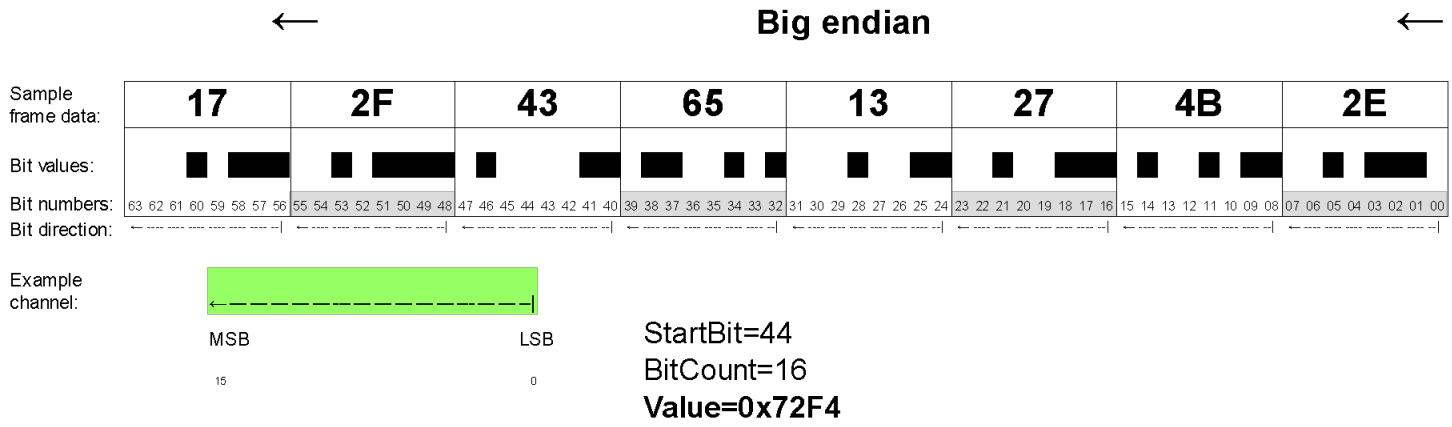
The ***Bit count*** parameter determines how many bits (1-16 bits) of information are present.
The ***Start bit*** parameter specifies the bit number at which the information in the CAN frame starts.

The next step is to scale /offset the values by ***Decimal places*** (4).
The "raw" value interpreted in Format (3) field can be scaled.
For example, Lambda in the EMU stream is saved as the value 0..255, where:

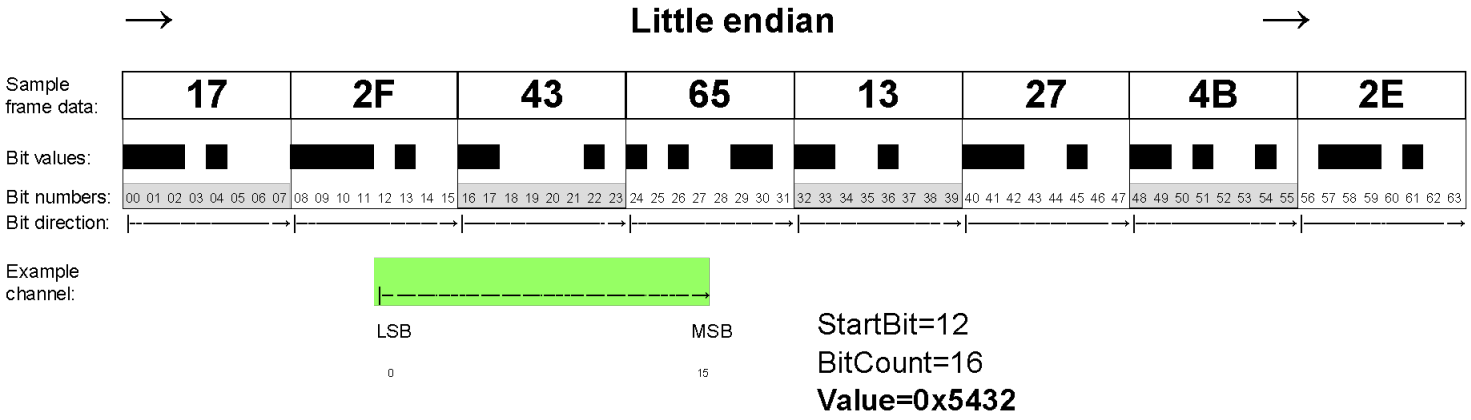
- raw value 0 means Lambda = 0.0,
- raw value 128 means Lambda = 1.0,
- raw value 255 means Lambda= approx. 2.0,

This value should be scaled. The following settings can be used: ***Multiplier*** =1000, ***Divider*** =128 and move decimal places using ***Decimal places*** =3. This way, you will add the end value of 1.000 to the raw value of 128.

Example for Big Endian, custom Data format:



Example for Little Endian, custom Data format:



Selection of a physical value and the unit (5). Typical SI units as well as units commonly used in the automotive industry are available. If a requested unit is not on the list, you can also use the ***User*** unit.

Once a unit has been selected, set the selection to the default value (6).
A default value is used from starting the device until receiving the first frame containing the channel.

Important:

Decimal places should be included in this constant. For example, if ***Decimal places*** = 2 has been selected, and the default value is to be 1.0, enter the value 100 into the field. This also applies to the following fields: ***Offset*** and ***Timeout value***.

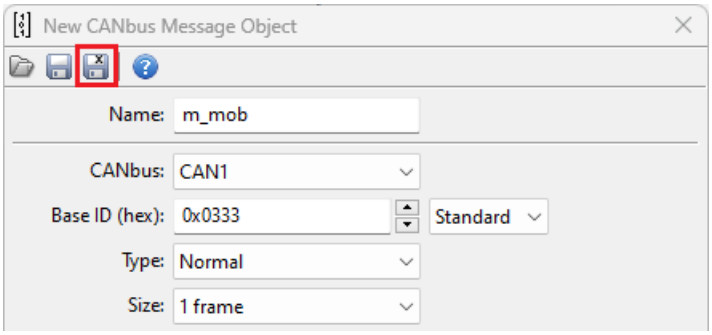
The definition of the behaviour in the event of a fade-out of frame reception on the CAN bus is carried out in the ***If message time out*** field (7).

If a given frame cannot be received for a time longer than that defined in ***Message Object*** configuration (***Timeout*** parameter in seconds), two options are available:
(a) the last value (possibly the default value if a frame was never received) may remain (***Use previous value***);
(b) a specific value can be set (***Set value***).

The last element of the ***CANbus Input***defining window are ***Test data*** fields(8). They are used only during editing. You can observe a received frame in real time (***Live capture*** on) or enter test data (***Live capture***off). In both cases the calculated final value is displayed, which accelerates configuration.

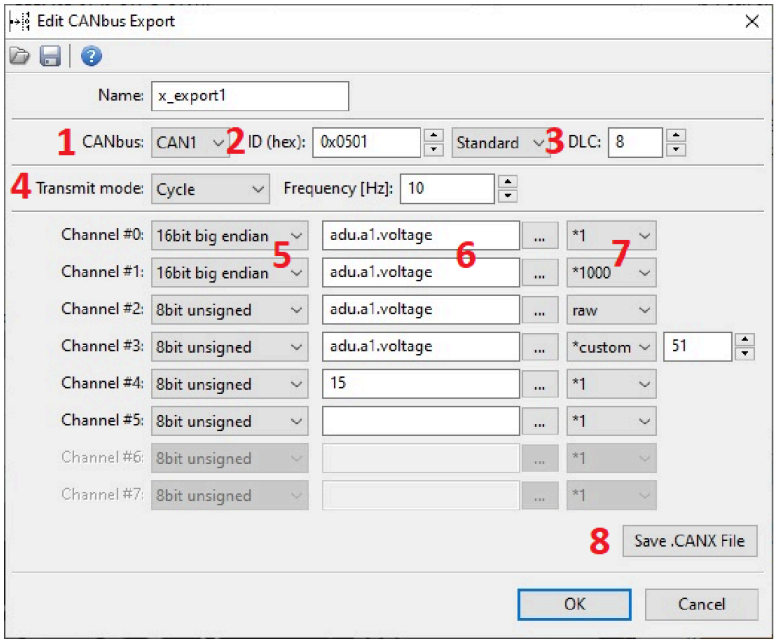
10.5. Own CAN streams – saving to a .CANX file

A configured ***Message Object*** together with all ***CANbus Input*** channels can be saved into a .CANX file using a toolbar button.



10.6. Sending frames using the CAN bus (CANbus Export)

Access to the CAN bus in the ADU device is completely open, which allows sending any available channel of the device. Frames with any CAN ID can be transmitted on one of two CAN buses.
The ***CANbus Export*** configuration window is comprised of the following sections:



CAN bus selection (1)

Selecting the CAN frame ID (2)

When selecting the CAN ID frame identifier, it is important to ensure that it does not come into conflict with other communications on the network. Recommended range of identifiers for the user: 0x500–0x57F. In this respect, ECUMASTER devices will never have their default CAN ID in the future.

Important:

In a CAN network, it is not permitted for two devices to transmit frames with the same CAN ID.

Determination of frame length *DLC* (3)

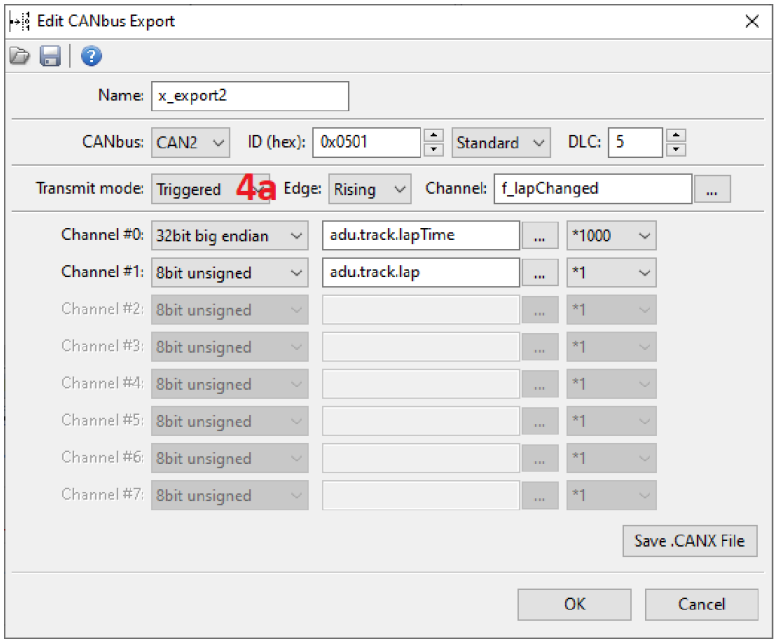
The *DLC* determines the length of a frame: from 0 to 8 bytes.

Selecting transmission type

Continuous transmission *Cycle* (4)

For continuous transmission, select a sending frequency (***Frequency***) in the range from 1 to 100 Hz (from 1 to 100 frames sent per second).

A maximum of 500 frames can be sent per second on CAN1 bus and 500 frames on CAN2 bus.



Triggered transmission *Triggered* (4a)

With ***Triggered*** transmission, a frame is sent when the appropriate ***Edge*** appears on the selected ***Channel***: (***Rising*** or ***Falling***).

Selection of the type of the data sent (5)

There are 6 options available:

- ***8bit unsigned*** – the value of the channel is limited to the range of 0..255 and sent as a single byte in a frame.
- ***8bit unsigned*** – the value of the channel is limited to the range of -128..127 and sent as a single byte in a frame.
- ***16bit big endian*** – the value is sent as the most significant byte, the least significant byte (for example the value 0x1234 will be sent as two consecutive bytes 0x12, 0x34).
- ***16bit little endian*** – the value is sent as the least significant byte, the most significant byte (for example the value 0x1234 will be sent as two consecutive bytes 0x34, 0x12).
- ***32bit big endian*** – 4-byte value, sent from the most significant byte to the least significant byte.
- ***32bit little endian*** – 4-byte value, sent from the least significant byte to the most significant byte.

4-byte values (***32bit***) are used, for example, to transmit the longitude and latitude for the ***gps.longitude*** and ***gps.latitude***channels. In order to maintain adequate accuracy of the transmitted data, the ***raw*** value may be used, which, for example for gps coordinates corresponds to the actual value (in degrees ***[deg]***) multiplied by $*10^7$.

Selected channels or constants (6)

You should select a channel from the list or enter a constant. In addition to the decimal notation, a constant can also be saved in a hexadecimal notation. To this end, the 0x prefix should be used (e.g. 0xE3 or 0xe3).

Selection of a multiplier or a raw value (7)

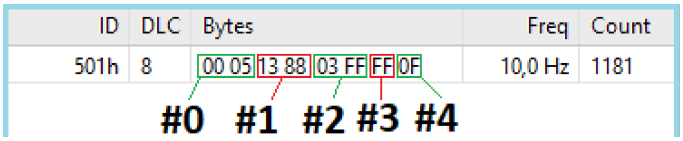
It is possible to multiply the actual value by a constant in the 1-1000 range (the fractional part is discarded) or alternatively to send the ***raw*** value.

Example:

From the drawing of the window *x_export1*:

- ***Channel #0*** – voltage value at input A1 will be sent as a number from the range: 0, 1, 2, 3, 4, 5 (in volts, but without the fractional part).
- ***Channel #1*** – voltage value at input A1 will be sent as a number from the range 0–5000 (in milivolts).
- ***Channel #2*** – voltage value at input A1 will be sent as a raw value from the ADC converter as a number from the range 0–1023.
- ***Channel #3*** – voltage value at input A1 will be sent as a number from the range 0–255.
- ***Channel #4*** – a constant value will be sent: 15 in the decimal system.

Below is a frame preview as seen in the ECUMASTER Light Client. At the analog input A1, the voltage is exactly 5 V. Accordingly, the channels ***Channel #0*** - ***Channel #4*** present themselves as in the example below:



- ***Channel #0*** – 0x0005, i.e. 5 [V]
- ***Channel #1*** – 0x1388, i.e. 5000 [mV]
- ***Channel #2*** – 0x03FF, i.e. 1023 [adc]
- ***Channel #3*** – value 0xFF, i.e. 255
- ***Channel #4*** – value 0x0F, i.e. 15

Saving to a .CANX file (8)

10.7. Reserved CAN ID

ID range	Default CAN bus	CAN bus configuration possible	Configurable ID	Description
0x012-0x017	CAN1	No, only CAN1	No, ID is fixed	Communication with ADU Client
0x032-0x035	CAN1	No, only CAN1	No, ID is determined	Communication with Light Client
0x400-0x407	CAN1	Yes, CAN1 or CAN2	Yes, ID must be a multiple of 8	ECUMASTER GPS
0x420-0x42F	CAN1	Yes, CAN1 or CAN2	Yes, ID must be a multiple of 16 (0x10)	ECUMASTER Tire Temp Camera (FL, FR, RL, RR) ECUMASTER Disk Temp Camera (FL, FR, RL, RR)

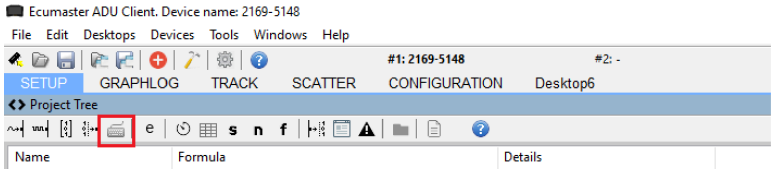
11. CAN bus keypad support

Software version 84.0 and later versions offer the option of connecting and operating a CAN bus keypad.

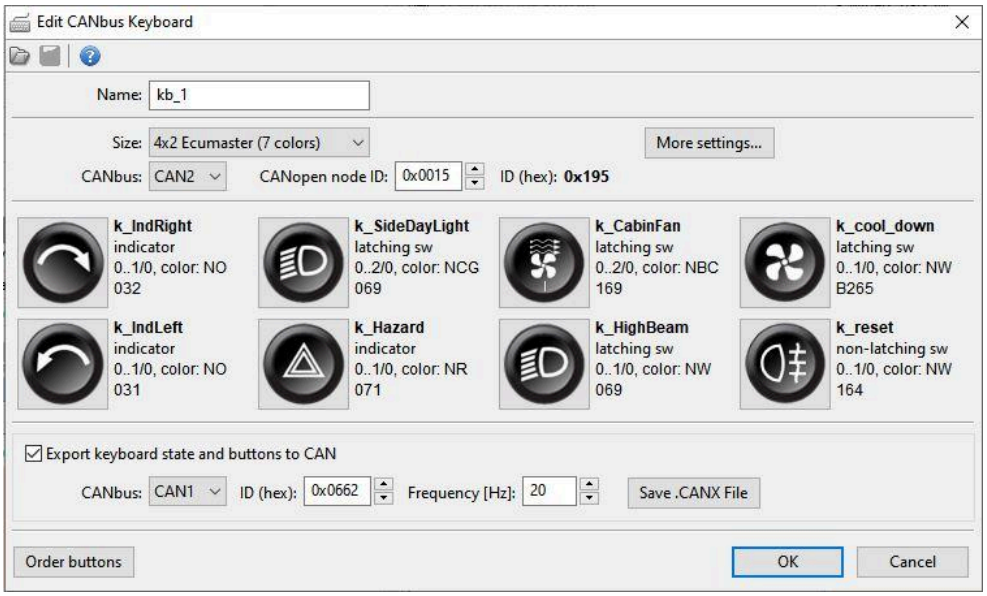
Keyboards from Ecumaster, Grayhill (using the CANopen protocol), MoTeC/RaceGrade, and Emtron are supported. If you're using the 5x3MT keyboard with two encoders, see Appendix E - How-to Set Up Keyboards with Encoders *(on page 233)* for more details.

One keyboard can be connected to the ADU.

To configure a keyboard, click ***Add*** in the ***Project Tree*** and then select ***CANbus Keyboard*** from the list or click the keyboard icon in the toolbar.

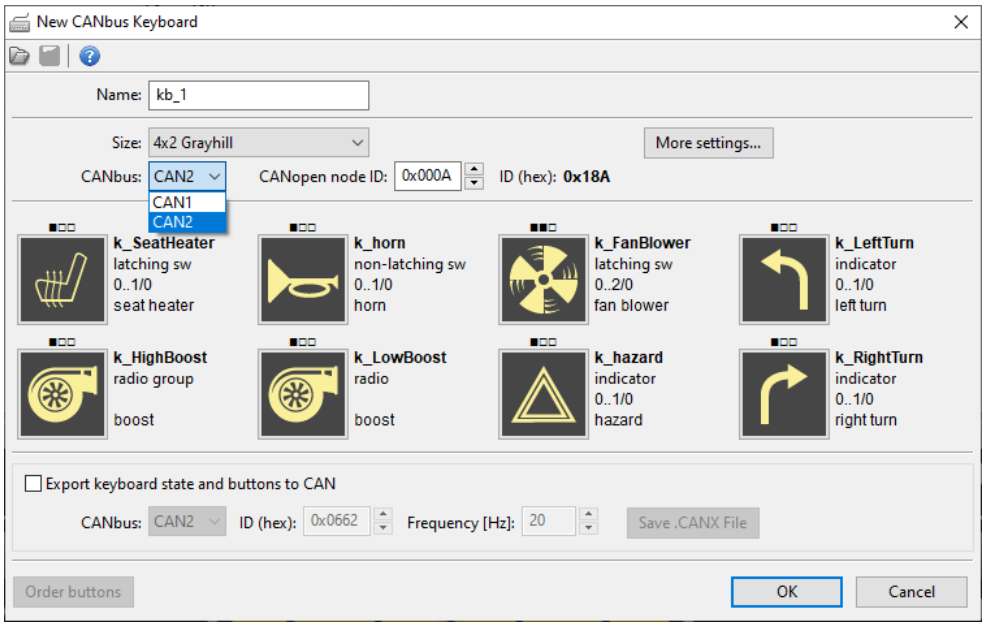


After opening the ***CANbus Keyboard*** window, individual parameters must be set:



- ***Name*** - name of the keyboard
- ***Size*** - size and type of keyboard to be configured.
- ***CANbus*** - CAN communication bus to which the keypad is connected

  The default speed of Ecumaster keyboards is 500 kbps, so it is recommended to select the CAN2 communication bus, which is set to the same speed by default. The CAN1 bus can only be selected for keyboards communicating at 1 Mbps.

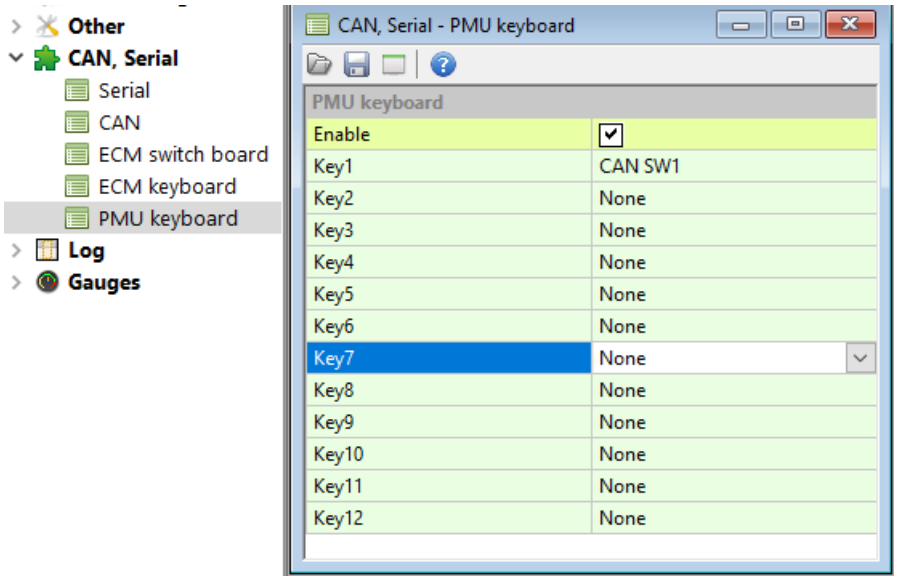


- ***More settings***

Settings	Description
Show open ceremony	Opening ceremony, i.e. key highlight animation when switching on the device (only for Ecumaster keyboards)
Master brightness channel [0-100]%	Defines any channel whose value can be used to regulate the brightness of the keyboard backlighting/keys in the range of 0-100%. The final brightness value is calculated using the formula: $finalKeyBrightness = \frac{Master\ brightness\ channel * Key\ LED\ brightness}{100}$ Similarly, the backlight brightness and alternative key LED brightness is calculated.
Key LED brightness	Backlighting intensity of the keys (Ecumaster keyboards) or LEDs above the keys (Grayhill keyboards)
Back light brightness	Backlighting intensity of the entire keyboard (night backlighting)
Back light color	Full keyboard backlighting color (only for Ecumaster keyboards)
Alternative brightness channel	Channel defining alternative backlighting
Alternative key LED brightness	Intensity of alternative key backlighting (Ecumaster keyboards) or LED above the keys (Grayhill keyboards)
Alternative back light brightness	Alternative backlighting intensity of the entire keyboard (night backlighting)
Alternative back light color	Alternative backlighting color of the entire keyboard (only for Ecumaster keyboards)

- ***CANopen node ID***- the factory setting for Ecumaster keyboards is 0x0015 (this item needs to be changed if there is a ***CANopen node ID*** conflict with another keyboard connected to the CAN bus).
  A detailed description of ***CANopen node ID*** changes can be found at: www.ecumaster.com/files/LightClient/LightClientManual_1_0.pdf in section 11.3.

- ***Export keyboard state and buttons to CAN*** - allows the export of keyboard status from the keyboard to the same or another CAN bus without the need to create and configure the ***CANbus export***. With this feature, the keyboard configured in the **ADU**will also work on the **PMUs**and **EMU BLACK**.
  When exporting key status to **EMU BLACK** the CAN ID address must be 0x662. To receive push button information, select the **PMU KEYBOARD** on **EMU BLACK**.

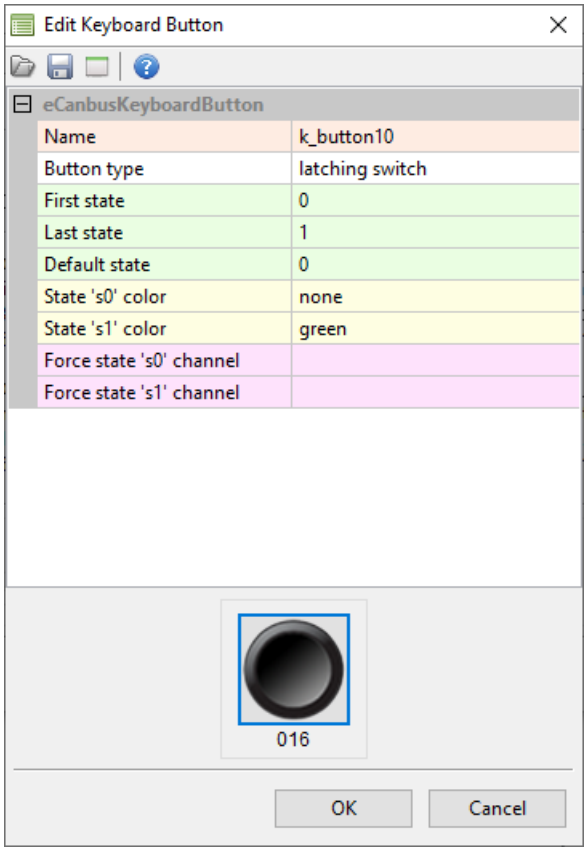


For the **PMU**, simply save the keyboard configuration settings to a ***.CANX*** file and then load the saved file (Import ***.CANX / .DBC*** file), which automatically configures the ***CANbus Message Objects*** and ***CANbus Inputs*** in the software.

- ***Edit Keyboard Button*** - edit individual keys:

Parameter	Description
Name	Button name
Button type:	Button mode of operation: ◦ ***non-latching switch*** - a non-latching button in which the user-defined backlight color of the button depends on one of the two possible states of the button. ◦ ***latching switch*** - a latching button in which two to four stable states can be defined. The color of the key's backlighting depends on the status of the button. ◦ ***radiobutton group***- button to open a group of radio buttons ◦ ***radiobutton*** - further buttons in the radio button group

Parameter	Description
	◦ ***indicator*** - a momentary pushbutton, with illumination to the specified color done by channel-assigned conditions (as opposed to a non-latching switch). Up to three different conditions can be set to change the state of illumination. ◦ ***indicator 3*** - in Grayhill keyboards, indicator mode 3 is available in addition to indicator mode. The difference between the two is that in indicator 3 mode, the independent illumination of each diode (in each possible configuration) is triggered by the fulfilment of three individual conditions assigned by separate channels. For the ***indicator*** mode on ***Grayhill***keyboards however, there is only one channel to assign conditions, the fulfilment of which allows the LEDs to be illuminated in up to three different states - one, two or three LEDs lit.
Default state	Default setting for the status of the button when the keyboard is activated
State 's#' color	Setting the color for a button state (only for Ecumaster keyboards)
Force state 's#' channel	Option available in ***latching switch*** mode. A channel for assigning a condition the fulfilment of which forces a change in the state of the button.
Source channel	Option available for ***indicator*** mode. Source channel for assigning a condition, the fulfilment of which triggers the illumination assigned to the button state.
Choose Button Image	Choice of button icons

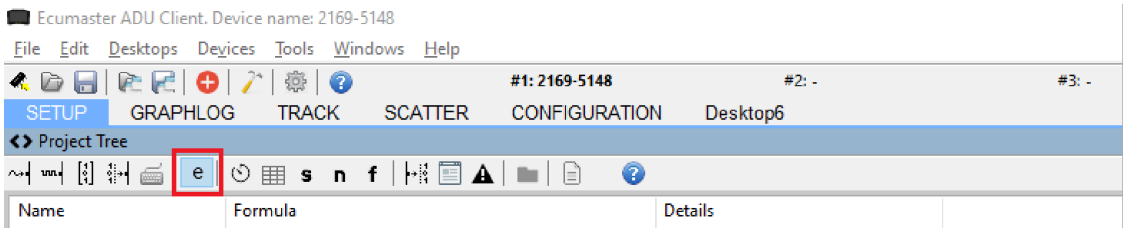


- ***Order buttons*** - prepares copy-ready information on the numbers of plastic key inserts, with selected laser-engraved icons, for order preparation.

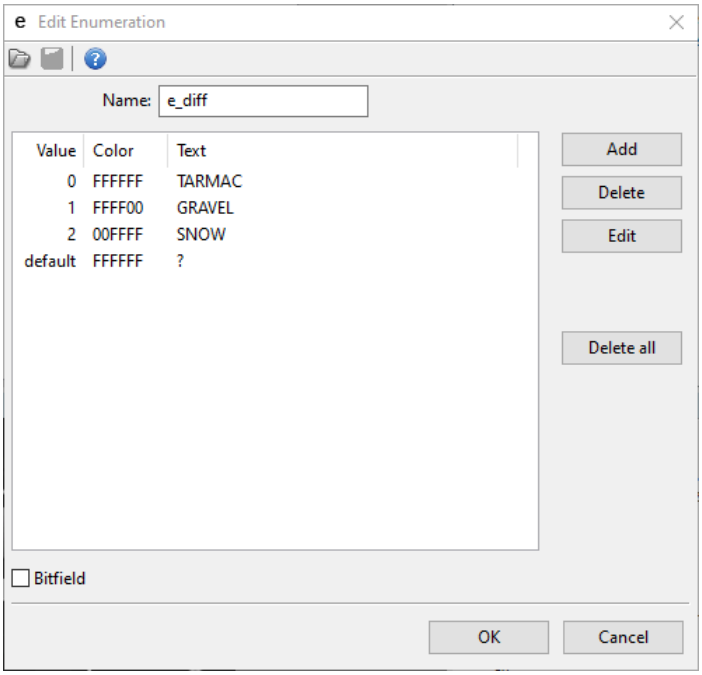
12. Enumeration

Enumeration allows assigning a selected text or color to a specific numerical value. This allows information from the channels to be displayed on a screen in the form of a text message.

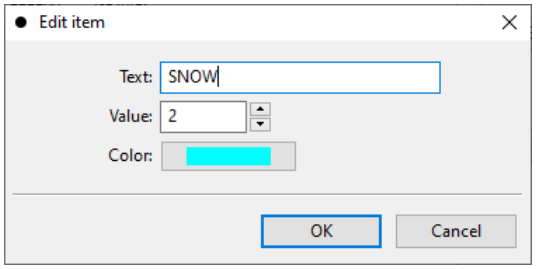
To define an enumeration, click the ***e*** icon in the ***Project Tree*** window on the toolbar or select ***Enumeration*** from the list via ***Add***.



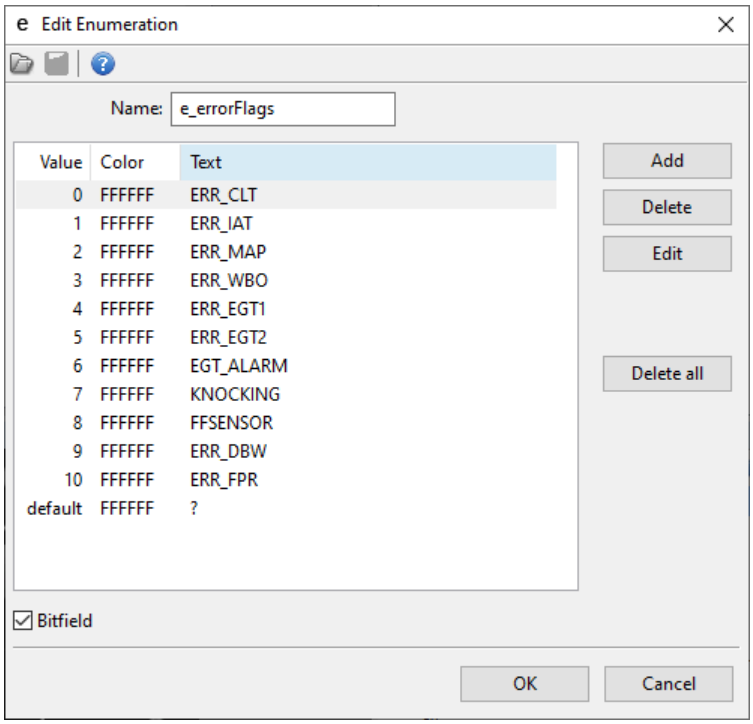
After opening the ***Edit Enumeration*** window, individual attributes must be defined:



- ***Name*** - enumeration name
- ***Add*** - opens the New item window, in which a numerical value (in the decimal system) must be entered, to which the selected text and color are assigned.



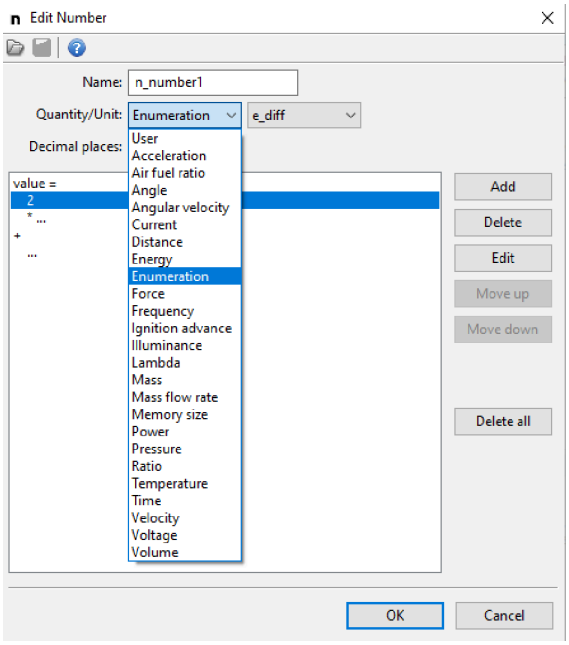
- ***Delete*** - removes the selected assignment
- ***Edit*** - edits the selected assignment
- ***Delete all*** - deletes all defined assignments
- ***Bitfield*** - an option that allows up to sixteen messages to be displayed simultaneously by assigning individual messages to the corresponding bit numbers (0 to 15).



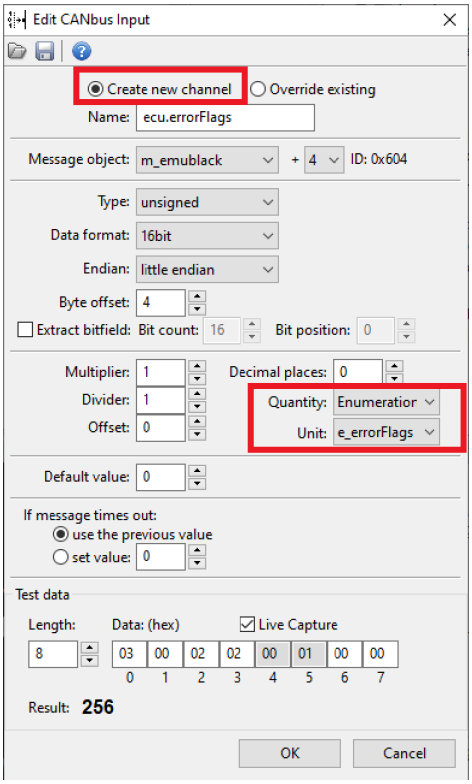
To use an enumeration and display a number in the form of an assigned text or color, it is not enough to simply define an assignment. The created enumeration should also be linked to the channel from which it will be taken. The channels in which enumeration can be used are as follows: ***Number***, ***Table*** and ***CANbus Input***.

Enumeration in the mathematical Number channel

In the ***Number*** channel, the result is a numerical value to which text or color is assigned by enumeration. For this purpose, in the ***Quantity / Unit*** field, select: ***Enumeration*** from the first list and the specific enumeration name previously defined by the user from the second list. In this way, enumeration will be associated with a channel. The association will be used when numerical information is displayed in the form of assigned text on the screen.



Enumeration in the *CANbus Input* channel

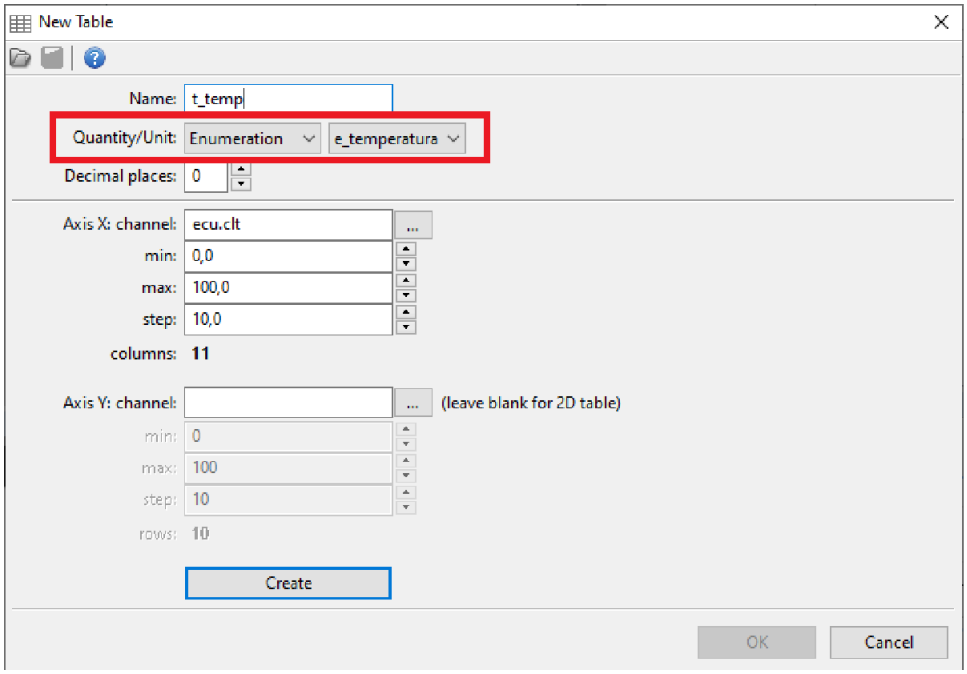


The assignment of Enumeration to ***CANbus Input*** is only possible with the ***Create new channel*** option selected. In the ***Quantity:*** field, set ***Enumeration*** and select the name of the previously created enumeration as the unit (***Unit:***).

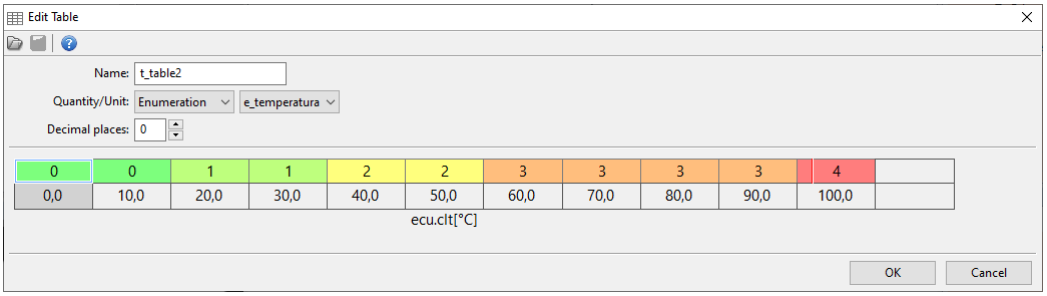
The value to which the text or color is assigned is ***Result***. This is the number resulting from scaling values from the indicated CAN frame by the selected coefficients: *Mulitplier*, *Divider*, *Offset* or *Decimal places*.

Enumeration in the *Table* channel.

The enumeration is assigned to the ***Table*** channel under ***Quantity / Unit*** item: Select ***Enumeration*** from the first list and the appropriate name of the previously created enumeration from the second list.



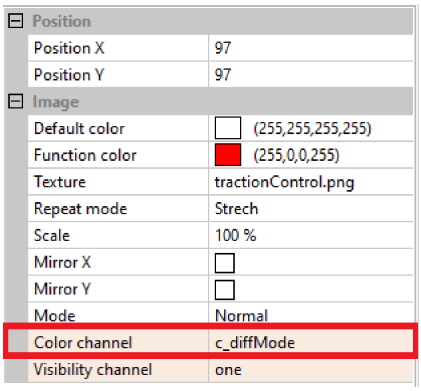
The individual arguments in the table are assigned values for which text / colour assignments are defined.



Display of enumeration on screen

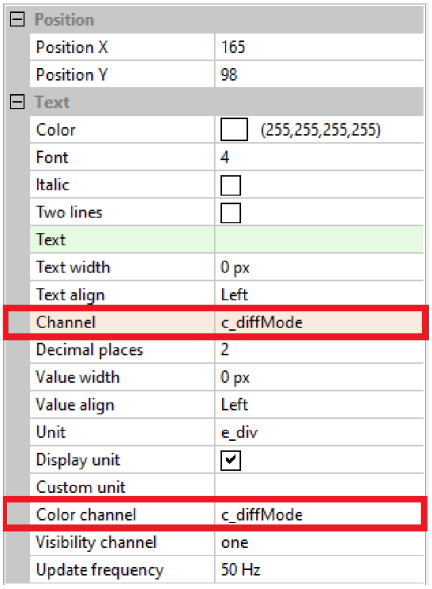
Once an enumeration has been defined and associated with the relevant channel, its result can be displayed on screen in the form of an assigned text / colour. Some page elements in the ***Page editor***, such as: ***Image***, ***Text*** and ***Grid***.

Image allows for the display of graphics, its color may be defined by a channel under the ***Color channel*** item.



Text allows the following values to be displayed:

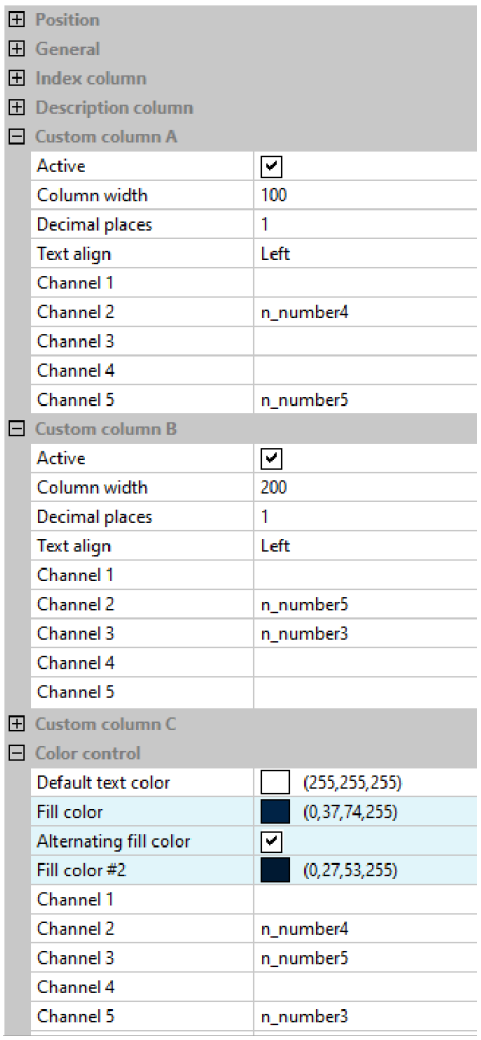
- using text - by selecting the channel with the assigned enumeration under ***Channel*** item;
- in an appropriate color - by selecting an enumerated channel in the ***Color channel item.***



An example of displaying an enumeration on the screen (*e_diff* connected to the channel *c_diffMode*) using *Image* and *Text*:

Example of on-screen display of enumeration with option *bitfield* (*e_errorFlags*). Depending on the value of the channel to which the sample enumeration is attached, between 0 and 10 messages can appear simultaneously.

Grid displays a table, which, in each of its cells, displays values defined by separate channels. The channel determining the colors of the cell values is defined separately for each row of the table.

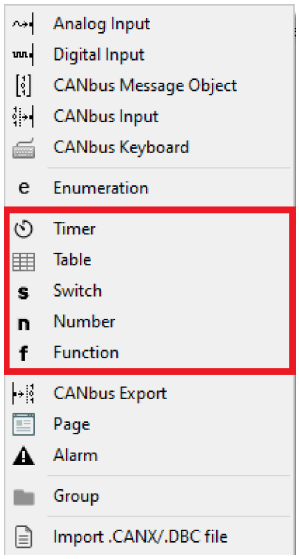


The color of the values on the screen can be assigned in ***Color Channel*** or defined by a primary color for a channel / condition value of 0 (***Default color*** - Image, ***Default text color*** - Grid, ***Color*** - Text) and an alternative color for a value different from zero (non-zero) (***function color*** - Image, red color - Text and Grid). When selecting an enumeration-related channel (without the bitfield option), the color assigned in the enumeration overrides the primary color. With bitfield enumeration, the color displayed on the screen is the primary or alternative color, depending on the condition met.

	Image	**Text**	**Grid**
Color defined directly	0 - Default color non-zero - function color	0 - Color non-zero - red	0 – default text color non-zero - red
Color defined by channel with enumeration	Color assigned by enumeration	Color assigned by enumeration	Color assigned by enumeration
Channel-defined color with bitfield enumeration	0 - Default color non-zero - function color	0 - Color non-zero - red	0 – default text color non-zero - red

13. Processing information in the ADU

The ADU has 5 information processing elements:

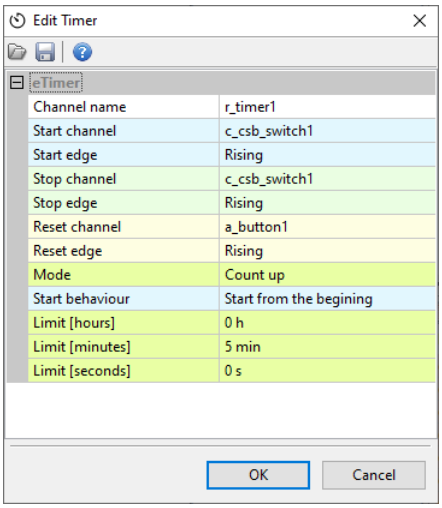


1. ***Timers*** – counting time
2. ***Tables*** – lookup tables
3. ***Switches*** – virtual switches, counters
4. ***Numbers*** – mathematical channels
5. ***Functions*** – logical functions

The above elements are processed at 500 Hz (every 2 ms) just like the other elements in the ADU. The order in which the elements are processed is shown above, i.e. ***Timers*** first, then ***Tables*** etc.

13.1. Timers – counting time

Timers are used for counting time.



Two ways of counting are available: the first from zero to the set value (*Count up*) and the second from the set value to zero (*Count down*).
Timers can store times up to 200 hours with an accuracy of up to 0.01 seconds.

Timers are started using a channel specified by the ***Start Channel*** and an edge defined by the ***Start Edge*** (*Rising* or *Falling*). Similarly, they can be stopped using the ***Stop Channel*** and a corresponding ***Stop Edge***.

The ***Reset Channel*** resets the timer to its initial value, whether it's running or stopped. Depending on whether the timer is in *Count Up* or *Count Down* mode, it resets to either zero or the set *Limit*.

When a timer is stopped, a starting edge detected in the Start Channel will trigger behavior defined by ***Start Behavior***:

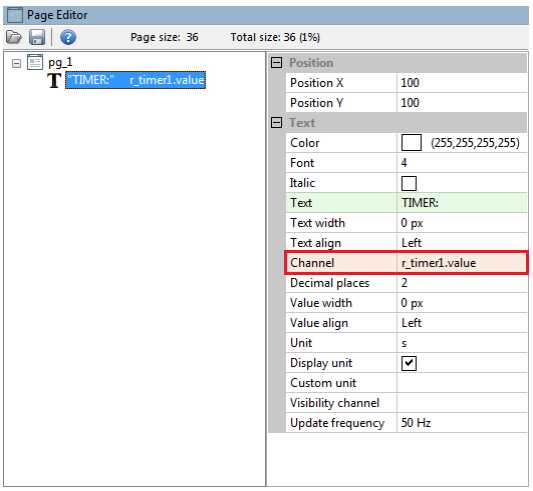
- ***Start from Beginning*** restarts the timer from the initial value.
- ***Resume if Not Yet Elapsed*** continues the timer if the limit was not reached before.

If the timer is already running, any further starting edges will be ignored. The timer will only react to a starting edge once it has been stopped or has finished counting. This allows you to start and stop the timer using the same channel and edge.

Each created timer has 3 subchannels:

- ***.value*** – value of time in seconds (up to 0.01 s).
- ***.elapsed*** – has a value of “1” when the time has elapsed; it will change to “0” after another start.
- ***.running*** – has a value of “1” when a timer is counting time.

A timer can be used on a page in the ***Text*** indicator.

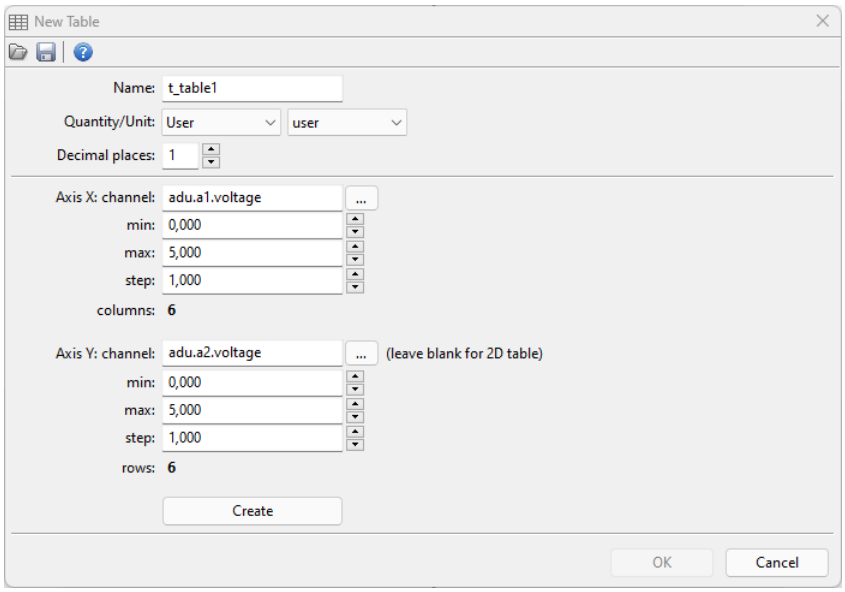


Use the ***.value*** subchannel for this purpose.

Time can be formatted using the Unit option e.g. using the hh:mm:ss format. Accuracy can be changed using the ***Decimal places*** parameter.

13.2. Tables – 2D / 3D lookup tables

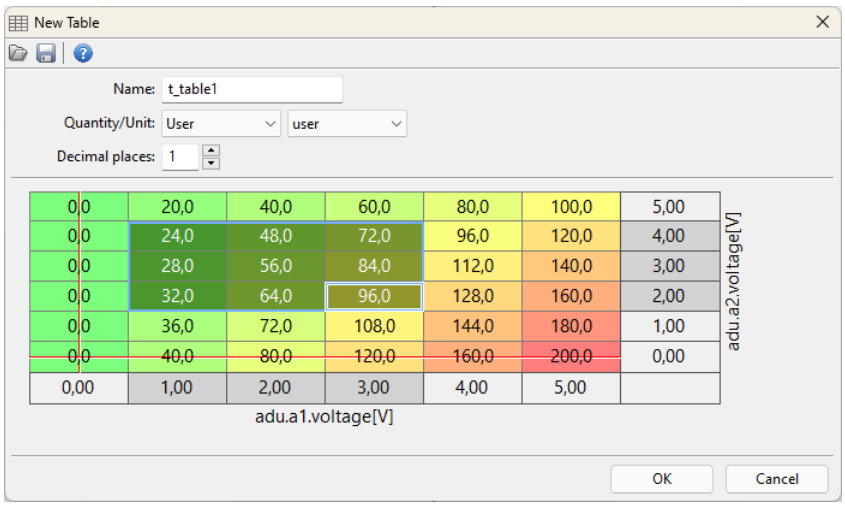
Configuration of a table starts with defining channels representing axes. Tables can be 2D or 3D. If a table is to be two-dimensional, leave the ***Axis Y: channel*** field empty.



You should also define the axis scope: ***min*** and ***max***. To change the number of elements in a table, change the step parameter which defines a step.

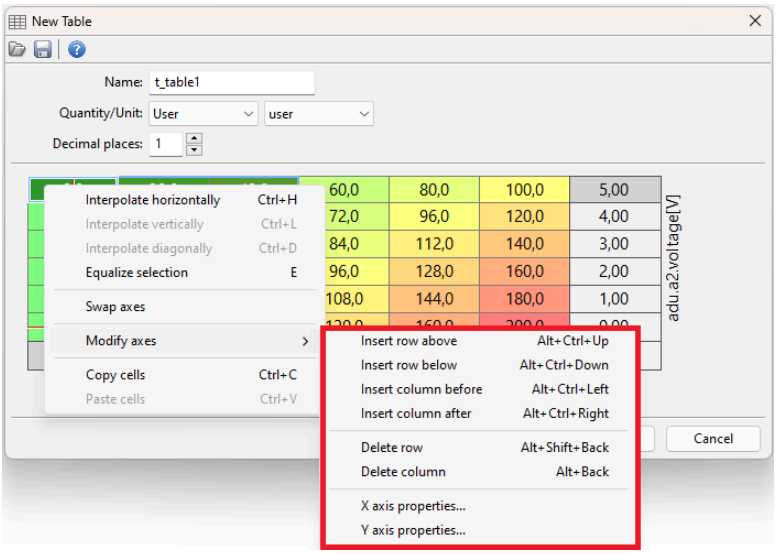
Table size can range from 2x1 (2D) or 2x2 (3D) to 21x1 (2D) or 21x21 (3D).

Next, fill the cells and axes with values. The values defined on axes are independent for each table.



You can select several cells using the Shift key. The Ctrl + arrow key copies to adjacent cells. You can also find the horizontal and vertical interpolation commands helpful.

The size of the table (number of columns or rows) can be changed at any time using the context menu available under the right mouse button.



Description of the commands in the context menu:

Command	Key shortcut	Description
Interpolate horizontally	Ctrl+H	Horizontal interpolation: the cell values in the selection area are calculated as a linear interpolation of the cells from the left and right edges of the selection.
Interpolate vertically	Ctrl+L	Vertical interpolation: the cell values in the selection area are calculated as a linear interpolation of the cells from the top and bottom edges of the selection.
Interpolate diagonally	Ctrl+D	Interpolation between apexes. Define the 4 corner points of the selection and the rest of the cells will be counted as bilinear interpolation. The command combines two commands: first the horizontal and then vertical interpolation.
Equalize selection	E	Smooths out selected cells
Insert row above	Alt+Ctrl+Up	Inserts a row above the selected cell
Insert row below	Alt+Ctrl+Down	Inserts a row below the selected cell
Insert column before	Alt+Ctrl+Left	Inserting a column to the left of the selected cell

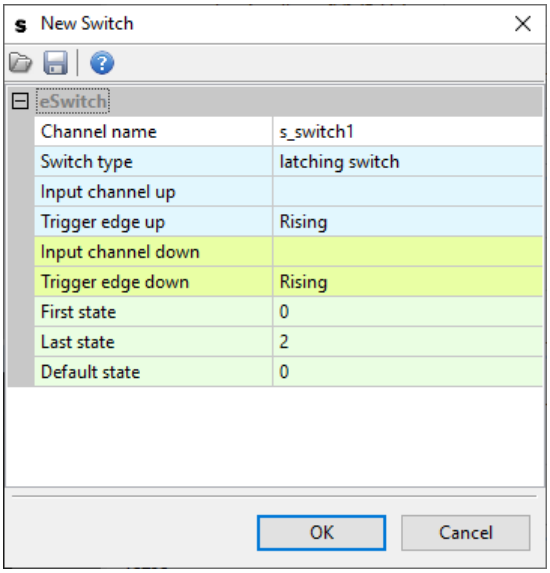
Command	Key shortcut	Description
Insert column after	Alt+Ctrl+Right	Inserts a column to the right of the selected cell
Delete row...	Alt+Shift+Back	Deletes a row containing the selected cell
Delete column...	Alt+Back	Deletes a column containing the selected cell
X Axis bins wizard		Starts the creator for the X axis allowing to define a new number of columns and to generate the Y axis cells according to the selected interpolation type
Y Axis bins wizard		Launches the Y-axis wizard to define a new number of rows and to generate Y-axis cells according to the selected type of interpolation

13.3. Switches virtual switches, counters

The main task of this element is to convert the momentary switch / non-latching switch available as an analog or CAN input into a Latching switch.
You should define the scope using the ***First state*** and ***Last state*** parameters, as well as the default value using the ***Default state*** parameter.

This element will operate as a counter. After each appearance of a ***Trigger edge up*** in the ***Input channel up*** element value will increase by 1. If the element already has a value equal to ***Last state*** and a ***Trigger edge up*** appears, the value will “roll back” and assume the value of the ***First state***.

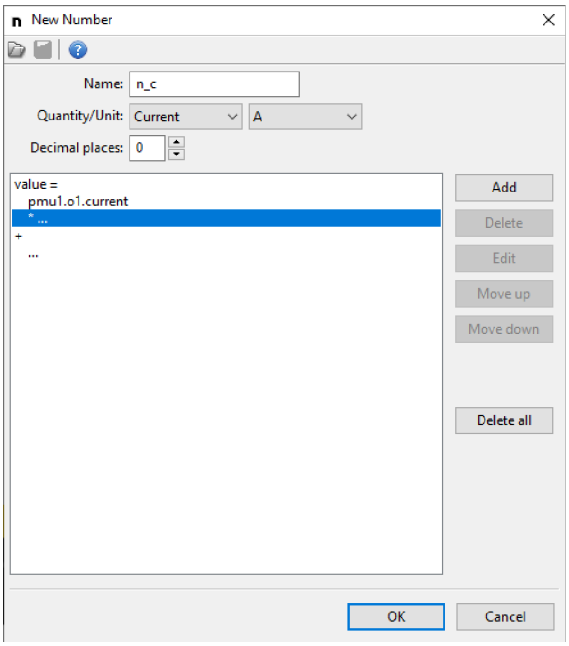
After each appearance of a ***Trigger edge down*** in the ***Input channel down*** the element value will increase by 1. If the element reaches a value equal to ***First state*** and it appears ***Trigger edge down***, it will change to the value of ***Last state***.


New Switch
eSwitch
Channel name
s_switch1
Switch type
latching switch
Input channel up
Trigger edge up
Rising
Input channel down
Trigger edge down
Rising
First state
0
Last state
2
Default state
0
OK
Cancel

13.4. Numbers – mathematical channels

A mathematical channel (*Number*) allows you to create a new channel resulting from the mathematical operations on selected values or channels. When you create a new channel, give it a relevant name (*Name*) and define the physical quantity and unit (*Quantity/Unit*) defining the created channel.

In the simplest form, the *Number* calculates the sum of the products of the selected values or channels.



value =

C1

*** C2**

*** C3**

*** ...**

+

C4

*** C5**

*** ...**

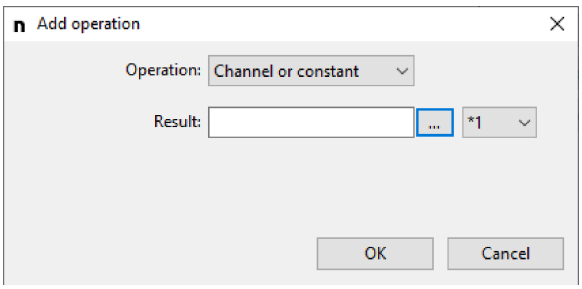
+

...

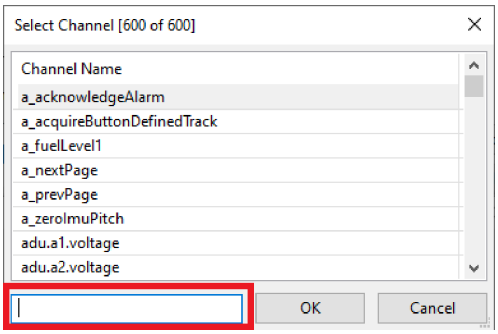
where C1, C2, C3... - is the selected channel or constant value (*Channel or Constant*)

To use the selected channel or constant value for calculations, click on the formula editor and select the field with the ellipsis '***...***'. Then click the ***Add*** button (located on the right).

The ***Add operation*** window will appear, in which you select the relevant operation. When selecting the ***Operation: Channel or constant***, enter a constant value in the ***Result:*** field or by clicking the button marked '***...***' select the relevant channel from the list.



For a faster search, you can enter the name of the channel you are looking for in the filtering field at the bottom of the ***Select Channel*** window.



After confirming with the ***OK*** button, the selected value or channel will appear in the formula field. Remember to select the relevant field (***...***) when assigning another channel, depending on whether it is multiplied (**** ...***) or added to the previously selected channel.
After marking channels in the formula editor, you can delete (*Delete*), edit (*Edit*) or move one place up (*Move up*) or down (*Move down*) in the formula.

You can also use other mathematical operations, including integer division - the *Divide* operation (“***/***”) or the remainder of the division - *Modulo* operation (***mod***).

e.g. value =

C1
* C2
/ C3

or value =

C1

mod C2

+

C3

* C4

* C5

List of operations available for mathematical channels.

FACTOR is a single multiplier (a constant or a channel) in the C1*C2*C3 notation.

RESULT is the calculated result of previous multiplications or divisions / residues.

Operation	Parameter	Pseudocode
Channel or constant	Result [1]	FACTOR = *Result*
Constant	Result [2]	FACTOR = *Result*
Choose	Condition channel Result if true Result if false	if *Condition_channel* ≠ 0 then FACTOR = *Result_if_true* else FACTOR = *Result_if_false*
Divide	Value	RESULT := RESULT DIV *Value* (DIV - integer division; e.g.: 9 DIV 2 = 4)
Modulo	Value	RESULT := RESULT MOD *Value* (MOD - division reminder; e.g.: 9 MOD 5 = 4)
Addition	Value 1 Value 2	FACTOR = *Value_1* + *Value_2*
Subtraction	Value 1 Value 2	FACTOR = *Value_1* - *Value_2*
Min	Value 1 Value 2	if *Value_1* < *Value_2* then FACTOR = *Value_1* else FACTOR = *Value_2*
Max	Value 1 Value 2	if *Value_1* > *Value_2* then FACTOR = *Value_1* else FACTOR = *Value_2*
Clamp	Input Min Max	if *Input* < *Min* then FACTOR = *Min* else if *Input* > *Max* then FACTOR = *Max* else FACTOR = *Input*

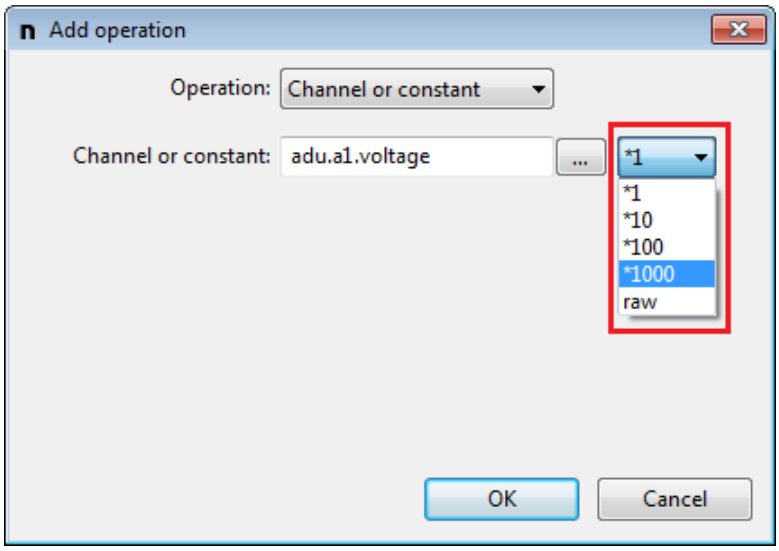
Operation	Parameter	Pseudocode
Lookup2	Channel First index Value[0], Value[1]	if *Channel* ≤ *First_index* then FACTOR = *Value[0]* else FACTOR = *Value[1]*
Lookup3	Channel First index Value[0], Value[1] Value[2]	if *Channel* ≤ *First_index* then FACTOR = *Value[0]* else if *Channel* = *First_index+1* then FACTOR = *Value[1]* else FACTOR = *Value[2]*
Lookup4	Channel First index Value[0], Value[1] Value[2], Value[3]	if *Channel* ≤ *First_index* then FACTOR = *Value[0]* else if *Channel* = *First_index+1* then FACTOR = *Value[1]* else if *Channel* = *First_index+2* then FACTOR = *Value[2]* else FACTOR = *Value[3]*
Get temperature element	Temperature x16 channel Element index	Channels: tire temperature camera ***ttc.tireTemp*** and brake disc temperature camera ***btc.brakeTemp*** Element index: 1-16 (temperature measurement point for the selected sensor)
Rate of change	Channel Sample period [ms] Filter strength [0-10]	derivative_raw = (*Channel* - previous_value) / *Sample period* derivative_filtered = ApplyFilter(derivative_raw) 0 - no filter, 10 - strongest filter RETURN derivative_filtered

[1] – The constant value for ***Chanel or constant*** operation may be in the [-16384, +16383] range

[2] – The constant value for ***Constant*** operation may be in the [-32768, +32767] range

Calculation of the channel value is done in raw values on integers. The fractional part is discarded. In order to obtain the needed accuracy of the created *Number* channel, each constant value or channel used in mathematical operations of the created channel should be multiplied by the appropriate value modifier (multiplier) and then take into account the *Decimal places* by "moving" the decimal point by the accuracy by which the individual channels/ values were multiplied.

Channel value modifiers.



The value of each channel available for mathematical operations can be modified. You may multiply by 1, 10, 100 or 1000 (the fractional part is discarded).

You may also choose to use the ***raw*** value with no modification. *Raw* applies to the voltage from analogue inputs, for example, when it is a value from an ADC converter with a range of 0-1023.

Decimal places

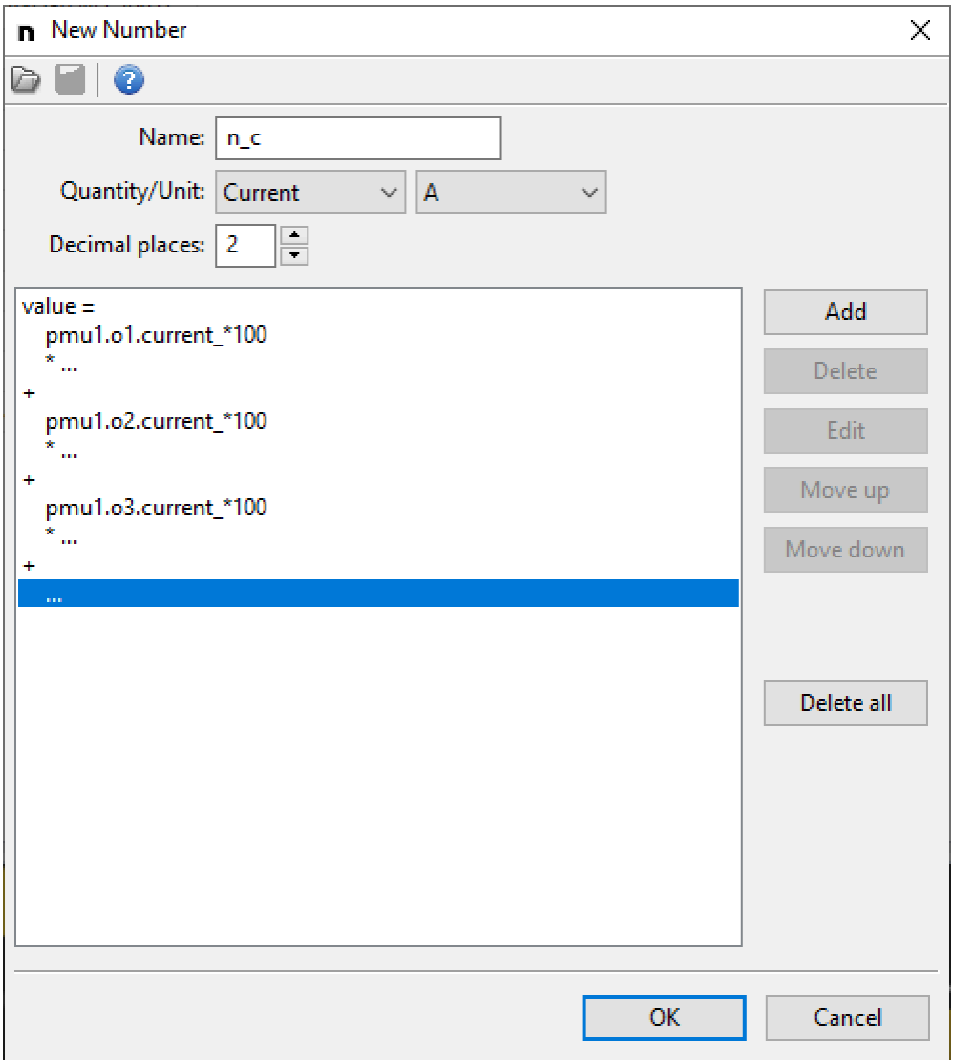
Each mathematical channel can store raw values within the range of [-32768, +32767].

You can additionally define decimal places by, “moving” the point by 0, 1, 2 or 3 places. For example, when ***Decimal places*** is set to 1, such channel can store real values in the [-3276.8, +3276.7] range.

Values are calculated in the raw value based on integers and then the point is “moved” by a defined number of decimal places.

Indirect calculations are performed using a broader range of numbers (ca. $\pm 2*10^9$). For example, calculations can be performed for the following values 1000*1000 / 123. At the end, the result is restricted (*clamp*) to the [-32768, +32767] range.

Calculating the sum of currents of 3 PMU current outputs



Three channels are given:

pmu_o1.current, *pmu_o2.current*, *pmu_o3.current*.

They contain the current sent from a PMU device.

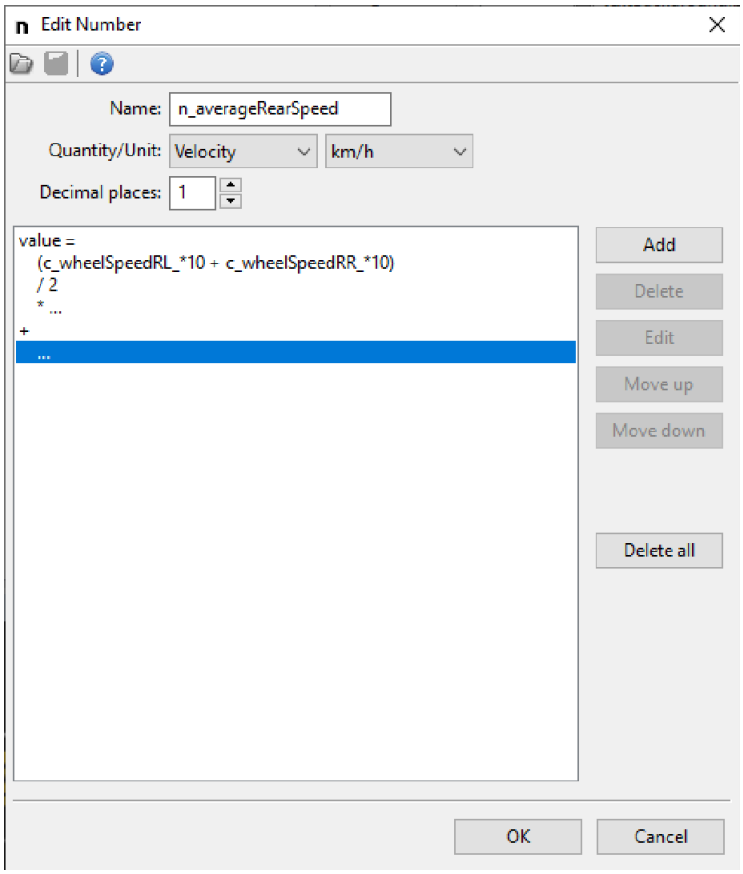
Let's assume that we want to obtain a result with an accuracy of up to 0.01 A.

For each channel, use the value modifier (multiplier) ****100*** and "move" the decimal point to the left by two decimal places (via ***Decimal places: 2***)

The following formula should be input:

n_c = (o1.current*100 + o2.current*100 + o3.current*100)

Calculating the average speed of the rear axle



Two channels are given: *c_speedRL* and *c_speedRR* with speed in km/h.
Let's assume that we want to obtain a result also with an accuracy of up to 0.1 km/h.

The following formula should be input:
n_averageRearSpeed = (c_speedRL*10 +c_speedRR*10) / 2

The decimal point should be moved by 1 place to the left. The "*/*" operation means integer division.

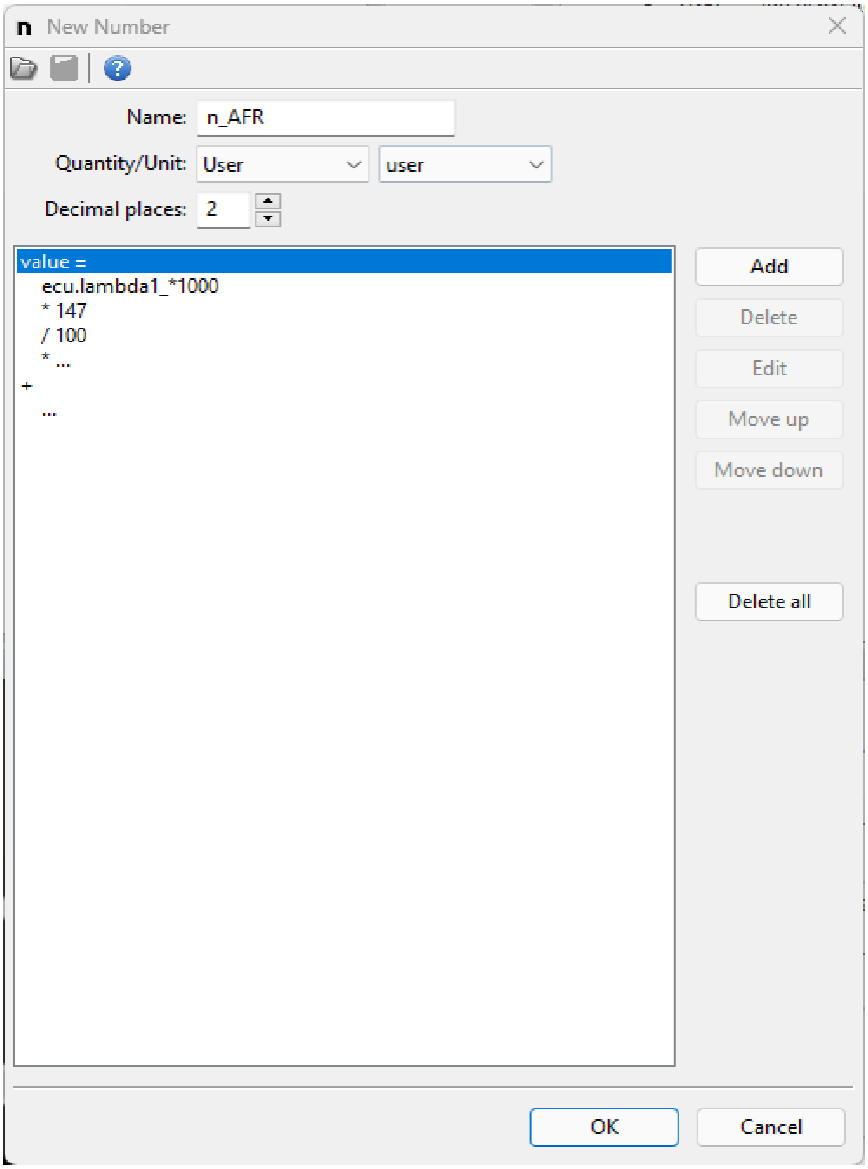
If you need to apply more complex mathematical operations (and it is difficult or impossible to keep the order of operations in one *Number* channel), you should break the operation into several stages by using multiple *Number* channels.

e.g. n_1 = C1+C2+C3
n_2 = C4+C5+C6
n_3 = n_1*n_2

If the final result of such operations should have the appropriate accuracy, remember that this accuracy must be taken into account at each stage of the calculation (in each intermediate channel created).

Changing the unit from lambda to AFR

n_AFR - channel displaying values with an accuracy of two decimal places



For *ecu.lambda* channel, use the value modifier (multiplier) ****1000*** and "move" the decimal point to the left by two decimal places (via ***Decimal places: 2***)

The following formula should be input:

n_AFR = (*ecu.lambda1000) * 147 / 100**

13.5. Functions – logical functions

Logical functions are used to define an extended behaviour of the display depending on the channel input values.

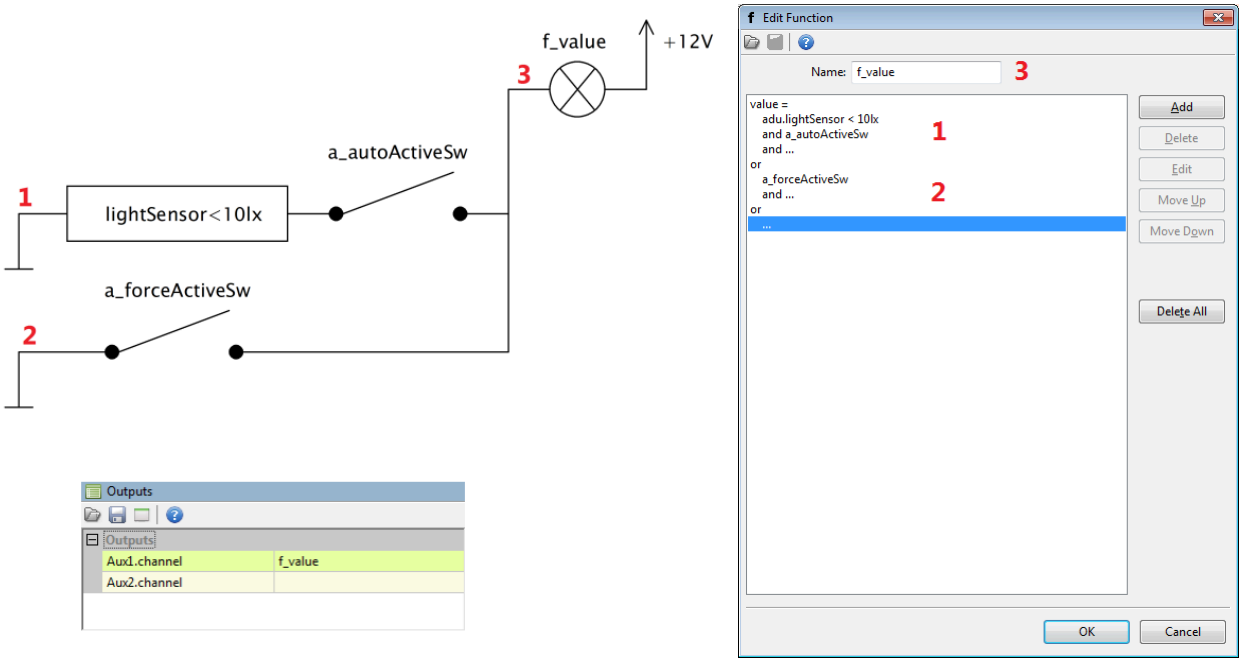
Controlling the bulb using a light sensor

The bulb lights up when at least one of the following conditions is met:

- when it is dark and the automatic mode is active (the *a_autoActiveSw* switch is on);
- when the manual lighting mode is active (the *a_forceActiveSw* switch is on).

Both switches are connected to the analog inputs of the device.

Below an analogy between a logical function and an electrical diagram is presented. A logical function will be real if at least one of the branches is true. A branch, in turn, is true when all operations in such branch are true.



Additionally, you should also set the Aux1 output, so that it can be controlled by the *f_value*:

List of operations available for logical functions.

Operations for logical functions can be divided into two groups: simple and special.

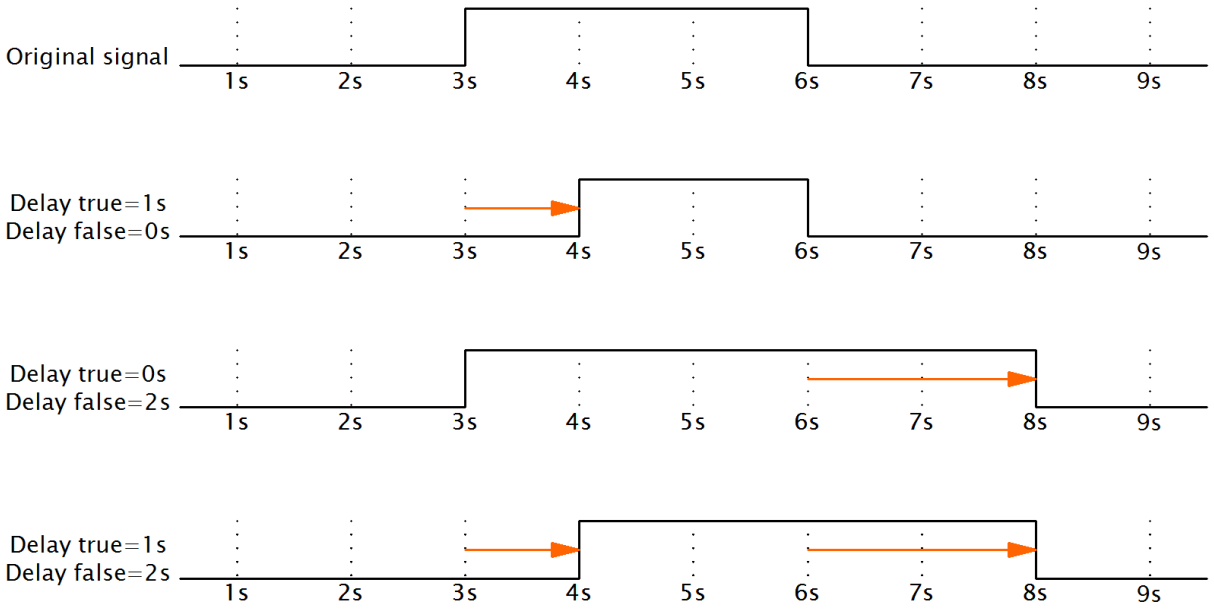
Simple operations are those whose result depends on the input state (alternatively a delay can be used for this result). Simple operations include: testing (*Is False*, *Is True*), (=, ≠, <, ≤, >, ≥) comparisons and logic operations (*And*, *Or*, *Xor*).

Important:

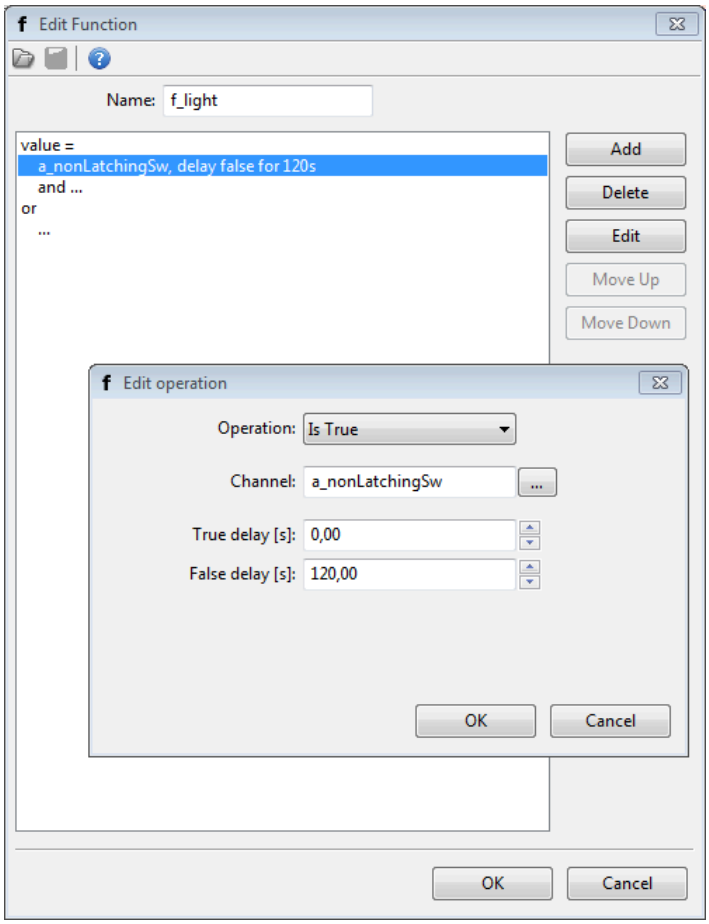
The following description contains **false** and **true** notions. **False** means a value of '**0**' (zero), and **true** means any value **other** than zero (e.g. '**1**').

Testing operations	
Is True	Returns **1** when the Channel value is **true** (non-zero); **0** otherwise.
Is False	Returns **1** when the Channel value is **false** (zero); **0** otherwise. (In electronics a NOT gate is analogous to this operation.)
Comparing operations	
Equal	Returns **1** when the Channel value **=** Constant; returns **0** otherwise.
Not Equal	Returns **1** when the Channel value **≠** Constant; returns **0** otherwise.
Less	Returns **1** when the Channel value **<** Constant; returns **0** otherwise.
Less or Equal	Returns **1** when the Channel value **≤** Constant; returns **0** otherwise.
Greater	Returns **1** when the Channel value **>** Constant; returns **0** otherwise.
Greater or Equal	Returns **1** when the Channel value **≥** Constant; returns **0** otherwise.
Logic operations	
And	Returns **1** when the values of both Channel #1 and Channel #2 are **true** (non-zero); returns **0** otherwise.
Or	Returns **1** when at least one of the channels, i.e. Channel #1 or Channel #2 is **true** (non-zero), returns **0** otherwise.
Xor	(Exclusive Or) Returns **1** only when exactly one of the channels Channel #1 or Channel #2, has a value of **true** (non-zero), returns **0** otherwise.

All simple operations allow to delay the switching on (***Delay true***) and the switching off (***Delay false***). The figure below shows the original signal and the following figures show how the ***Delay true*** and ***Delay false*** parameters modify with this signal.



The bulb goes on following pressing the button and remains on for another 120 s after releasing it.



This functionality can be achieved by using the *Is True* operation with the parameter *Delay false* = 120 s.

Special operations

<table>
<tr><th colspan="2">Signal generating</th></tr>
<tr><td>Flash</td><td>This operation generates impulses so long as the Channel is true (non-zero).
When the Channel value assumes false (zero), the operation returns the value 0.
When high state appears on the Channel channel (non-zero value), the Flash operation starts cycling between the value of 1 (duration defined by Time on) and the value 0 (duration defined by Time off). When the Channel value is false (zero), the operation will immediately start returning 0, thus interrupting the cycle.</td></tr>
<tr><td>Pulse</td><td>This operation generates N impulses following appearance of a trigger edge.
When the selected edge appears (Rising or Falling) on the Channel impulse generation will start. The number of impulses is determined by the Count parameter. Each impulse has an active phase (then the operation returns 1) and a non-active phase (the operation returns 0).
The Retrigger parameter determines if the appearance of a trigger edge during impulse generation will cause the process to restart or if it will be ignored.</td></tr>
<tr><th colspan="2">State-storing operations</th></tr>
<tr><td>Set-Reset Latch</td><td>The operation sets a new or returns the previous one according to the settings of the two input channels: Set Channel and Reset Channel.

<table>
<tr><th>Set channel value</th><th>Reset channel value</th><th>Operation value</th></tr>
<tr><td>true (non-zero)</td><td>false (0)</td><td>1</td></tr>
<tr><td>false (0)</td><td>true (non-zero)</td><td>0</td></tr>
<tr><td>true (non-zero)</td><td>true (non-zero)</td><td>0</td></tr>
<tr><td>false (0)</td><td>false (0)</td><td>previous value</td></tr>
</table>

An analogous operation is performed in the electronic SR latch. SR latch): https://en.wikipedia.org/wiki/Flip-flop_(electronics)
The initial value of this operation following starting the device can be defined using the Default State parameter.</td></tr>
</table>

Toggle	***Toggle*** changes the state between **0** and **1** each time the selected ***Edge*** (***Rising*** or ***Falling***) appears on the ***Channel***. ***Set channel*** allows setting the value to **1**, and ***Reset channel*** resets the value to **0**. The initial value of this operation following starting the device can be defined using the ***Default State***.

Toggle channel	***Set channel*** **value**	***Reset channel*** **value**	**Operation value**
Rising	false (0)	false (0)	state change
Falling	false (0)	false (0)	previous state
x	true (non-zero)	false (0)	1
x	x	true (non-zero)	0

x - regardless of condition

The table uses the ***Toggle*** channel with an ***Edge: Rising***.

Detecting changes	
Changed	When the value of ***Channel*** changes by a predefined ***Threshold***, the operation will initiate an active state (it will return the value 1) for the amount of seconds defined using the parameter ***Time on***. If, during this time, the channel value changes by the set threshold once again, the active state will be extended again by the number of seconds specified by the parameter ***Time on***. After the end of the active state, the operation will begin returning the value **0**.
Hysteresis	
Hysteresis	a) For the ***Polarity=Above*** parameter If the value of ***Source channel*** is greater than the predefined ***Upper value*** threshold, the value of the operation will be **1**. If it is lower than the ***Lower value*** threshold, the value of the operation will be **0.** If it is within **[*Lower value, Upper value*]** range, the value of the operation will be the previous value. b) For the ***Polarity=Below*** parameter If the value of ***Source*** channel is lower than the predefined ***Lower value*** threshold, the value of the operation will be **1**. If it is greater than the ***Upper value*** threshold, the value of the operation will be **0.** If it is within **[*Lower*

	value, Upper value] range, the previous value will be the value of the operation.

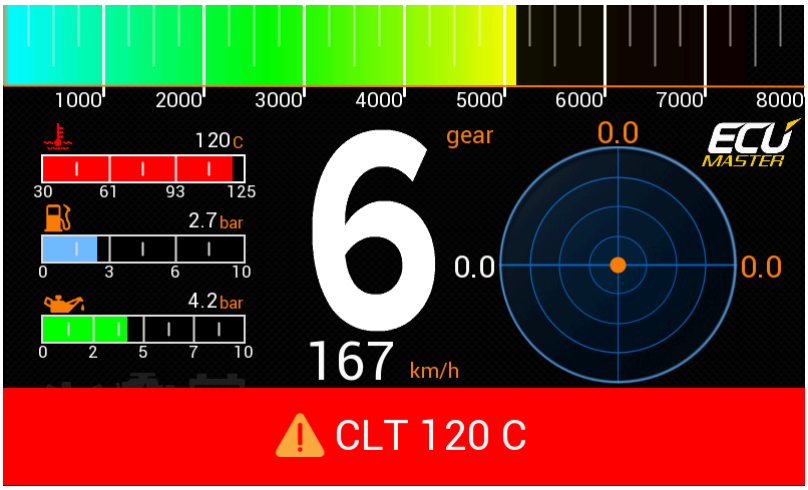
Important:

For ***Pulse*** , ***Flash*** and ***Changed*** operations setting the ***Time on*** parameter to 0 s will result in generation of a 2 ms impulse.

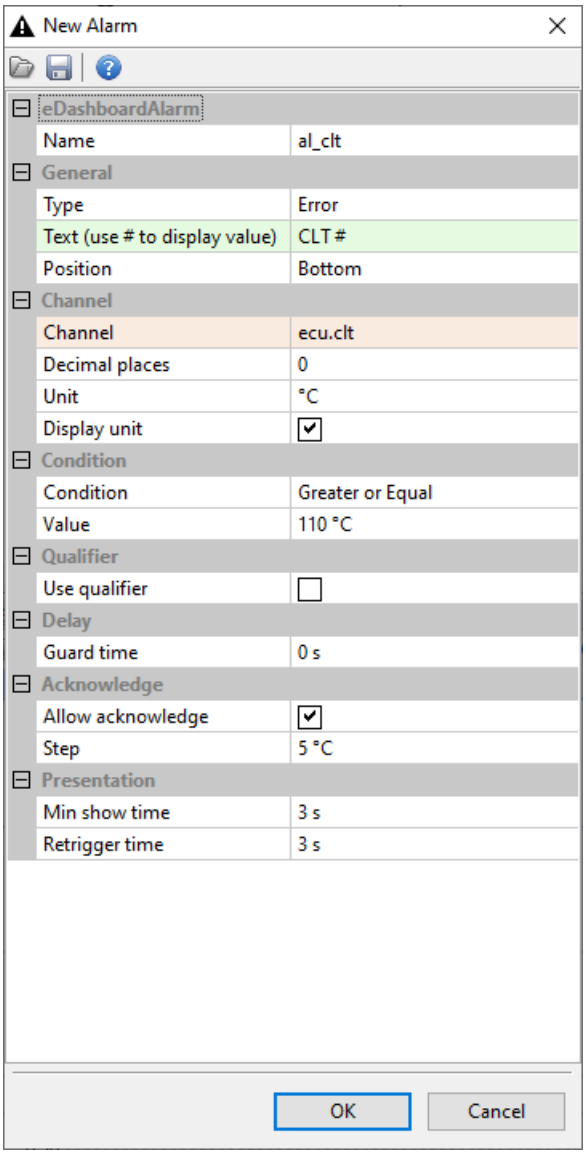
14. Alarms

The role of alarms is to display information about emergency states detected by the device.

A screen displaying an example alarm



1. **Basic alarm configuration**



Configuration consist in defining the ***Channel*** and the ***Condition / Value*** that will activate an alarm.You should also define the text (***Text*** parameter).

The alarm text can show the current channel value. To do this, you should put the special symbol # where the channel value with a unit is to appear.

If you need to display the # character alone, enter two # characters (##).

2. ***Qualifier***

   A precondition, the so-called ***Qualifier***, can additionally be entered into the alarm configuration. For example, when an alarm is to be checked only at a high speed, you can use the following precondition: ***ecu.rpm > 4000***.

3. ***Guard time***

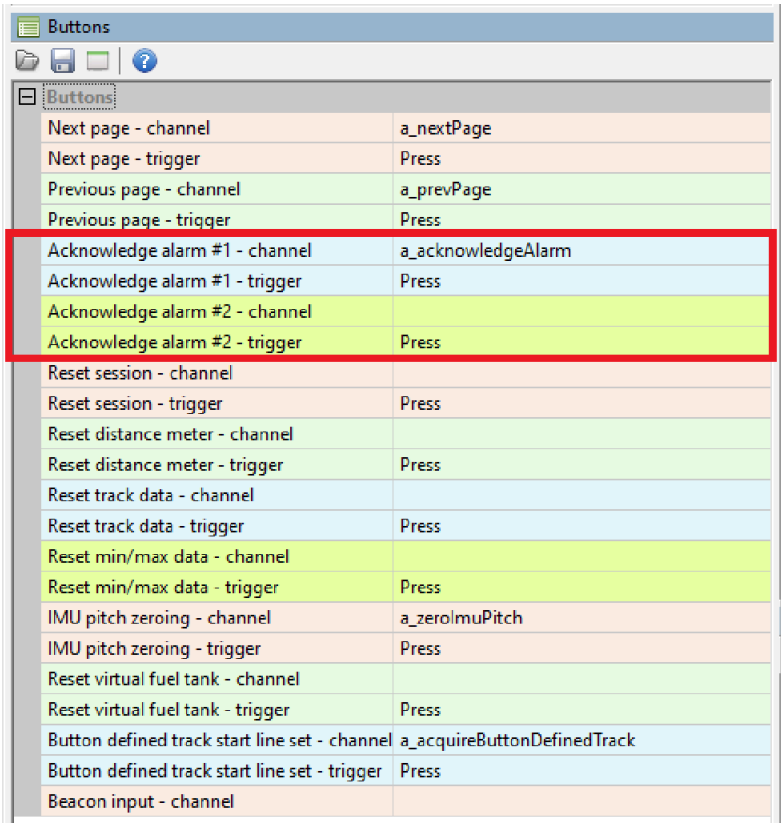
   Specifies the minimum time for which the condition must be met for the alarm to be displayed.

4. ***Presentation***

   To prevent an alarm from appearing and disappearing and thus causing confusion, it is worth defining two additional parameters:

   - ***Min show time*** - defines the minimum time during which an alarm will be visible. An alarm will be visible during this time even if the alarm condition is no longer met.
   - ***Retrigger time*** - defines the minimum time during which an alarm will not reappear after it disappears.

5. ***Acknowledge***

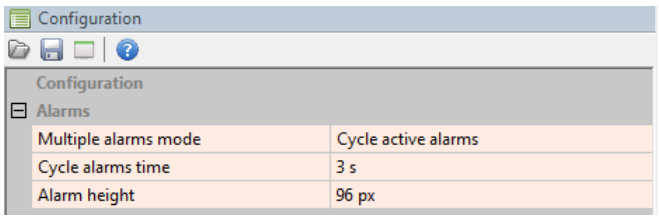


   After reading the alarm message, the driver can confirm it (the ***Allow acknowledge*** option). The alarm will disappear and the threshold value will move by a predefined ***Step***.

   In the ***Buttons*** section of the ***Configuration*** panel one can define one or two buttons for acknowledging an alarm.

6. **Global settings**

   It is possible for two alarms to occur simultaneously.



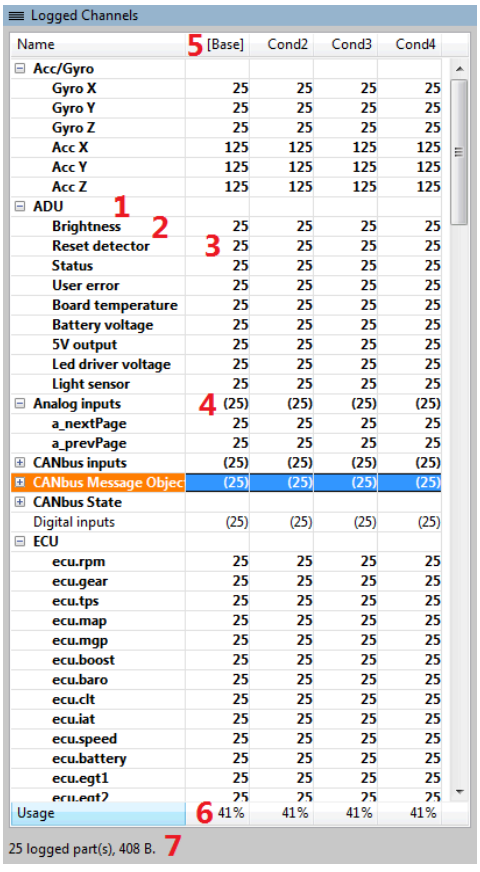
There are two ways to solve this problem (depending on the settings of the ***Multiple alarms mode***):

- ***Cycle active alarms*** causes the active alarms to be displayed consecutively, each for a specified period of time: ***Cycle alarms time***.
- ***Only one active with highest priority*** (***last in Project Tree***) displays only the alarm with the highest priority. The highest priority is understood to be in the ***Project Tree*** - alarm, which is at the end, has the highest priority.

In addition, there is a channel called ***adu.anyAlarmActive*** , which indicates whether any defined alarms are active. A list of active alarms can be displayed in a separate logging panel: ***Alarms / active*** , where you can also check their status.

15. Logging channels

The ***Logged Channels*** window defines the logging frequencies for particular channels. These values are expressed in Hz.



It is worth noting that the same frequencies are used for both logging to the USB storage device and for logging directly to the ADU Client programme on a PC.

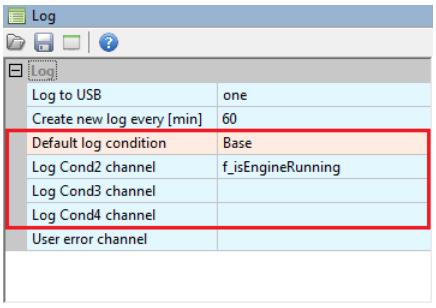
In the configuration window, we can distinguish the following elements:

- Groups (1) containing channels associated with a particular group
- Channels (2) containing data corresponding to their names
- Channel logging frequencies (3)
- Default logging frequencies for new elements (4) When a new element is created (e.g. Analog Input), one of these default frequencies will be assigned to its channel.
- ***Log condition*** (5), after which channels from a given column are logged.
- The bandwidth usage (6), expressed in [%] for particular Log conditions.
- The bandwidth usage expressed in bytes (7).

Configuration can be carried out in the context menu or by using the shortcut keys listed below. If a given command or key is used on an entire group, the frequency will change for all channels within it. However, if they are used in a single channel, they will change the frequency of that channel only. ***Log condition*** values may be changed individually or all at once depending on the column selected.

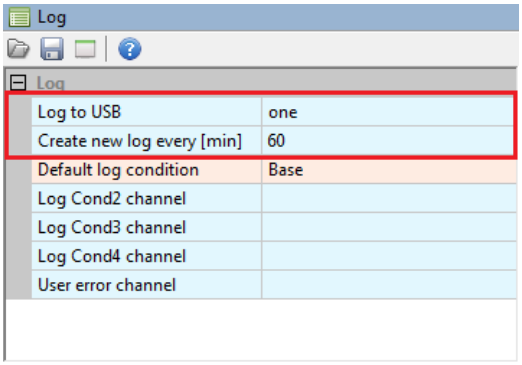
Key:	**Logging frequency:**
Alt+~	Deactivation of channel / group logging
Alt+1	1 Hz
Alt+2	5 Hz
Alt+3	25 Hz
Alt+4	50 Hz
Alt+5	125 Hz
Alt+6	250 Hz
Alt+7	500 Hz

A given log condition is selected 25 times per second. If the values of two condition channels are non-zero (e.g. the one selected in the ***Log Cond2 channel*** field and the one selected in the ***Log Cond3 channel*** field) the first one will be selected.

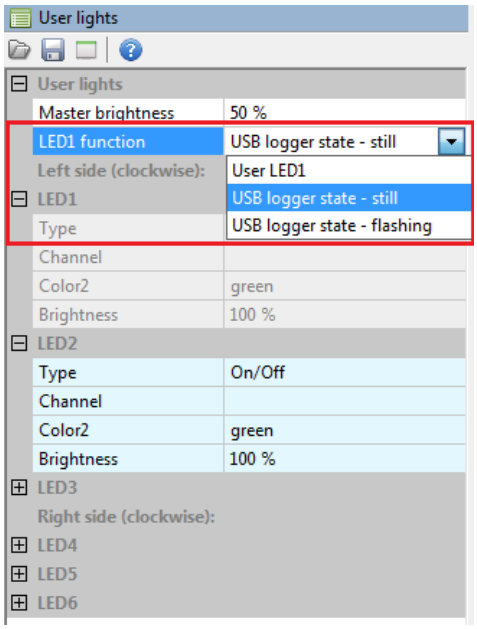


16. Logging to a USB storage device

Logging to a USB storage device is activated by default. Immediately after detecting a USB storage device connected to the device, ADU starts to save the log file in the .ADULOG format. It saves the ADU client logs in the same format. Files are created cyclically at defined time intervals (default 1 h, maximum 5 h). The shorter the intervals, the smaller the files and the easier their subsequent analysis. For a complete guide, see Appendix H - How-to Log Data to a USB drive *(on page 253)*.



The logging state can be shown using the LED1 diode.

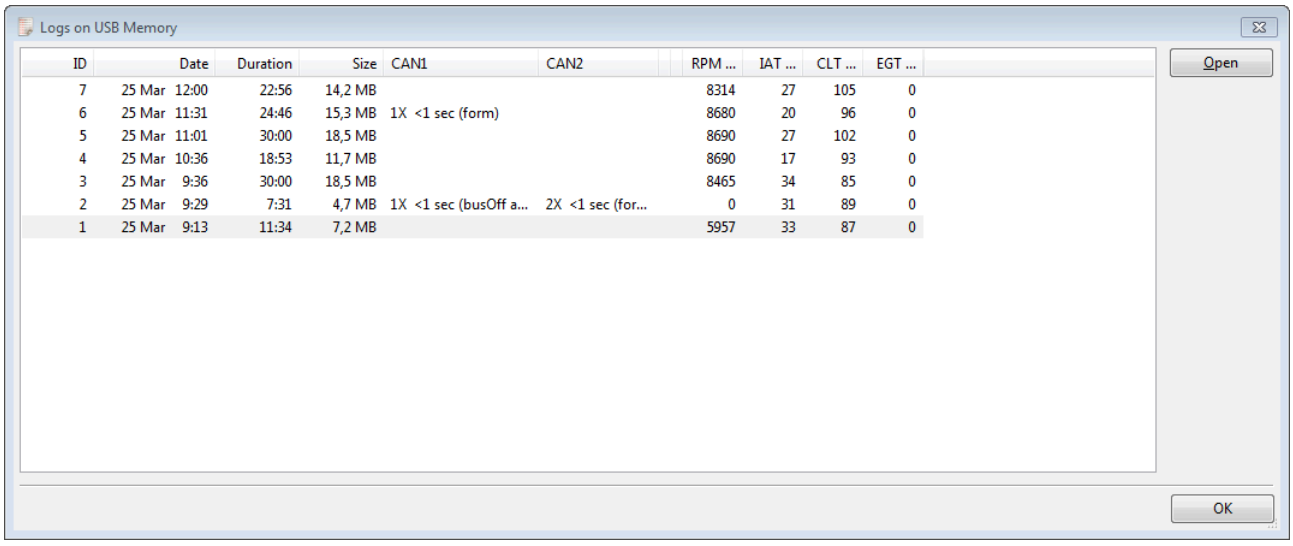


This is done using the ***LED1 function*** parameter, which has two options available:

- ***User LED1*** (the diode works just like the other ***User light LED***)
- ***USB logger state - still*** (the green diode stays on throughout the entire log saving time)
- ***USB logger state - flashing*** (the green diode flashes each time a log is saved - option useful for a diagnosis)

Color:	Description:
Blue	USB storage device is not connected to the ADU device.
Orange	The process of recognising the USB storage device stick and the process of loading the file system information is in progress. A state usually lasts a few seconds.
Green	Log file saving in progress.
Red	Error: the USB storage device is not compatible with the ADU device. The reason could be, for example, that the ADU does not support the exFAT file system on the USB stick.
Red and Green interchangeably.	USB storage device is full.
Orange and blue interchangeably	The USB stick is of poor quality - it saves logging data slower than the ADU generates it.

IMPORTANT: ***The USB logger state - flashing*** option can be used to diagnose problems with poor-quality USB storage devices. After starting recording a new log, the green diode will be lit continuously for a few seconds, but later, during operation, green should not be visible longer than 1 second. Otherwise, it means that the memory does not guarantee the correct write speed.
Once the data has been saved, the USB storage device should be removed from the port connected to the ADU and connected to a PC. Then, in the ADU Client, from the ***Devices*** menu, select the ***Receive logs*** command (Shift+F4 key) to view the logs stored in memory.

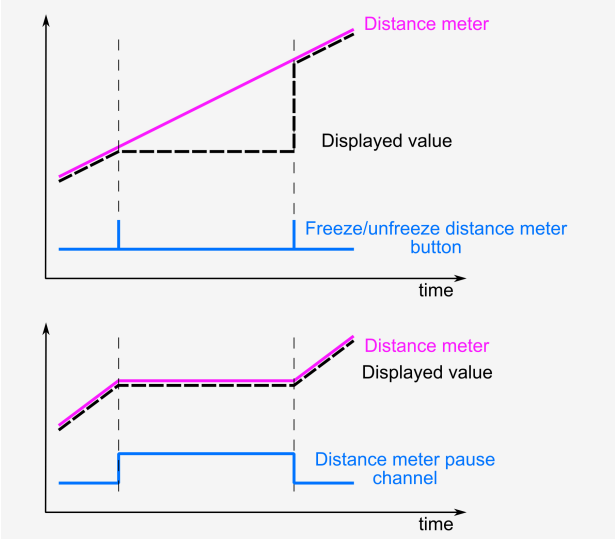


17. Permanent meters

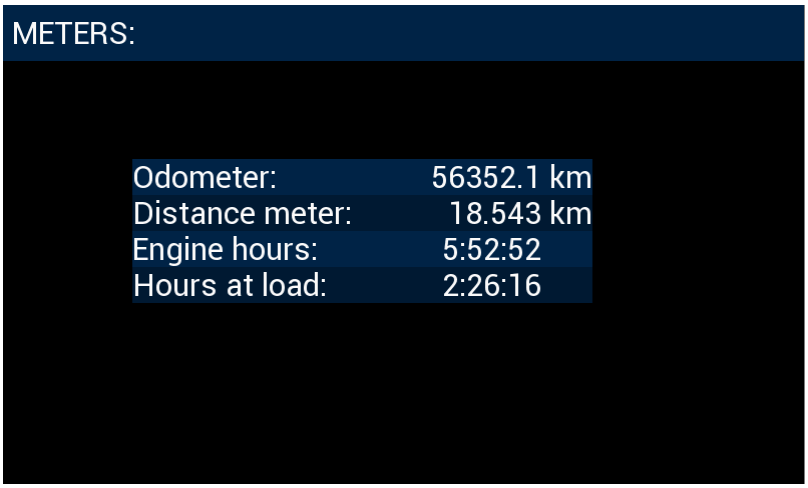
The ADU is equipped with 4 permanent meters, i.e. meters the value of which is kept in memory even after switching the device off.

Meter name	Channel	Description
Odometer	*adu.odometer*	An odometer, a travel distance meter
Distance meter	*adu.distanceMeter* *adu.distanceMeter2*	An additional, resettable odometer. Reset is performed via the button defined in the panel *Buttons > Reset distance meter - channel > trigger*. These channels can also be frozen/unfrozen (as described in the section *Buttons > Freeze/unfreeze distance meter*) or paused (as described in the section *Configuration > Distance meter pause channel*).[1]
Engine hour meter	*adu.engineHours*	It measures the number of hours the engine has run. (*ecu.rpm* > 0 channel)
Hours at load meter	*adu.hoursAtLoad*	It measures the number of hours the engine ran under a big load. The parameters for a big engine load can be defined in the *Configuration > Hours at load meter*: • *Minimal ecu.rpm* • *Minimal ecu.tps* • *Minimal ecu.map*

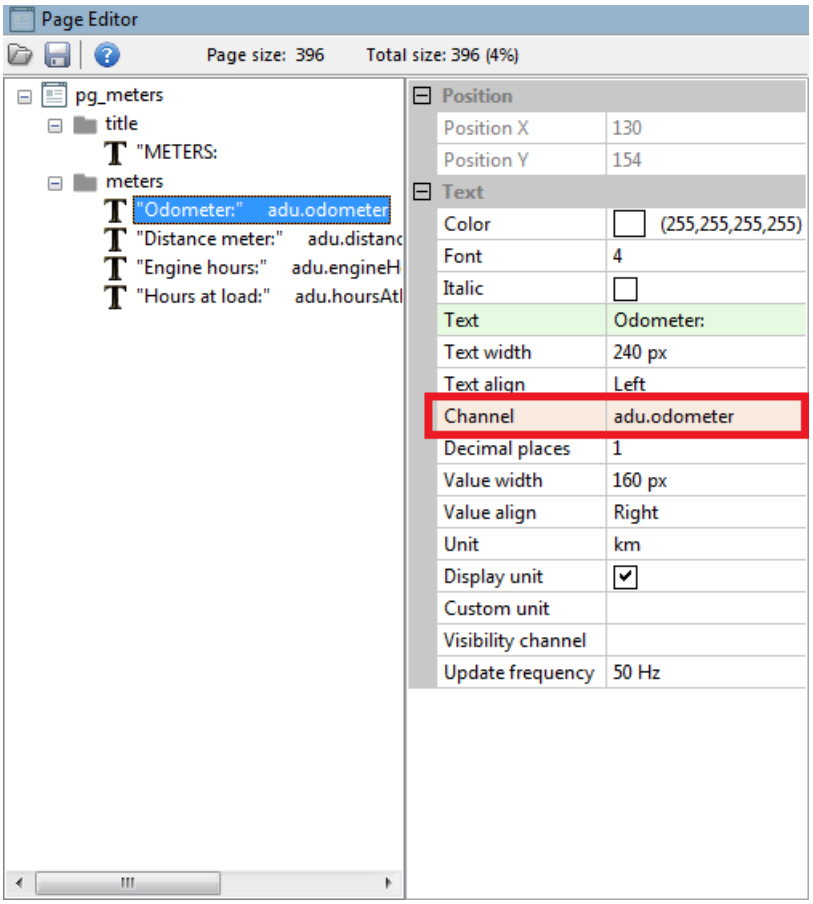
[1] Freeze vs. Pause – Visual Explanation:



A screen presenting meters with example values

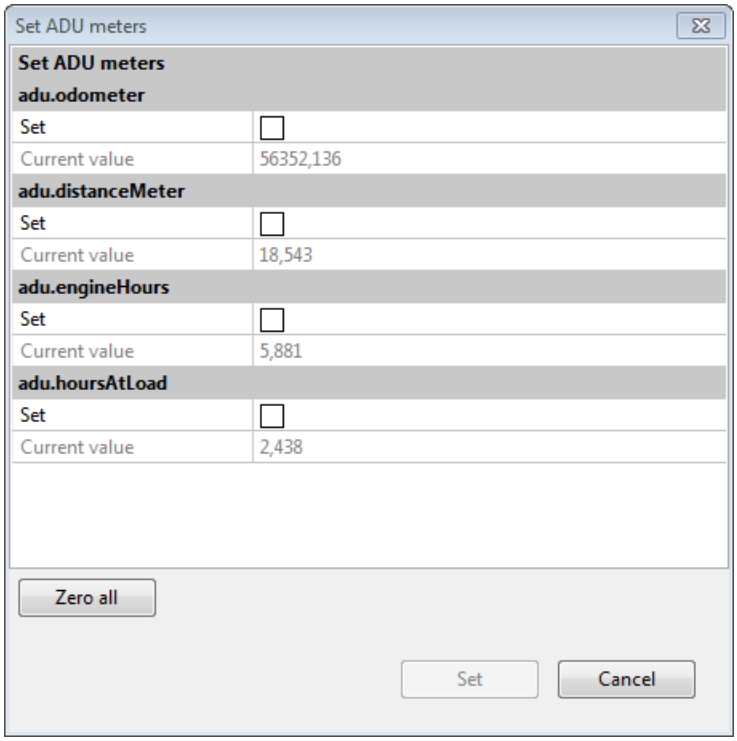


To add a meter to a page, insert the *Text* object and select a channel from the above table.



17.1. Resetting / changing meter status

Changes to the status of the meters can be carried out from the ADU Client. It is worth setting the actual vehicle mileage as the initial odometer mileage.



This can be done using the menu command ***Tools > Set meters...***

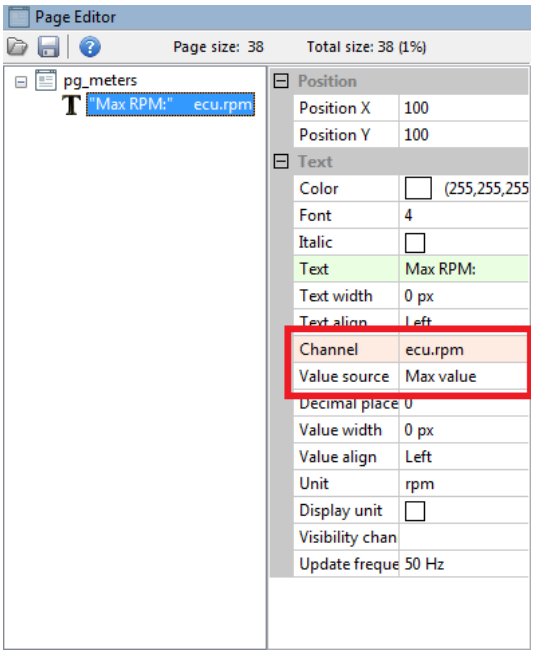
To set a given meter value, check the Set selection field for the meter and enter the new value.

To set all values at zero, you can use the *Zero all* button.

After entering the desired values accept them by using the *Set* button at the bottom of the window.

18. Min / max values for ECU channels

Each channel from the ECU group (e.g. *ecu.rpm*) records its minimum and maximum value. These values can be displayed on a page using the ***Text*** object.



Choose the correct channel from the ECU group and set the ***Value source*** parameter to ***Min value*** or ***Max value***.
The ***Value source*** parameter appears only for ECU group channels.
If a channel has not recorded any value yet (e.g. there is no CANbus Input element recording data in this channel), the ***Text*** object will display a "?" (question mark).

Just like for permanent meters, the min / max values of ECU channels are recorded in non-volatile memory of the device and remain stored even after the device has been switched off.
However, the process of registering the min / max values can be started again (by deleting the previous values). For this purpose, use the ***Reset min / max mode*** option in the ***Min / Max reset*** section of the ***Configuration*** menu. The option offers two options to choose from:

- ***Every firmware upgrade*** – resetting the min / max values with each replacement of the device software
- ***Every power off*** – resetting the min/max values each time the device is turned on

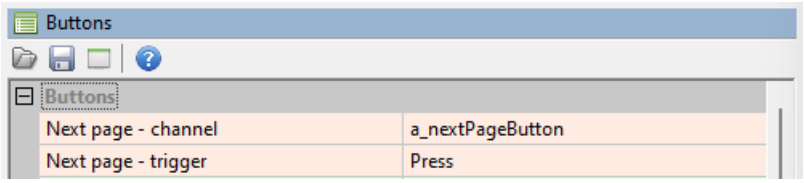
Independently of the above options, a manual delete button can also be defined and made available to the driver. The button can be defined in the ***Buttons > Reset min / max data channel > trigger***panel

19. Panels

Panels offer additional device configuration options not included in the main menu or the ***Project Tree***.

19.1. Buttons

The ***Buttons*** panel makes it possible to assign buttons to different internal functions of the device, such as toggling between the pages, cancelling alarms, cancelling times for a given track, etc.
A given function (e.g. opening the next page) is configured by selecting a channel / function representing a button and the button interpretation methods (trigger).



There are 4 channel interpretation types (triggers):
Press – function activated by pressing a button
Release – function activated by releasing a button
Click – function activated by quickly pressing and releasing a button
Hold – function activated by pressing and holding a button

Position	Description
Next page	Toggling to the next page
Previous page	Toggling to the previous page
Acknowledge alarm#1	Cancelling the alarm displayed on the screen
Acknowledge alarm#2	Cancelling the alarm displayed on the screen
Reset session	Deletes: internal clock ***Session time***, current lap time ***CurrentLapTime***, last lap time ***LastLapTime***, predicted lap time ***PredictiveLapTime***, the channel recording lap number ***adu.track.lap***, the channel recording the time difference between the best lap and the current lap ***adu.track.gainLoss***, the channel recording the current lap time ***adu.track.lapTime*** and the channel recording the distance travelled from the start line ***adu.track.lapDistance***

Position	Description
Reset distance meter	Resets the distance meter (the ***adu.distanceMeter*** channel)
Reset distance meter 2	Resets the distance meter (the ***adu.distanceMeter2*** channel)
Freeze / unfreeze distance meter	Temporarily stops updating the value of the adu.distanceMeter channel. When you freeze the meter, the channel shows the value from the moment the button was pressed, while the internal distance measurement continues. When unfreezing, the channel value updates to account for the distance covered while the display was frozen.
Freeze / unfreeze distance meter 2	Temporarily stops updating the value of the adu.distanceMeter2 channel. When you freeze the meter, the channel shows the value from the moment the button was pressed, while the internal distance measurement continues. When unfreezing, the channel value updates to account for the distance covered while the display was frozen.
Reset track data	Cancels the list of the best times and the reference lap for the current racing track
Reset min / max data	Deletes the min / max values for the ecu.* channels.
IMU pitch zeroing	Calibrates the overload sensor (accelerometer)
Reset virtual fuel tank	Resets the virtual fuel tank, a function used at the end of refuelling
Button defined track start line/finish line set	Determines the start / finish line when defining a new track
Button defined track delete	Deletes the *Button defined track*
Mode	Used for reading J1939 errors
Beacon input	Channel input from an external beacon. It informs the ADU about the completion of another lap on a track.

19.2. Shift light

Shift light uses the upper LEDs of the device to indicate the optimal gear change time. Three methods of LEDs control are available. The first of them is a parametric method where for each

gear you can define the moment when the first LED lights up. The second method involves a 3D table where for each LED and a given gear it is possible to input the engine speed at which the LED will light up. The third method consists in sending information about the shift light status from the ECU, which shall result in the diodes lighting up.

Parametric control

To activate the parametric shift light, select Parametric in the Control type field.

Parameter	Description
Brightness	LED brightness
Maximum RPM	The maximum RMP at which the upper LEDs begin to flash red
Neutral and reverse	The RPM at which the first LED for the reverse and neutral gear lights up
1 to 8	The RPM at which the first LED for gears 1 to 8 lights up
RPM channel	Channel with the current engine speed.
Gear channel	Channel with the current gear. If not defined, neutral gear will be selected by default.

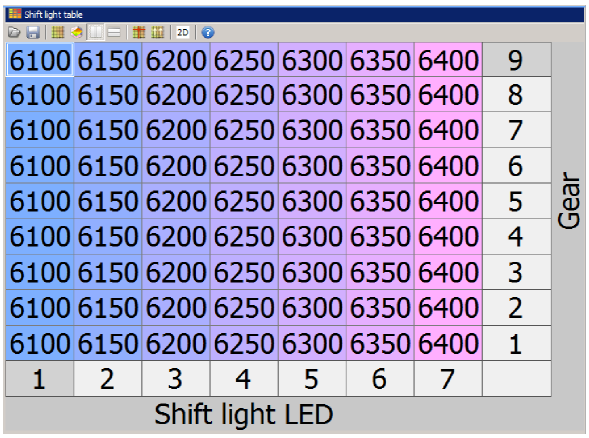
Below you can find the RMP at which the individual diodes will light up when the Maximum RPM setting is defined at 7400, and the initial RPM for a given gear is at 7000. Above 7400 RMP all diodes begin to flash red.

Control using a 3D table

To activate control using a 3D table, select *Table 3D* from the *Control type* field.

Parameter	Description
Brightness	LED brightness
RPM channel	Channel with the current engine speed.
Gear channel	Channel with the current gear. If not defined, the neutral gear will be selected by default.
Shift light color #n	Color assigned to a given diode
Flash when all leds on	Activation of the LED flash mode. When active, the LEDs flash instead of being on continuously.
Flash color	Selects flashing LED color. If you choose *Use default colors*, the color of the flashing diodes will correspond to the colors assigned in the *Shift light color* fields. Otherwise, the diodes will change color to the one selected in this field.

The following map lights up individual LEDs between 6100 and 6400 RMP with at 50 step.



First gear values re used for reverse and neutral gear.

Control using an external channel (CAN bus)

To activate control using a channel / variable, select ***CAN bus*** in the ***Control type*** field.

Parameter	Description
Brightness	LED brightness
Num states	Number of the shift light (4-6) states
Shift light channel	Channel with information about the number of diodes currently on

This control lights up respective diodes depending on the value of the variable assigned to the ***Shift light channel*** field.

Pit limiter indicator

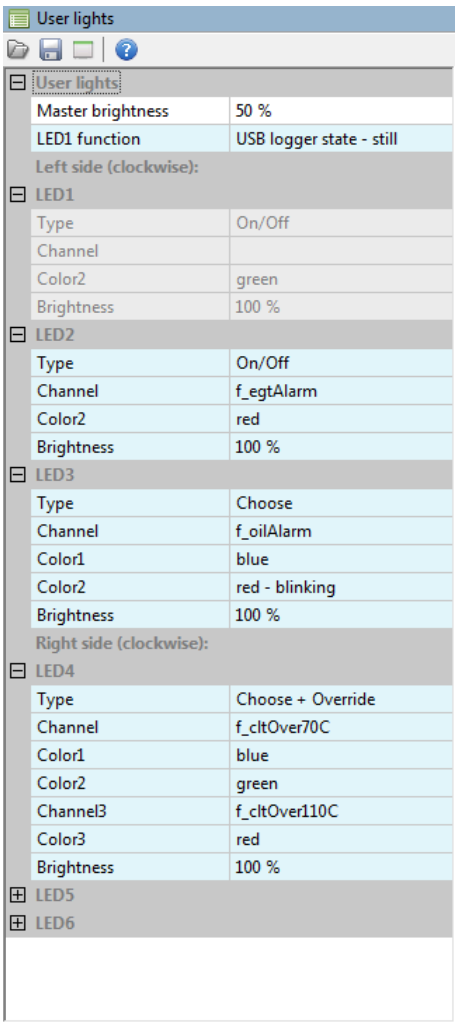
The following settings allow you to configure the behavior and appearance of LED indicators for the pit limiter. These options include channel activation, color customization for odd and even LEDs, and the cycle time for alternating between colors.

Parameter	Description
Activation channel	The pit limiter indicator remains on while this channel has a value of 1.
Odd LEDs colour 1	Primary color for odd-numbered LEDs (1, 3, 5, ...).
Even LEDs colour 1	Primary color for even-numbered LEDs (2, 4, 6, ...).
Odd LEDs colour 2	Alternate color for odd-numbered LEDs (1, 3, 5, ...).
Even LEDs colour 2	Alternate color for even-numbered LEDs (2, 4, 6, ...).
Cycle time	Duration for switching between the primary and alternate colors of the LEDs.

If the pit limiter is active, the LEDs change colour according to the parameters set.

19.3. User lights

The ***User lights*** panel allows you to control the color of the LEDs located on the left and the right hand side of the device. The LEDs are numbered clockwise LED1 to LED6.



The LEDs are used to inform about the condition of the vehicle or emergency situations. Colors and channels can be assigned for each LED that will affect them.

The LED1, depending on the selection of ***LED1 function***, can be assigned a user control method (***User LED1***) or used as a USB log state indicator (***USB logger state***). For more information, see *Logging on to a USB stick*.

The ***Master brightness*** parameter allows you to set the brightness jointly for all LEDs on a scale of 0-100%.

The LED#/***Brightness*** parameter sets the brightness of an individual LED.

We can select the control method for each LED:

Type	Parameters	Description
On / Off	***Channel*** ***Color2***	When ***Channel*** value is ***true*** (non-zero), the LED color is as defined by ***Color2***; otherwise it is off.
Choose	***Channel*** ***Color1*** ***Color2***	When ***Channel*** value is ***true*** (non-zero), the LED color is as defined by ***Color2***; otherwise it has the ***Color1***.
Choose + Override	***Channel*** ***Color1*** ***Color2*** ***Channel3*** ***Color3***	When ***Channel3*** value is ***true*** (non-zero), the LED color is as defined by ***Color3*** (irrespective of the value of the variable in the ***Channel*** field). Otherwise, the color is defined by the ***Channel*** value. When ***Channel*** value is ***true*** (non-zero), the LED color is as defined by ***Color2***; otherwise it is ***Color1***.

19.4. User tracks

The ADU device includes a built-in database of car racing tracks popular all over the world. This database continues growing with each new version of the software. The currently available tracks can be found in Appendix no.1. If a track is not available or if a track configuration differs from that assumed in the device, you can define a track of your own. You should enter the coordinates (latitude and longitude) of the centre of the start / finish line.
You can define up to 5 tracks of your own. If the track area coincides with a track area defined in the device, the one defined by the user will have priority.

Parameter	Description
Enable	Enables a given track.
Name	Track name
Latitude	Latitude of the centre of the start / finish line
Longitude	Longitude of the centre of the start / finish line
Track length	Track length in meters
Track width	Track width
Track radius	Radius of the circle defining the track area. The centre of the circle is defined as the centre of the start / finish line.

The following drawings are a graphical representation of the track in Poznań.

19.5. User track defined by the button

It is possible to define an own race track using an external button connected to the ADU.

Buttons

Buttons

Next page - channel	
Next page - trigger	Press
Previous page - channel	
Previous page - trigger	Press
Acknowledge alarm #1 - channel	
Acknowledge alarm #1 - trigger	Press
Acknowledge alarm #2 - channel	
Acknowledge alarm #2 - trigger	Press
Reset session - channel	
Reset session - trigger	Hold
Reset distance meter - channel	
Reset distance meter - trigger	Press
Reset distance meter 2 - channel	
Reset distance meter 2 - trigger	Press
Freeze/unfreeze distance meter - channel	
Freeze/unfreeze distance meter - trigger	Press
Freeze/unfreeze distance meter 2 - channel	
Freeze/unfreeze distance meter 2 - trigger	Press
Reset track data - channel	
Reset track data - trigger	Click
Reset min/max data - channel	
Reset min/max data - trigger	Release
IMU pitch zeroing - channel	0
IMU pitch zeroing - trigger	Hold
Reset virtual fuel tank - channel	0
Reset virtual fuel tank - trigger	Press
Button defined track start/finish line set - channel	
Button defined track start/finish line set - trigger	Press
Button defined track delete - channel	
Button defined track delete - trigger	Press
Mode - channel	
Mode - trigger	Press
Beacon input - channel	0

The button should be defined in the Buttons panel. When the driver presses the button for the first time, the GPS coordinates of the vehicle are downloaded. For a track to be defined, the driver must complete a full lap. If the driver presses the button a second time during a lap, the track definition process will be interrupted.

Information about the process of defining a track can be displayed on a specially prepared Overlay page. To load its template, create a new page, press the Open file icon in the window defining the page layout and load the ov_buttonDefinedTrackOverlay.aduepg file.

When the button is first pressed, the following page will be displayed.

It will disappear automatically at the end of the definition process - when the driver has completed a full lap. The track defined in this way will be given the name ***Button defined track*** and can be displayed using the $(TRACK) command in the *Text* control.
The overlay (shown above) uses the *adu.track.userButtonDefinedTrackAcquiring* channel as a display condition (the channel assumes value 1 when defining a new track).
The *Button defined track delete* button deletes the *Button defined track*.

Note:
In firmware versions prior to **105.0**, a double press of the *Button defined track start/finish line set* button also deleted the *Button defined track*. After upgrading from **FW < 105.0**, the settings from *Button defined track start/finish line set* will be copied to *Button defined track delete* to maintain previous functionality.

The ADU supports 3 types of track definition. The first type is the tracks defined in the device's internal software (built-in), the second type is user-defined tracks in the ADU software (5 tracks in the User tracks panel) and the third type is button-defined tracks. The priority of these tracks is as follows:

1. Button-defined tracks
2. User tracks (up to 5 tracks in the *User tracks* panel)
3. Embedded tracks (built-in)

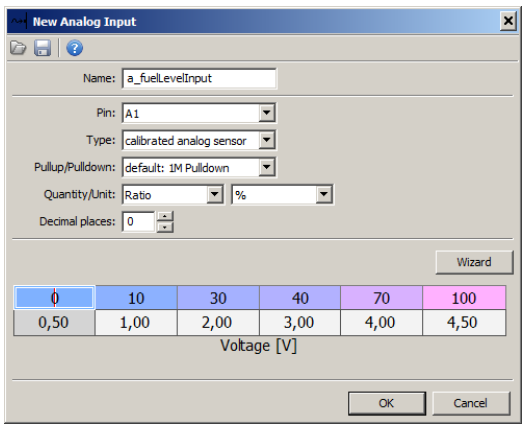
19.6. Fuel level filter

The ***Fuel level filter*** function is used for filtering the signal from the tank fuel level sensor.
The result of operation of this function is saved in the ***ecu.fuelLevel***. channel.
In ***two fuel level sensor mode (average)*** mode the average of both sensors is stored on the ***ecu.fuelLevel1*** channel and the ***ecu.fuelLevel2*** channel assumes the 0.0 value. In ***two fuel level sensors (separate level)*** mode the signal from each sensor is recorded in a separate channel - the first one in ***ecu.fuelLevel1*** and the second one in ***ecu.fuelLevel2***.

Parameter	Description
Fuel level mode	Fuel level mode: ***one fuel level sensor*** ***two fuel level sensor (average)*** - average of two fuel level sensors stored in the channel ecu.fuelLevel1 ***two fuel level sensors (separate level)***

Parameter	Description
Fuel level 1 – source channel	Channel / variable that determines the amount of fuel in the tank based on the signal from the first sensor (for ***one fuel level sensor*** mode) or from the average of two sensors (for ***two fuel level sensor (average) mode)***. A 2D table with calibration is most often used to indicate the fuel level.
Fuel level 2 – source channel	Channel / variable indicating the amount of fuel in the tank on the basis of the signal from the second sensor (for ***two fuel level sensor mode (separate level)***)
Maximum change step	The maximum allowed fuel level change during one second.
Filter time	The time when samples are to be collected and averaged. The longer the time, the more robust the filtering.
Disable filter when no RPM	Switching off the fuel level filter when the engine is not revving (useful for refuelling); window checked by default
RPM channel	Engine speed information channel

In a typical car the fuel level sensor is a potentiometer that requires an additional pull-up resistor (these are usually small values ranging between 100–200 ohm). Connect the signal to an analog input and select Calibrated analog sensor. In this way, a calibration of the fuel quantity for a given voltage can be carried out in the table.



If the size of the table needs to be changed, right-click the table cells and select Modify Bins / Insert cell (to add a cell) or Modify Bins / Delete cell (to delete it).

Then enter the name of the (***a_fuelLevelInput***) channel in the Source channel field. After filtering the value is available in the ecu.fuelLevel channel.

19.7. OBD 2

The OBD 2 panel is used for configuring the communication using the OBD 2 communication protocol. The read channels are divided into two groups. Fast channels (such as RPM, TPS) are marked green, while slow channels (such as CLT, Fuel Level) are marked pink. The more channels that are selected from a given group, the lower their log-in frequency will be. The read channels are mapped to ecu.* channels. It should be pointed out that some engine management do not allows all of the specified channels to be read.

Parameter	Description
Enable	Activates support for the OBD 2 protocol on CAN bus 2.
Mode	***Off*** - the device is inactive and not engaged with the vehicle's systems. ***Passive*** - the device reads data only when another active device requests it, preventing communication disruptions when multiple devices are connected. Ideal for monitoring without initiating data queries. ***Active*** - the device actively requests real-time data from the vehicle's systems, providing continuous and direct access to selected parameters. Only one device should operate in this mode at a time to maintain efficient and reliable communication.
RPM	Engine speed readout. Parameter shown in the log as *ecu.rpm*
Intake manifold absolute pressure	The absolute pressure in the intake manifold. Parameter shown in the log as ***ecu.map***
Vehicle speed	Vehicle speed. Parameter shown in the log as ***ecu.speed***
Throttle position	Throttle position. Parameter shown in the log as ***ecu.tps***
Timing advance	Ignition timing angle. Parameter shown in the log as ***ecu.ignAngle***
Fuel pressure	Fuel pressure. Parameter shown in the log as ***ecu.fuelPress***
Engine coolant temperature	Engine coolant temperature. Parameter shown in the log as ***ecu.clt***
Intake air temperature	Temperature in the intake manifold. Parameter shown in the log as ***ecu.iat***
Fuel level	Fuel level. Parameter shown in the log as ***ecu.fuelLevel***
Barometric pressure	Barometric pressure. Parameter shown in the log as ***ecu.baro***

Parameter	Description
Ambient air temperature	Temperature outside the vehicle. Parameter shown in the log as ***ecu.ambientAirTemp***
Ethanol fuel %	Content of ethanol in fuel. Parameter shown in the log as ***ecu.ethanolContent***
Engine oil temperature	Engine oil temperature. Parameter shown in the log as ***ecu.oilTemp***
Update rate	A parameter determining the frequency of sending inquiries to the engine control unit. The higher the frequency of sending requests, the higher the frequency of logging can be obtained. Unfortunately, not all control units are capable of responding with a frequency of 100 Hz and may require a lower frequency. The lowest possible frequency is 1 Hz.

19.8. J1939

The ADU supports J1939 Diagnostic Message 1 (DM1), which is used to report active Diagnostic Trouble Codes (DTC) for vehicle systems. This functionality enables reading active DTCs and their associated warning lights as defined in the J1939 standard.

To enable this basic functionality, select the correct CANbus (CAN 1 or CAN 2) and activate the ***Active diagnostic trouble codes (DM1)*** option. Any additional custom configurations enhance the functionality but are not required for basic operation.

The J1939 standard defines four diagnostic warning lamps:

- **MIL** – Malfunction Indicator Lamp
- **RSL** – Red Stop Lamp
- **AWL** – Amber Warning Lamp
- **PL** – Protect Lamp

J1939 DM1 Diagnostic Channels

The following channels are used to read and interact with active diagnostic trouble codes:

Channel	Description
adu.j1939.dtc.activeCount	The number of active diagnostic trouble codes

Channel	Description
adu.j1939.dtc.current.fmi	Fault Mode Identifier for the currently selected fault
adu.j1939.dtc.current.spn	Suspect Parameter Number for the current fault
adu.j1939.dtc.current.index	Index of the currently selected fault
adu.j1939.dtc.current.occurrenceCount	The number of occurrences of the selected fault
adu.j1939.dtc.current.sourceAddress	The module address from which the current fault originated (e.g., 0x00 for ECU#1, 0x01 for ECU#2, etc.)
adu.j1939.dtc.lamps.MIL	Lamp status (0 = inactive, 1 = active)
adu.j1939.dtc.lamps.AWL	
adu.j1939.dtc.lamps.RSL	
adu.j1939.dtc.lamps.PL	

Configuration in J1939 Panel

The J1939 configuration panel includes the following key options:

Parameter	Description
General	
CANbus	Select the CAN bus (CAN 1 or CAN 2) for communication
DTC reader	
Active diagnostic trouble codes (DM1)	Activates the DM1 functionality
Activation timeout	Time delay before the J1939 DTC Reader starts functioning after the device is powered on
Vehicle preset	Predefined settings for selected vehicles (Polaris RZR Turbo R 2023)
DTC reader custom settings (optional, for custom behavior)	
Source address #1-7	Define the source addresses of the modules from which DM1 messages are received. These addresses can be extracted from the CAN frame ID (e.g., 0x18FECA00, where the least significant byte LSB is the source address). A value of 0xFF indicates no defined module. If a module reports multiple

Parameter	Description
	faults, DM1 messages may be received on transport layer frames (e.g., 0x1CECFF00, 0x1CEBFF00), but the source address remains in the LSB of the frame ID.
Lamp timeout behaviour #1-7	Defines custom behaviors for the selected warning light if no DM1 messages are received from a defined module within the specified time.[1] Available options for lamp behavior upon DM1 message timeout: **Turn on** – Turns on the warning light **Slow flash** – Causes the warning light to flash slowly
Custom warning lamp (CWL)	Activates an additional warning lamp, which can override any other warning lamp from any selected module.
Source address for CWL	Specifies the source address of the module for which the lamp is to be overridden
Overriden lamp	Defines which lamp (from PL, AWL, RSL, MIL) should be overridden by the CWL

[1]Note: Each module capable of reporting faults in the J1939 standard sends DM1 messages (frame FECA) at a frequency of once per second, even if no faults are reported. Therefore, the absence of such frames may indicate a module failure or a bus issue, which is why custom light behaviors can be defined for when messages stop arriving.

Displaying Active Faults

To display active faults, press the *Mode* button (temporary name until a dedicated menu is implemented in ADU). The *adu.j1939.dtc.activeCount* channel must be greater than 0 for the active faults to be displayed. When *adu.j1939.dtc.activeCount* > 0 and the *Mode* button is pressed, channels from the *adu.j1939.dtc.current* group will be populated with data about the selected fault.

The *adu.j1939.dtc.current.index* will update to reflect the index of the currently selected fault. When *adu.j1939.dtc.current.index* > 0, the *Next/Prev page* buttons will change functionality to navigate through the list of active faults.

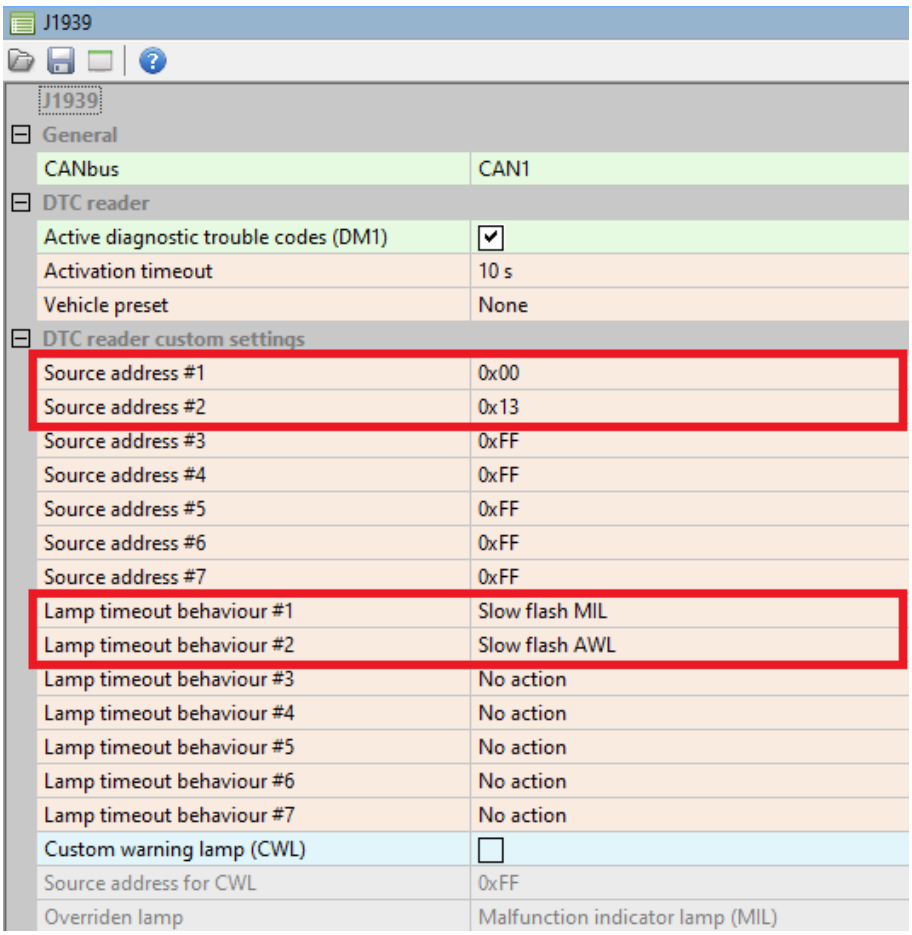
A second press of the *Mode* button will reset the index to 0, and the *Next/Prev page* buttons will return to their previous functionality.

If *adu.j1939.dtc.activeCount* is 0, pressing the *Mode* button will have no effect.

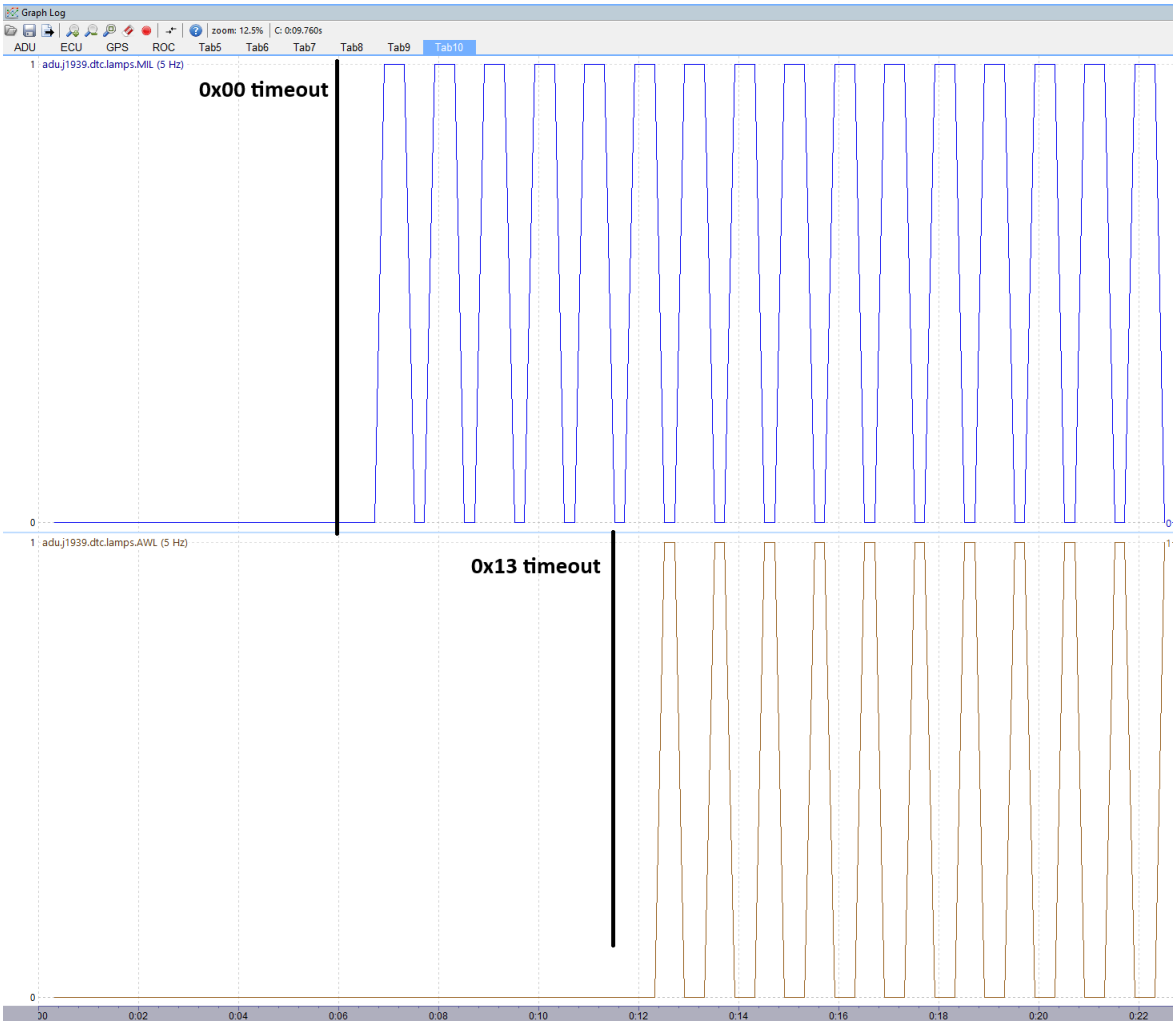
Example of fault display usage

When adu.j1939.dtc.current.index > 0, an overlay page or an overlay with a background can be activated to display active faults and navigate through them using the *Next/Prev page* buttons. Pressing *Mode* again will exit fault browsing.

Example: Lamp timeout behavior



If the MIL light starts flashing when a DM1 message from ECU#1 (source address 0x00) stops arriving, and the AWL light starts flashing when a DM1 message from the EPS module (source address 0x13) stops arriving, this behavior can be customized for each module as needed.



Example: Custom warning lamp (CWL)

Custom warning lamp (CWL)	☑
Source address for CWL	0xFF
Overriden lamp	Malfunction indicator lamp (MIL)

If the EPS module (source address 0x13) reports an error and requests the MIL light to turn on, instead of the MIL lamp, the CWL (Custom Warning Lamp) will be activated.

This allows you to quickly distinguish whether the fault was reported by Module X or Module Y since different modules can report faults by triggering the same light. This feature is particularly useful when you need to prioritize faults from specific modules. For example, if both the ECU and EPS modules report faults at the same time, both triggering the MIL light, the Custom Warning Lamp can be used to differentiate them. In this case, you can set the EPS module to trigger the CWL instead of the MIL, so you immediately know the issue is with the EPS module.

19.9. Outputs

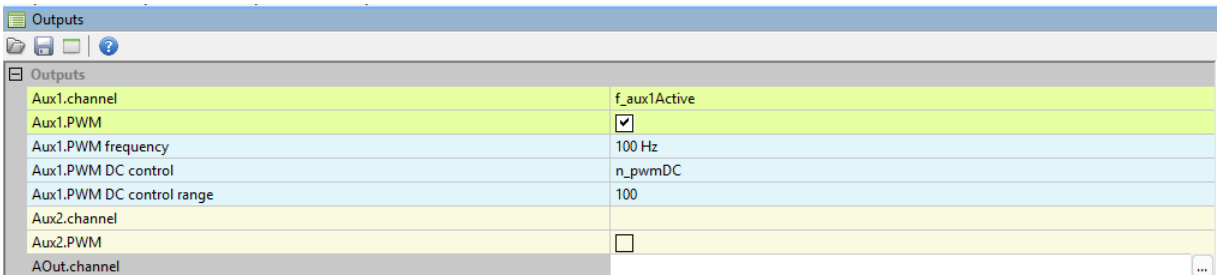
The Outputs panel is used for configuring the ADU outputs control (AUX1, AUX2, Analog out).

Parameter	Description
Aux#.channel	A channel / variable controlling the AUX # output. 0 means an inactive output, 1 means an output is switched to ground.
Aux#.PWM	When enabled, the PWM configuration becomes visible. Otherwise, the Aux# output state depends only on the value of the channel assigned in *Aux#.channel*.
Aux#.PWM frequency	The frequency of the generated PWM signal, adjustable from 10 Hz to 1000 Hz.
Aux#.PWM DC control	The DC control value is taken from any assigned channel or a fixed value.
Aux#.PWM control range	The range of the DC control (1–5000).
AOut.channel	A channel / variable controlling the AUX2 output. 0 corresponds to 0 V voltage, and 5000 corresponds to 5 V (mV scale).

The **duty cycle (DC)** is calculated using the following formula:

$$adu.outputs.aux\#.dc = \left(\frac{Aux\#.\ PWM\ DC\ control}{Aux\#.\ PWM\ DC\ control\ range} \right) * 100\%$$

Example Configuration for Aux1 Output



- The **Aux1 output** will be active when the **f_aux1Active** channel value is different from 0.
- A **PWM signal** will be generated with a frequency of **100 Hz**, and the duty cycle (DC) will depend on the value of the **n_pwmDC** channel.
- If **n_pwmDC** ≥ **Aux1.PWM control range**, then **DC = 100%**.

Important:
If the **PWM signal from Aux# outputs** is used with an **inductive load**, a **flyback diode** must be connected in parallel with the inductive load to protect the circuit.

19.10. Autobrightness

The *Autobrightness* table defines the brightness of the screen and the leds as a function of the external light intensity.
The method for calculating the brightness of the screen and the leds (all values are comprised within the range 0% and 100%) is shown below.
LCD brightness = Autobrightness
User led brightness = Autobrightness * User led master brightness * Led brightness
Shift light brightness = Autobrightness * Shift light master brightness

The automatic brightness is additionally controlled by a *Configuration* panel parameter named *Adaptation rate*. It determines the maximum permitted brightness changes per second.

19.11. Virtual fuel tank

The Virtual fuel tank function allows the user to calculate the amount of fuel remaining in the tank. Usually, this method of calculation is much more accurate than a calculation using a fuel level sensor. The function is based on the fuel consumption information from the ECU (ecu.usedFuel channel). It calculates not only the amount of fuel remaining in the tank, but also the number of laps the vehicle can run on that fuel (based on the fuel consumption information from the previous lap). This information allows adapting the race strategy to the amount of fuel remaining in the tank.

Parameters

Parameter	Description
Fuel tank size definition	***Fixed volume*** - tank is always filled to its maximum volume (defined by ***Fuel tank size (fixed)***) ***Defined by channel*** - volume of fuel in the tank after the refill is defined with a channel (***Fuel tank size channel [L]***)
Fuel tank size (fixed)	Capacity of fuel tank in litres. This parameter is active when the *Fuel tank size definition* is set to ***Fixed Volume***

Parameter	Description
Defined by channel	Channel defines total volume of fuel in the tank after refill (what was left + what was filled). This parameter is active when the *Fuel tank size definition* is set to ***Defined by channel***
Used fuel channel	The channel contains used fuel information sent by the ECU By default, this is the ecu.usedFuel channel.
Max used fuel change	Maximum permissible difference between two consecutive readings of fuel consumption information. If the change is greater than this parameter, it will be ignored. The parameter also enables the function to work correctly if the fuel consumption channel in the ECU is reset.
Fuel usage correction	Correction factor when there is a difference between the fuel consumption information provided by the ECU and the actual fuel consumption. This parameter is selected empirically based on the measured actual fuel consumption.
Stage mode channel	The Virtual Fuel Tank can monitor fuel consumption and distance traveled simultaneously in two modes: stage and road. Each mode has two channels for distance and used fuel. The ***Stage mode channel*** determines which mode is active. If the value of the channel assigned in this field is 0 - used fuel and covered distance is assigned to the road channel. If the value of the channel assigned in this field is 1 - used fuel and covered distance is assigned to the stage channel.

In addition, in the Buttons panel a ***Reset virtual fuel tank*** button has been defined. This button is crucial to the whole method, as the driver has to reset the function during refuelling. The value of the fuel remaining in the tank is then set to the value of the tank capacity, and all data is counted anew. All parameters are stored in the Flash memory of the device, so that they will also be remembered by the ADU when the ignition is switched off. It should be emphasised that the correct connection of the power supply to the device is required for this function to work correctly, i.e. a separate connection of the *battery* and a separate *ignition switch*. In this way, when the ignition is switched off, the parameters are saved in the Flash memory.

Logging channels

Channel	Description
adu.vft.fuelTankSize	Currently used fuel tank size
adu.vft.remainingFuel	Amount of fuel remaining in the tank
adu.vft.numLapsOnRemainingFuelAvg	Number of laps that can be completed using the fuel remaining in the tank, based on the average fuel consumption (Fuel used per lap avg)
adu.vft.numLapsOnRemainingFuelLast	Number of laps that can be completed using the fuel remaining in the tank, based on the fuel consumption of the last lap (Fuel used per last lap)
adu.vft.usedFuelPerLap	Average fuel consumption per lap (since resetting the fuel tank)
adu.vft.usedFuelPerLastLap	Fuel consumption on the last lap
adu.vft.usedFuelFromLastReset	Fuel used since last virtual fuel tank reset
adu.vft.isStageMode	Shows whether fuel is assigned to stage or road
adu.vft.stage.distanceFromLastReset **adu.vft.road.distanceFromLastReset**	Distance traveled in stage/road mode since the last refueling
adu.vft.stage.usedFuelFromLastReset ***adu.vft.road.usedFuelFromLastReset***	Fuel used in stage/road mode since the last refueling

For more details and an example configuration, see Appendix F - How-to Use Virtual Fuel Tank *(on page 239)*.

19.12. Configuration

The *Configuration* panel contains the configuration parameters of elements such as the startup screen, tire/ brake temperature cameras, as well as internal accelerometer calibration, screen rotation, etc.

Parameter	Description
Alarms	
Multiple alarm mode	Alarm system operation mode. If a number of alarms are activated simultaneously, the parameter determines their behaviour. ***Only one active with highest priority*** – only the highest-priority alarm will be displayed. Priority is determined based on the order of the alarms in the project. The lower the position in the list, the higher the priority of the alarm. ***Cycle active alarms*** – toggling between all active alarms will be activated.
Cycle alarm time	If the ***Cycle active alarms option is*** set in the ***Multiple alarm mode*** parameter, this parameter defines the length of time the alarm will be displayed. (in seconds).
Alarm height	The height of the rectangle displayed as an alarm background
Screen	
Rotate screen	It allows to rotate the screen 180 degrees. Option used if the screen is installed turned 180 degrees (with the shift light leds at the bottom)
Brightness	
Adaptation rate	A parameter handling the automatic brightness of the screen and the LEDs determining the maximum allowed brightness change during one second
IMU	
Lat/Long G-Force source	Defines the source for longitudinal and lateral G-force measurements: ***Internal IMU*** – uses the built-in IMU ***GPS2CAN IMU*** – uses the IMU integrated into the GPStoCAN module

Parameter	Description
	Custom IMU – allows defining custom X and Y acceleration channels (e.g., High Rate IMU from GPStoCAN2) *Acceleration X/Y channel* – becomes visible when G-Force source is set to Custom IMU. The user must specify custom IMU channels for acceleration in the X and Y axes.
Pitch	The internal IMU pitch calibration
Speed source for ecu.speed	
Source	The speed source for the ecu.speed channel can be obtained from: CAN bus (***requires CANbus Input with override***); ***GPS***; digital Inputs (***Wheels***).
Wheel #1 [km/h]	When selecting the Wheels speed source, select the Digital Input channel in which the sensor is defined.
Wheel #2 [km/h]	For Wheels source channel for the second digital input with a defined sensor
Odometer / Distance meter	
Speed source	The source of vehicle speed information for the odometers can be obtained from the ***ecu.speed*** or ***gps.speed*** channels.
Distance meter pause channel	Specifies the channel that pauses the distance meter counting when the channel's value is different from 0.
Distance meter 2 pause channel	Specifies the channel that pauses the distance meter 2 counting when the channel's value is different from 0.
Hours at load meter	
Minimal ecu.rpm	Minimum rpm defining engine load
Minimal ecu.tps	Minimum throttle angle defining the engine load
Minimal ecu.map	Minimum intake manifold pressure defining the engine load
Tire temperature cameras	
Tire temp. range min	Minimum tire temperature displayed on the graph in blue
Tire temp. range max	Maximum tire temperature displayed on the graph in red
Brake temperature cameras	

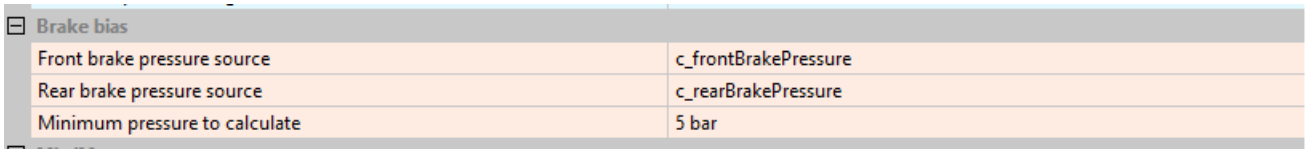
Parameter	Description
Brake temp. range min	Minimum brake disk temperature displayed on the graph in blue
Brake temp. range max	Maximum brake disk temperature displayed on the graph in red
Brake bias	
Front brake pressure source	Defines the channel containing front brake pressure data. The following pressure units can be used: MPa, kPa, Pa, bar, mbar, and psi. If an incorrect unit is used, the value will be interpreted as bar.
Rear brake pressure source	Defines the channel containing rear brake pressure data. The following pressure units can be used: MPa, kPa, Pa, bar, mbar, and psi. If an incorrect unit is used, the value will be interpreted as bar.
Minimum pressure to calculate	Minimum brake pressure required for brake bias calculation. If the pressure is below this value, the brake bias channel will be set to zero.
Min / Max reset	
Reset min / max mode	This parameter defines the behaviour of the min / max values for the ecu channels*. ***Every power off*** – the min / max values are deleted each time the device is activated. ***Every firmware upgrade*** – the min / max values are deleted when the internal device software is replaced.
Startup screen	
Enable	It activates the startup screen.
Texture	Texture that will be displayed on the startup screen (centred)
Scale	Scale of the displayed texture
Duration	Start screen display time
Color	Color of the displayed texture
Background color	Color of the background
Lap timing	
Lap timing mode	Lap time countdown:

Parameter	Description
	Circuit - for lap timing with a single start/finish line ***Bracket racing*** - from the time set on the first lap down to zero ***Hill climb*** - for lap timing with separate start and finish points

Parameter	Description
Reset track data button function	Defines the reset function for track data: ***Reset all track data*** – deletes all track data, including best lap times for all tracks. ***Reset current track data*** – deletes best lap times and other recorded data only for the currently active track.
Start type	Becomes visible when ***Lap timing mode*** is set to ***Hill Climb***. Determines the start method: ***Standing start*** – Requires the vehicle's speed to be below the *Start speed threshold* before starting. ***Flying start*** – No need to stop before starting.
Start speed threshold	When in ***Hill Climb*** mode with *Start type* set to ***Standing start***, this parameter defines the maximum speed at which the Start point can be recognized.
Changing pages	
Page selector channel	Pages can be changed directly from the CAN input or via a rotary switch connected to the analog input. ***Page selector*** takes control of the ***Next page*** and ***Previous page*** keys defined in the Buttons panel.
Page selector first value	Setting the first value of the page selector
Delayed turn off	
Turn off minimum time	Minimum time after turning off the ignition that must elapse for the ***Turn off channel*** to turn off the ADU
Turn off maximum time	Maximum time after turning off the ignition that the ADU will remain on (provided that the ***Turn off channel*** has not been triggered beforehand)
Turn off channel	ADU shutdown control channel. When the defined channel has a non-zero value between ***minimum time*** and ***maximum time***, the ADU will switch off.

19.12.1. Brake Bias

Brake bias refers to the distribution of braking force between the front and rear brakes. The ADU can automatically calculate the brake bias based on pressure data from the channels defined in the ***Configuration > Brake Bias Front / Rear brake pressure channels*** window.



The ***Minimum pressure to calculate*** parameter specifies the minimum pressure in the braking system from which the brake bias calculation will be performed. This pressure is determined as the lower value between the *Front brake pressure source* and *Rear brake pressure source*. If the pressure falls below the *Minimum pressure to calculate* value, the brake bias channel will be set to zero.

The *Front/Rear brake pressure* channels can use the following pressure units: MPa, kPa, Pa, bar, mbar, and psi. The system will automatically convert the values. If an incorrect unit is used, the channel value will be interpreted as bar.

The brake bias is calculated using the following formula:

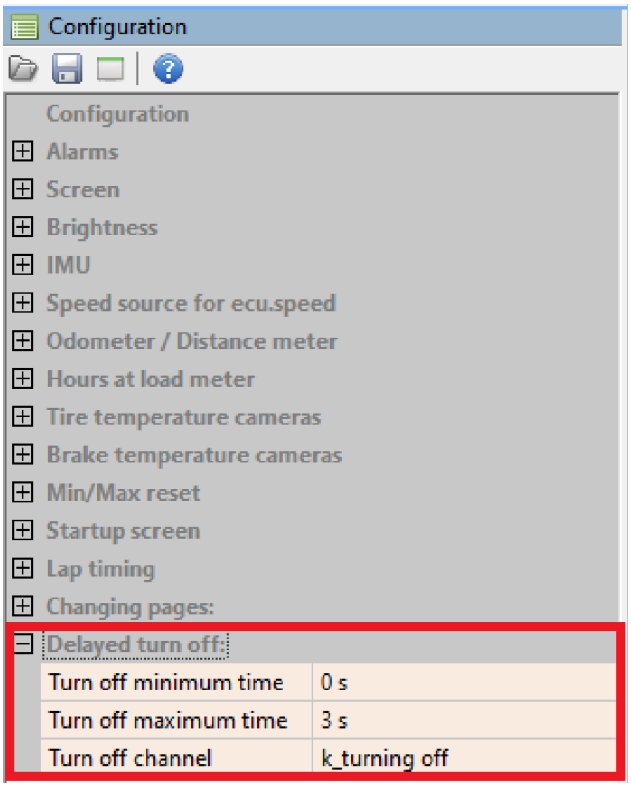
$$Brake\ Bias\ = \left(\frac{Front\ Brake\ Pressure}{Front\ Brake\ Pressure\ +\ Rear\ Brake\ Pressure} \right) * 100$$

19.12.2. Delayed turning off

By default, the ADU switches off immediately when the ignition is switched off. It is possible to configure an ADU switch-off delay. This requires separate battery (Battery 12V) and ignition switch (Switched 12V) wiring.

The ADU enters a ***Delayed turning off*** state when the ignition is switched off and remains in this state until it is completely switched off. When the device is operating in ***Delayed turning off*** mode, the value of the ***adu.isTurningOff*** channel is 1.

To configure the shutdown delay, select ***Desktop / Add new panel*** from the menu or use the **F9** key and select ***Configuration*** from the list.



Define the following parameters in the ***Configuration*** panel under ***Delayed turning off:***

- ***Turn off minimum time:*** the minimum time after the ignition is switched off that must elapse for the additional factor (***Turn off channel***) to take effect, causing the ADU to switch off.
- ***Turn off maximum time:*** maximum time after turning off the ignition for which the ADU will remain switched on (provided that the turn-off channel has not been triggered beforehand).
- ***Turn off channel:*** when the defined channel has a non-zero value between minimum time and maximum time, the ADU will turn off.

Turning the ignition back on while in ***Delayed turning off*** mode on the unit will restart the ADU 0.5 seconds after the ignition is switched on.

19.13. Protection

The Protection panel contains options related to the password protection of the device. A password must be provided before any changes can be made to the device if protected. If no password is provided the only way to remove the protection is by restoring the default settings.

Parameter	Description
Enable password protection	Activates password protection of the device.

Parameter	Description
Copyrights	Information displayed when attempting to connect to the device if protected
Contact info	Information including contact details (e-mail, telephone) displayed when attempting to connect to the device if protected

19.14. Log

The *Log* panel contains the logging configuration.

Parameter	Description
Log to USB	A channel / variable determining if the device is to log to a USB. The value *one* means that logging is to be active the whole time. Using this parameter allows to formulate the condition determining when logging is to be active / inactive.
Create new log every [min]	It determines the maximum size of a single file in minutes. When the size of the logged data exceeds the predefined time, a new file will be created.
Default log condition	The ADU device allows to create 4 logging profiles. These profiles are defined in *Menu > Logged channels.* They allow to record to a file different channels with different frequencies depending on the current condition. The *Default log condition* parameter defines the default profile that is to be used if none of the conditions for the profiles will be met.
Log Cond2 channel	A channel / function activating logging of the *Cond 2* profile. Log profile may change every 40ms (25 times per second).
Log Cond3 channel	A channel / function activating logging of the *Cond 3* profile
Log Cond4 channel	A channel / function activating logging of the *Cond 4* profile

19.15. CANbus / Serial Setup

The CANbus / Serial Setup panel is used for configuring the CAN bus and for RS232 serial communication.

Parameter	Description
General:	
CAN2 terminator	Activation of the terminator on the CAN2 bus.
CAN2 speed	The speed of the CAN2 bus
GPS:	
CANbus	Selection of the CAN bus to which the GPS module is connected.
Base ID	The base CAN ID for the GPS module is set to 0x400 by default. This value can be changed, but it must be a multiple of 8.
Static hold	GPS position freeze (when the channel with vehicle speed information, e.g. ecu.speed, has a value of zero)
Tire / Brake temperature cameras:	
CANbus	Selection of the CAN bus to which the thermal cameras measuring the tire / brake disks temperature are connected
Base ID	The base ID of the first thermal imaging camera (in the hexadecimal notation).
Serial:	
Serial protocol	Selection of a serial protocol: **Ecumaster serial protocol -** serial protocol supported by EMU Classic and EMU Black **Ecumaster Classic EDL protocol -** serial protocol sent by the Ecumaster EMU Classic to the Ecumaster EDL1 device **AIM serial** – a serial protocol consistent with the AIM protocol **Hondata 9600** - a serial protocol consistent with the Hondata PRO protocol (speed of 9600 bps). A signal inverter should be used between the Hondata and the ADU. **Hondata 115200** – a serial protocol consistent with the Hondata PRO protocol (speed of 115 kbps). A signal inverter should be used between the Hondata and the ADU. **Autronic SM 4** – a serial protocol consistent with the Autronic SM 4 protocol. This version refers to the SM 4 software version. **AEM –** a serial protocol compatible with AEM protocol **GEMS –** a serial protocol compatible with GEMS protocol **Athena GET –** a serial protocol compatible with Athena GET protocol

19.16. Autosaved channels

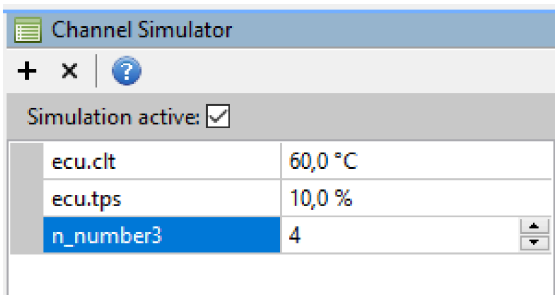
The Autosave Channels feature allows users to store values for up to 20 channels, which are loaded when the device starts. These values are saved automatically. Full description can be found in: Appendix G - How-to Configure Autosaved Channels *(on page 250)*.

19.17. Channel Simulator

Channel Simulator is used to simulate channel values in order to check the correct operation of established strategies or functions. Any existing channel or multiple channels can be simulated simultaneously.
Information flowing from the simulator has the highest priority, with the result that data flowing over the CAN bus is bypassed. The *Channel Simulator* works only when the device is powered on and connected.

To open ***Channel Simulator***, select ***Desktop > Add new panel*** from the menu bar or use the **F9** key and then select ***Channel Simulator*** from the list in the ***Select panel*** window.

Adding channels to the simulator is done by clicking the **+** icon in the top left corner of the window or using the ***Insert*** shortcut, and then in the ***Select Channel*** window selecting the appropriate channel from the list.
The channel value to be simulated is shown in the cell next to the channel name.
The simulator is active when the ***Simulation active*** box is checked.



To delete a selected channel, select it and then click the **x** icon in the top left corner of the window or use the ***Delete*** key.

19.18. Channel report

Channel report is a panel displaying the minimum, maximum and average values for the selected engine channels. The data is continuously updated.

Channel report

Channel	Min	Max	Avg
ecu.rpm	0	8551	4881
ecu.oilPress	0,00	9,00	3,13
ecu.oilTemp	0,0	0,0	0,0
ecu.clt	86,0	109,0	93,2
ecu.battery	9,96	14,28	13,99
ecu.lambda1	0,562	1,992	1,037
ecu.lambda2	0,000	0,000	0,000
ecu.egt1	0	0	0
ecu.egt2	0	21760	0
ecu.ignAngle	-4,5	37,0	20,1

20. Timing

The ADU may operate as a Lap timer, i.e. a device for measuring time on a racing track. The time can be measured using a GPS or a beacon.

Note:
The ADU only supports the **Ecumaster GPS to CAN** module and does not support other GPS modules or interfaces.

The beacon is placed near the start / finish line and then it sends an infra-red beam, which is registered by a receiver inside a car. The receiver “informs” the ADU when the start / finish line has been crossed.

The lap timing system supports both closed circuits (single start/finish line) and ***Hill Climb*** mode (separate start and finish points). Detailed information about Hill Climb mode can be found in: Appendix B - How-to Configure the Hill Climb *(on page 220)*. Circuit mode is described in the following topics.

In addition to measuring time, the GPS module allows to display the anticipated lap time (Predictive timing - *adu.track.predictiveLapTime*) and a change in the anticipated lap time (gain / loss) in the form of a graph. Once the data from the GPS module or other data, such as sensor values or engine parameters, have been logged onto an external USB storage device stick, the user is able to carry out further data analysis using Ecumaster Data Master software package: https://www.ecumaster.com/products/data-master/.

20.1. Configuration of time measurement with a beacon

Depending on the beacon system used, the output of the receiver should be connected to an analog or digital input of the ADU and an appropriate channel should be created.
The channel should be assigned in the ***Buttons*** panel in the ***Beacon input - channel*** field. When the channel value changes from 0 to 1, the last lap time counter (*last lap*) will assume the value of the lap time counter (*lap time*), which will be reset. If a current lap time is better than the previous best lap time (*best lap*), it will be saved as the best. Additionally, the ***adu.track.lap*** channel will be increased by 1. To display the times on the screen, use the Time object. To display the current lap number, use the Text object and enter the ***adu.track.lap*** channel in the ***Channel*** field. If a *beacon* system is used, the *predictive timing* function is not available.

20.2. Configuration of time measurement with a GPS

A GPS module offers more possibilities than a beacon system. Based on GPS coordinates, the ADU automatically detects the track the vehicle is currently on. A list of automatically detected tracks can be found in attachment 1. If a track is not listed, it can be added as an user track (see *Panels / User tracks* for more information).
The ADU is capable of storing information from 20 tracks in its memory. 8 best times are recorded, including their date, time and maximum speed. This data can be displayed using the *Track record table* object.
The best time from a given track is used as a reference time for the algorithm determining the predicted lap time. This algorithm assumes the current time of the run for a given place on the track and the time from the best run for the rest of the track.
By displaying the Predictive time graph, it is possible to check in real time if a given part of the track has been completed faster or slower than during the best lap.
To reset a given track data you can use the *Tools> Reset track* data menu or connect a button and assign it to the Reset track data channel function in the Buttons panel. It is recommended to use a *Hold* type trigger to prevent accidental data deletion.
In the *Configuration > Lap timing* window, you can select the action to be performed when pressing the ***Reset track data button***:

- **Reset all track data** – Default setting. Pressing the Reset track data button with this setting will delete all stored track data, including best lap times.
- **Reset current track data** – Deletes data only for the currently detected track. If no track is detected, pressing the button will have no effect.

To display the name of the track the car is currently on, use the ***Text*** object and enter $(TRACK) in the text field.

> **Important:**
> The GPS module should be installed using anti-vibration rubber pads included in the module kit.

21. Data analysis

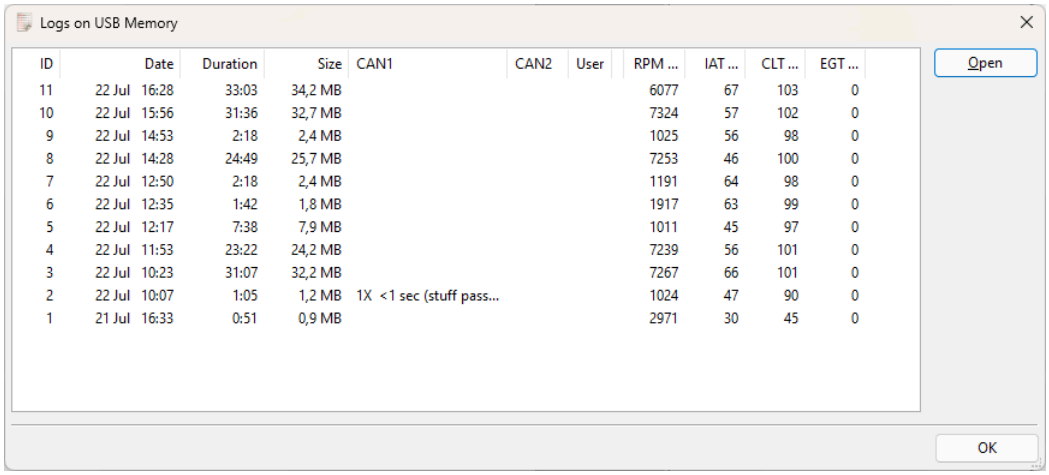
By saving racetrack data on a memory stick, users can later perform more advanced analysis using Windows software.

ADU software provides basic data analysis capabilities, allowing users to check the accuracy of custom functions, assess GPS data quality, and validate lap timing precision. This software equips users with key panels: ***Graph Log***, ***Scatter Plot***, ***Track Preview***, ***Section Times***, ***Histogram***, and ***Lap Time Plot***.

For more comprehensive analysis, ECUMaster offers a free, fully-capable data analysis tool called Data Master, which provides additional advanced features and functionalities.

The log is loaded via the ***Open log*** icon in the ***Graph Log*** or ***Scatter Plot*** panel.

Another way to read data from a memory stick is to use the ***log file*** option in the ***Devices / Receive*** menu or use the **SHIFT+F4** keyboard shortcut. A window with a selection of log files will appear.

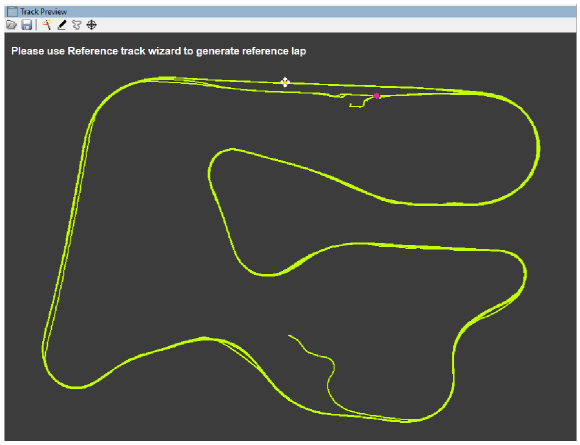


ID	Date	Duration	Size	CAN1	CAN2	User	RPM ...	IAT ...	CLT ...	EGT ...
11	22 Jul 16:28	33:03	34,2 MB				6077	67	103	0
10	22 Jul 15:56	31:36	32,7 MB				7324	57	102	0
9	22 Jul 14:53	2:18	2,4 MB				1025	56	98	0
8	22 Jul 14:28	24:49	25,7 MB				7253	46	100	0
7	22 Jul 12:50	2:18	2,4 MB				1191	64	98	0
6	22 Jul 12:35	1:42	1,8 MB				1917	63	99	0
5	22 Jul 12:17	7:38	7,9 MB				1011	45	97	0
4	22 Jul 11:53	23:22	24,2 MB				7239	56	101	0
3	22 Jul 10:23	31:07	32,2 MB				7267	66	101	0
2	22 Jul 10:07	1:05	1,2 MB	1X <1 sec (stuff pass...			1024	47	90	0
1	21 Jul 16:33	0:51	0,9 MB				2971	30	45	0

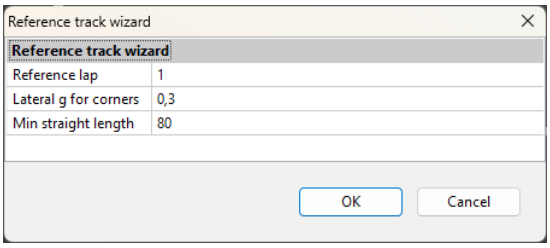
To upload the respective log, double click on it with the left button.

The next step is to go to the ***Track*** tab. The ***Track Preview*** window should display the entire track covered by the car and saved in the file. The figure shows the saved route from the example log. Depending on the quality of the signal, the raw GPS data points are shown in colors ranging from red (poor signal quality) to green (for the best signal quality). The channel indicating the quality of

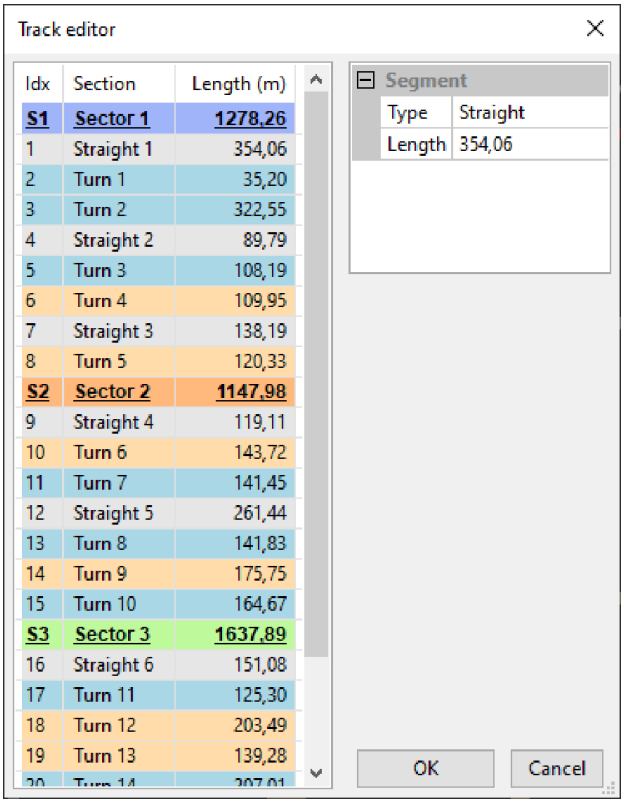
the signal is ***gps.status***, where status with a value of 0 means no signal, a value of 1 a very weak signal and successively up to values 4 and 5, which indicate the best signal quality. The start / finish line of the track is marked with a white circular marker.



In order to be able to analyse a run, a track outline must be generated from the collected data. To do this, press the magic wand icon (*create reference track*) in the tool bar of the ***Track Preview*** window. The ***Reference track wizard*** window should appear. From the ***Reference lap*** field select the lap to be used for creating your reference track. When switching laps, the track drawing changes in real time. The ***Lateral g for corners*** parameter defines the lateral g-force above which a given track part is interpreted as a turn. The ***Min straight length*** parameter defines the value of the minimum section of a track where the g-force was lower than Parameters ***Lateral g for corners*** for the distance to be considered a straight section.



After pressing OK, a reference track divided into segments (left turn, right turn, straight line) and their constituent sectors will be generated. By default the track is divided into three sectors. The start / finish line is marked as a chequered flag with an arrow indicating the direction of travel on the track. This line is also a fixed boundary of the first sector –, i.e. it cannot be removed or moved. Other sector boundaries can be created at any given point. These boundaries are marked with cross lines and described by the number of the sector they begin, e.g. S2.
To edit segments and their constituent track sectors, select the pencil icon in the toolbar (*Edit reference track*). The ***Track editor*** window will appear.



In this window, segments and sectors can be deleted or divided, and the type and length of segments can be changed. When a segment or sector is selected, it is automatically highlighted on the preview screen. To edit the division of the reference track, click on the corresponding segment with the right mouse button.

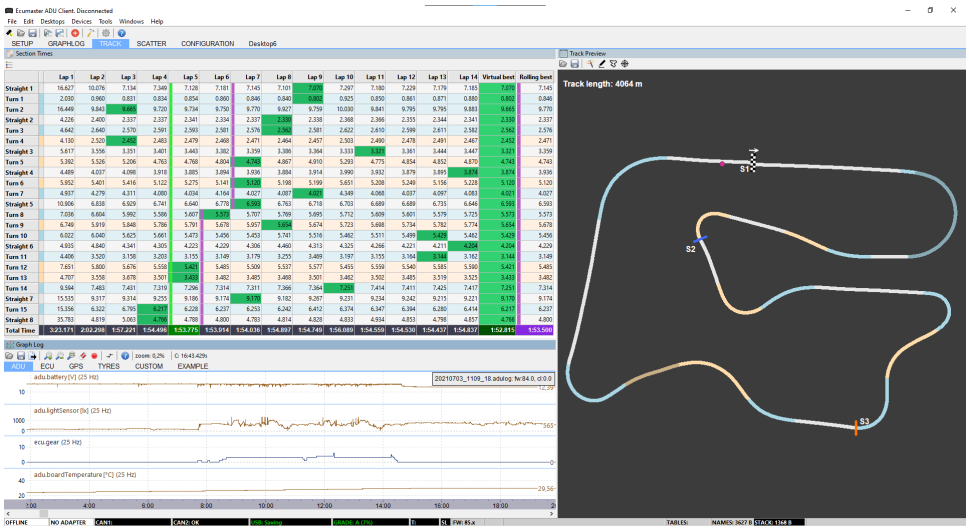
A window will appear with the following options:

Option	Description
Remove segment	Deletes the selected segment by attaching it to the segment immediately following it.
Split segment	Divides the marked segment in the middle of its length.
Split sector	Adds a sector border at the end of the selected segment.
Remove sector	Deletes a sector in which the selected segment is located by adding it to the next sector. The exception is the last sector (whose end is the start / finish line)– deleting the last sector adds it to the previous sector.
Move current sector end	Moves the end border of the sector in which the selected segment is located to the end of the selected segment. If the end of the selected sector is the start / finish line, an additional sector border will be added at the end of the selected segment (additional sector will be created).

Option	Description
Move current sector start	Moves the initial boundary of the sector in which the selected segment is located to the beginning of the selected segment. If the beginning of the selected sector is the start / finish line, an additional sector border will be added at the beginning of the selected segment (additional sector will be created).

Press OK after you finish editing. The created track can be saved by pressing the floppy disk icon. The next time the programme is started, the reference track that was last saved or read will be loaded.

Once the reference track has been created, the times in the sections should recalculate automatically. A sample application screen is presented below.



The purple point on the reference track determines the position of the vehicle for the position selected in the logging window. This way, by moving along the log window, you can preview the set of parameters for particular vehicle positions.

The segments into which a track is divided are colour-coded (white, light orange and light blue), depending on the type of segment (straight, left turn, right turn). To make the data easier to read, the same colors were used to mark the corresponding segments in the table in the *Section Times*. The ***Section Times*** panel displays times in defined segments for all laps. The best times are marked green. The green color of the individual cells shows the best times achieved for all runs in each section of the track, while the vertical green line indicates the best time for the entire lap. Additionally, this window comprises of two columns: ***Virtual best*** and ***Rolling best***. The ***Virtual best*** is the sum of the best times in individual sections, while ***Rolling best*** shows the best continuous time of a single lap. It is marked with a purple vertical line and, in the example

in question, was obtained in the 5th and 6th laps. *Rolling best* shows a realistic time that could have been achieved.

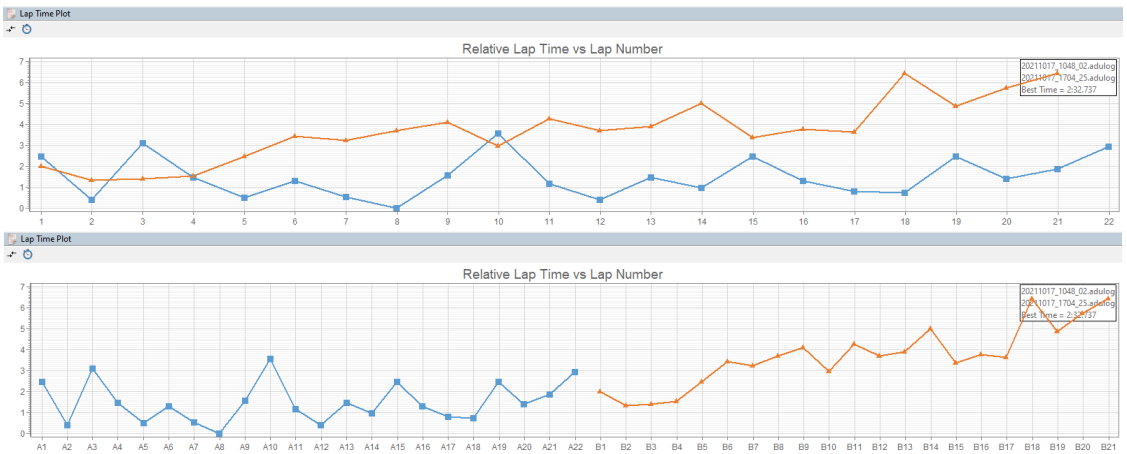
The ***Section Times*** window can also be used to display the run times of all laps divided into sectors. This form of data presentation can be clearer and easier to analyse. Use the ***Sectors / Segments*** icon at the top left of the *Section Times* panel to switch the view between tables.

You can use the graph of lap times displayed in the ***Lap Time Plot*** panel to analyse the runs. For correct operation of the ***Lap Time Plot*** visualisation it is necessary to generate a reference lap in the *Track Preview*.

In the graph, the X axis represents the lap numbers and the Y axis represents the lap times. In the top right corner, a box shows which logs are currently being analysed and what the best lap time was. Several logs can be analysed simultaneously on a single chart.

When analysing two logs simultaneously, by default the graphs from both logs are plotted side by side, so that one is a continuation of the other. Using the arrow icon *Compare / Append Mode* they can be overlaid on top of each other. By clicking again on the *Compare / Append Mode* icon, you return to the previous view. When the graphs of the two logs are lined up side by side, the laps are described on the X-axis by a letter denoting the log in question and the lap number, e.g. A1, A2 etc. for the first log and B1, B2 etc. for the second log.

In addition, for better clarity of the graph, the Y axis can be scaled using the icon with the stop watch symbol *Absolute / Relative Mode*. By default, the times achieved on the individual laps are marked on the Y axis. Clicking on the aforementioned icon will cause the Y-axis to show the time difference to the best lap, i.e. the best lap will be marked as zero on the axis and any other lap will show the loss to that best race.



The channels that may be helpful in analysing data:

Channel	Description
adu.track.lap	Displays the current lap.
adu.track.lapTime	Displays the current lap time.
adu.track.gainLoss	Displays the gain / loss relative to the best recorded lap.
adu.latG	Lateral overload acting on the vehicle (negative values for a right turn, positive values for a left turn)
adu.longG	Longitudinal overload acting on the vehicle (negative values when accelerating, positive values when braking)
gps.speed	Vehicle speed read from the GPS
gps.status	GPS module status If signal problems occur, the analysed data may be inaccurate.
ecu.rpm	Engine speed
ecu.tps	Throttle position.
ecu.clt	Engine coolant temperature
ecu.map	Pressure in the intake manifold
ecu.oilPress	Engine oil pressure
ecu.oilTemp	Engine oil temperature

Incidence of a given phenomenon can be analysed using the ***Histogram*** panel. To define a suitable channel, click on the pencil icon which opens the ***Define channels*** window.

Option	Description
Channel X	The channel defining the X axis
Filter channel	The filter channel introducing an additional condition on a separate channel, the fulfilment of which determines whether (at a particular point in time) data from channel X will be collected for analysis or rejected
Discard samples above	Rejects samples above the set value for the filter channel
above value	Upper limit value for the filter channel
Discard samples below	Rejects samples below the set value for the filter channel

Option	Description
below value	Lower limit value for the filter channel
Autoscale X axis and bins	Automatic scaling of the X axis range and the number of bins
Autoscale X axis	Automatic scaling of the X axis range
Min	Initial value of the X axis
Max	Final value of the X axis
Set bin width	Sets the bin width
Bin width	Width of a bin
Bin count	Number of bins
Exclude outliers	Rejects outliers (deviating from normal distribution)
Y axis unit	Y-axis unit defining the frequency of occurrence of the defined phenomenon ***Percent*** - percentage ***Time*** - time frame
Entered bins	Sets the description of the X axis values at the centre of the bins.

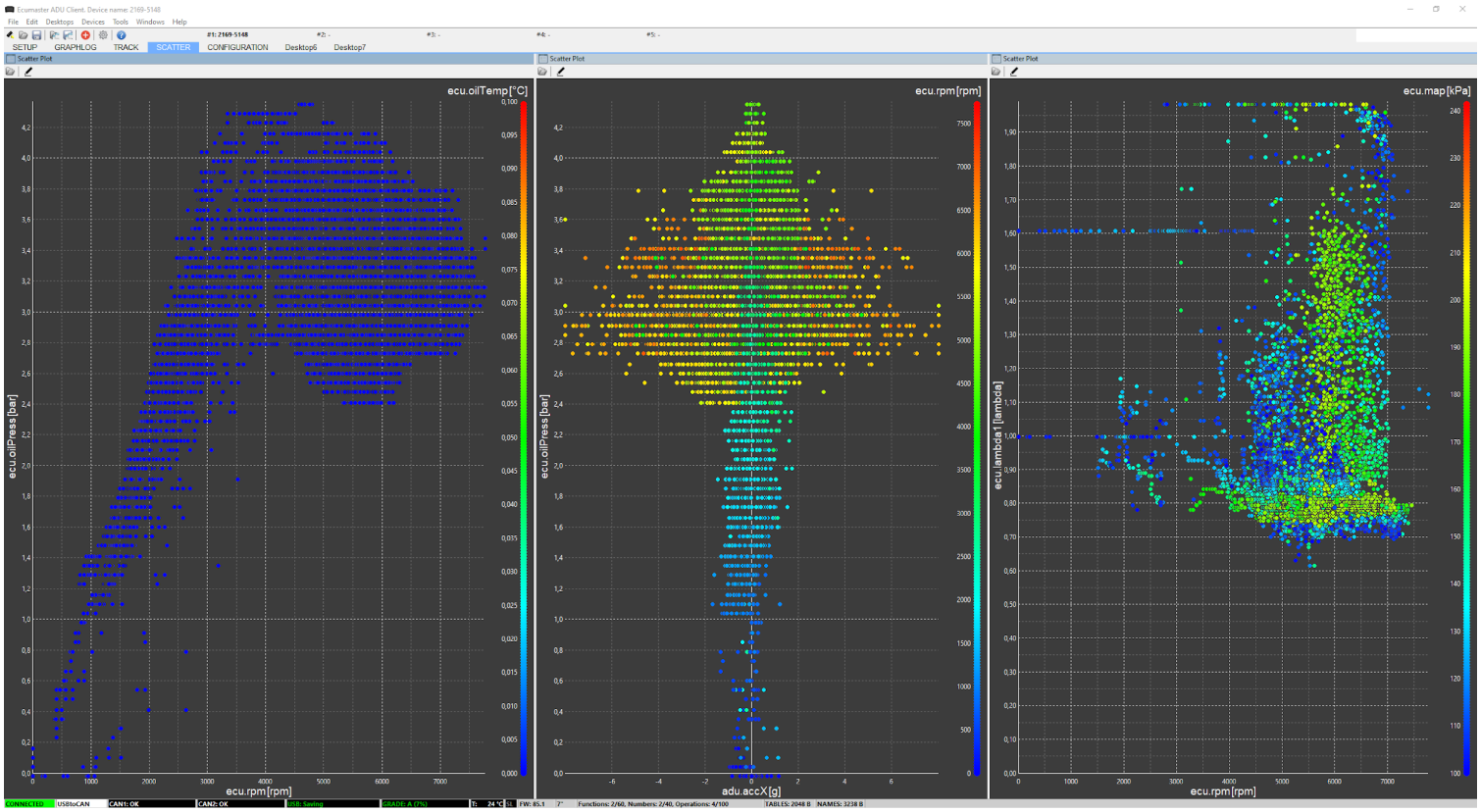
The ***Histogram*** panel allows data analysis from several logs at the same time. They can be analysed merged into one long log on one histogram (default setting) or separately on separate axes (one below the other). Use the arrow icon to switch between these views ***Compare / Append*** mode. In the split log analysis mode we can plot the logs onto a single histogram, where each log will be marked with a different color. The ***Tiled / Overlay*** icon is used for this purpose.

Another way of visualising data is A scattered plot (with three axes: X, Y and color axes). This chart can be configured independently, while several pre-defined configurations can be used once the ***Scatter Plot*** panel has been added:

- *Scatter Plot: oilPress vs rpm (color oil Temp)*
- *Scatter Plot: oilPress vs adu.latG (color rpm)*
- *Scatter Plot: oilPress vs gps.latG (color rpm)*
- *Scatter Plot: lambda vs rpm (color map)*

To define axes independently, click the pencil icon, which opens the ***Define channels*** window.

Option	Description
Channel X	Channel defining the X axis
Channel Y	Channel defining the Y axis
Color channel	Channel defining the color axis
Discard color samples above	Rejects samples above the set value for the Color channel
above value	Upper limit value for the Color channel
Discard color samples below	Rejects samples below the set value for the Color channel
below value	Lower limit value for the Color channel



22. Appendix A - List of built-in tracks

Tracks stored in the device's memory (automatically detected):

Australia

Mount Panorama Circuit (Bathurst), Winton Motor Raceway, Sydney Motorsport Park, Phillip Island Grand Prix Circuit, The Bend Motorsport Park

Belgium

Circuit de Spa-Francorchamps, Circuit Zolder

Croatia

Rijeka Grobnik

Czech Republic

Automotodrom Brno

France

Circuit de Nevers Magny-Cours

Germany

Eurospeedway Lausitzring, Hockenheim, Motorsport Arena Oschersleben, Nurburgring, Schleiz

Hungary

Pannonia Ring

Italy

ACI Vallelunga Circuit, Autodromo Nazionale Monza, Fanciacorta International, Imola, Misano World Circuit, Mugello Circuit, Racalmuto

Latvia

Bikernieki

Netherlands

Circuit Zandvoort, TT Circuit Assen

Poland

Tor Poznań, Tor Słomczyn

Portugal

Autódromo Internacional do Algarve

Scotland

Knockhill Racing Circuit

Slovakia

Slovakia ring

Spain

Circuit de Barcelona-Catalunya, Circuit de la Comunitat Valenciana Ricardo Tormo, Circuito de Albacete, Circuito de Navarra, Circuito Permanente de Jerez, Ciudad del Motor de Aragón

Sweden

Anderstorp Raceway, Falkenbergs Motorbana, Gelleråsen Arena, Gotland Ring, Kinnekulle Ring, Linköpings Motorstadion, Mantorp Park, Mittsverigebanan, Ring Knutstorp, Sturup Raceway, Tierp Arena

UK

Anglesey International, Brands Hatch GP, Cadwell Park, Donington Park, Oulton Park, Silverstone Natl., Snetterton Circuit, Thruxton Motorsport Centre.

USA

APEX Motor Club, Arizona Park, Barber Motorsport Park, Buttonwillow, Chuckwalla, Circuit of the Americas, Eagles Canyon Raceway, Hallett Motor Racing Circuit, High Plains, Laguna Seca Raceway, Michelin Raceway Road Atlanta, Miller Motorsports Park, Motorsport Ranch Cresson, MSR Houston, New Jersey Motorsport Park, Pittsburgh International Race Complex, Pueblo Park, Road America, Sears Point Raceway, Thermal Blue, Thunderhill, Virginia International Raceway, Willow Springs

Qatar

Lusail Circuit

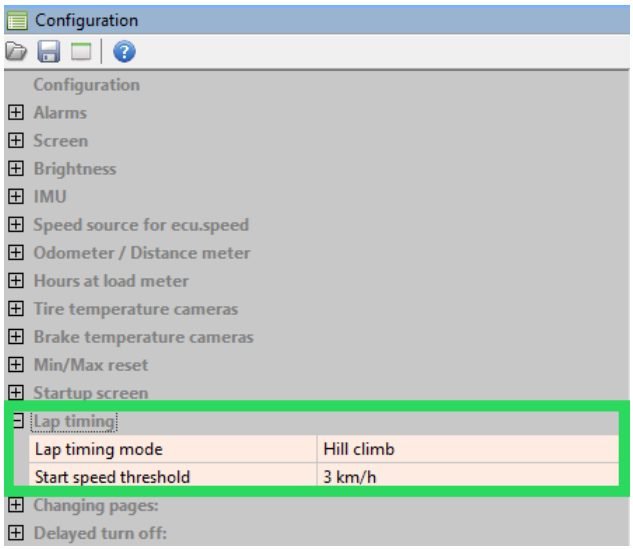
23. Appendix B - How-to Configure the Hill Climb

23.1. Description

Details about version 105.0 or later are provided at the end of the document.

Upgrade the firmware of your ADU to version 94.0 or later.

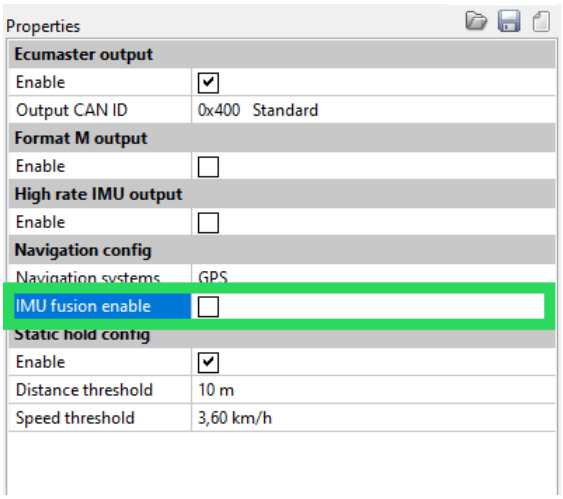
In the *Configuration* panel, change the *Lap timing mode* to the 'Hill climb'.



A start of an event is triggered when the vehicle speed (‘ecu.speed’) exceeds the *Start speed threshold*. The vehicle speed channel (‘ecu.speed’) is required.

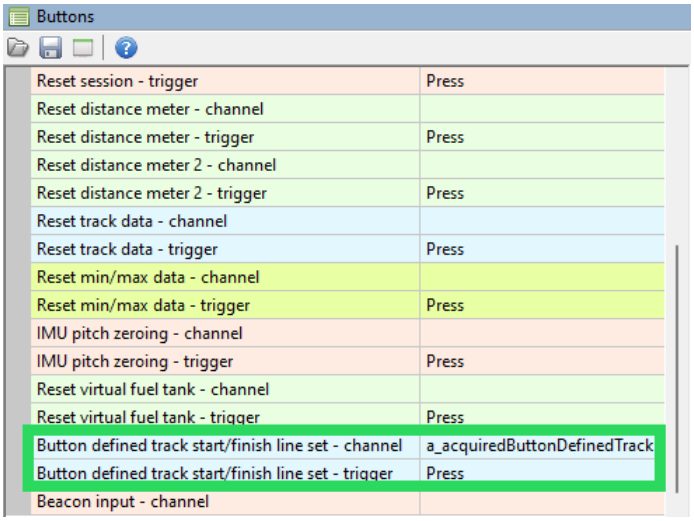
Note:
We are recommending to use vehicle speed sensor (VSS) for this channel. You can still use a speed from the GPStoCAN module, but you may need to increase the *Start speed threshold* (*Lap timing* section of the *Configuration* panel).

GPStoCAN V1 users:
If you are using GPStoCAN V1 as a speed source, you can consider to untick the *IMU fusion enable* option in the *Light Client* app with your GPS connected. In this way, the GPStoCAN module will use only the GPS signal (and NOT a GPS+IMU). It decrease an update frequency of your GPStoCAN V1 module to 10 Hz, but it will help to prevent jitter on the GPS speed when your car is standing at the start line.



How to configure the track in ADU memory:

1. The Button to define a track is wired to analog input A5 in the default project. In the Buttons panel, you need to define the *Button defined track start/finish line set* parameter (referred to as 'Button' in the remainder of this document).



2. Make sure your GPS is working correctly (has a fix).
3. You need to stop at the start line and press the Button
4. Drive to the finish line and press the Button again. And now your track will be defined.

How to start the lap timing:

1. Make sure your GPS is working correctly (has a fix).
2. You need to stop at the start line (the radius around the start point is 15 m).
3. The timer will start when the speed will be higher than *Start speed threshold* (default: 3 km/h).
4. Drive to the finish line and when you cross the finish line the timer will stop automatically.

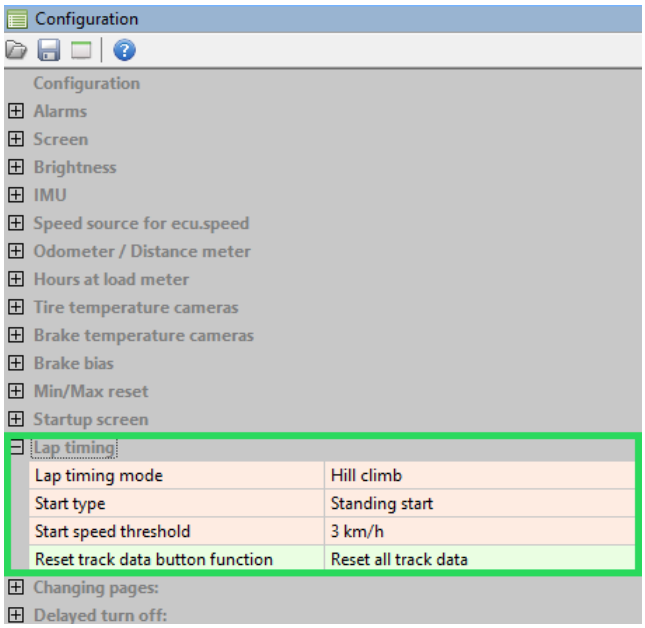
There are two diagnostics channel available: enumeration 'adu.track.lapTimingState' and 'adu.track.distanceToStart' in meters.

All racing features in the ADU are functional in Hill Climb - predicted time, gain/loss, gain/loss graph, record table, and session times.

In version 105.0 or later:

Upgrade the firmware of your ADU to version 105.0 or later.

In the *Configuration* panel, change the *Lap timing mode* to the 'Hill climb'.



Select the appropriate *Start type* mode: 'Standing start' or 'Flying start'

The vehicle speed channel (‘ecu.speed’) is required.

Start type: Standing start:
A start of an event is triggered when the vehicle speed (‘ecu.speed’) exceeds the *Start speed threshold*.

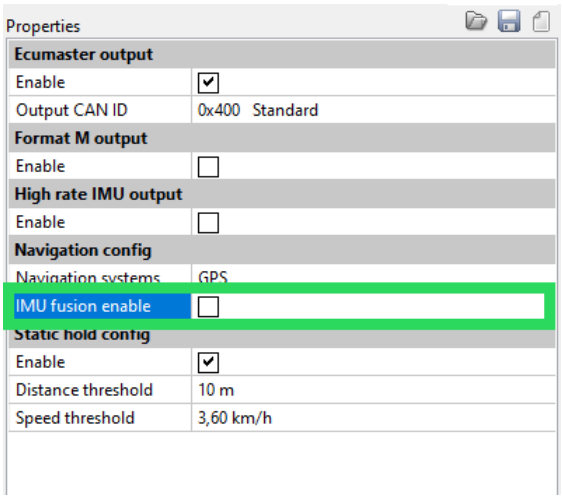
Start type: Flying start:
A start of an event is triggered when the vehicle crosses the start line.

Note:

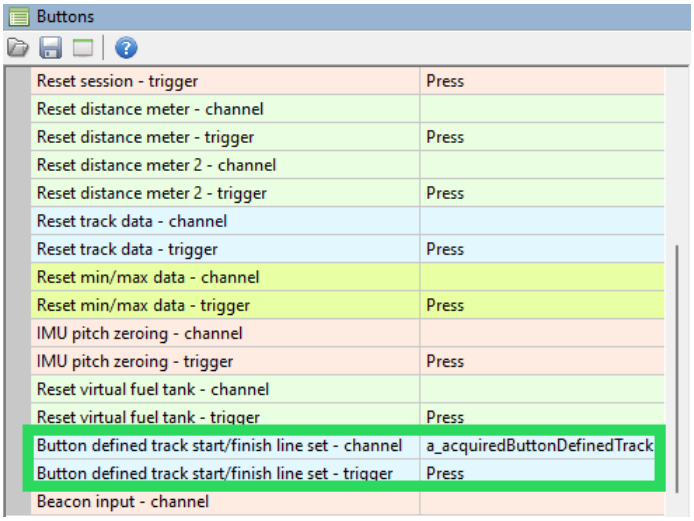
We are recommending to use vehicle speed sensor (VSS) for this channel. You can still use a speed from the GPStoCAN module, but you may need to increase the *Start speed threshold* (*Lap timing* section of the *Configuration* panel).

GPStoCAN V1 users:

If you are using GPStoCAN V1 as a speed source, you can consider to untick the *IMU fusion enable* option in the *Light Client* app with your GPS connected. In this way, the GPStoCAN module will use only the GPS signal (and NOT a GPS+IMU). It decrease an update frequency of your GPStoCAN V1 module to 10 Hz, but it will help to prevent jitter on the GPS speed when your car is standing at the start line.



The Button to define a track is wired to analog input A5 in the default project. In the Buttons panel, you need to define the *Button defined track start/finish line set* parameter (referred to as 'Button' in the remainder of this document).



Standing start

How to configure the track in ADU memory:

1. Make sure your GPS is working correctly (has a fix).
2. You need to stop at the start line and press the Button.
3. Drive to the finish line and press the Button again. And now your track will be defined.

How to start the lap timing:

1. Make sure your GPS is working correctly (has a fix).
2. You need to stop at the start line (the radius around the start point is 15 m).
3. The timer will start when the speed will be higher than *Start speed threshold* (default: 3 km/h).
4. Drive to the finish line and when you cross the finish line the timer will stop automatically.

Flying start

How to configure the track in ADU memory:

1. Make sure your GPS is working correctly (has a fix).
2. You need to press the button as you cross the start line.
3. Drive to the finish line and press the Button again. And now your track will be defined.

How to start the lap timing:

1. Make sure your GPS is working correctly (has a fix).
2. The timer will start when you cross the start line.
3. Drive to the finish line and when you cross the finish line the timer will stop automatically.

There are two diagnostics channel available: enumeration 'adu.track.lapTimingState' and 'adu.track.distanceToStart' in meters.

All racing features in the ADU are functional in Hill Climb - predicted time, gain/loss, gain/loss graph, record table, and session times.

23.2. Document history

Version	Date	Changes
1.0	2023.10.10	Initial release
1.1	2024.01.19	Screenshots changed
1.2	2024.09.03	Additions in version 105.0 - 'Standing' or 'Flying' Start option

24. Appendix C - How to Configure PMU CAN Stream

24.1. Description

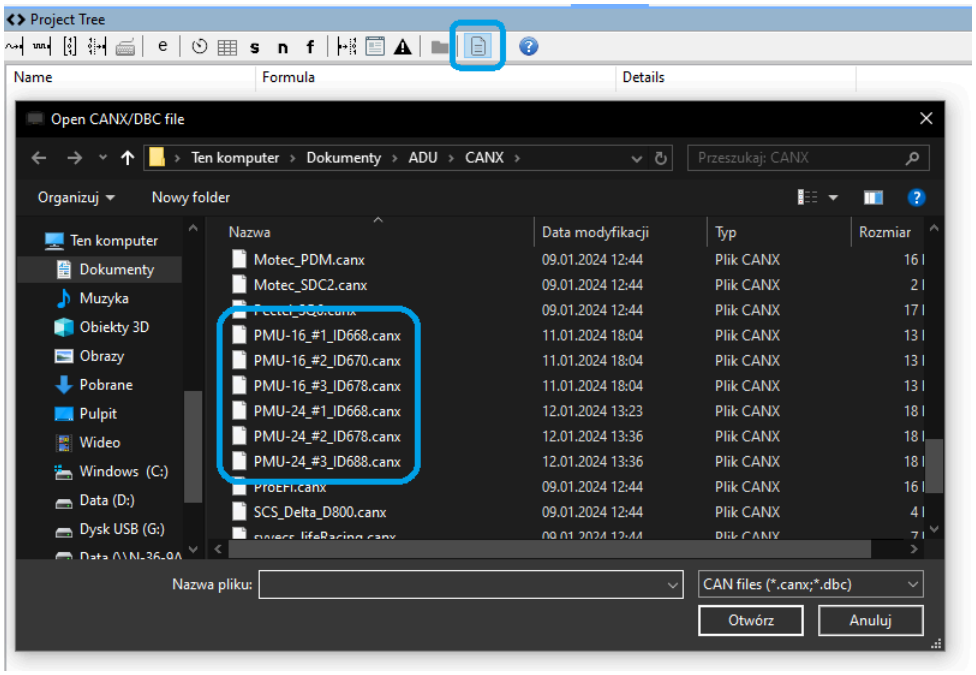
ADU firmware 100.0 addes the native support of the PMU-24 to the previously supported PMU-16. The ADU supports up to 3 PMUs, each of them being 16 or 24 version.

To add the PMU CAN stream, import one of the .CANX files. There are three files to choose from, each containing the PMU device number and default base address in its name:

- PMU-24_#1_ID668.canx
- PMU-24_#2_ID678.canx
- PMU-24_#3_ID688.canx

Additionally, the names of the CANX files for PMU-16 have changed. These are three files:

- PMU-16_#1_ID668.canx
- PMU-16_#2_ID670.canx
- PMU-16_#3_ID678.canx



It's important to notice that importing the PMU .CANX file creates only CANbus Message objects in the Project Tree. In the case of PMU-24, for PMU number #1, these are two CANbus Message Objects of the following types:

- PMU1 [1-16]
- PMU1 [17-24]

In the case of PMU-16 for PMU number #1 there is only one CANbus Message Object of type:

- PMU1 [1-16]

The CANbus Inputs are not necessary, because the channels decode automatically. This saves the user resources.

The scheme below shows the default CAN stream locations of the PMUs. Please note, that if using PMU 24 and PMU 16 on the same CANbus, the default CAN stream locations may need to be adjusted to avoid conflict.

ID	Default CAN stream locations for PMU-16	Default CAN stream locations for PMU-24
0x668	PMU-16 #1 at 0x668	PMU-24 #1 at 0x668
0x669		
0x66A		
0x66B		
0x66C		
0x66D		
0x66E		
0x66F		
0x670	PMU-16 #2 at 0x670	
0x671		
0x672		
0x673		
0x674		
0x675		
0x676		
0x677		
0x678	PMU-16 #3 at 0x678	PMU-24 #2 at 0x678
0x679		
0x67A		
0x67B		
0x67C		
0x67D		
0x67E		
0x67F		
0x680		
0x681		
0x682		
0x683		
0x684		
0x685		
0x686		
0x687		
0x688		PMU-24 #3 at 0x688
0x689		
0x68A		
0x68B		
0x68C		
0x68D		
0x68E		
0x68F		
0x690		
0x691		
0x692		
0x693		
0x694		
0x695		
0x696		
0x697		

To keep track of PMU channel status information, you can open a separate Log panel for each PMU. Below is a list of channels for each of the three built-in PMU devices:

- pmuX.totalCurrent
- pmuX.battery
- pmuX.boardTemperatureL
- pmuX.boardTemperatureR
- pmuX.boardTemperatureMax
- pmuX.status
- pmuX.userError

- pmuX.oY.status – states for each of the 24 outputs
- pmuX.oY.active – activity flags for each of the 24 outputs
- pmuX.oY.current (*) – current values for each of the 24 outputs
- pmuX.oY.voltage (**) – voltage values for each of the 24 outputs
- pmuX.aY.voltage – voltages in the range of 0-5V for each of the 16 inputs

(*) – For outputs O1-O16, the current value resolution is 0.25A, and for outputs O17-O24, it is 0.1A.
(**) – For outputs O1-O16, the measurement range is 0-16V, and for outputs O17-O24, it is 0-20V.

Inputs A9-A16 have the capability to measure voltage in the range of 0-20V, but the information sent on the CAN channels pmuX.aY.voltage is limited to the range of 0-5V. To read the voltage value on these analog inputs across the entire measurement range, you can utilize channels pmuX.oY.voltage. For instance, the voltage value on input A9 is also available on pmuX.o17.voltage. Similarly, the voltage value on input A16 is also available on pmuX.o24.voltage.

24.2. Document history

Version	Date	Changes
1.0	2024.01.19	Initial release

25. Appendix D - How-to Merge Projects

25.1. Description

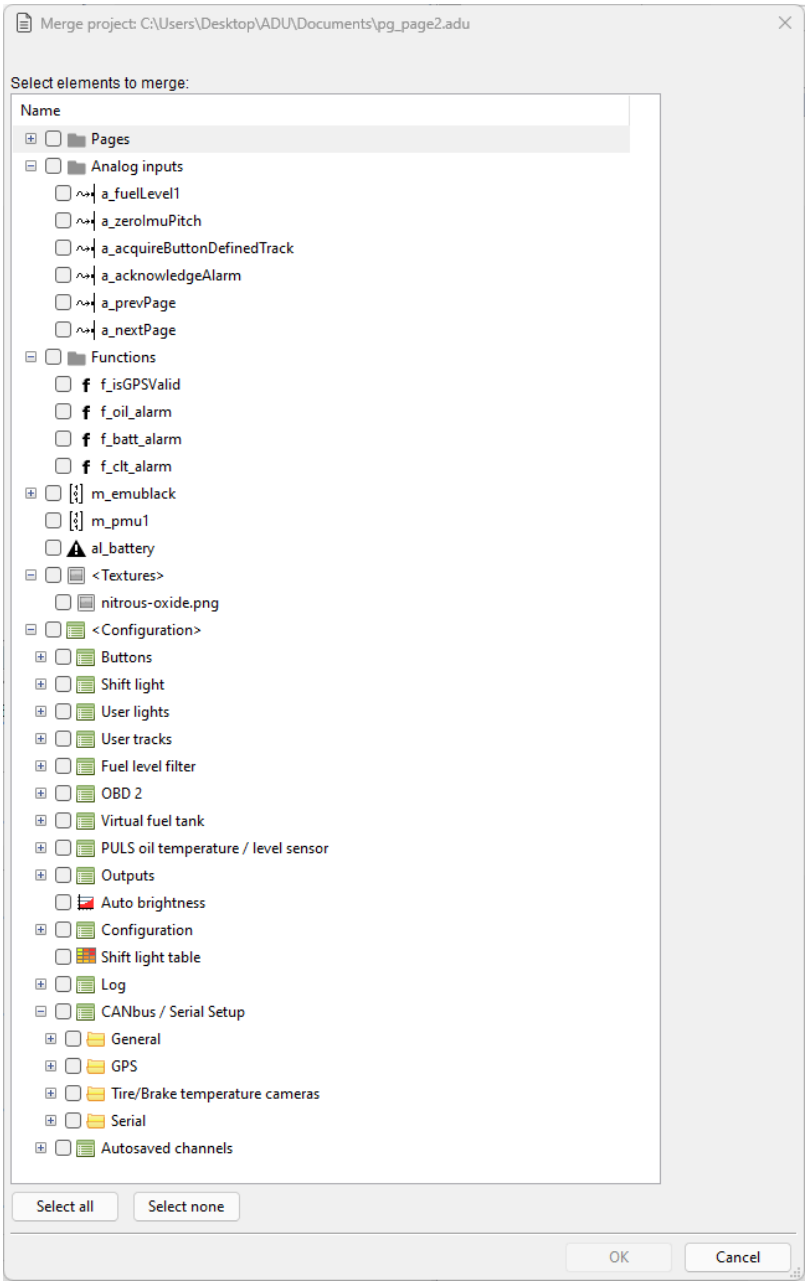
Merging projects enables you to pick specific elements from one project and integrate them into another.

In the document, the project you're currently working on will be named 'base'. The project you want to load the elements from, will be called 'incoming'.

All items from the Project Tree and the configurations can be merged between projects. This means all of the following:

- Analog inputs
- Digital inputs
- CANbus message objects
- CANbus inputs
- CANbus keyboards
- Enumerations
- Timers
- Tables
- Switches
- Numbers
- Functions
- CANbus exports
- Pages
- Alarms
- Textures
- Groups
- Configurations

To merge project select ***File/Merge project*** or press **Ctrl+M**. After selecting the incoming project, a dialog window will appear, displaying all elements from the incoming project available for merging:



After selecting individual elements or choosing all using the 'Select All' button, confirm your selection by clicking 'OK.' The selected elements will then be added to the Project Tree, and the chosen configuration settings will be applied.

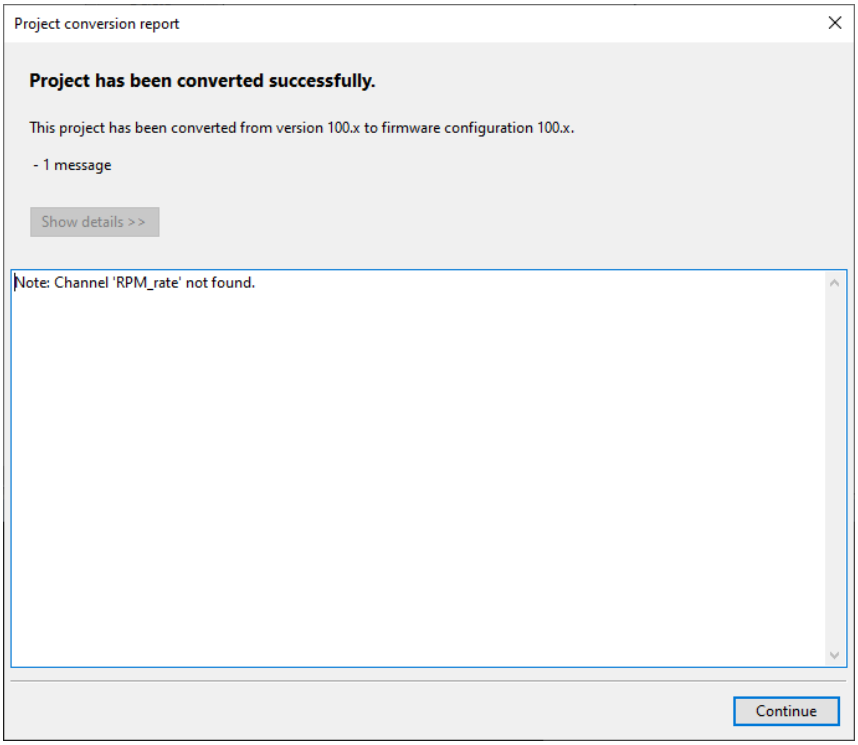
25.2. Resolving potential errors

Name conflict

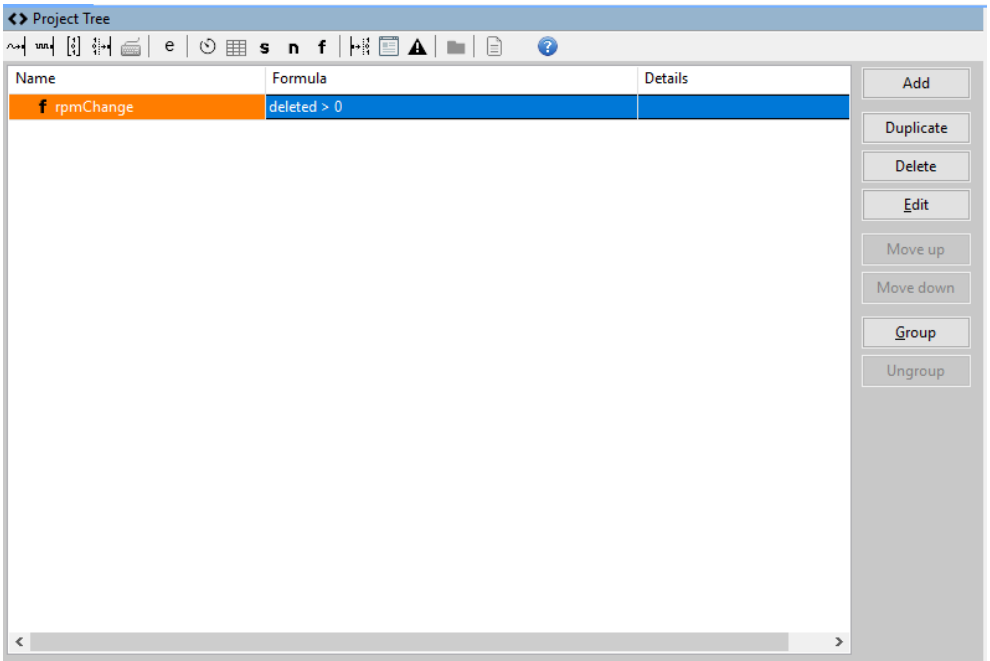
Name conflict is a situation where an incoming element has the same name as any element from the base project. In this case, the incoming element will be assigned a name with a '2' appended at the end.

Missing dependency

During the project merging process, you may encounter a missing dependency error. This occurs an incoming element (e.g., a calculation) relies on another item not present in the base project. If such an error arises, you will be notified through the following window.



Despite this error, the affected element will still be loaded, and you have the option to manually resolve the missing dependency.



To prevent the missing dependency error, organize your projects using groups. Grouping related elements together makes sharing across projects easier. Choosing the entire group from the incoming project simplifies the process and reduces the chance of errors.

25.3. Document history

Version	Date	Changes
1.0	2024.01.19	Initial release
1.1	2024.02.29	Add the list of mergeable elements Separated a section on potential errors

26. Appendix E - How-to Set Up Keyboards with Encoders

26.1. Description

This guide explains how to configure CANbus keyboards with rotary encoders in the Ecumaster ADU. From firmware version 120.0, two types of such keyboards are supported:

- **5x3MT Ecumaster Keyboard** – equipped with 13 standard buttons and 2 rotary encoders.
- **RacePad** – equipped with 8 standard buttons and 4 encoders. Mounted on a steering wheel.

This document focuses mainly on the configuration options for the encoders. For more general information on CAN bus keyboard support, see the ADU user manual https://www.ecumaster.com/files/ADU/adu_manual_en.pdf.

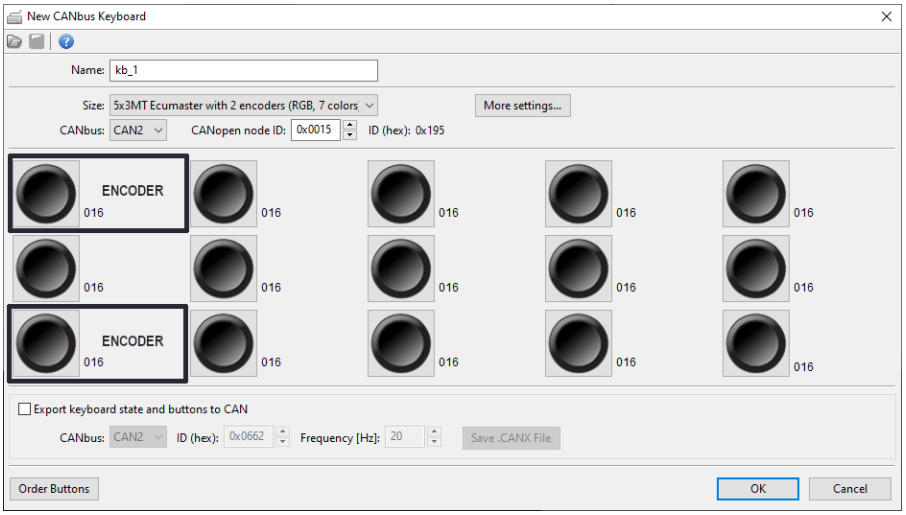
Basic Configuration

To configure any keyboard, click ***Add*** in the ***Project Tree*** and select ***CANbus Keyboard*** from the list, or click the keyboard icon in the toolbar.

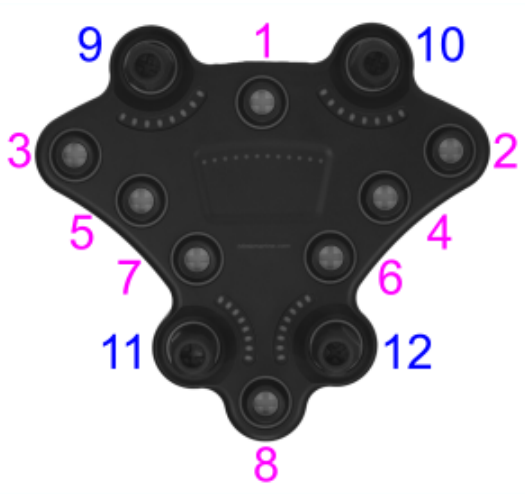
Then, select the appropriate keyboard from the ***Size*** dropdown menu:

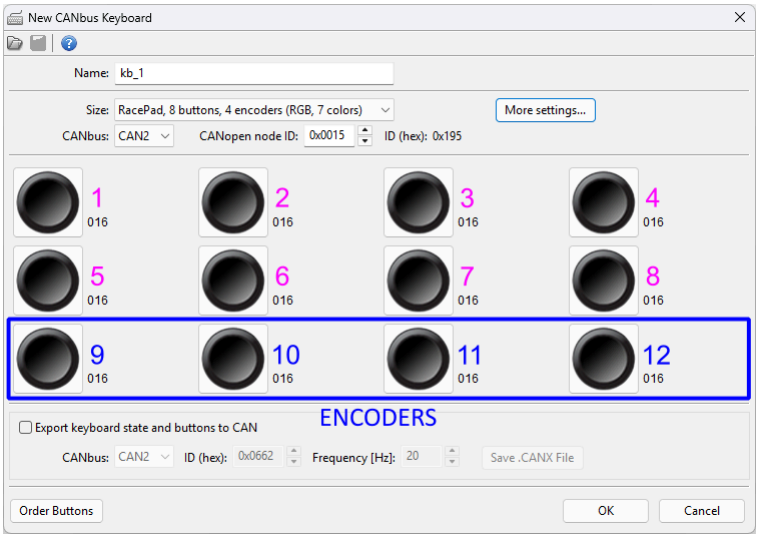
- 5x3MT Ecumaster with 2 encoders (RGB, 7 colors)
- RacePad, 8 buttons, 4 encoders (RGB, 7 colors)

5x3MT Ecumaster Keyboard configuration window:



RacePad configuration window:





Keyboard-Specific Settings

5x3MT Ecumaster Keyboard

Surrounding each encoder are 16 LEDs that visually represent the current state of the encoder. The default starting LED is at 9 o'clock, numbered clockwise.

To change this, go to ***More settings*** and adjust the ***Start offset for encoder LEDs*** from 0 to 15, where 0 corresponds to the 9 o'clock position.

RacePad

Each encoder features 8 arc-shaped LEDs. Additionally, a central LED bar with 12 LEDs is positioned in the middle of the device.

RacePad has several dedicated settings available in ***More settings***:

Parameter	Description
Invert LEDs encoder #1–4	Reverses the direction in which the LEDs light up for each encoder.
Invert encoder direction #1–4	Reverses the rotation logic. By default, turning clockwise increases value.
Enable central LEDs	Enables the center 12-LED strip.

Parameter	Description
Bi-directional	Enables symmetric (centered) visualization, e.g., for G-force or time gain/loss channels. In default mode, LEDs progressively light up from left to right. In bi-directional mode, LEDs light up outward from the center to the left or right, depending on the channel value.
Central LEDs channel	Assigns a channel to control the center LED strip, based on the channel's value.
Channel min value	Lower bound of the channel value range.
Channel max value	Upper bound of the channel value range.
Channel central value	Midpoint value used for symmetric LED visualization.

Encoder Operation Modes

Knowing the three types of encoder operations is essential for getting the best configuration:

1. ***encoder***

   Rotating the encoder adjusts the associated channel's value. Unlike a rotary switch, you have the flexibility to configure the encoder's range of values.

   - *First state* - the lowest encoder state
   - *Last state* - the highest encoder state
   - *Default state* - After startup, the encoder will adopt its default state unless the *Autosaved channels* feature is used. (Refer to the how-to guide document for detailed instructions on using *Autosaved channels* https://www.ecumaster.com/files/ADU/HowTo/How_to_Configure_Autosaved_Channels_in_ADU.pdf)

2. ***page changer***

   Turning the encoder clockwise moves to the next page, and turning counter-clockwise goes back to the previous page. In this mode, the LEDs illuminate in a fan-like pattern, offering a visual indication of the operating mode.

   - *Wrap pages* decide if the encoder should jump from the last page to the first and vice versa.

3. ***parameter controller*** and ***parameter selector***

   You can use the keyboard encoder to control multiple settings. This is possible by setting it to *parameter controller* mode. For each setting you want to manage, select a button and set

its mode to *parameter selector* mode. These selector buttons function like radio buttons, allowing you to choose one active parameter at a time that is modified by the knob.

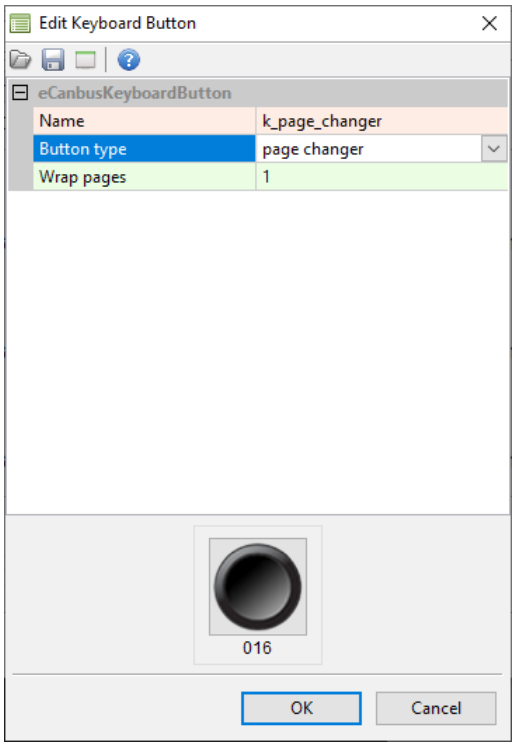
Each *parameter selector* comes with its own range and state, updating the *parameter controller* when pressed.

a. If there are no button presses defined as a *parameter selector*, *parameter controller* remains neutral and rotation has no effect.
b. Pressing the button defined as a *parameter selector*, conveys the parameter information to the encoder. Encoder can change the state of the *parameter selector*.
c. After pressing the currently selected *parameter selector* button again or pressing the *parameter controller*, the encoder returns to the neutral behaviour. (Pressing any other button (not defined as a *parameter selector*) will not cause the encoder to lose control of the *parameter selector*.)

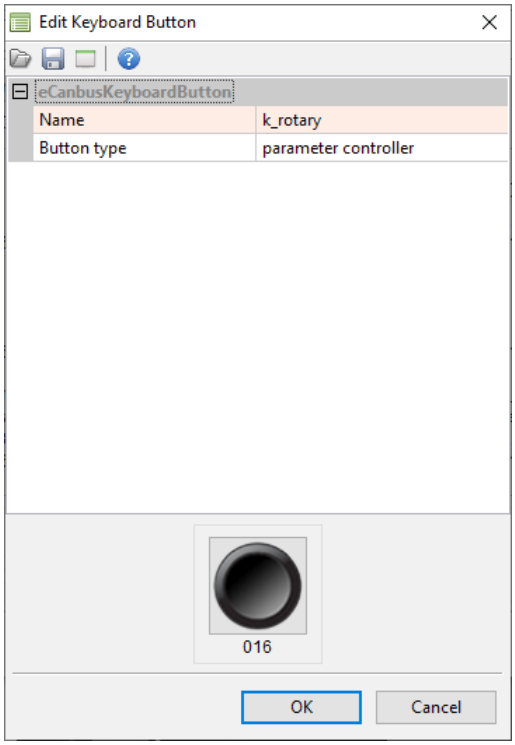
26.2. Example

We're configuring the 5x3MT to finely adjust four parameters and quickly navigate pages with the two rotary encoders. In this example we'll use the top encoder for pages and the bottom one to control parameters.

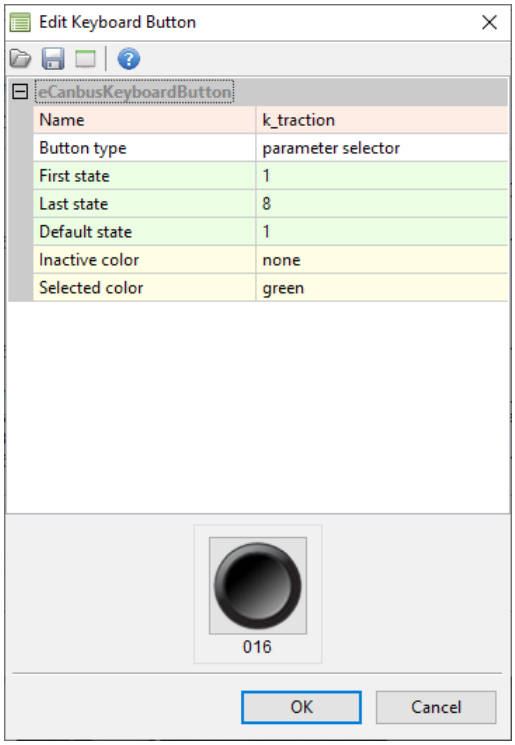
Let's start with the pages. In the keyboard settings, select the top-left box for the rotary encoder. Set it as a '*page changer*' and determine whether pages should wrap around.



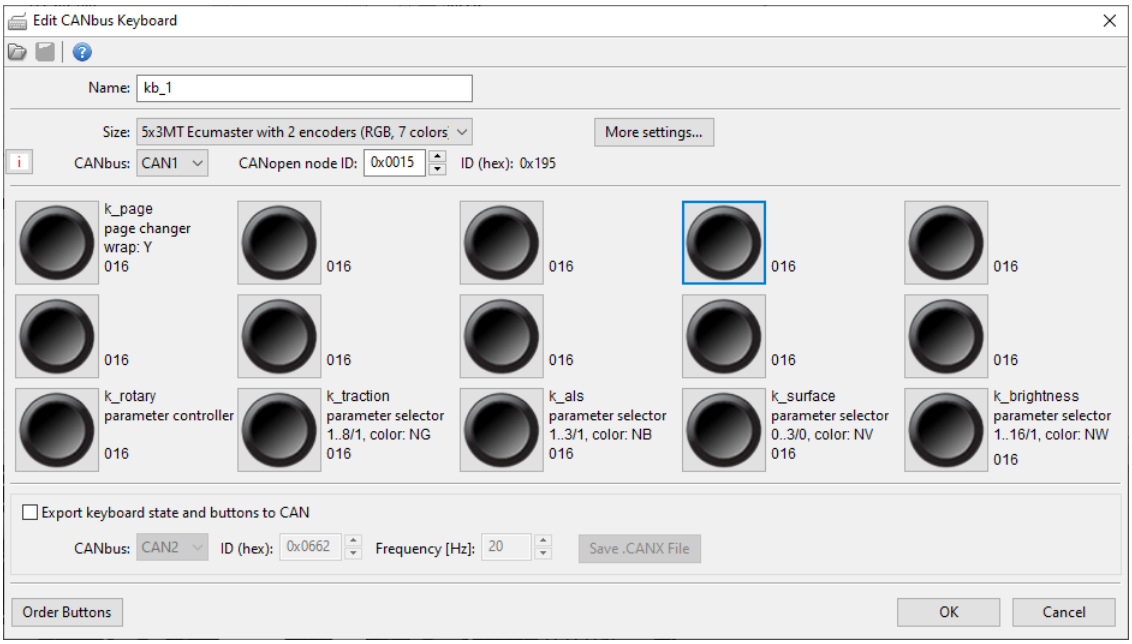
Moving on to our second goal, we aim to use the other encoder for controlling four parameters. Begin by selecting the bottom-left box and designate it as a '*parameter controller*'.



Now proceed to configure each parameter. For each of the four, choose one button, change its type to '*parameter selector*' and set the first, last, and default states to adjust their value range.



The final setup of our example keyboard looks as follows:



When the driver wishes to adjust a setting, they can press the corresponding button. The rotary encoder will activate, enabling them to change the parameter by turning it.

26.3. Document history

Version	Data	Changes
1.0	2025.04.30	Initial release

27. Appendix F - How-to Use Virtual Fuel Tank

27.1. Description

The Virtual Fuel Tank feature helps you know how much fuel is left in your tank. By analyzing fuel usage data from the ECU (*ecu.usedFuel* channel), it not only tells you the remaining fuel but also estimates how many laps you can do with that fuel. This helps you plan your race strategy based on your fuel levels.

Parameter	Description
Fuel tank size definition	- Fixed volume – tank is always filled to its maximum volume (defined by *Fuel tank size (fixed)*) - Defined by channel – volume of fuel in the tank after the refill is defined with a channel (*Fuel tank size channel [L]*)
Fuel tank size (fixed)	Capacity of fuel tank in litres. This parameter is active when the *Fuel tank size definition* is set to "Fixed Volume"
Fuel tank size channel [L]	Channel defines total volume of fuel in the tank after refill (what was left + what was filled). This parameter is active when the *Fuel tank size definition* is set to "Defined by channel"
Used fuel channel	The channel contains used fuel information sent by the ECU. By default, this is the *ecu.usedFuel* channel
Max used fuel change	Maximum acceptable difference between two consecutive readings of fuel consumption information. If the change is greater than this parameter, it will be ignored. The parameter also enables the function to work correctly if the fuel consumption channel in the ECU is reset
Fuel usage correction	Correction factor when there is a difference between the fuel consumption information provided by the ECU and the actual fuel consumption. This parameter is adjusted empirically based on the measured actual fuel consumption
Stage mode channel	The Virtual Fuel Tank can monitor fuel consumption and distance traveled simultaneously in two modes: stage and road. Each mode has two channels for distance and used fuel. The ***Stage mode channel*** determines which mode is active.

Parameter	Description
	If the value of the channel assigned in this field is 0 - used fuel and covered distance is assigned to the road channel. If the value of the channel assigned in this field is 1 - used fuel and covered distance is assigned to the stage channel.

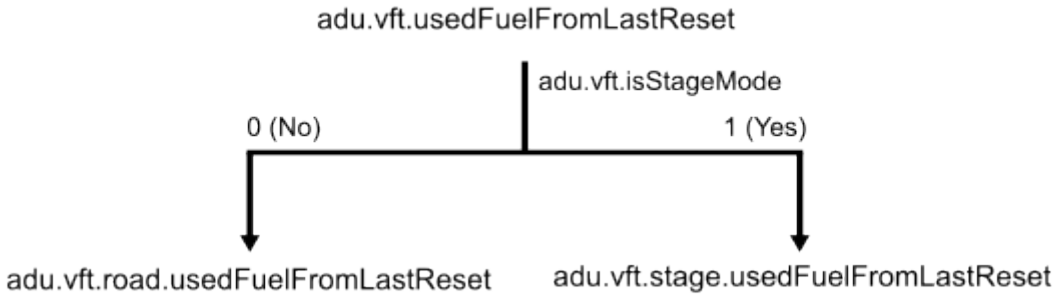
Channel	Description
adu.vft.fuelTankSize	Currently used fuel tank size
adu.vft.remainingFuel	Amount of fuel remaining in the tank
adu.vft.numLapsOnRemainingFuelAvg	Number of laps that can be completed using the fuel remaining in the tank, based on the average fuel consumption (Fuel used per lap avg)
adu.vft.numLapsOnRemainingFuelLast	Number of laps that can be completed using the fuel remaining in the tank, based on the fuel consumption of the last lap (Fuel used per last lap)
adu.vft.usedFuelPerLap	Average fuel consumption per lap (since resetting the fuel tank)
adu.vft.usedFuelPerLastLap	Fuel consumption on the last lap
adu.vft.usedFuelFromLastReset	Fuel used since last virtual fuel tank reset
adu.vft.isStageMode	Shows whether fuel is assigned to stage or road
adu.vft.stage.distanceFromLastReset **adu.vft.road.distanceFromLastReset**	Distance traveled in stage/road mode since the last refueling
adu.vft.stage.usedFuelFromLastReset ***adu.vft.road.usedFuelFromLastReset***	Fuel used in stage/road mode since the last refueling

Additionally, it's important to set up a button to reset the Virtual Fuel Tank. In the *Buttons* panel, there's a button defined as "Reset virtual fuel tank." This button is crucial because the driver needs to press it when refueling. When pressed, it resets the fuel level to full and begins a fresh count of all the data.

The Virtual Fuel Tank data is saved in the device's memory, so even when you turn off the ignition, it's retained. Properly connecting the device to power is crucial for this function to work correctly, requiring separate connections for the battery and ignition switch. This ensures that even when the ignition is off, the settings remain saved.

Using *Stage mode channel* If there's a significant difference in fuel use, like when driving to a special stage compared to driving the stage itself, you can measure fuel use for two separate situations: on the road and during the stage. With this setup, the overall fuel use channel, *adu.vft.usedFuelFromLastReset*, gets additionally divided into two separate channels: *adu.vft.stage.usedFuelFromLastReset* and *adu.vft.road.usedFuelFromLastReset*. This lets you

analyze and predict fuel use in more detail. The channel determining which of the two channels to assign the data is the one defined in the setting: *Stage mode channel*.



Similarly, with the distance channels: *adu.vft.stage.usedFuelFromLastReset* and *adu.vft.road.usedFuelFromLastReset*.

Using *Fuel tank size channel [L]* You can adjust what is the value of the Virtual Fuel Tank when you reset it. This is possible when the *Fuel tank size definition* is set to "Defined by channel". Set the *Fuel tank size channel [L]* to the channel indicating the amount of fuel in the tank after refill, which includes the remaining amount plus what was added. For example, this can be adjusted using a knob during refill. Then, pressing the "Reset virtual fuel tank" button will set the remaining fuel value to the current channel's value.

27.2. Example

In this example we will cover a step-by-step basic configuration and use the Virtual Fuel Tank (VFT) for a real-time analysis of the consumption data.

To properly configure the VFT, it's very important to set up the button *Reset virtual fuel tank* in the *Buttons* panel:

Each time the car is refueled, the driver has to press the button to let know the system, the tank is full once again. This action will reset the remaining fuel.

In this example, we will cover the basic mode of operation, with the fuel tank being filled to full every time. Hence, we will leave the *Fuel tank size definition* as "Fixed volume". Next, we have to input the *Fuel tank size*.

To enable the calculation of the fuel, it's necessary to configure the *ecu.usedFuel* channel. Most ECUs are able to export this data based on injectors flow and injection times.

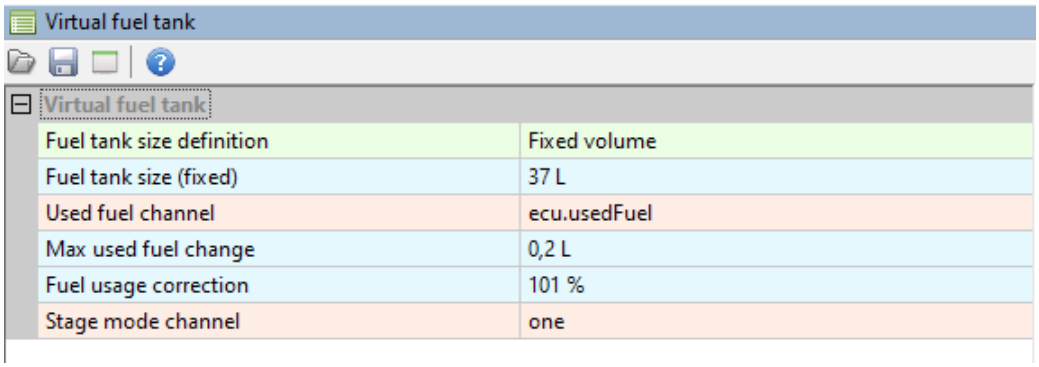
Max used fuel, as defined before, is maximum acceptable difference between two consecutive readings of fuel consumption information. If the change is greater than this parameter, it will be ignored. For most cases, the default parameter 0,2 l should be correct. A typical race car uses about 0,5 l of fuel per minute of race drive.

The last parameter, *Fuel usage correction*, can be used for precise adjustment of the calculations. We recommend testing it before the event. To determine the value of this parameter, measure the actual volume of fuel used during a session and the last value of the *adu.vft.fuelUsedFromLastReset*:

*Fuel usage correction = (Actual volume of fuel / adu.vft.fuelUsedFromLastReset) * 100%*.

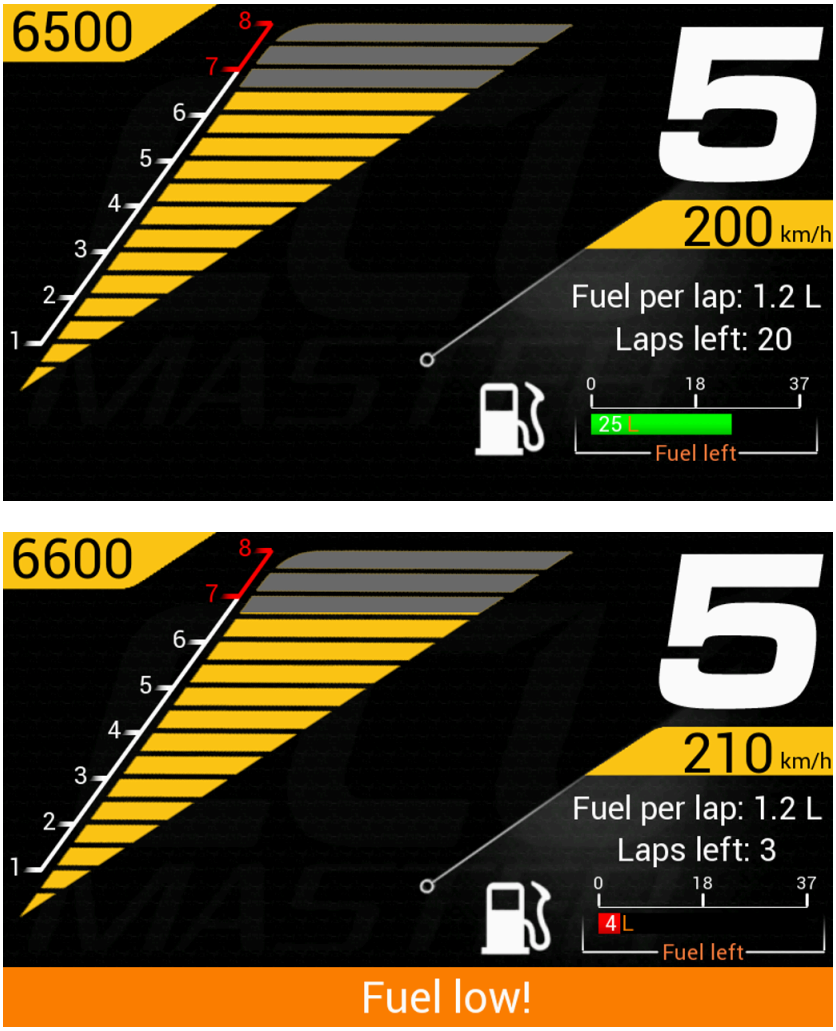
If the real value was greater than the VFT value, the *Fuel usage correction* value will be greater than 100%, and vice versa.

In this case, we will utilize only one used fuel meter, hence the *Stage Mode Channel* will be left to default channel one. For more details on the *Stage Mode Channel*, see: Description *(on page 239)*.



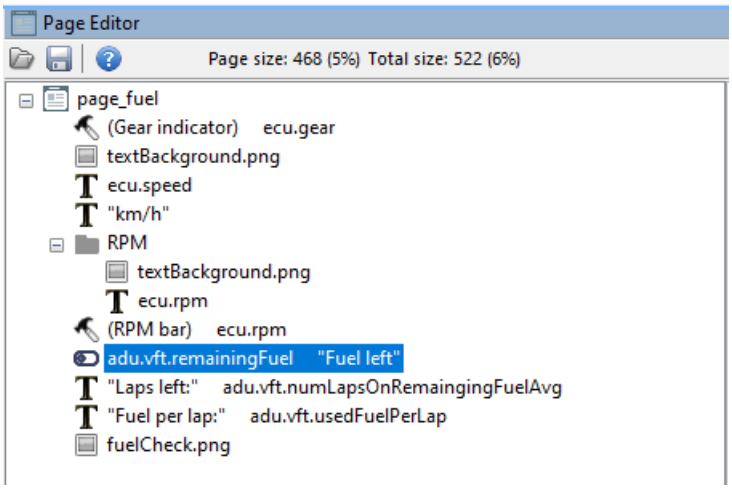
Now, with the VFT configured, let's utilize its abilities.

We'll display the remaining fuel, average fuel usage, and laps left on one page, with a warning if fuel drops below 5 liters. You can easily change an existing page or add a new one using the *Project Tree* panel. We'll use the *Page Editor* and *Dashboard Preview* for simple layout.



The current document covers practical use of the ADU features. For more details on using the *Page Editor*, refer to the ADU Manual.

These are all the elements of our example page:



Firstly add the *Bar graph* to a page.

Position	
Position X	557
Position Y	331
Channel and unit	
Channel	adu.vft.remainingFuel
Decimal places	0
Min	0
Max	37
Display value	☑
Update frequency	50 Hz
Display unit	☑
Unit	L
Custom unit	
Description	
Text	Fuel left
Icon	
Use icon	☐
Icon texture	
Icon scale	130 %
Alarm	
Alarm channel	
General	
Bar type	Horizontal
Bar style	Style 1
Length	200
Width	20
Border width	1
Number of ticks	3
Decimal places for tick	0
Colors	
Color	(255,255,255)
Background color	(0,0,0)
Bar color	(0,255,0)
Alarm color	(255,0,0)
Icon color	(255,255,255)
Value color	(255,255,255)
Text color	(255,128,64)
Unit color	(255,128,0)
Fonts	
Tick font	1
Value font	2
Text font	2
Functions	
Redline when	When value below
Redline start	5
Visibility channel	one

The suggested configuration of the ***Bar graph***:

- In the *Channel* field select *adu.vft.remainingFuel*.
- Set *Max* to the value previously input as the *Fuel tank size*.
- You can add some description in *Description / Text*, we entered "Fuel left"

For some extra information, you can define when the bar turns red:

- *Redline when* set to "When value below"
- *Redline start* in our case it's 5 liters
- Leave *Visibility channel* with it's default value "one"

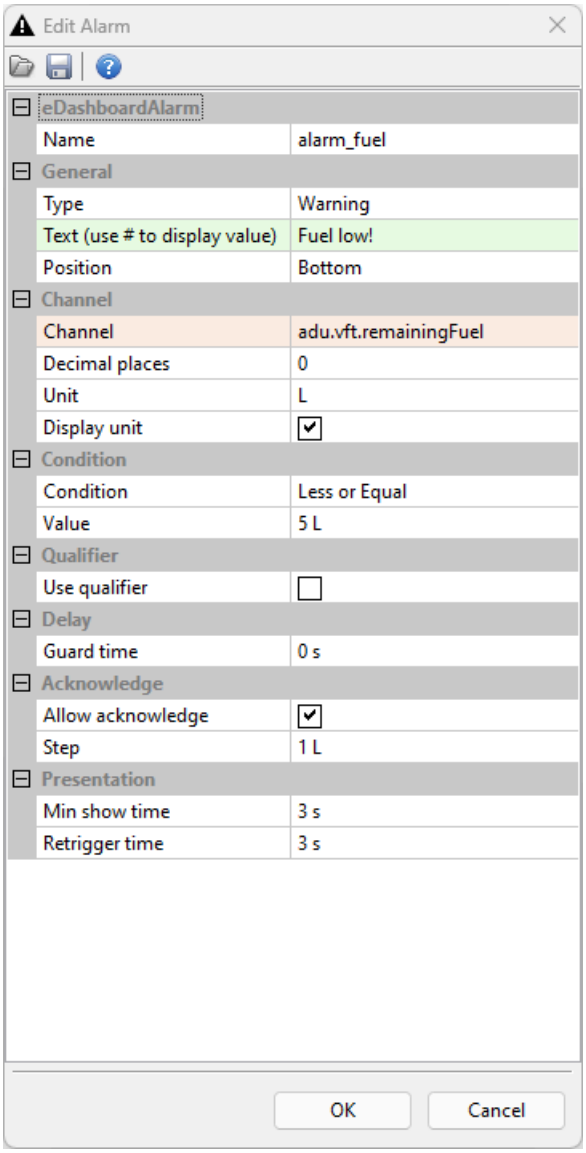
Next, we'll add a ***Text*** element informing the driver about remaining laps on fuel. The configuration is straightforward, requiring only two parameters to be filled.

- Add text description in field *Text*. We set it to "Laps left: "
- In the *Channel* field select *adu.vft.numLapsOnRemainingFuelAvg*.

Position	
Position X	588
Position Y	304
Text	
Color	(255,255,255,255)
Font	4
Italic	☐
Two lines	☐
Text	Laps left:
Text width	0 px
Text align	Left
Channel	adu.vft.numLapsOnRemaingingFuelAvg
Decimal places	0
Value width	0 px
Value align	Left
Unit	user
Display unit	☑
Custom unit	
Color channel	
Visibility channel	one
Update frequency	50 Hz

Similarly, for the "Fuel per lap" element, follow the same steps.

The final step is to set up an alarm that triggers when the fuel level drops below 5 liters. To do this, navigate to the *Project Tree* and add a new *Alarm*. For further instructions on using the *Project Tree*, refer to the ADU Manual.



In the example, we defined the following parameters, leaving the rest as default:

- Add a meaningful *Name*
- Add an informative *Text* for the driver, such as "Fuel low!".
- In the *Channel* field select *adu.vft.remainingFuel*.
- Set the *Condition* to "Less or equal"
- Set the *Value* to the trigger point for the alarm. In our case that's 5 L.
- Set the *Step* to 1 L. Even if the driver acknowledges the alarm, it will reappear if the fuel level drops by another liter.

27.3. Document history

Version	Date	Changes
1.0	2024.04.19	Initial release

28. Appendix G - How-to Configure Autosaved Channels

28.1. Description

The Autosave Channels feature allows users to store values for up to 20 channels, which are loaded when the device starts. These values are saved automatically.

Below is a list of elements from Project Tree that can save their state:

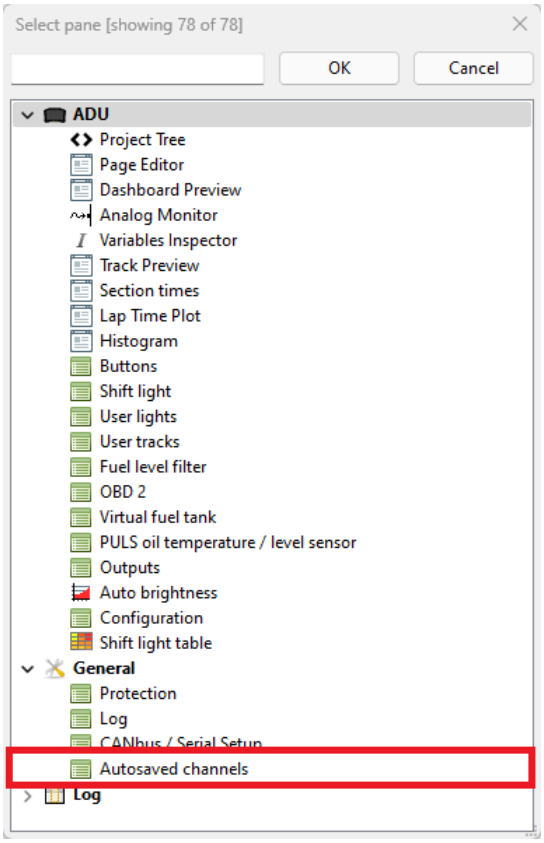
1. Keyboards buttons
2. Switches
3. Logical functions
4. Numbers

The Autosave Channels prove useful in various scenarios, such as preserving the state of any encoder. By doing so, users can easily recover the encoder's state after the device is turned on.

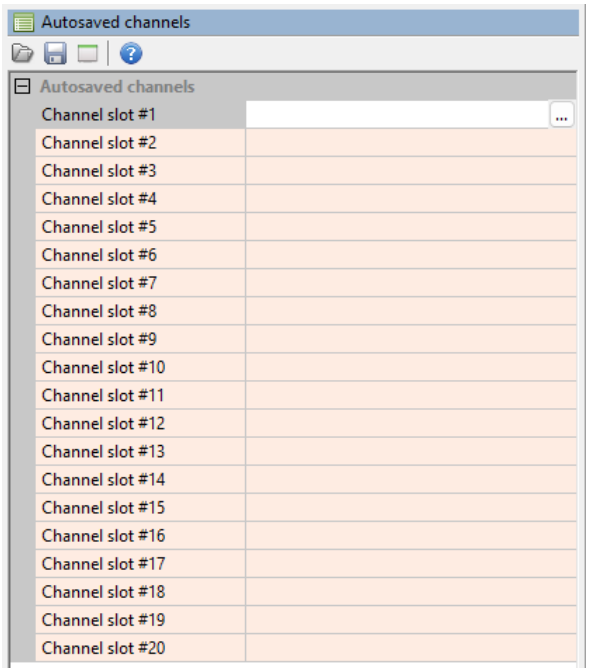
Auto-save operates based on two independent paths. These paths are pre-programmed and not configurable by the user:

- **Ignition switch** – the save of all channels occurs when the ignition switch +12V is disconnected, but the constant battery +12V remains powered (requires separate wiring for battery and ignition switch). This method provides most reliable results.
- **On change** – after detecting a change in any of the autosaved channels, a save occurs, followed by a 2-minute period without further saving. If none of the selected channels change their values, no save occurs, and the system awaits a channel change.
  The 2-minute intervals without saving are implemented to prevent memory wear as it is not designed to handle rapid changes.

To open the *Autosave Channels* panel, press F9 and select it from the *General* category.



In the panel, select the channels you want to be autosaved.



Upon turning on the device, the autosaved channel values will be loaded into their respective channels. Subsequently, the evaluation proceeds as usual, meaning the loaded value will be replaced with any new data.

28.2. Document history

Version	Data	Changes
1.0	2024.01.19	Initial release
1.1	2024.05.13	Improved description

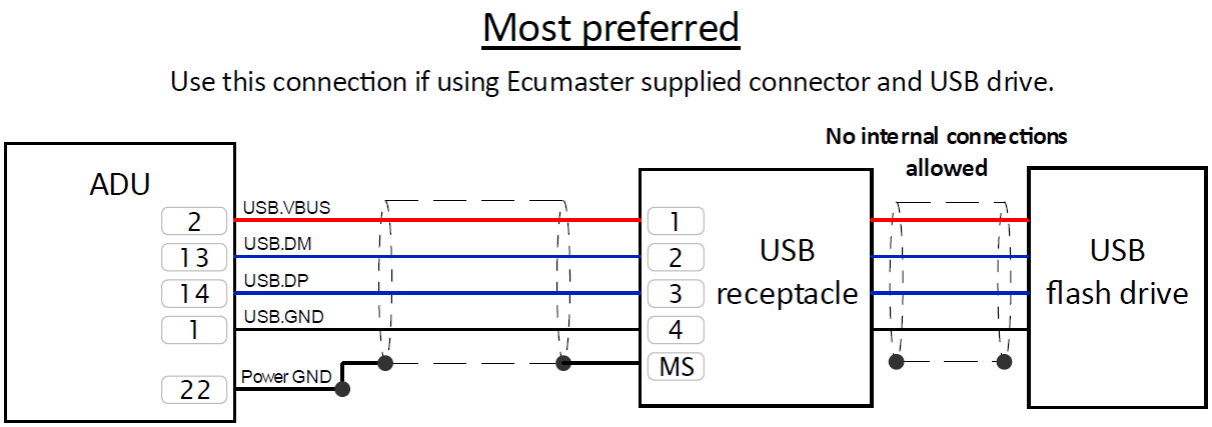
29. Appendix H - How-to Log Data to a USB drive

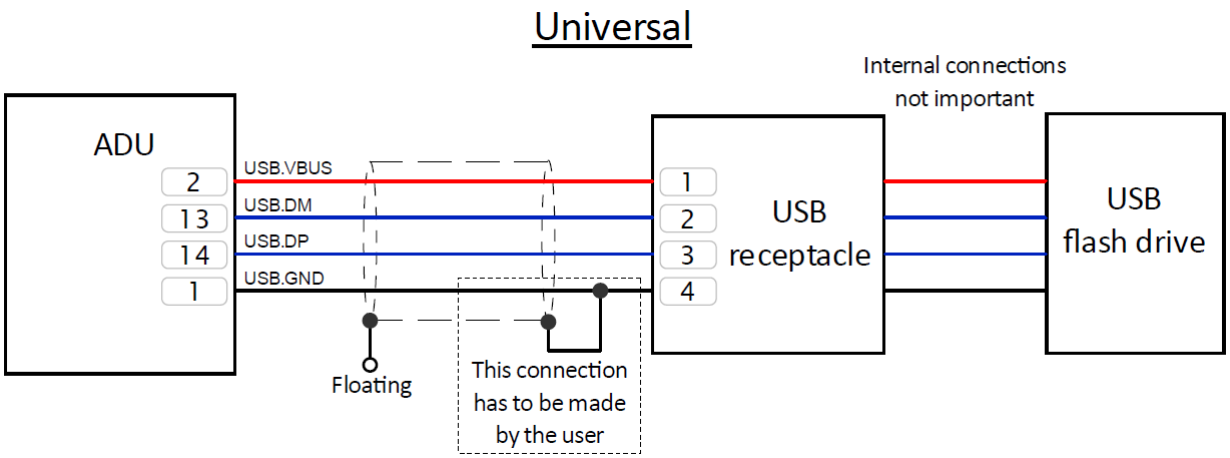
29.1. Introduction

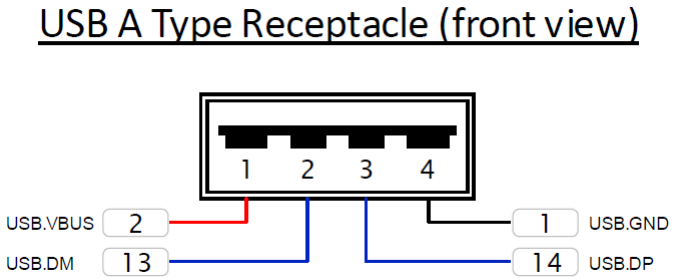
The ADU can log data at up to 500 Hz, either to a USB drive or directly to a connected PC.
To log without a PC, the USB drive must be properly configured. This guide covers how to prepare the system, ensure proper wiring, and troubleshoot common issues.
With correct wiring and USB drive, the system works out of the box—adjustments to the logging configuration are optional and explained here. For more information refer to full ADU manual https://www.ecumaster.com/files/ADU/adu_manual_en.pdf.

29.2. Wiring

Proper wiring is essential for reliable USB logging.
Ensure good shielding and connect the USB shield and ground at only one end to prevent ground loops and maintain stable data transfer.
If possible, use twisted pair wiring for the D+ and D− (DP and DM) lines. Keep the USB cable as short as possible.







Recommended shielding configurations (see diagrams on previous page):

- **Most preferred:** connect the shield to a power ground on the ADU side. No internal connections between USB ground and shield are allowed.
- **Universal:** connect the shield on the USB drive side. In this case, other connections between shield and ground are not important.
- **Not good:** shield connected at both ends.
- **Not good:** shield not connected to ground at either end.

Internal USB connections

The USB shield and ground may be internally connected. This is often not considered and can create shield/ground issues. Internal connections can occur:

- inside the USB drive (only if the casing is metal, not present in any plastic USB drive),
- inside the USB socket (always the case for sockets with 4 connections/wires).

How to check your setup

1. Remove the shield and ground terminals from the ADU connector.
2. Perform a continuity test (or measure resistance):

- If continuity exists between shield and ground: do not connect the shield to any ground on the ADU side.
- If there is no continuity: connect the shield to a power ground (preferred), or to a ground.

3. If you're using a cut USB cable or one with unknown wire assignment, use a continuity test to properly identify each wire by mapping it to the correct USB pin on the connector.
   Do not rely solely on wire colors—always verify connections with a multimeter.

Ecumaster-supplied connectors and USB drives

Ecumaster supplies two types of USB connectors and a USB drive. None of them have an internal connection between the USB shield and ground. Therefore, it is essential to use the "Most Preferred" wiring configuration described above.

29.3. Logging Configuration & Management

Selecting Channels to Log

Open the ***Logged Channels*** window from the *Select panel* menu (F9). This is where you configure:

- Which groups or individual channels should be logged
- Logging frequency per channel or per group
- Bandwidth usage per condition

To quickly enable or change logging frequency, use keyboard shortcuts:

Shortcut	Frequency
Alt + 1	1 Hz
Alt + 2	5 Hz
Alt + 3	25 Hz
Alt + 4	50 Hz
Alt + 5	125 Hz
Alt + 6	250 Hz
Alt + 7	500 Hz
Alt + ~	Deactivation of channel / group logging

You can also right-click on a group or channel and choose frequency from the context menu.

Conditional Logging Profiles

ADU supports up to four logging profiles (*Cond1–Cond4*), each with different sets of channels and frequencies.

- A condition (e.g., engine running, track armed, GPS fix) activates a specific profile
- If no condition is met, the *Default log condition* is used

To configure:

- Assign condition channels under the *Log* panel → *Log Cond2/3/4 Channel* settings
- Prioritize important data under higher-frequency profiles

Conditions can be based on any internal or custom channel.

Each logging condition is evaluated 25 times per second. If multiple conditions are active, the one with the highest priority (*Cond2* > *Cond3* > *Cond4*) is used.

USB Logging & File Storage

ADU automatically logs to a connected USB stick (more on the USB can be found in the next chapter):

- Files are saved in `.ADULOG` format, split by time (default: every 1 hour)
- By default, logging starts automatically upon USB detection
- LED1 can indicate logging status (green = logging; red = error)

Logs can be downloaded from the USB via **Devices > Receive Logs** in the ADU Client (Shift+F4) or simply open/copied from the USB drive.

Real-Time Clock (RTC)

To ensure meaningful file name, the ADU must have the correct date and time. This setting does not update automatically.

Common situations requiring RTC setup:

- Initial installation - when configuring the device for the first time
- Time zone changes - after moving the device to a different time zone
- Daylight saving adjustments - when transitioning between standard time and daylight saving time

Setting the RTC:

- Connect to the device via **ADU Client**
- From the top menu, go to ***Devices / Set Real Time Clock***

The ADU maintains RTC settings using its internal backup supply under normal conditions.

29.4. Preparing a USB drive

To enable reliable data logging via USB, the storage device must be correctly prepared. Improper formatting or usage can prevent logging or cause data loss.

Warning:
The ADU supports only the FAT32 file system. Drives formatted as exFAT or NTFS are not compatible.

Formatting the Drive as FAT32

To format a USB drive in Windows:

1. Insert the USB drive into your PC.
2. Open **File Explorer**, right-click the drive, and select **Format**.
3. In the **File System** dropdown, select **FAT32**.
4. Ensure **Quick Format** is checked.
5. Click **Start** to begin formatting.

> **Note:**
> On some versions of Windows, FAT32 may not appear for drives larger than 32 GB. You may need to use a third-party tool (e.g. Rufus) to format larger drives.

If you are formatting the drive in the third-party tool the most important settings are:

- Type: *Non-bootable drive*
- Partition scheme: *MBR*
- File system: *Large FAT32*
- Extended label and icon files: *Disabled*

Recommended Drive Size

For best compatibility and reliability, use USB drives between 8 GB and 32 GB. Logging capacity is approximately 11h of logging per 1GB.

Use an Empty, Freshly Formatted Drive

Always use a USB stick that has been freshly formatted and contains no other files or folders. Unrelated files may interfere with logging.

Folder and File Structure

The ADU saves log files directly in the root folder of the USB drive. Files follow this naming scheme:
YYYYMMDD_HHmm_ss.adulog
Example: 20250617_1458_32.adulog
This naming depends on the device's real-time clock (RTC). An incorrectly set RTC will result in misleading timestamps, making log management difficult.

29.5. Reviewing and Exporting Logs

Using the *Graph Log* panel

The ***Graph Log*** panel is often used to verify which channels are being logged and at what frequency. It also allows for basic editing of the logging configuration.

You can:

- Add or remove channels from the graph
- Right-click on a channel to change the logged channel or adjust Log frequency

Logging frequency set in *Graph Log* is reflected in the main *Logged Channels* configuration.

Exporting & Reviewing Logs

In ADU Client:

- Use *Open Log* to load files
- Compare laps or channels using *Graph Log*
- Export data as: CSV, VBO (for VBOX tools), PNG (graph snapshot)

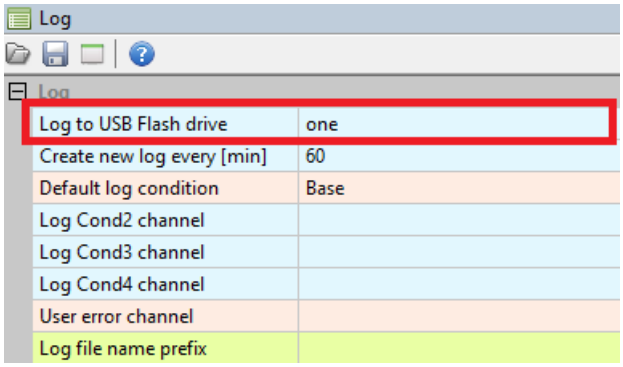
For advanced data analysis, Ecumaster provides a free, professional analysis tool – **Data Master**. Download it from: https://www.ecumaster.com/products/data-master/.

29.6. Troubleshooting

If logging to USB is not working as expected, the following steps will help identify and resolve common issues.

1. **Confirm Logging is enabled**

Ensure that USB logging is properly activated in the software. *Log / Log to USB* should be set to '*one*' for most cases. Change this value only if you're an advanced user. (Logging to USB is enabled by default, so in most cases, it should start automatically once a USB drive is inserted.)



If a conditional log profile (*Cond2–Cond4*) is used, make sure the associated condition channel is active when logging should start.

2. **Check USB Status in the Software**

The ADU Client displays USB logging information in the bottom status bar. The most relevant messages include:

- **USB logger state** – indicates the current USB activity:
  Disconnected – USB drive not connected
  Accessing USB – device is recognizing the USB
  Saving – logging is active and working
  Memory Is Full – USB storage is full; logging will stop
  Memory Too Slow – USB cannot keep up with the logging rate
  Unrecognized File System – USB is not formatted in a supported format (e.g., exFAT)

> **Note:**
> **Expected sequence when USB works correctly:**
> Disconnected → Accessing USB → Saving
> **If misconfigured (with exchanged DP and DM pins):**
> Disconnected → Accessing USB → Disconnected (stable, not changing again)

- **USB buffer usage** – includes **Grade (A–F)** indicating USB quality:
  A or B – recommended
  C or lower – may cause instability

3. **Use a reliable Power Source**

Unstable or fluctuating power can interrupt logging or corrupt log files. Use a stable power supply and ensure good electrical connections.

4. **Use Proper Shielding**

Proper shielding of all connections is crucial to prevent electromagnetic interference, which can disrupt the data logging process. Use shielded cables, particularly for the USB connection, and verify that the shielding is properly grounded. Poor shielding can lead to gaps or incomplete logs by allowing noise to interfere with data transmission. For more details, see chapter Wiring *(on page 253)*.

5. **Use a Suitable USB Drive**
   - Use short, compact USB sticks to avoid disconnections due to vibration.
   - Avoid low-quality or physically large drives.
   - Format the USB as FAT32. For more information, see chapter Preparing a USB drive *(on page 257)*.
   - Monitor the *Grade value* in the status bar to evaluate performance.

If issues persist after following these steps, refer to the ADU Manual for detailed diagnostics and hardware recommendations.

29.7. Document history

Version	Date	Changes
1.0	2025.07.28	Initial release

30. Document history

Version	Date	Changes
1.206	2020.04.09	Virtual fuel tank added Button defined track added Updated list of automatically detected race tracks Correction of "Signal (transmit) for RS232 serial bus"
92.2	2023.09.12	Keyboard configuration added Enumerations added Added specification of ADU 5 Rev.2 and ADU 7 Rev.2 CAD technical drawings added USBtoCAN error descriptions added Channel Simulator description added DBC Importer added Updated list of mathematical / logical operations Added description of Delayed Turning Off functionality Added description of built-in support for Ecumaster PMU Added description of default project "pg_page1.adu" Description of new controls added: Session Results, Grid The possibility of changing pages with a variable / rotary switch is described Added description for PULS Oil Level and Temperature Sensor Added description of the "Time Range" parameter Predictive time graph Added description of gradients for the Bargraph control Added channel description for Alarm objects Descriptions of visualisations related to data analysis have been expanded: Channel Report, Histogram, Lap Time Plot View, Scatter Plots, Section Times, Track Preview Added description of Memory Report window Added channel description for Ecumaster Brake Temperature Cameras (btc. channels) Added table of reserved CAN IDs
104.1	2024.08.23	New Ecumaster standard layout applied Added description of *Autosaved Channels* panel Updated description of *Timers*

Version	Date	Changes
		Updated description of *Virtual fuel tank* Updated description of *Buttons* Updated description of *Shift light* Added How-to documents as appendices: "How-to Configure the Hill Climb" "How-to Configure PMU CAN Stream" "How-to Merge Projects" "How-to Configure 5x3MT Encoders" "How-to Use Virtual Fuel Tank"
104.1.1	2024.09.19	Corrected description of *Freeze / unfreeze distance meter* and *Freeze / unfreeze distance meter2* parameters in the *Buttons* chapter
104.1.2	2024.09.26	Added note about 'GPS to CAN' module support in the *GPS module* and *Timing* chapters.
104.1.3	2024.10.03	Added 11 built-in tracks from Sweden to *Appendix A - List of built-in tracks*.
104.1.4	2024.10.08	Added information about CAN1 and CAN2 bus termination.
104.1.5	2025.02.05	Detailed information about CAN added to the *Characteristics* table
111.0	2025.04.29	Updated description of *Buttons* Updated description of *Permanent meters* Updated description of *CAN bus keypad support* Added description of *J1939* Added description of *Brake Bias* *Rate of change* operator described in *Numbers - mathematical channels* Added description of new PWM functionality in *Outputs* Added information about different acceleration sources Added 9 built-in tracks from Australia and the US to *Appendix A - List of built-in tracks*
120.2.2	2025.07.28	Added description of the new *Project Tree* view selector and available view modes Added How-to document as appendix: *"How-to Log Data to a USB drive in ADU"* Appendix E changed from *"How-to Configure 5x3MT Encoders"* to *"How-to Set Up Keyboards with Encoders"*

Version	Date	Changes
120.2.3	2025.11.21	Minor correction in pinout Added drawings for ADU 5 AS and ADU 7 AS Rev.2 Added corresponding information to the characteristics table
120.2.4	2026.03.30	Fixed incorrect internal pull-up resistor value